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STUDY MATERIAL

(CHEMISTRY)
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UNIT 1
SOME BASIC CONCEPTS OF CHEMISTRY

Chemistry: Chemistry is the branch of science that deals with the composition,structure and properties of matter. Chemistry is called the science of atoms and molecule

Branches of Chemistry

- Organic Chemistry -This branch deals with study of carbon compounds especially hydrocarbons and their derivatives.
- Inorganic Chemistry-This branch deals with the study of compounds of all other elements and compounds except carbon compounds. It largely concerns itself with the study of minerals found in the Earth's crust.
- Physical Chemistry-The explanation of fundamental principles governing various chemical phenomena is the main concern of this branch. It is basically concerned with laws and theories of the different branches of chemistry.
- Industrial Chemistry-The chemistry involved in industrial processes is studied under this branch.
- Analytical Chemistry-This branch deals with the qualitative and quantitative analysis of various substances.
- Biochemistry-This branch deals with the chemical changes going on in the bodies of living organisms; plants and animals.
- Nuclear Chemistry-Nuclear reactions, such as nuclear fission, nuclear fusion, transmutation processes etc. are studied under this branch.

PROPERTIES OF MATTER AND THEIR MEASUREMENT--Every substancehas unique or characteristic properties. These properties can be classified into two categories – **physical properties** and **chemical properties**.

Physical properties are those properties which can be measured or observed withoutchanging the identity or the composition of the substance. E.g. colour, odour, melting point, boiling point, density etc.

The measurement or observation of **chemical properties** requires a chemical change to occur. e.g. Burning of Mg-ribbon in air

Chemical properties are characteristic reactions of different substances; theseinclude acidity or basicity, combustibility etc.Many properties of matter such as length, area, volume, etc., are quantitative in nature.

Metric System was based on the decimal system.

The International System of Units (SI)

The International System of Units (in French Le Systeme International d'Unites– abbreviated as SI) was established by the 11th General Conference on Weights and Measures (CGPM from *ConferenceGenerale des Poids at Measures*). The SI system has seven *base units*

Quantity	Unit	Symbol
Length	metre	m
Mass	kilogram	kg
Time	second	S
Temperature	kelvin	K
Amount of substance	mole	mol
Electric current	ampere	A
Luminous intensity	candela	Cd

Prefixes in SI system

Multiple	Prefix	Symbol
10^{-12}	pico	p
10^{-9}	nano	n
10^{-6}	micro	μ
10^{-3}	milli	m
10^{-2}	centi	c
10^{-1}	deci	d
10	deca	da
10^{2}	hecto	h
10^{3}	kilo	k
10^{6}	mega	M
10^{9}	giga	G
10^{12}	tera	T

Mass and Weight-- Mass of a substance is the amount of matter present in it while **weight** is the force exerted by gravity on an object. The mass of a substance isconstant whereas its weight may vary from one place to another due to change in gravity. The mass of a substance can be determined very accurately by using an analytical balance

Volume-- Volume has the units of $(\text{length})^3$. So volume has units of m^3or cm^3ordm^3.A common unit, litre (L) is not an SI unit, is used for measurement of volume of liquids. 1 L = 1000 mL, 1000 cm^3 = 1 dm^3

Density: Density of a substance is its amount of mass per unit volume.SI unit ofdensity = SI unit of mass/SI unit of volume = kg/m^3 or kg m^{-3}This unit is quite large and a chemist often expresses density in g cm^{-3}.

Temperature--There are three common scales to measure temperature—°C (degreecelsius), °F (degree Fahrenheit) and K (kelvin). Here, K is the SI unit.

$$°F = \frac{9}{5}(°C) + 32$$

K = °C + 273.15

Note—Temperature below 0 °C (i.e. negative values) are possible in Celsius scale but in Kelvin scale, negative temperature is not possible.

Scientific Notation

In which any number can be represented in the form $N \times 10^n$ (Where n is an exponent having positive or negative values and N can vary between 1 to 10).

e.g. We can write 232.508 as 2.32508 $x10^2$ in scientific notation. Similarly, 0.00016 can be written as 1.6×10^{-4}.

Precision refers to the closeness of various measurements for the same quantity. **Accuracy** is the agreement of a particular value to the true value of the result

Significant Figures

The reliability of a measurement is indicated by the number of digits used to represent it. To express it more accuratelywe express it with digits that are known with certainty. These are called as Significant figures. They contain all thecertain digits plus one doubtful digit in a number.

Rules for Determining the Number of Significant Figures

- All non-zero digits are significant. For example, 6.9 has two significant figures, while 2.16 has three significantfigures. The decimal place does not determine the number of significant figures.
- A zero becomes significant in case it comes in between non zero numbers. For example, 2.003 has four significantfigures, 4.02 has three significant figures.
- Zeros at the beginning of a number are not significant. For example, 0.002 has one significant figure while 0.0045has two significant figures.
- All zeros placed to the right of a number are significant. For example, 16.0 has three significant figures, while 16.00has four significant figures. Zeros at the end of a number without decimal point are ambiguous.
- In exponential notations, the numerical portion represents the number of significant figures. For example, 0.00045 isexpressed as 4.5×10^{-4} in terms of scientific notations. The number of significant figures in this number is 2, while inAvogadro's number (6.023×10^{23}) it is four.
- The decimal point does not count towards the number of significant figures. For example, the number 345601 has sixsignificant figures but can be written in different ways, as 345.601 or 0.345601 or 3.45601 all having same number ofsignificant figures.

Retention of Significant Figures - Rounding off Figures

The rounding off procedure is applied to retain the required number of significant figures.

1. If the digit coming after the desired number of significant figures happens to be more than 5, the precedingsignificant figure is increased by one, 4.317 is rounded off to 4.32.
2. If the digit involved is less than 5, it is neglected and the preceding significant figure remains unchanged, 4.312 isrounded off to 4.31.

   If the digit happens to be 5, the last mentioned or preceding significant figure is increased by one only in case ithappens to be odd. In case of even figure, the preceding digit remains unchanged. 8.375 is rounded off to 8.38 while8.365 is rounded off to 8.36.

Dimensional Analysis During calculations generally there is a need to convert unitsfrom one system to other. This is called **factor label method** or **unit factor method** or **dimensional analysis**.

For example- 5 feet and 2 inches (height of an Indian female) is to converted in SI unit

1 inch = 2.54×10^{-2} m

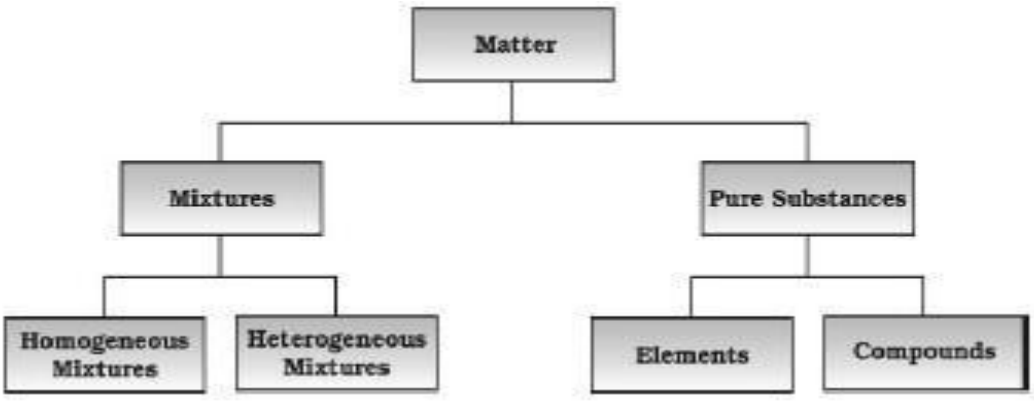
$1 = \frac{2.54 \times 10^{-2}\ m}{1\ inch}$ then, 5 feet and 2 inch = 62 inch

$= 62\ inch \times \frac{2.54 \times 10^{-2}\ m}{1\ inch} = 1.58\ m$

Physical Classification of Matter

Properties	**Solid**	**Liquid**	**Gas**
1. volume	Definite	Definite	Indefinite
2. Shape	Definite	Indefinite	Indefinite
3.Inter molecular force of attraction	Very high	Moderate	Negligible / Very low
4. arrangement of molecules	Orderly arranged	Free to move within the volume	Free to move every where
5.Inter molecular space	Very small	Slightly greater	Very great
7. Compressibility	Not compressible	Not compressible	Highly compressible
8. Expansion on heating	Very little	Very little	Highly expand
9. Rigidity	Very rigid	Not rigid knownas fluid	Not rigid and known as fluid
9. Fluidity	Can't flow	Can flow	Can flow
10. Diffusion	They can diffuse due to kinetic energy of liquid/gases	Can diffuse And rate of diffusion is very fast	Can diffuse And rate of diffusion is very fast

Chemical Classification of matter---



Elements

An element is the simplest form of matter that cannot be split into simpler substances or built from simpler substances by any ordinary chemical or physical method. There are 114 elements known to us, out of which 92 are naturally occurring while the rest have been prepared artificially.

Elements are further classified into metals, non-metals and metalloids.

Compounds

A compound is a pure substance made up of two or more elements combined in a definite proportion by mass, which could be split by suitable chemical methods.

Characteristics of compound

- Compounds always contain a definite proportion of the same elements by mass.
- The properties of compounds are totally different from the elements from which they are formed.
- Compounds are homogeneous.
- Compounds are broadly classified into inorganic and organic compounds. Inorganic compounds are those, which areobtained from non-living sources such as minerals. For example, common salt, marble and limestone. Organiccompounds are those, which occur in living sources such as plants and animals. They all contain carbon. Commonorganic compounds are oils, wax, fats etc.

Mixtures

A mixture is a combination of two or more elements or compounds in any proportion so that the components do not lose their identity. Air is an example of a mixture Mixtures are of two types, homogeneous and heterogeneous.

Homogeneous mixtures have the same composition throughout the sample. Thecomponents of such mixtures cannot be seen under a powerful microscope. They are also called solutions. Examples of homogeneous mixtures are air, seawater, gasoline, brass etc.

Heterogeneous mixtures consist of two or more parts (phases), which have differentcompositions. These mixtures have visible boundaries of separation between the different constituents and can be seen with the naked eye e.g., sand and salt, chalk powder in water etc.

LAWS OF CHEMICAL COMBINATIONS

Law of Conservation of Mass (Given by Antoine Lavoisier in 1789).Itstates that matter (mass) can neither be created nor destroyed**.**

Law of Definite Proportions or Law of Constant Composition:

This law was proposed by Louis Proust in 1799, which states that:

'A chemical compound always consists of the same elements combined together in the same ratio, irrespective of the method of preparation or the source from where it is taken'.

Law of Multiple Proportions Proposed by Dalton in 1803, this law states that:

'When two elements combine to form two or more compounds, then the different masses of one element, which combine with a fixed mass of the other, bear a simple ratio to one another'.

Gay Lussac's Law of Gaseous Volumes (Given by Gay Lussac in 1808.)According to this law when gases combine or are produced in a chemical reaction they do so in a simple ratio by volume provided all gases are at same temperature and pressuree.g.$H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$ 1V 1V 2V

All reactants and products have simple ratio 1:1:2.

Avogadro Law (In 1811, Given by Avogadro)

According to this law equal volumes of gases at the same temperature and pressure should contain equal number of molecules.

Dalton's Atomic Theory

All substances are made up of tiny, indivisible particles called atoms.

Atoms of the same element are identical in shape, size, mass and otherproperties.

Atoms of different elements are different in all respects.

Atom is the smallest unit that takes part in chemical combinations.

Atoms combine with each other in simple whole number ratios to form compound atoms called molecules.

Atoms cannot be created, divided or destroyed during any chemical or physical change.

Atoms and Molecules

The smallest particle of an element, which may or may not have independent existence, is called an atom, while the smallest particle of a substance which is capable of independent existence is called a molecule.

Molecules are classified as homoatomic and heteroatomic. Homoatomic molecules are made up of the atoms of the same element and heteroatomic molecules are made up of the atoms of the different element have different atomicity (number of atoms in a molecule of an element) like monoatomic, diatomic, triatomic and polyatomic.

Atomic Mass Unit

One atomic mass unit is defined as a mass exactly equal to one twelfth the mass of one carbon -12 atom. And 1 amu = 1.66056×10^{-24} g.

Today, 'amu' has been replaced by 'u' which is known as unified mass.

Atomic Mass

Atomic mass of an element is defined as the average relative mass of an atom of an element as compared to the mass of an atom of carbon -12 taken as 12.

$$\text{Atomic mass} = \frac{\text{mass of an atom}}{1/12 \text{ mass of a carbon atom}(^{12}C)}$$

Gram Atomic Mass

The quantity of an element whose mass in grams is numerically equal to its atomic mass. In simple terms, atomic mass of an element expressed in grams is the gram atomic mass or gram atom.

For example, the atomic mass of oxygen = 16 amu
Therefore gram atomic mass of oxygen = 16 g

Molecular Mass

Molecular mass of a substance is defined as the average relative mass of its molecule as compared to the mass of an atom of C-12 taken as 12. It expresses as to how many times the molecule of a substance is heavier than 1/12th of the mass of an atom of carbon.

For example, a molecule of carbon dioxide is 44 times heavier than 1/12th of the mass of an atom of carbon. Therefore the molecular mass of CO_2 is 44 amu.
It is obtained by adding the atomic masses of all the atoms present in one molecule.

Gram Molecular Mass

A quantity of substance whose mass in grams is numerically equal to its molecular mass is called gram molecular mass. In simple terms, molecular mass of a substance expressed in grams is called gram molecular mass.

e.g., the molecular mass of oxygen = 32 amu
Therefore, gram molecular mass of oxygen = 32 g

Formula Mass-

Sum of atomic masses of the elements present in one formula unit of a compound. It is used for the ionic compounds.

Mole Concept.

Mole is defined as the amount of a substance, which contains the same number of chemical units (atoms, molecules, ions or electrons) as there are atoms in exactly 12 grams of pure carbon-12.

A mole represents a collection of 6.022 $x10^{23}$ (Avogadro's number) chemical units..

The mass of one mole of a substance in grams is called its molarmass. Molar Volume

The volume occupied by one mole of any substance is called its molar volume. It is denoted by Vm. One mole of all gaseous substances at 273 K and 1 atm pressure occupies a volume equal to 22.4 litre or 22,400 mL. The unit of molar volume is litre per mol or millilitre per mol

PERCENTAGE COMPOSITION—

The mass percentage of each constituent element present in any compound is called its percentage composition

$$\text{Mass \% of the element} = \frac{Mass\ of\ element\ \ in\ one\ molecule}{Molar\ \ mass\ \ of\ \ the\ compound} \text{x } 100$$

Empirical Formula and Molecular Formula—

An **empirical formula** represents the simplest whole number ratio of various atoms present in a compound. e.g. CH is the empirical formula of benzene.

The **molecular formula** shows the exact number of different types of atoms present in a molecule of a compound. E.g. C_6H_6 is the molecular formula of benzene.

Relationship between empirical and molecular formulae

The two formulae are related as Molecular formula = $(\text{empirical formula})_n$

$$N = \frac{Molecular\ \ Mass}{Empirical\ \ formula\ \ Mass}$$

Chemical Equation-

Shorthand representation of a chemical change in terms ofsymbols and formulae of the substances involved in the reaction is called chemical equation.

The substances that react among themselves to bring about the chemical changes are known as reactants, whereas the substances that are produced as a result of the chemical change, are known as products.

Limiting Reagent- The reactant which gets consumed first or limits the amount ofproduct formed is known as **limiting reagent**

Reactions in Solutions-- The concentration of a solution can be expressed in any ofthe following ways.

1. **Mass Percent** is the mass of the solute in grams per 100 grams of the solution.

   $$\text{Mass \% of the solute} = \frac{\text{Mass of the solute}}{\text{Mass of the solution}} \times 100$$

   A 5 % solution of sodium chloride means that 5 g of NaCl is present in 100g of the solution.

2. **Volume percent** is the number of units of volume of the solute per 100 unitsof the volume of solution.

   $$\text{Volume \% of the solute} = \frac{\text{Volume of the solute}}{\text{Volume of the solution}} \times 100$$

   A 5 % (v/v) solution of ethyl alcohol contains 5 cm^3 of alcohol in 100 cm^3 of the solution

3. **Molarity** of the solution is defined as the number of moles of solute dissolvedper litre (dm^3) of the solution. It isdenoted by the symbol M. Measurements in Molarity can change with the change in temperature because solutionsexpand or contract accordingly.

   $$\text{Molarity of the solution} = \frac{\text{No. of moles of the solute}}{\text{Volume of the solution in litre}} = \frac{n}{V}$$

The Molarity of the solution can also be expressed in terms of mass and molar mass

$$\text{Molarity of the solution} = \frac{\text{Mass of the solute}}{\text{Molar mass of the solute X volume of the solution in liter}}$$

In terms of weight, molarity of the substance can be expressed as:

$$\text{Molarity} = \frac{W_g}{M \text{ g mol}^{-1} \times V \text{ litre}} = \frac{W}{M \times V} \text{ mol/L}$$

Molarity equation

To calculate the volume of a definite solution required to prepare solution of other molarity, the following equation is used:

$M_1V_1 = M_2V_2$, where M_1= initial molarity, M_2= molarity of the new solution, V_1= initial volume and V_2= volume of the new solution.

4. **Molality**- Molality is defined as the number of moles of solute dissolved per 1000 g (1 kg) of solvent. Molality is expressed as 'm'.

   $$\text{Molality} = \frac{\text{Moles of the solute}}{\text{Wt. of Solvent (in gm)}} \times 1000$$

5. **Mole Fraction** is the ratio of number of moles of one component to the total number of moles (solute and solvents) present in the solution. It is expressed as 'x'.

$$\text{Mole fraction of the solute} = \frac{\text{Moles of the solute}}{\text{Moles of solute} + \text{Moles of solvent}}$$

$$\text{Mole fraction of the solvent} = \frac{\text{Moles of the solvent}}{\text{Moles of solute} + \text{Moles of solvent}}$$

Mole fraction of the solute + Mole fraction of solvent = 1

One Mark questions with answers

1. What is the significant figures in 1.050×10^4?

Ans. Four

2. What is the S.I. unit of Density?

Ans. Kg m^{-3}

3. What do mean by Mole fraction?

Ans. Mole Fraction is the ratio of number of moles of one component to the total number of moles (solute and solvents) present in the solution. It is expressed as 'x'.

4. Round off up to 3 significant figure (a) 1.235 (b) 1.225

Ans. (a) 1.24 (b) 1.22

5. What is AZT?

Ans. Azidothymidine.

6. What is limiting reagent?

Ans.The reactant which gets consumed first or limits the amount ofproduct formed is known as **limiting reagent**

7. What is the relation between temperature in degree Celsius and degree fahrenheit?

Ans.

$$°F = \frac{9}{5}(°C) + 32$$

8. Define one mole?

Ans. One mole is the amount of a substance that contains as many particles as there are atoms in exactly 12 g of the carbon-12.

9. Calculate the formula mass calcium chloride.

Ans. Formula mass of $CaCl_2$= 40+2 x35.5=40+71 = 111 u

10. What is the law called which deals with the ratios of the volumes of the gaseous reactants and products?

Ans. Gay Lussac's law of gaseous volumes.

Two Marks questions with answers

1. Give the two points of differences between homogeneous and heterogeneous mixtures.

Ans.

Homogeneous mixture	Heterogeneous mixture
1. Homogeneous mixtures have the same composition throughout the sample.	1. Heterogeneous mixtures consist of two or more parts (phases), which have different compositions.
2. The components of such mixtures cannot be seen under a powerful microscope.	2. These mixtures have visible boundaries of separation between the different constituents and can be seen with the naked eye

2. Copper oxide obtained by heating copper carbonate or copper nitrate contains copper and oxygen in the same ration by mass. Which law is illustrated by this observation? State the law.

Ans.**Law of Definite Proportions** This law states that: A chemical compound always consists of the same elements combined together in the same ratio, irrespective of the method of preparation or the source from where it is taken.

3. Write the empirical formula of the following:

(a) N_2O_4 (b) $C_6H_{12}O_6$ (c) H_2O (d) H_2O_2

Ans. (a)NO_2 (b) CH_2O (c) H_2O (d) HO

4. Briefly explain the difference between precision and accuracy.

Ans. Precision refers to the closeness of various measurements for the same quantity. However, accuracy is the agreement of a particular value to the true value of the result.

5. Define the law of multiple proportions. Explain it with one example.
Ans.When two elements combine to form two or more compounds, then the different masses of one element, which combine with a fixed mass of the other, bear a simple ratio to one another. For example- carbon combines with oxygen to form two compounds CO and CO_2.

Compound	CO	CO_2
Mass of C	12	12
Mass of O	16	32

Masses of oxygen which combine with a fixed mass of carbon (12g) bear a simple ratio of 16:32 or 1:2.

6. Chlorine has two isotopes of atomic mass units 34.97 and 36.97. The relative abundance of the isotopes is 0.755 and 0.245 respectively. Find the average atomic mass of chlorine.

Ans. Average atomic mass = 34.97 x 0.755 +36.97 x 0.245 = 35.46 u

7. Calculate the percentage composition of water.

Ans. Mass % of an element = $\frac{\text{mass of that element in the compound} \times 100}{\text{molar mass of the Comd}}$

Molar mass of water = 18.02 g

Mass % of hydrogen = $\frac{2 \times 1.008 \times 100}{18.02}$

$= 11.18$

Mass % of oxygen = $\frac{16.00 \times 100}{18.02} = 88.79$

8. State the number of significant figures in each of the following:

(i) 208.91 (ii) 0.00456 (iii) 453 (iv) 0.346

Ans.

(i) 208.91 has five significant figures.

(ii) 0.00456 has three significant figures.

(iii) 453 has three significant figures.

(iv) 0.346 has three significant figures.

- Express the results of the following calculations to the appropriate number of significant figures.

(i) $\frac{3.24 \times 0.08666}{5.006}$ (ii) $\frac{(1.36 \times 10^{-4})(0.5)}{2.6}$

Ans.

(i) $\frac{3.24 \times 0.08666}{5.006}$ = 0.05608 = 0.0561

` (ii) $\frac{(1.36 \times 10^{-4})(0.5)}{2.6}$

$= 0.2615 \times 10^{-4} = 0.3 \times 10^{-4}$

9. How are 0.50 mol Na_2CO_3 and 0.50 M Na_2CO_3 different?

Ans. Molar mass of Na2CO3= 2 x 23 +12 + 3 x 16 = 106 g / mol

0.50 mol Na_2CO_3means 0.50 x 106 = 53 g

0.50 M Na_2CO_3 means 0.50 mol i.e. 53 g of Na_2CO_3 are present in I L of the solution.

Three Marks questions with answers-

- What is unit factor method? Express the following in SI units - 93 million miles (distance between earth and sun)

Ans. Method to convert units fromone system to other is called unit factor method.

93 million miles = 93×10^6 miles

1 mile = 1.60934 km = 1.60934×10^3 m

$1 = \frac{1.60934 \times 10^3 \text{ m}}{1 \text{ mile}}$

$93 \text{ million mile} = 93 \times 10^6 \text{ mile} \times \frac{1.60934 \times 10^3 \text{ m}}{1 \text{ mile}}$

$= 1.5 \times 10^{11}$ m

2. Write the three points of difference between compound and mixture. Ans.

Compound	Mixture
Constituents are always present in a fixed ratio by mass	Constituents may be present in any ratio
May or may not be homogeneous in nature	Always homogeneous in nature
Constituents can be easily separated	Constituents cannot be easily

by simple mechanical means	separated by simple mechanical means
Properties are midway between those of its constituents.	Properties are entirely different from those of its constituents.

3. What do mean bygram atomic mass.One million silver atoms weigh 1.79 x 10^{16} g. Calculate the gram atomic mass of silver.

Ans.atomic mass of an element expressed in grams is the gram atomic mass Number of silver atoms = 1 million = 1 x 10^{6}

Mass of one million Ag atoms = 1.79 x 10^{16} g

Mass of 6.023 x 10^{23} atoms of silver = $\frac{1.79 \times 10^{16} \text{g} \times 6.023 \times 10^{23}}{1 \times 10^{6}}$

= 107.8 g

- What is the percentage of carbon, hydrogen and oxygen in ethanol?

Ans. Molecular formula of ethanol is : C_2H_5OH

Molar mass of ethanol is : (212.01 + 61.008 + 16.00) g = 46.068 g Mass per cent of carbon = (24.02g / 46.068g) ×100 = 52.14%

Mass per cent of hydrogen = (6.048g / 46.068g)×100 = 13.13% Mass per cent of oxygen = (16.00 g / 46.068g)×100 = 34.73%

5. What do mean by molarity .Calculate the molarity of NaOH in the solution prepared by dissolving its 4 g in enough water to form 250 mL of the solution. Ans.The number of moles of solute dissolved per litre (dm^3) of the solution is called molarity

   Since molarity (M) = No. of moles of solute /Volume of solution in litres =(Mass of NaOH/Molar mass of NaOH)/0.250 L

   =(4 g / 40 g 0.1 mol)/0.250L =0.1 mol/0.250 L

   = 0.4 mol L^{-1}

   = 0.4 M

6. Classify the following as pure substances or mixture-

   (a) ethyl alcohol (b) oxygen (c) blood (d) carbon (e) steel (f) distilled water

   Ans. Pure substance- ethyl alcohol ,oxygen , carbon, distilled water
   Mixture- blood, steel

7. What are the rules for rounding off?

Ans.1. If the digit coming after the desired number of significant figures happens to be more than 5, the precedingsignificant figure is increased by one,

2. If the digit involved is less than 5, it is neglected and the preceding significant figure remains unchanged,

3. If the digit happens to be 5, the last mentioned or preceding significant figure is increased by one only in case it happens to be odd. In case of even figure, the preceding digit remains unchanged.

- 8.Define –(a) Average atomic mass (b) Molecular mass (c) Formula mass

  Ans. (a) Average atomic mass- Atomic mass of an element is defined as the average relative mass of an atom of an element as compared to the mass of an atom of carbon -12 taken as 12.

  (b) Molecular mass- it is sum of atomic masses of the elements present in a molecule.

  (c) Formula mass- it is sum of atomic masses of the elements present in a formula unit of a compound.

9. Express the following in the scientific notation with 2 significant figures-
   (a) 0.0048 (b) 234,000 (c) 200.0

   Ans. (a) 4.8×10^{-3} (b) 2.3×10^{5} (c) 2.0×10^{2}

10. Calculate the number of atoms in each of the following (i) 52 moles of Ar (ii) 52 u of He (iii) 52 g of He

    Ans. **(i)** 1 mole of Ar = 6.022×10^{23} atoms of Ar

    $\therefore$ 52 mol of Ar = $52 \times 6.022 \times 10^{23}$ atoms of Ar = 3.131×10^{25} atoms of Ar

    (ii)1 atom of He = 4 u of He Or,

    4 u of He = 1 atom of He

    1 u of He = 1/4 atom of He

    52u of He = 52/4 atom of He

    = 13 atoms of He

    (iii) Molar mass of He = 4 g/mol

    4 g of He contains = 6.022×10^{23} atoms of He

    52 g of He contains = $\frac{6.022 \times 10^{23} \times 52}{4} = 78.286 \times 10^{23}$ atoms of He

Five Marks questions with answers-

1. What is the difference between empirical and molecular formula? A compound contains 4.07 % hydrogen, 24.27 % carbon and 71.65 % chlorine. Its molar mass is98.96 g. What are its empirical and molecular formulas?

Ans. An empirical formula represents the simplest whole number ration of various atoms present in a compound whereas the molecular formula shows the exact number of different types of atoms present in a molecule of a compound.

Name of element	Percentage of elements	Step-1Conversion of mass per cent to grams.	Step 2. number moles of each element	Step 3. Divide the mole value by the smallest number
C	24.27%	24.27g	24.27/12 = **2.0225**	**2.0225/2.018 = 1**
H	4.07%	4.07g	4.07/1= **4.07**	**4.07/2.018 =2**
Cl	71.65%	71.65g	71.65/35.5 = **2.018**	**2.018/2.018 = 1**

The empirical formula of the above compound is CH_2Cl. empirical formula mass is 12 + (1x2) + 35.5 = 49.5

n= molecular mass/ empirical formula mass =98.96/49.5

= 2 Hence molecular formula is $C_2H_4Cl_2$

2. Dinitrogen and dihydrogen react with each other to produce ammonia according to the following chemical equation:

$$N_{2(g)} + H_{2(g)} \rightarrow 2NH_{3(g)}$$

(i) Calculate the mass of ammonia produced if 2.00×10^3 g dinitrogen reacts with 1.00×10^3 g of dihydrogen.

(ii) Will any of the two reactants remain unreacted?

(iii) If yes, which one and what would be its mass?

Ans. (i) Balancing the given chemical equation, $N_{2(g)} + 3H_{2(g)} \longrightarrow 2NH_{3(g)}$

From the equation, 1 mole (28 g) of dinitrogen reacts with 3 mole (6 g) of dihydrogen to give 2 mole (34 g) of ammonia.

$\Rightarrow 2.00 \times 10^3$g of dinitrogen will react with $\frac{6\text{ g}}{28\text{ g}} \times 2.00 \times 10^3\text{ g}$ dihydrogen i.e., 2.00×10^3 g of dinitrogen will react with 428.6 g of dihydrogen.

Given,

Amount of dihydrogen = 1.00×10^3 g Hence, N_2 is the limiting reagent.

$\therefore$ 28 g of N_2 produces 34 g of NH_3.

Hence, mass of ammonia produced by 2000 g of N_2 $= \frac{34\text{ g}}{28\text{ g}} \times 2000\text{ g}$

= 2428.57 g

(ii) N_2 is the limiting reagent and H_2 is the excess reagent. Hence, H_2 will remain unreacted.

(iii) Mass of dihydrogen left unreacted = 1.00×10^3 g – 428.6 g

= 571.4 g

3. A welding fuel gas contains carbon and hydrogen only. Burning a small sample of it in oxygen gives 3.38 g carbon dioxide, 0.690 g of water and no

other products. A volume of 10.0 L (measured at STP) of this welding gas is found to weigh 11.6 g. Calculate (i) empirical formula, (ii) molar mass of the gas, and (iii) molecular formula.

Ans. **(i)** 1 mole (44 g) of CO_2 contains 12 g of carbon.

3.38 g of CO_2 will contain carbon $= \frac{12\text{ g}}{44\text{ g}} \times 3.38\text{ g}$

= 0.9217 g

18 g of water contains 2 g of hydrogen.

0.690 g of water will contain hydrogen

$= \frac{2\text{ g}}{18\text{ g}} \times 0.690$ =

0.0767 g

Since carbon and hydrogen are the only constituents of the compound, the total mass of the compound is:= 0.9217 g + 0.0767 g= 0.9984 g

Percent of C in the compound $= \frac{0.9217\text{ g}}{0.9984\text{ g}} \times 100$ = 92.32%

Percent of H in the compound $= \frac{0.0767\text{ g}}{0.9984\text{ g}} \times 100$ = 7.68%

Moles of carbon in the compound $= \frac{92.32}{12.00}$ = 7.69

Moles of hydrogen in the compound = $\frac{7.68}{1}$ = 7.68

Ratio of carbon to hydrogen in the compound = 7.69: 7.68= 1: 1 Hence, the empirical formula of the gas is CH.

(ii) Given,

Weight of 10.0L of the gas (at S.T.P) = 11.6 g

Weight of 22.4 L of gas at STP $= \frac{11.6\text{ g}}{10.0\text{ L}} \times 22.4\text{ L}$

= 25.984 g≈ 26 g

Hence, the molar mass of the gas is 26 g.

(iii) Empirical formula mass of CH = 12 + 1 = 13 g

$$n = \frac{\text{Molar mass of gas}}{\text{Empirical formula mass of gas}}$$

$$= \frac{26\,\text{g}}{13\,\text{g}}$$

$n = 2$

$\therefore$ Molecular formula of gas = $(CH)_n = C_2H_2$

HOTS (Higher Order Thinking Skills)

1. What is the difference between 160 cm and 160.0 cm

Ans. 160 has three significant figures while 160.0 has four significant figures. Hence, 160.0 represents greater accuracy.

2. In the combustion of methane, what is the limiting reactant and why?

Ans. Methane is the limiting reactant because the other reactant is oxygen of the air which is always present in excess. Thus, the amounts of CO_2 and H_2O formed depend upon the amount of methane burnt.

3. A compound made up of two elements A and B has A= 70 %, B = 30 %. Their relative number of moles in the compound is 1.25 and 1.88. calculate
   a. Atomic masses of the elements A and B
   b. Molecular formula of the compound , if its molecular mass is found to be 160

Ans. Relative no. of moles of an element = $\frac{\%\text{ of the element}}{\text{Atomic mass}}$

Or atomic mass = $\frac{\%\text{ of the element}}{\text{Relative no. of moles}} = \frac{70}{1.25} = 56$

Atomic mass of B = 30/1.88 = 16

Calculation of Empirical formula

Element	Relative no. of moles	Simplest molar ratio	Simplestwhole no. molar ratio
A	1.25	1.25/1.25 = 1	2
B	1.88	1.88/1.25 = 1.5	3

Empirical formula = A_2B_3 Calculation of molecular formula-

Empirical formula mass = 2 x 56 + 3x 16 = 160

n= molecular mass / Empirical formula mass = 160/160 = 1 Molecular formula = A_2B_3

Value based questions

Q. 1. Chemistry plays a very important role in our everyday life. Beginning from morning when we use toothpaste to brush our teeth and throughout the day we use a large number of materials which have been provided to us by R & D (Research and development) carried out in the field of chemistry.

Now, answer the following questions :

(i) Why should we use iodized salt ?

(ii) Why should our toothpaste contain fluoride?

(iii) What are the relative advantages and disadvantages of using soaps and detergents?

(iv) Why earlier leaded petrol was used but now a days unleaded petrol is used?

Ans. (i) Iodine is needed to protect us from thyroid disorders.

(ii) Fluoride protects our gums and prevents tooth decay.

(iii) Soap is a good cleansing agent & is 100% biodegradable. As a result, it does not cause any pollution problem. However, soaps have two disadvantages. Firstly, they cannot be used in hard water. Secondly, they cannot be used in acidic solution. However, the detergents have one disadvantage that they are not completely biodegradable and hence cause water pollution.

(iv) Earlier, tetraethyl lead was added to petrol by the oil companies to act as anti-knocking agent. But on decomposition, it produces lead which is poisonous. Hence, its use has been stopped. Quality of petrol has been improved by other methods.

Q. 2. Chemistry has made a significant contribution in the medical field by providing a large number of drugs and medicines to fight our diseases. It is an endless effort by pharmaceutical companies to develop better and better medicines. The day is not far off when treatment of disease like cancer will be considered as common as bad cold.

Now, answer the following questions :

(i) Why should we avoid the use of drugs?

(ii) What for are analgesics used?

(iii) What for are antibodies used?

(iv) What for are anaesthetics used?

(v) What for are tranquilizers used?

(vi) What for Dettol and savlon are used?

(vii) What for do we use Gammexane?

(viii) Which compunds are generally present in the life saving drugs used for curinjg cancer and AIDS?

(ix) What is the role of vitamins in our life?

(x) Which vitamin is produced in our body when we sit in the sun or which vitamin helps in the absorption of calcium in our body?

Ans. (i) A medicine is a chemical substance which cures the disease, is safe to use, has negligible toxicity and does not cause addiction. In contrast, a drug is a chemical substance which also cures the disease but is habit forming, causes addiction and has serious side effects. Hence, its use should be avoided.

(ii) Analgesics are medicines used to get relief from pain.

(iii) Antibiotics are medicines used to curb infection.

(iv) Anaesthetics are medicines given during surgery (operation) to stop you feeling pain in the whole of your body when you are unconscious or in a part of your body when you are awake.

(v) Tranquilizers are drugs used for the treatment of mental stress/disease.

(vi) Dettol and savlon are antiseptics used to stop infection of wounds.

(vii) Gammexane is an insecticide used to kill rats, mosquitos and flies.

(viii) cis-platin and taxol are used for cancer and AZT (Azido- thymidine) is used for AIDS.

(ix) Vitamins are chemical substances, the deficiency occurs, they have to be taken in the form of pills or capsules.

(x) Vitamin D.

Q. 3. Chemistry has played a vital role in the field of agriculture. It has provided us with chemical fertilizers which have helped to increase the yield of fruits, vegetables & other crops. It has given us insecticide and pesticides which protect the plants from insects and pests. It has given us preservation to preserve food products like jam, butter etc. for a longer period.

Now, answer the following questions :

(i) How is the use of chemical fertilizers harming the aquatic life and mankind? what alternative do you suggest?

(ii) How is the use of pesticides and insecticides harming the mankind? What alternative do you suggest?

(iii) Huge amounts of synthetic milk and milk products are flowing into the market. Suggest what should be done to check it?

Ans. (i) During rainfall, nitrogenous fertilizers wash away into rivers, lakes & oceans. It help the algae to bloom. When short live algae die, decomposing bacteria consume dissolved oxygen. As a result, aquatic life begins to perish. Secondly, artificial fertilizers deplete the soil of organic matter. As a result, it loses ability to hold water and is more subject to erosion. Moreover, these fertilizers tavel from soil to plant and then from plant to human being causing health hazards. The solution to the problem is that natural fertilizers like cow-dung should be encouraged, i.e., production of organic food should be increased.

(ii) Pesticides have neurological effects on humans such as headache and hand tremors. A number of them are responsible for cancer. The solution to the problem is that their use should be minimized and such chemicals should be produced which kill pests but have no effect on humans.

(iii) The supply of milk and milk products should not flow direct to the consumer through the vendor. It should be regulated through a centre (set up by the Govt. or licenced by the Govt. to private agency) where all milk and milk products are received and tested before they are sold to consumer.

Q. 4. India is a fast developing nation when a very large number of chemical industries are coming evry year. These include industries like those of glass, paper, textile, plastics, leather, dye paints, sugar, pharmaceuticals etc. The rat- race for earning more profits and also due to

competition of rates in the market, some industries are producing poor quality products and chemicals which are harming the society at large.

Now, answer the following questions:

(i) Production of spurious drugs is harming the mankind. What steps should be taken to check it.

(ii) How do paints sometimes prove to be harmful?

(iii) Why should the production of polythene bags be banned?

(iv) Name a few industrial products which are recyclable. What should we do with their products?

(v) What are the relative advantage and disadvantages of using synthetics fibres versus natural fibres (cotton, silk, wool etc.)?

Ans. (i) Strict quality control law should be enforced. Not only such factories should be closed, people running such factories should be severely punished. These factories should be black-listed and photographs of owners should be published in all leading newspapers.

(ii) Many paints contain lead which is poisonous.

(iii) Use of polythene bags should be banned because it is not biodegradable. It may, therefore, pile up as waste and swallowed by animals causing death or choke the sewer.

(iv) Advantages of synthetic fibres.

(a) Strong (b) Easy to wash and wear (need no ironing)

(c) can be stretched easily to different sizes like nylon socks

(d) Available in variety of colours

(e) Generally cheaper than natural fibres

(f) do not shrink.

Disadvantages of synthetic fibres.

(a) does not absorb moisture/sweat keeping the heat trapped in the body.

(b) Give uncomfortable feel for being used as underwears

(c) Catch fire or melt more easily than natural fibres.

Advantages of natural fibres.

(a) Clothes made from natural fibres are more comfortable because they can absorb moisture/sweat.

(b) As they are natural materials, they are harmless to environment.

(c) They are fire-resistant.

Disadvantages of natural fibres,

(a) More expensive

(b) Need ironing as they are not crease-resistant.

(c) Might shrink on aggressive washing.

CHAPTER 2

STRUCTURE OF ATOM

• Atom is the smallest indivisible particle of the matter. Atom is made ofelectron,proton and neutrons.

PARTICLE	ELECTRON	PROTON	NEUTRON
Discovery	Sir. J. J. Thomson (1869)	Goldstein (1886)	Chadwick (1932)
Nature of charge	Negative	Positive	Neutral
Amount of charge	1.6×10^{-19}Coloumb	1.6×10^{-19}Coloumb	0
Mass	9.11×10^{-31}kg	1.672614×10^{-27}kg	1.67492×10^{-27}kg

- Electrons were discovered using cathode ray discharge tube experiment. Nucleus was discovered by Rutherford in 1911.
- Cathode ray discharge tube experiment: A cathode ray discharge tube madeof glass is taken with two electrodes. At very low pressure and high voltage,current starts flowing through a stream of particles moving in the tube fromcathode to anode. These rays were called cathode rays. When a perforatedanode was taken, the cathode rays struck the other end of the glass tube atthe fluorescent coating and a bright spot on the coating was developed
- **Results:**
- Cathode rays consist of negatively charged electrons.
- Cathode rays themselves are not visible but their behavior can be observed with help of fluorescent or phosphorescent materials.
- In absence of electrical or magnetic field cathode rays travel in straight lines
- In presence of electrical or magnetic field, behaviour of cathode rays is similar to that shown by electrons
- The characteristics of the cathode rays do not depend upon the material of the electrodes and the nature of the gas present in the cathode ray tube.
- Charge to mass ratio of an electron was determined by Thomson. The chargeto mass ratio of an electron as 1.758820×10^{11} C kg^{-1}
- Charge on an electron was determined by R A Millikan by using an oildropexperiment. The value of the charge on an electron is -1.6×10^{-19}C.
- The mass on an electron was determined by combining the results ofThomson's experiment and Millikan's oil drop experiment. The mass of anelectron was determined to be 9.1094×10^{-31}kg.
- **Discovery of protons and canal rays:** Modified cathode ray

tubeexperimentwas carried out which led to the discovery of protons. Characteristics of positively charged particles:

- Charge to mass ratio of particles depends on gas from which these originate
- The positively charged particles depend upon the nature of gas present in the cathode ray discharge tube
- Some of the positively charged particles carry a multiple of fundamental of electrical charge.
- Behaviour of positively charged particles in electrical or magnetic field is opposite to that observed for cathode rays
- **Neutrons** were discovered by James Chadwick by bombarding a thin sheet ofberyllium by α- particles. They are electrically neutral particles having a mass slightly greater than that of the protons.
- Atomic number (Z): the number of protons present in thenucleus (Moseley1913).
- **Mass Number (A) :**Sum of the number of protons and neutrons present inthenucleus.
- **Thomson model of an atom:**This model proposed that atom is considered asauniform positively charged sphere and electrons are embedded in it.An important feature of Thomson model of an atom was that mass of atom isconsidered to be evenly spread over the atom.Thomson model of atom is also called as Plum pudding, raisin pudding orwatermelon modelThomson model of atom was discarded because it could not explain certainexperimental results like the scattering of α- particles by thin metal foils.
- **Observations from α- particles scattering experiment by Rutherford**:
- Most of the α- particles passed through gold foil undeflected
- A small fraction of α- particles got deflected through small angles
- Very few α- particles did not pass through foil but suffered large deflection nearly180^{o}they were bounced back to the same medium.
- **Conclusions Rutherford drew from α- particles scattering experiment:**
- Since most of the α-particles passed through foil undeflected, it means most of the space in atom is empty
- Since some of the α-particles are deflected to certain angles, it means that there is positively mass present in atom
- Since only some of the α-particles suffered large deflections, the positively charged mass must be occupying very small space
- Strong deflections or even bouncing back of α-particles from metal foil were due to direct collision with positively charged mass in atom
- **Rutherford's model of atom:**This model explained that atom consistsofnucleus which is concentrated in a very small volume. The nucleus comprisesof protons and neutrons. The electrons revolve around the nucleus in fixedorbits. Electrons and nucleus are held together by electrostatic forces ofattraction.
- **Drawbacks of Rutherford's model of atom**:
- According to Rutherford's model of atom, electrons which are negatively

charged particles revolve around the nucleus in fixed orbits. Thus, theelectrons undergo acceleration. According to electromagnetic theory of Maxwell, a charged particle undergoing acceleration should emitelectromagnetic radiation. Thus, an electron in an orbit should emitradiation. Thus, the orbit should shrink. But this does not happen.

- The model does not give any information about how electrons aredistributed around nucleus and what are energies of these electrons
- **Isotopes:** These are the atoms of the same element having the sameatomicnumber but different mass number.e g H^1,$_1H^2$,$_1H^3$
- **Isobars:** Isobars are the atoms of different elements having the samemassnumber but different atomic number.e g $_{18}Ar^{40}$,$_{20}Ca^{40}$
- **Isoelectronic species**: These are those species which have the same numberofelectrons.
- **Electromagnetic radiations**: The radiations which are associatedwithelectrical and magnetic fields are called electromagnetic radiations. When anelectrically charged particle moves under acceleration, alternating electricaland magnetic fields are produced and transmitted. These fields aretransmitted in the form of waves. These waves are called electromagneticwaves or electromagnetic radiations.
- **Properties of electromagnetic radiations:**
- Oscillating electric and magnetic field are produced by oscillating charged particles. These fields are perpendicular to each other and both areperpendicular to the direction of propagation of the wave.
- They do not need a medium to travel. That means they can even travel in vacuum.
- **Characteristics of electromagnetic radiations**:
- **Wavelength:** It may be defined as the distance between twoneighbouring crests or troughs of wave as shown. It is denoted by λ.
- **Frequency (ν):** It may be defined as the number of waves whichpassthrough a particular point in one second.
- **Velocity (v):** It is defined as the distance travelled by a wave inonesecond. In vacuum all types of electromagnetic radiations travel with thesame velocity. Its value is $3 X10^8$m $sec^{-1.}$ It is denoted by v
- Wave number: Wave number $(\bar{\nu})$ is defined as the number of wavelengths per unit length.
- Velocity = frequency x wavelength $c = \nu\lambda$
- **Planck's Quantum Theory-**
  - The radiant energy is emitted or absorbed not continuously but discontinuously in the form of small discrete packets of energy called 'quantum'. In case of light , the quantum of energy is called a 'photon'
  - The energy of each quantum is directly proportional to the frequency of the radiation, i.e. $E \alpha \upsilon$ or $E= h\upsilon$ where h= Planck's constant = 6.626×10^{-27} Js
  - Energy is always emitted or absorbed as integral multiple of this quantum. $E=nh\upsilon$ Where n=1,2,3,4,.....

- **Black body**: An ideal body, which emits and absorbs all frequencies, is calledablack body. The radiation emitted by such a body is called black bodyradiation.
- **Photoelectric effect**: The phenomenon of ejection of electrons from thesurfaceof metal when light of suitable frequency strikes it is calledphotoelectric effect. The ejected electrons are called photoelectrons.
- Experimental results observed for the experiment of Photoelectric effect-
- When beam of light falls on a metal surface electrons are ejectedimmediately.
- Number of electrons ejected is proportional to intensity or brightness of light
- Threshold frequency (v_o): For each metal there is a characteristicminimum frequency below which photoelectric effect is not observed. Thisis called threshold frequency.
- If frequency of light is less than the threshold frequency there is noejection of electrons no matter how long it falls on surface or how high isits intensity.
- Photoelectric work function (W_o): The minimum energy required to ejectelectrons is called photoelectric work function.$W_o = hv_o$ Energy of the ejected electrons :

$$h(\nu - \nu_0) = \frac{1}{2} m_e v^2$$

- Dual behavior of electromagnetic radiation- The light possesses both particle and wave like properties, i.e., light has dual behavior, whenever radiation interacts with matter, it displays particle like properties.(Black body radiation and photoelectric effect) Wave like properties are exhibited when it propagates(interference an diffraction)
- When a white light is passed through a prism, it splits into a series ofcoloured bands known as spectrum.
- Spectrum is of two types: continuous and line spectrum
- The spectrum which consists of all the wavelengths is called continuous spectrum.
- A spectrum in which only specific wavelengths are present is known as a line spectrum. It has bright lines with dark spaces between them.
- Electromagnetic spectrum is a continuous spectrum. It consists of a range ofelectromagnetic radiations arranged in the order of increasing wavelengths ordecreasing frequencies. It extends from radio waves to gamma rays. Spectrum is also classified as emission and line spectrum.
- Emission spectrum: The spectrum of radiationemitted by a substancethat has absorbed energy is called an emissionspectrum.
- Absorption spectrum is the spectrum obtained when radiation is passedthrough a sample of material. The sample absorbs radiation of certainwavelengths. The wavelengths which are absorbed are missing and comeas dark lines.

The study of emission or absorption spectra is referred asspectroscopy.

Spectral Lines for atomic hydrogen:

Series	n_1	n_2	Spectral Region
Lyman	1	2, 3, 4, 5 ...	Ultraviolet
Balmer	2	3, 4, 5 ...	Visible
Paschen	3	4, 5 ...	Infrared
Brackett	4	5, 6 ...	Infrared
Pfund	5	6, 7...	Infrared

Rydberg equation

$$\bar{\nu} = 109{,}677 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{ cm}^{-1}$$

R = Rydberg's constant = 109677 cm^{-1}

Bohr's model for hydrogen atom:

4. An electron in the hydrogen atom can move around the nucleus in a circular path of fixed radius and energy. These paths are called orbits orenergy levels. These orbits are arranged concentrically around thenucleus.
5. As long as an electron remains in a particular orbit, it does not lose or gain energy and its energy remains constant.
6. When transition occurs between two stationary states that differ inenergy, the frequency of the radiation absorbed or emitted can becalculated

$$\nu = \frac{\Delta E}{h} = \frac{E_2 - E_1}{h}$$

7. An electron can move only in those orbits for which its angularmomentum is an integral multiple of h/2π

$$m_e \, v r = n . \frac{h}{2\pi} \qquad n = 1,2,3.....$$

The radius of the *n*th orbit is given by $r_n = 52.9\text{pm} \times \frac{n^2}{Z}$

energy of electron in *n*th orbit is :

$$E_n = -2.18 \times 10^{-18} \left(\frac{Z^2}{n^2} \right) J$$

Limitations of Bohr's model of atom:

a. Bohr's model failed to account for the finer details of the hydrogen spectrum.
b. Bohr's model was also unable to explain spectrum of atoms containing more than one electron.

Dual behavior of matter: de Broglie proposed that matter exhibitsdualbehavior i.e. matter shows both particle and wave nature. de Broglie's relation is $\lambda = \frac{h}{mv} = \frac{h}{p}$

Heisenberg's uncertainty principle: It states that it is impossible todetermine simultaneously, the exact position and exact momentum (or velocity) of an electron.The product of their uncertainties is always equal to or greater than h/4π.

Mathematically $\Delta x \times \Delta p \geq \frac{h}{4\pi}$

where Δx = uncertainty in position,

Δp = uncertainty in momentum

Heisenberg's uncertainty principle rules out the existence of definite pathsor trajectories of electrons and other similar particles

Failure of Bohr's model:

a. It ignores the dual behavior of matter.

b. It contradicts Heisenberg's uncertainty principle.

Classical mechanics is based on Newton's laws of motion. It successfullydescribes the motion of macroscopic particles but fails in the case ofmicroscopic particles.

Reason: Classical mechanics ignores the concept of dual behaviour of matter especially for sub-atomic particles and the Heisenberg's uncertainty principle.

Quantum mechanics is a theoretical science that deals with the study ofthemotions of the microscopic objects that have both observable wavelike andparticle like properties.

Quantum mechanics is based on a fundamental equation which is calledSchrodinger equation.

Schrodinger's equation: For a system (such as an atom or a molecule whoseenergy does not change with time) the Schrödinger equation is written as:

$$\hat{H}\Psi = E\Psi$$

$\hat{H}$ is the Hamiltonian operator

E is the total energy of the system

Ψ represents the wave function which is the amplitude of the electron Wave

When Schrödinger equation is solved for hydrogen atom, the solution givesthe possible energy levels the electron can occupy and the correspondingwave function(s) of the electron associated with each energy level.Out of the possible values, only certain solutions are permitted. Eachpermitted solution is highly significant as it corresponds to a definite energy state. Thus, we can say that energy is quantized.

ψ gives us the amplitude of wave. The value of ψhas no physicalsignificance.

Ψ^2gives us the region in which the probability of finding an electronismaximum. It is called probability density.

Orbital: The region of space around the nucleus where the probability offinding an electron is maximum is called an orbital.

Quantum numbers: There are a set of four quantum numbers which specifythe energy, size, shape and orientation of an orbital. To specify an orbital only

three quantum numbers are required while to specify an electron all four quantum numbers are required.

Principal quantum number (n): It identifies shell, determines sizes and energy of orbitals

N	1	2	3	4
Shell no.:	K	L	M	N
Total number of orbitals in a shell = n^2	1	4	9	16
Maximum number of electrons = $2n^2$	2	8	18	32

Azimuthal quantum number (l): Azimuthal quantum number. 'l' is alsoknown as orbital angular momentum or subsidiary quantum number. l. It identifies sub-shell, determines the shape of orbitals, energy of orbitals in multi-electron atoms along with principal quantum number and orbital angularmomentum, *i.e.*, $\sqrt{l(l+1)}\frac{h}{2\pi}$. The number of orbitals in a subshell = $2l + 1$. For a given value of *n*, it can have *n* values ranging from 0 to n-1. Total number of subshells in a particular shell is equal to the value of n.

Subshell notation	s	p	d	f	g
Value of '*l*'	0	1	2	3	4
Number of orbitals	1	3	5	7	9

Magnetic quantum number or Magnetic orbital quantum number (m_l): Itgives information about the spatial orientation of the orbital with respect tostandard set of co-ordinate axis.For any sub-shell (defined by 'l' value) 2l+1 values of ml are possible.For each value of l, $m_l = -l, -(l-1), -(l-2) \ldots 0, 1 \ldots (l-2), (l-1), l$

Electron spin quantum number (m_s): It refers to orientation of the spin oftheelectron. It can have two values $+^1/_2$ and $-^1/_2$. If $+^1/_2$ identifies the clockwisespin and $-^1/_2$ identifies the anti- clockwise spin.

The region where this probability density function reduces to zero iscallednodal surfaces or simply nodes.

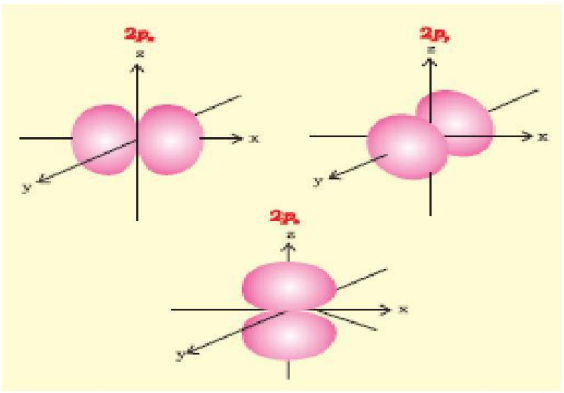
Radial nodes: Radial nodes occur when the probability density of wavefunctionfor the electron is zero on a spherical surface of a particular radius. Numberof radial nodes = n – l – 1

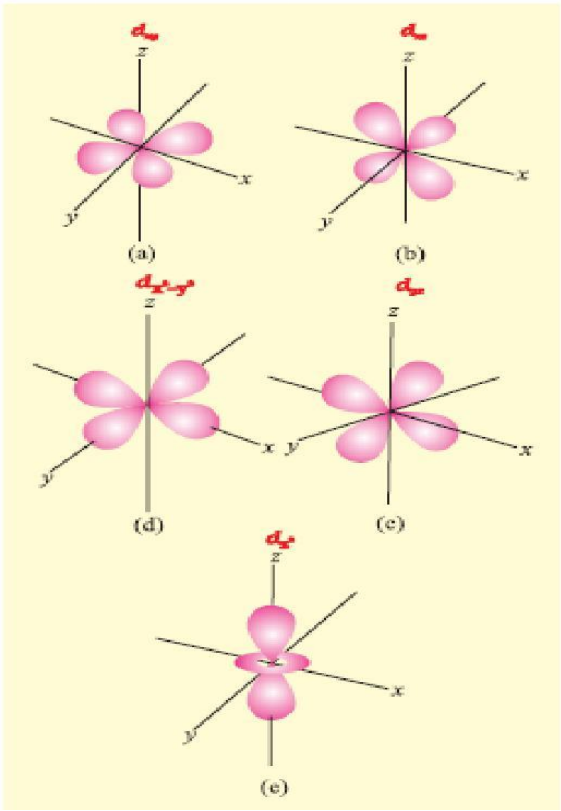
Angular nodes: Angular nodes occur when the probability density wavefunction for the electron is zero along the directions specified by a particularangle. Number of angular nodes = l

Total number of nodes = n – 1

Degenerate orbitals: Orbitals having the same energy are calleddegenerateorbitals.

Shape of p and d-orbitals





Shielding effect or screening effect: Due to the presence of electrons intheinner shells, the electron in the outer shell will not experience the full positivecharge on the nucleus.

So, due to the screening effect, the net positive charge experienced by theelectron from the nucleus is lowered and is known as effective nuclearcharge. Effective nuclear charge experienced by the orbital decreases with increase of azimuthal quantum number (l).

Aufbau Principle: In the ground state of the atoms, the orbitals are filledinorder of their increasing energies

(n+l) rule-Orbitals with lower value of (n+l) have lower energy. If two orbitals have the same value of (n+l) then orbital with lower value of nwill have lower energy.

The order in which the orbitals are filled isas follows: 1s,2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, 5p, 4f, 5d, 6p, 7s...

Pauli Exclusion Principle: No two electrons in an atom can have the samesetof four quantum numbers. Only two electrons may exist in the same orbitaland these electrons must have opposite spin.

Hund's rule of maximum multiplicity: Pairing of electrons in theorbitalsbelonging to the same subshell (p, d or f) does not take place until eachorbital belonging to that subshell has got one electron each i.e., it is singlyoccupied.

Electronic configuration of atoms: Arrangement of electrons in differentorbitals of an atom. The electronic configuration of differentatoms can be represented in two ways.

- $s^a p^b d^c$..... notation.
- Orbital diagram:, each orbital of the subshell is represented by a box and the electron is represented by an arrow (↑) a positive spin or an arrow (↓) a negative spin.

Stability of completely filled and half filled subshells:

- Symmetrical distribution of electrons- the completely filled or half filled sub-shells have symmetrical distribution of electrons in them and are more stable.
- Exchange energy-The two or more electrons with the same spin present in the degenerate orbitals of a sub-shell can exchange their position and the energy released due to this exchange is called exchange energy. The number of exchanges is maximum when the subshell is either half filled or completely filled. As a result the exchange energy is maximum and so is the stability.

ONE MARK QUESTIONS

- Neutrons can be found in all atomic nuclei except in one case. Which is this atomic nucleus and what does it consists of?

Ans. Hydrogen atom. It consists of only one proton.

2. Calculate wave number of yellow radiations having wavelength of 5800 A^0.

Ans. Wave number = 1/ wavelength

Wavelength = 5800 A^0 = 5800 x 10^{-10} m

Wave number = 1/5800 x 10^{-10} m = 1.72 x 10^6 m^{-1}

- What are the values of n and l for 2p orbital? Ans. n=2 and l= 1
- Which of the following orbitals are not possible? 1p, 2s, 3f and 4d

Ans. 1p and 3f are not possible.

- Write the electronic configuration of the element having atomic number 24. Ans. $1s^2$ $2s^2$ $2p^6$ $3s^2$ $3p^6$ $3d^5$ $4s^1$
- What atoms are indicated by the following electronic configurations?

a. $1s^2\ 2s^2\ 2p^1$ b. $[Ar]4s^2\ 3d^1$

Ans. a. Boron b. Scandium

7. What is the relationship between frequency and wavelength of light?

Ans. velocity of light = frequency x wavelength. Frequency and wavelength are inversely proportional to each other.

8. State Pauli Exclusion Principle.

Ans. No two electrons in an atom can have the same set of four quantum numbers or an orbital can have maximum two electrons and these must have opposite spin.

9. When α- rays hit a thin foil of gold, very few α- particles is deflected back. What does it prove?

Ans. There is a very small heavy body present within the atom.

10. What is the difference between a quantum and a photon?

Ans. The smallest packet of energy of any radiation is called a quantum whereas that of light is called photon.

TWO MARKS QUESTIONS

1. Write the complete symbol for the atom with the given atomic number (Z) and mass number(A). (a) Z = 17, A = 35 (b) Z = 92 , A = 233

Ans. (a) $^{35}_{17}Cl$ (b) $^{233}_{92}U$

- Using s,p,d and f notation, describe the orbital with the following quantum numbers-
  (a) n=1,l=0 (b) n=3, l=1 (c) n=4, l=2 (d) n=4, l=3

Ans. (a) 1s (b) 3p (c)4d (d) 4f

3. How many electrons in an atom have the following quantum numbers?

a. n=4, m_s= -1/2 b. n =3 , l=o

Ans. (a) 16 electrons (b) 2 electrons.

4. An element with mass number 81 contains 31.7 % more neutrons as compared to protons. Assign the atomic symbol.

Ans. Mass number = 81, i.e., p + n = 81

If protons = x, then neutrons = $x + \frac{31.7}{100} X\ x = 1.317\ x$

$x+1.317x = 81$ or $2.317x = 81$ $x=35$

Thus proton = 35, i.e., atomic no. = 35 Hence symbol is $^{81}_{35}Br$

2. (i) The energy associated with the first orbit in the hydrogen atom is -2.18×10^{-18} J/atom. What is the energy associated with the fifth orbit
   (ii) Calculate the radius of Bohr's fifth orbit for hydrogen atom.

Ans. (i) $E_n = -2.18 \times 10^{-18} / n^2$ $E_5 = -2.18 \times 10^{-18} / 5^2 = -8.72 \times 10^{-20}$ J

(ii) For H atom, $r_n = 0.529 \times n^2$ $r_5 = 0.529 \times 5^2 = 13.225\ A^0 = 1.3225$ nm

6. Explain , giving reasons, which of the following sets of quantum numbers are not possible.

(a) n=0, l=0; m_l = 0, m_s= + ½ (c)n=1, l=0; m_l = 0, m_s= - ½

(b) n=1, l=1; m_l =- 0, m_s= + ½ (d) n=2, l=1; m_l = 0, m_s= + ½

Ans. (a) Not possible because $n \neq 0$ (b) Not possible because when n=1, $l \neq 1$

(c) possible (d) Possible

3. (a)What is the lowest value of n that allows g orbitals to exist?

(b)An electron is in one of the 3d orbitals, Give the possible values of n,l and $m_l$for this electron.

Ans.(a) minimum value of n= 5

(b)n=3, l=2, m_l = -2, -1, 0, +1, +2

8. Calculate the total number of angular nodes and radial nodes present in 3p orbitals.

Ans. For 3p orbitals, n=3, l= 1

Number of angular nodes = l= 1

Number of radial nodes = n-l-1 = 3-1-1= 1

9.Mention the draw backs of Rutherford's atomic model.

Ans. 1. It could not explain the stability of an atom.

= It could not explain the line spectrum of H- atom.

10.State de-Broglie concept of dual nature of matter. How do dual nature of electron verified?

Ans. Just as light has dual nature, every material particle in motion has dual nature (particle nature and wave nature). The wave nature has been verified by Davisson and Germer'sexperiment whereas particle nature by scintillation experiment.

THREE MARKS QUESTIONS

1. State (a)Hund's Rule of maximum Multiplicity (b) Aufbau Principle (c) n+l rule

Ans.(a) Pairing of electrons in the orbitals belonging to the same subshell (p, d or f) does not takeplace until each orbital belonging to that subshell has got one electron each i.e., it is singly occupied.

(b)In the ground state of the atoms, the orbitals are filled inorder of their increasing energies

(c) Orbitals with lower value of (n+l) have lower energy.If two orbitals have the same value of (n+l) then orbital with lower value of n will have lower energy.

2. Write down the quantum numbers n and l for the following orbitals

a. 2p b. 3d c. 5f

Ans. a. n=2, l= 1 b. n= 3, l=2 c. n= 5, l=3

3. Write the 3 points of difference between orbit and orbital.

Ans.

Orbit	Orbital
1. An orbit is a well defined circular path around the nucleus in which the electrons revolve	1. An orbital is the three dimensional space around the nucleus within which the probability of finding an electron is maximum(upto 90 %)
2. It represents the planar motion of an electron around the nucleus	2. It represents the three dimensional motion of an electron around the Nucleus
3. All orbits are circular and disc like	3. Different orbitals have different shapes, i.e., s-orbitals are spherically symmetrical, p-orbitals are dumb-bell shaped and so on.

4. State Heisenberg's uncertainty principle. Calculate the uncertainty in the position of an electron if the uncertainty in its velocity is 5.7×10^5 m/s.

Ans. It states that it is impossible to determine simultaneously, the exact position and exact momentum (or velocity) of an electron. The product of their uncertainties is always equal to or greater than $h/4\pi$.

$$\Delta x \times (m \times \Delta v) = h/4\pi$$

$$\Delta x = h/4\pi \times m \times \Delta v = \frac{6.6 \times 10^{-34}}{4 \times 3.14 \times 9.1 \times 10^{-31} \times 5.7 \times 10^5} = 1.0 \times 10^{-10} \text{ m}$$

5. Write 3 points of differences between electromagnetic waves and matterwaves.

Electromagnetic waves	Matter waves
1. These are associated with electric and magnetic fields	1. These are not associated with electric and magnetic field.
2. They do not require any medium for propagation.	2. They require medium for Propagation
3. They travel with the same speed a that of light	3. They travel with lower speeds not constanr for all matter Waves

6. (i) Calculate the number of electrons which will together weigh one gram. (ii) Calculate the mass and charge of one mole of electrons

Ans. (i) Mass of one electron = 9.10939×10^{-31} kg

$\therefore$ Number of electrons that weigh 9.10939×10^{-31} kg = 1

Number of electrons that will weigh 1 g = $(1 \times 10^{-3}$kg)

$$= \frac{1}{9.10939 \times 10^{-31} \text{ kg}} \times \left(1 \times 10^{-3} \text{ kg}\right)$$

$= 0.1098 \times 10^{-3 + 31}$

$= 0.1098 \times 10^{28}$

$= 1.098 \times 10^{27}$

(ii) Mass of one electron = 9.10939×10^{-31} kg

Mass of one mole of electron = $(6.022 \times 10^{23}) \times (9.10939 \times 10^{-31}$ kg) = 5.48×10^{-7} kg

Charge on one electron = 1.6022×10^{-19}coulomb

Charge on one mole of electron = $(1.6022 \times 10^{-19}\text{C})\ (6.022 \times 10^{23})$

$= 9.65 \times 10^{4}$ C

7. Find energy of each of the photons which
   (i) Correspond to light of frequency 3×10^{15}Hz.
   (ii) (ii) have wavelength of 0.50 Å.

   Ans.(i) Energy (E) of a photon is given by the expression,

   $E = h\nu$
   Where,
   h = Planck's constant = 6.626×10^{-34}Js ν = frequency of light = 3×10^{15}Hz
   Substituting the values in the given expression of*E:*
   $E = (6.626 \times 10^{-34})\ (3 \times 10^{15})$ $E = 1.988 \times 10^{-18}$J

   (ii) Energy (E) of a photon having wavelength (λ)is given by the expression,

$$E = \frac{hc}{\lambda}$$

   h = Planck's constant = 6.626×10^{-34}Js
   c = velocity of light in vacuum = 3×10^{8}m/s
   Substituting the values in the given expression of *E:*

$$E = \frac{(6.626\times10^{-34})(3\times10^{8})}{0.50\times10^{-10}} = 3.976\times10^{-15}\ \text{J}$$
$$\therefore E = 3.98\times10^{-15}\ \text{J}$$

8. What is the wavelength of light emitted when the electron in a hydrogen atom undergoes transition from an energy level with n = 4 to an energy level with n = 2?
   Ans.Then_i= 4 to n_f= 2 transition will give rise to a spectral line of the Balmer series. The energy involved in the transition is given by the relation,

$$E = 2.18\times10^{-18}\left[\frac{1}{n_i^2} - \frac{1}{n_f^2}\right]$$

Substituting the values in the given expression of *E:*

$$E = 2.18\times10^{-18}\left[\frac{1}{4^2}-\frac{1}{2^2}\right]$$

$$= 2.18\times10^{-18}\left[\frac{1-4}{16}\right]$$

$$= 2.18\times10^{-18}\times\left(-\frac{3}{16}\right)$$

E =–(4.0875 × 10^{-19}J)

The negative sign indicates the energy of emission.

Wavelength of light emitted $(\lambda)=\frac{hc}{E}$

$\left(\text{since } E=\frac{hc}{\lambda}\right)$

Substituting the values in the given expression of λ:

$$\lambda = \frac{\left(6.626\times10^{-34}\right)\left(3\times10^{8}\right)}{4.0875\times10^{-19}}$$

$$\lambda = 4.8631\times10^{-7}\ \text{m}$$

$$= 486.3\times10^{-9}\text{m}$$

$$= 486\ \text{nm}$$

9. An atom of an element contains 29 electrons and 35 neutrons. Deduce **(i)**the number of protons and **(ii)** the electronic configuration of the element (iii) Identify the element.

Ans.**(i)**For an atom to be neutral, the number of protons is equal to the number of electrons.

∴ Number of protons in the atom of the given element = 29

(ii) The electronic configuration of the atom is $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10}$(iii) Copper

10. Give the number of electrons in the species H_2^+, H_2 and O_2^+

Ans. Number of electrons present in hydrogen molecule (H_2) = 1 + 1 = 2

∴ Number of electrons in H_2^+ = 2 – 1 = 1

Numberof electrons in H_2 = 1 + 1 = 2

Number of electrons present in oxygen molecule (O_2) = 8 + 8 = 16

∴ Number of electrons in O_2^+ = 16 – 1 = 15

FIVE MARKS QUESTIONS WITH ANSWERS

1.What are the draw backs of Bohr's atomic model? Show that the circumference of the Bohr orbit for the hydrogen atom is an integral multiple of the de Broglie wavelength associated with the electron revolving around the orbit.

Ans.1.Bohr's model failed to account for the finer details of the hydrogen spectrum.

2. Bohr's model was also unable to explain spectrum of atoms containing more than one electron.
3. Bohr's model was unable to explain Zeeman effect and Stark effect
4. Bohr's model could not explain the ability of atoms to form molecules by chemical bonds

Since a hydrogen atom has only one electron, according to Bohr's postulate, the angular momentum of that electron is given by:

$$mvr = n\frac{h}{2\pi} \quad \ldots\ldots\ldots(1)$$ Where, n = 1, 2, 3, …

According to de Broglie's equation:

$$\lambda = \frac{h}{mv}$$

$$\text{or } mv = \frac{h}{\lambda} \quad \ldots\ldots\ldots(2)$$

Substituting the value of 'mv' from expression (2) in expression (1):

$$\frac{hr}{\lambda} = n\frac{h}{2\pi}$$

$$\text{or } 2\pi r = n\lambda \quad \ldots\ldots\ldots(3)$$

Since'$2\pi r$'represents the circumference of the Bohr orbit (r), it is proved by equation (3) that the circumference of the Bohr orbit of the hydrogen atom is an integral multiple of de Broglie's wavelength associated with theelectron revolving around the orbit.

2. State photoelectric effect. The work function for caesium atom is 1.9 eV. Calculate (a) the threshold wavelength and (b) the threshold frequency of the radiation. If the caesium element is irradiated with awavelength 500 nm, calculate the kinetic energy and the velocity of the ejected photoelectron.

Ans. Photoelectric effect: The phenomenon of ejection of electrons from the surface of metal when light of suitable frequency strikes it is called photoelectric effect. The ejected electrons are called photoelectrons. It is given that the work function (W_0) for caesium atom is 1.9 eV.

(a) From the expression, $W_0 = \frac{hc}{\lambda_0}$, we get:

$$\lambda_0 = \frac{hc}{W_0}$$

Where,

λ_0 = threshold wavelength

h = Planck's constant

c = velocity of radiation

Substituting the values in the given expression of (λ_0):

$$\lambda_0 = \frac{(6.626\times10^{-34}\ \text{Js})(3.0\times10^{8}\ \text{ms}^{-1})}{1.9\times1.602\times10^{-19}\ \text{J}}$$

$\lambda_0 = 6.53 \times 10^{-7}$ m

Hence, the threshold wavelength λ_0 is 653 nm.

(b) From the expression, $W_0 = h\nu_0$, we get:

$$\nu_0 = \frac{W_0}{h}$$

Where,

ν_0= threshold frequency
h = Planck's constant

Substituting the values in the given expression of ν_0:

$$\nu_0 = \frac{1.9\times1.602\times10^{-19}\text{J}}{6.626\times10^{-34}\ \text{Js}}$$

(1 eV = 1.602×10^{-19}J)

$\nu_0 = 4.593 \times 10^{14}\text{s}^{-1}$

Hence, the threshold frequency of radiation (ν_0) is $4.593 \times 10^{14}\text{s}^{-1}$.

(c) According to the question:

Wavelength used in irradiation (λ) = 500 nm Kinetic energy = h ($\nu - \nu_0$)

$$= hc\left(\frac{1}{\lambda} - \frac{1}{\lambda_0}\right)$$

$$= (6.626\times10^{-34}\ \text{Js})(3.0\times10^{8}\ \text{ms}^{-1})\left(\frac{\lambda_0 - \lambda}{\lambda\lambda_0}\right)$$

$$= (1.9878\times^{-26}\ \text{Jm})\left[\frac{(653-500)10^{-9}\text{m}}{(653)(500)10^{-18}\ \text{m}^2}\right]$$

$$= \frac{(1.9878\times10^{-26})(153\times10^{9})}{(653)(500)}\text{J}$$

$= 9.3149 \times 10^{-20}$ J

Kinetic energy of the ejected photoelectron = 9.3149×10^{-20}J Since

K.E $= \frac{1}{2}mv^2 = 9.3149\times10^{-20}$ J

$$v = \sqrt{\frac{2(9.3149\times10^{-20}\ \text{J})}{9.10939\times10^{-31}\ \text{kg}}}$$

$$= \sqrt{2.0451\times10^{11}\ \text{m}^2\text{s}^{-2}}$$

$\nu = 4.52 \times 10^{5}\text{ms}^{-1}$

Hence, the velocity of the ejected photoelectron (ν) is $4.52 \times 10^{5}\text{ms}^{-1}$.

3. (a)The quantum numbers of six electrons are given below. Arrange them in order of increasing energies. If any of these combination(s) has/have the

same energy lists:

1. n= 4, l = 2, m_l=–2 , m_s=–1/2
2. n= 3, l = 2, m_l= 1 , m_s= +1/2
3. n= 4, l = 1, m_l= 0 , m_s= +1/2
4. n = 3, l= 2, m_l=–2 , m_s=–1/2
5. n = 3, l= 1, m_l=–1 ,m_s= +1/2
6. n = 4, l= 1, m_l= 0 , m_s= +1/2

(b)Among the following pairs of orbitals which orbital will experience the larger effective nuclearcharge? (i) 2*s* and 3*s*, (ii) 4*d* and 4*f*, (iii) 3*d* and 3*p*

Ans.(a)For$n$ = 4 and l = 2, the orbital occupied is 4*d*.

For n = 3 and l = 2, the orbital occupied is 3*d*.

For n = 4 and l = 1, the orbital occupied is 4*p*.

Hence, the six electrons i.e., 1, 2, 3, 4, 5, and 6 are present in the 4*d*, 3*d*, 4*p*, 3*d*, 3*p*, and 4*p* orbitals respectively.

Therefore, the increasing order of energies is 5(3*p*) < 2(3*d*) = 4(3*d*) < 3(4*p*) = 6(4*p*) < 1 (4*d*).

(b)Nuclear charge is defined as the net positive charge experienced by an electron in the orbital of a multi-electron atom. The closer the orbital, the greater is the nuclear charge experienced by the electron (s) in it.

(i) The electron(s) present in the 2*s* orbital will experience greater nuclear charge (being closer to the nucleus) than the electron(s) in the 3*s* orbital.

(ii) 4*d* will experience greater nuclear charge than 4*f*since 4*d* is closer to the nucleus.

(iii) 3*p* will experience greater nuclear charge since it is closer to the nucleus than 3*f*.

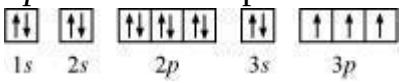
4. (i) The unpaired electrons in Al and Si are present in 3p orbital. Which electrons will experience more effective nuclear charge from the nucleus?

(ii) Indicate the number of unpaired electrons in: (a) P, (b) Si, (c) Cr, (d) Fe
Ans. (i) the electrons in the 3*p* orbital of silicon will experience a more effective nuclear charge than aluminium.

(ii) **(a)** Phosphorus (P):

Atomic number = 15

The electronic configuration of P is:$1s^2\ 2s^2\ 2p^6 3s^2\ 3p^3$ The orbital picture of P can be represented as:

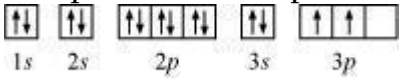


From the orbital picture, phosphorus has **three** unpaired electrons.

(b) Silicon (Si):

Atomic number = 14

The electronic configuration of Si is:$1s^2\ 2s^2\ 2p^6\ 3s^2 3p^2$ The orbital picture of Si can be represented as:

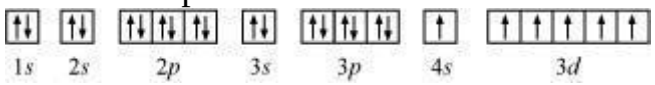


From the orbital picture, silicon has **two** unpaired electrons.

(c) Chromium (Cr):Atomicnumber = 24

The electronic configuration of Cr is:$1s^2\ 2s^2\ 2p^6\ 3s^2 3p^6\ 4s^1\ 3d^5$

The orbital picture of chromium is:

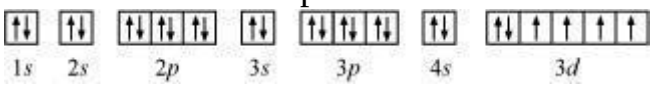


From the orbital picture, chromium has **six** unpaired electrons.

(d) Iron (Fe):

Atomic number = 26

The electronic configuration is:$1s^2\ 2s^2\ 2p^6\ 3s^2 3p^6\ 4s^2\ 3d^6$ The orbital picture of chromium is:



From the orbital picture, iron has **four** unpaired electrons.

HOTS QUESTIONS WITH ANSWERS

1. Give the name and atomic number of the inert gas atom in which the total number of d-electrons is equal to the difference between the numbers of total pand total s electrons.

Ans. electronic configuration of Kr (atomic no.=36) $=1s^2\ 2s^2\ 2p^6\ 3s^2 3p^6 3d^{10} 4s^2 4p^6$ Total no. of s-electrons = 8, total no. of p-electrons = 18. Difference = 10

No. of d- electrons = 10

2. What is the minimum product of uncertainty in position and momentum of an electron?

Ans.$h/4\pi$

3. Which orbital is non-directional ?

Ans. s- orbital

4. What is the difference between the notations *l* andL?

Ans. l represents the sub-shell and L represents shell.

5. How many electrons in an atom can have $n + l = 6$?

Ans. 18

6. An anion A^{3-} has 18 electrons. Write the atomic number of A.

Ans.15

7. Arrange the electron (*e*), protons (*p*) and alpha particle (α) in the increasing order for the values of *e/m* (charge/mass).

Ans.. $\alpha < p < e$

Value Based questions

Q1. The use of fluorescent light tubes for lighting homes, offices etc are very common. Also , in the advertisements displayed in front of offices , shops etc., you have quiet often seen use of colures light tubes.To reduce the consumption of electricity , use of CFLs (compact fluorescent lamps light bulbs) is becoming very popular . To further cut consumption, now a days, we have LED (light emitting diodes) in the form of very efficient solid bulbs grouped in clusters .

Now, answer the following question:

(i) How do differ from fluorescent light tubes? What is the basic principles? What harm they do to the atmosphere after they are discarded?

(ii)How do LED bulbs differ from CFL bulbs?

(iii) A laymen asks you how different colors are obtained in the light tubes of advertisements? How would you explain this to him?

ANS:(i) Fluorescent light tubes are cathode ray tubes coated with a suitable material (called phosphor) which produces visible light when hit by cathode rays. CFLs are simply miniature versions of full-sized fluorescent. They screw into standard lamp sockets and give off light similar to the fluorescent light tubes.They are more efficient. They contain mercury vapuors. When, broken mercury vopours enter into atmosphere and cause health hazards.

(ii) LED bulbs do not contain mercury and hence are eco-friendly. They consume less electricity than CFLs. Moreover, they last longer than CFLs.

(iii) The light tubes emitting different colors are basically discharge tubes. The color of the light emitted depends upon the gases taken inside the discharge tube and the color of the glass. For example, neon in colorless tubes produces orange red light and deep red in light red tube. Similarly, a mixture of neon, argon and mercury vapor emit light blue, dark blue and light green light depending upon weather the tube colorless, purple or green.

Q2. All sources of energy such as sun, electric bulb , heater etc. emit energy which travels in the form of waves called electromagnetic wave because they are associated with electric and magnetic fields. Depending upon the nature of the source, we have different types of electromagnetic waves such as microwaves, radiowaves, X-rays, У-rays, cosmic rays etc differing from each other their wavelength or frequency.

Now, answer the following question :

(i) Should microwave oven be used for heating or cooking the food at home? Comment with points in favour and against.

(ii) Why is it advised not to keep your cellphone in the breast pocket of your shirt?

(iii) Why a person should go for minimum X-rays and a pregnant woman should not go for X-rays at all?

(iv) Give one important application of ultrasound waves in medical field. How is it being misused in medical field ?

ANS:(i)Microwaves are high energy electromagnetic waves. It is believe that they agitate the molecules to move faster and faster. It destroys vitamins, minerals and protein. It changes them into products which may cause cancer, memory loss and decrease in intelligence.However, it is believed that a limited use has no harmful effects. In a modern fast life, it saves a lot of time and is a clean method of warming the food.

(ii)It is believed that microwaves or radio waves of cellophane could interact with our heart. This problem is more serious for people using cardiac pace makers. Hence it is advised not to keep cell phone in breast pocket.

(iii) X-ray is a type of high energy radiation. This can be destructive to all living cells and can cause DNA damage and mutation. They have bad effects on pregnancy and child birth. They can deform the body of infant and could be fatal to his life. They can harm the tissues of the bones called "bone marrow'' .Thus, there can be congenital disorders (birth defects).

(iv)Ultrasound waves are high frequency sound waves. They are used to treat pain conditions and promote tissue healing in physiotherapy. The ultrasound machine is also used in the medical field to follow the growth of baby in the womb of the pregnant women. However, it is being misused to know the sex of the unborn baby.

Q3. Nobody has seen electrons, protons and neutrons. However, the discovery of Rutherford model, Bohr model and further modification have revolutionized their application in different fields.

Now, answer the following questions:

(I) Give one application where electron beam is used.

(ii) Give one application of proton in medical field.

(iii) Like electrons, protons in the nuclei also spin. This property of protons has helped to develop machine called nuclear magnetic Resonance (PNMR or simply NMR). What for this machine is used?

(iv) How has the discovery of neutron been useful as well as harmful to the society?

Ans. (i) Cathode rays (beam of electrons) when focused on a heavy metal like tungsten, molybdenum etc. produce X-rays which find wide applications in medical field. Also, electron microscope which uses a beam of electrons to illuminate a specimen (an extremely small piece) produces a magnified image.

(ii)Proton therapy is a type of particle therapy which uses a beam of protons to irradiate disease tissue, most often in the treatment of cancer.

(iii)NMR spectroscopy is used to determine the structure of an organic compound by revealing the hydrogen and carbon skeleton.

(iv)When a neutron hits uranium, a chain reaction is set up accompanied with evolution of tremendous amount of energy. The energy can be used in the atom bomb for destruction or it can be used in the nuclear reactor to produce electricity.

Q4. When a light or some other form of electromagnetic radiation strike the surface of metal, or certain kinds of matter, electrons are ejected. This phenomenon is called photoelectric

effect. Two or most important applications of the photoelectric effect is the photochemical cell and solar cells. A photocell consist of a vacuum fitted with 2 metal plates, on one of which light falls and act as cathode and the other receives the electrons and act as anode(connected to a source of positive charge). A solar cell converts light energy into electrical energy and generally uses silicon as an emitter.

(i) Give 3 important applications of everyday life where photocells are used.

(ii) Give one important application of photocell in factory.

(iii) Give 3 applications where solar cells are used or can be used. Why there use is not so common?

Ans. (i) (a) In automatic door openers:- As a approaches the door, light between the emmiter and the receiver is cut off. The current stops flowing. It activates the switch and the door is opened automatically.

(b) In intrusion alarm(burglar alarm):- When a family is not at home and they lock the door, they switch on the photocell. When a thief enters the house, the switch is activated and the bell starts ringing .Photo cell thus used sometimes is also called electric eye. This is also used in bank security system.

(c)In turning street lights on and off during dusk and dawn:- This is done by installing photo cells on the light poles.

(ii)In a factory, the items on a conveyer belt pass between a beam of light and photocell. As each time passes the beam, it intersceptsthe light. The current in the photocellstops and counter is turned on.With this method,the exact numberof items living the factory can be counted.

(iii) (a)Solar cells can be used to provide small amounts of electricity in remote location where other sources are not available.

(b)In solar lantern in place of torches.

(c) In solar geysers for heating water.

There use is not so common because:-

(1) Individual solar cells produce a low voltage of about 0.6 volt.

(2) They are quite expensive.

CHAPTER-3

CLASSIFICATION OF ELEMENTS AND PERIODICITY IN PROPERTIES OF ELEMENTS

Mandeleev's Periodic Law:- The properties of the elements are the periodicfunction of their atomic masses.

Moseley, the English physicist showed that atomic number is more fundamental property of an element than its atomic mass. Therefore, the position of an element in the periodic table depends on its atomic number than its atomic mass.

Modern Periodic Law: The physical and chemical properties of elements are theperiodic functions of their atomic numbers.

Types of Elements: s-, p-, d- and f- blocks.

MAIN GROUP ELEMENTS/ REPRESENTATIVE ELEMENTS:

The s- and p- block elements are called main group elements or representative elements.

s- block elements: Group-1 (Alkali metals) and Group-2 elements (Alkalineearth metals) which respectively have ns^1 and ns^2 outermost electronic configurations.

p- Block elements: They belongs to group- 13 to 18. The outer most electronicconfiguration is $ns^2\ np^{1-6}$. He ($1s^2$) is a s- block element but is positioned with the group 18 elements ($ns^2\ np^6$) because it has completely filled valence shell and as a result, exhibits properties characteristic of other noble gases.

d- block elements (Transition elements) are the elements of group 3 to 12having outer electronic configuration (n-1) $d^{1-10}\ ns^{1-2}$. Four transition series are 3d, 4d, 5d and 6d. The 6d- series is incomplete. Atomic radius generally decreases across a period and increases as we descend the group.

f-Block elements (Inner- transition Series)

Lanthanoids charecterised by the filling of4f-orbitals, are the elements following lanthanum from $_{58}$Ce to $_{71}$Lu. Actinoids characterised by filling of 5f-orbitals, are the elements following actinium from $_{70}^{Th}$ to $_{103}$Lr. Characteristic outer electronic configuration is (n-2) f^{1-14} (n-1) $d^{0-1}\ ns^2$.

Noble Gases: The gaseous elements of group 18 are called noble gases. Thegeneral outermost electronic configuration of noble gases (except He) is $ns^2 np^6$. He exceptionally has $1s^2$ configuration. Thus the outermost shell of noble gases is completely filled.

PERIODICITY: The repetition of similar properties after regular intervals iscalled periodicity.

Cause of Periodicity: The properties of elements are the periodic repetition ofsimilar electronic configuration of elements as the atomic number increases.

ATOMIC PROPERTIES: The physical characteristics of the atom of anelement are called atomic properties. The properties such as atomic radius, ionic radius, ionisation energy, electro-negativity, electron affinity and valence etc., called atomic properties.

ATOMIC RADIUS- The distance from the centre of the nucleus to theoutermost shell of the electrons in the atom of any element is called its atomic radius.

Periodicity- (a) In period- Atomic radius of elements decreases from left to rightin a period.

(b) In Group- Atomic radius of elements increases on moving top to bottom in a group.

COVALENT RADIUS- Half the inter-nuclear distance between two similaratoms of any element which are covalently bonded to each other by a single covalent bond is called covalent radius.

VAN DER WAALS' RADIUS: Half the inter-nuclear separation between twosimilar adjacent atoms belonging to the two neighbouring molecules of the same substance in the solid state is called the van der waals'radius of that atom.

METALLIC RADIUS: Half the distance between the nuclei of the two adjacentmetal atoms in a close packed lattice of the metal is called its metallic radius.

Van der Waals'radius > Metallic radius > Covalent radius

IONIC RADIUS: The effective distance from the centre of the nucleus of an ionupto which it has an influence on its electron cloud is called its ionic radius. A cation is smaller but the anion is larger than the parent atom. In case of iso-electronic species, the cation with greater positive charge has smaller radius but anion with greater negative charge has the larger radii.

IONISATION ENTHALPY: The ionisation enthalpy is the molar

enthalpychange accompanying the removal of an electron from a gaseous phase atom or ion in its ground state. Thus enthalpy change for the reaction; $M_{(g)} \rightarrow M^{+}_{(g)} + e^{-}$

is the ionisation enthalpy of the element M. Like ionisation energies for successive ionisation, the successive ionisation enthalpy may also be termed as 2nd ionisation enthalpy ($\Delta_r H_2$), third ionisation enthalpy ($\Delta_r H_3$) etc. The term ionisation enthalpy is taken for the first ionisation enthalpy, ($\Delta_r H_1$) is expressed in kj mol^{-} or in eV.

Periodicity:

i) Generally the ionisation enthalpies follow the order (there are few exceptions):

$(\Delta_r H_1) < (\Delta_r H_2) < (\Delta_r H_3)$

3. The ionisation enthalpy decreases on moving top to bottom in a group.
4. The ionisation enthalpy increases on moving from left to right in a period.

ELECTRON GAIN ENTHALPY: The electron gain enthalpy (($\Delta_{eg}H$) is themolar enthalpy change when an isolated gaseous atom or ion in its ground state adds an electron to form the corresponding anion thus the enthalpy change for the reaction; $X_{(g)} + e^{-} \rightarrow X^{-}_{(g)}$

is called the electron gain enthalpy (Δ_{eg} H) of the element X. TheΔ_{eg} H may be positive or negative.

The successive values for the addition of second, third etc. Electron, these are called second, third etc. electron gain enthalpies. For example,

$X_{(g)} + e^{-} \rightarrow X^{-}_{(g)}$ $\Delta H = \Delta_{eg} H_1$ is called first electron gain enthalpy

$X^{-}_{(g)} + e^{-} \rightarrow X^{2-}_{(g)}$ $\Delta H = \Delta_{eg} H_2$ is called second electron gain enthalpy

$X^{2-}_{(g)} + e^{-} \rightarrow X^{3-}_{(g)}$ $\Delta H = \Delta_{eg} H_3$ is called third electron gain enthalpy Usually the term electron gain enthalpy ($\Delta_{eg}H$) means the first electron gain enthalpy.

Periodicity:

1. In period- The electron gain enthalpy increases from left to right in a period.
2. In group- The electron gain enthalpy decreases from top to

bottom in a group.

ELECTRONEGATIVITY: "The relative tendency of an atom in a molecule toattract the shared pair of electrons towards itself is termed as its electro-negativity."

Periodicity:

1. In period- The electro-negativity increases from left to right in a period.
2. In group- The electro-negativity decreases from top to bottom in a group.

VALENCE ELECTRONS: The electrons present in outermost shell are calledas valence electron. Because the electrons in the outermost shelldetermine the valency of an element.

VALENCY OF AN ELEMENT: The number of hydrogen or halogen atom ordouble the number of oxygen atom, which combin with one atom of the element is taken as its valency. According to the electronic concept of valency, " the number of electrons which an atom loses or gains or shares with other atom to attain the noble gas configuration is termed as its valency."

Periodicity:

6. In period- The valency first increases then decreases from left to right in a period.
7. In group- The valency remains constant from top to bottom in a group.

ELECTROPOSITIVE OR METALLIC CHARACTER: The tendency of anelement to lose electrons and forms positive ions (cations) is called electropositive or metallic character. The elements having lower ionisation energies have higher tendency to lose electrons, thus they are electropositive or metallic in their behaviour.

Alkali metals are the most highly electropositive elements.

Periodicity: In period- The electropositive or metallic characters decreases fromleft to right in a period.

In group- The electropositive or metallic characters increases from top to bottom in a group.

ELECTRO-NEGATIVE OR NON- METALLIC CHARACTERS: Thetendency of an element to accept electrons to form an anion is called its non metallic or electronegative character. The elements having high electro-

negativity have higher tendency to gain electrons and forms anion. So, the elements in the upper right hand portion of the periodic table are electro-negative or non-metallic in nature.

Periodicity:

3. In period- The electro-negative or non- metallic characters increases from left to right in a period.
4. In group- The electro-negative or non-metallic characters decreases from top to bottom in a group.

REACTIVITY OF METALS:

Periodicity:

1. In period- The tendency of an element to lose electrons decreases in a period. So the reactivity of metals decreases from left to right in a period.
2. In group- The tendency of an element to lose electrons increases in a period. So the reactivity of metals increases from top to bottom in a group.

REACTIVITY OF NON- METALS:

1. In period- The tendency of an element to gain electrons increases in a period. So the reactivity of non-metals increases from left to right in a period.
2. In group- The tendency of an element to gain electrons decreases in a group. So the reactivity of non-metals increases from top to bottom in a group.

SOLUBILITY OF ALKALI METALS CARBONATES AND BICARBONATES:

PERIODICITY IN GROUP: The solubility of alkali metal carbonates andbicarbonates in water increases down the group (From Lithium to Caesium).

SOLUBILITY OF ALKALINE EARTH METAL HYDROXIDES AND SULPHATES:

PERIODICITY IN GROUP: The solubility of alkaline earth metal hydroxideand sulphates in water increases down the group (From Beryllium to Barium).

BASIC STRENGTH OF ALKALINE EARTH METAL HYDROXIDES:

PERIODICITY IN GROUP: The basic strength of alkaline earth metalhydroxide in water increases down the group (From Beryllium to Barium), i.e.,

$$Be(OH)_2 < Mg(OH)_2 < Ca(OH)_2 < Sr(OH)_2 < Ba(OH)_2$$

Basic strength increases

THERMAL STABILITY OF CARBONATES OF ALKALI AND ALKALINE EARTH METALS:

Except lithium carbonate, ($LiCO_3$), the carbonates of all other alkali metals are stable towards heat, i.e., carbonates of alkali metals (except $LiCO_3$) do not decompose on heating. $LiCO_3$ decomposes on heating to give lithium oxide ($LiCO_3$).

The carbonates of alkaline earth metals are relatively less stable. On heating, they decompose to give corresponding oxide and CO_2 gas. The decomposition temperature for alkaline earth metal carbonates increases as we go down the group.

Anomalous Properties of Second Period Elements

Their anomalous behaviour is attributed to their small size, large charge/radius ratio, high electro negativity, non- availability of d- orbitals in their valence shell. The first member of each group of p-Block elements displays greater ability to form pπ-pπ multiple bonds to itself (e.g. C=C, C≡C O=O, N≡N) and to other second period elements (e.g. C=O, C≡N, N=O) compared to subsequent member of the group.

ONE MARK QUESTIONS

Q1. Select the species which are iso-electronic (same number of electron) with each other.

(1) Ne (2) Cl^- (3) Ca^{2+} (4) Rb^+

Ans-TheCl^- and Ca^{2+}. Both have 18 e^- each.

Q.2. What the elements of a group have common among them?

Ans- They have same number of electrons in the valence shell.

Q.3. What the s- and p- block elements are collectively called?

Ans- Representative elements.

Q.4. Define atomic radius.

Ans- The one-half the distance between the nuclei of two covalently bonded atoms of the same element in a molecule is called as atomic radius.

Q.5. State the modern periodic law.

Ans- The physical and chemical properties of the elements are theperiodic function of their atomic numbers.

Q.6. Name the groups of elements classified as s-, p- and d- blocks.

Ans- s- block= 1,2 (including He), p- block= 13 to 18 (except He), d- block= 3 to 12.

Q.7. Define the term ionisation enthalpy.

Ans- The energy required to remove the outer most electron from the valence shell of an isolated gaseous atom is called as ionisation enthalpy.

Q.8.In how many groups and periods the elements in modern periodic table are classified?

Ans- In 18 groups and 7 periods.

Q.9. What do you mean by electronic configuration of the elements?

Ans- The systematic distribution of the electrons among the orbitals of an atom of an element according to increasing order of their energies is called as electronic configuration of that element.

TWO MARKS QUESTIONS

Q.1. Describe the two merits of long form periodic table over the Mendeleev's periodic table?

Ans- 1. It removed the anomalies about the position of isotopes which existed in the Mendeleev's table.

2. It relates the position of an element in the periodic table with its electronic configuration.

Q.2. What is a period in the periodic table? How do atomic sizes change in a period with an increase in atomic number?

Ans- The horizontal rows in periodic table are called as periods. The atomic sizes decrease in a period with an increase in atomic number.

Q.3. The outer electronic configuration of some elements are:

(a) $3s^2\ 3p^4$ (b) $3d^{10}4s^2$ (c) $3s^2\ 3p^6\ 4s^2$ (d) $6s^2\ 4f^3$

To which block of elements in the periodic table each of these belongs?

Ans- (a) p- Block (b) d- Block (c) s- Block (d) f- Block

Q.4. What is meant by periodicity in properties of elements? What is the reason behind this?

Ans- The repetition of similar properties after regular intervals is called as periodicity. It is due to the similarity in the outer electronic configurations which gives rise to the periodic properties of the elements.

Q.5. How do atomic radii vary in a group and a period?

Ans- In group- Atomic size increases on moving from top to bottom.

In period- Atomic size decreases on moving left to right in a period.

Q.6. Arrange the following in the order of increasing radii:

(a) I, I^+, I^- (b) O, P, N

Ans- (a) $I^+ < I < I^-$ (b) $O < N < P$

Q.7. Name the factors which affect the ionisation enthalpy of an element.

Ans- (i) Size of atom or ion (ii) Nuclear charge (iii) Electronic configuration

(iv) Screening effect (v) Penetration effect of the electrons

Q.8. How does ionisation enthalpy vary in a group and a period?Ans- In Period- It increases from left to right

In group- It decreases down the group.

Q.9. Noble gases have positive electron gain enthalpy values. Explain.

Ans- Because the outer most shell of noble gases is completely filled and no more electrons can be added.

Q.10. Elements in the same group have equal valency. Comment on it.

Ans- Because the general outer most electronic configurations of the elements of a group remain same and they contain equal number of electrons in their respective outer most shells.

THREE MARKS QUESTIONS

Q.1. The first ionisation enthalpy of magnesium is higher than that of sodium. On theother hand, the second ionisation enthalpy of sodium is very much higher than that of magnesium. Explain.

Ans- The 1st ionisation enthalpy of magnesium is higher than that of Na due to higher nuclear charge and slightly smaller atomic radius of Mg than Na. After the loss of first electron, Na^+ formed has the electronic configuration of neon (2,8). The higher stability of the completely filled noble gas configuration leads to very high second ionisation enthalpy for sodium. On the other hand, Mg^+ formed after losing first electron still has one more electron in its outermost (3s) orbital. As a result, the second ionisation enthalpy of magnesium is much smaller than that of sodium.

Q.2. What are the major differences between metals and non-metals?Ans-

Property	Metal	Non- Metal
Nature	Electropositive	Electronegative
Type of ion formed	Cation (Positively Charged)	Anion (Negatively Charged)
Reaction with acids	Active metals displace hydrogen	Do not displace hydrogen
Oxides	Basic	Acidic

Q.3. Among the elements of the second period Li to Ne pick out the element:

(i) with the highest first ionisation energy(ii) with the highest electronegativity

(iii) with the largest atomic radius Give the reason for your choice.

Ans- (i) The ionisation energy increases on going from left to right. Therefore, the element with the highest ionisation energy is Ne.

ii)The electro negativity is electron- accepting tendency. This increases on going from left to right and decreases down the group. Therefore, the element with the highest electro- negativity is F.

iii)The atomic radius decreases across a period on going from left to right.

Thus, the first element of any period should have the largest atomic radii. Here, Li has the largest atomic radii.

Q.4. Arrange the following as stated:

(i) N_2, O_2, F_2, Cl_2 (Increasing order of bond dissociation energy)
(ii) F, Cl, Br, I (Increasing order of electron gain enthalpy)
(iii) F_2, N_2, Cl_2, O_2 (Increasing order of bond length)

Ans- (i) $F_2 < Cl_2 < O_2 < N_2$

(ii) I < Br < F < Cl

(iii) $N_2 < O_2 < F_2 < Cl_2$

Q.5. Why does the first ionisation enthalpy increase as we go from left to right through a given period of the periodic table?

Ans- In a period, the nuclear charge (the number of protons) increases on going from left to right. The electron added to each element from left to right enters the same shell. This results in an increase of the effective nuclear charge across the period on moving from left to right. As a result, the electron get more firmly bound to the nucleus. This causes an increase in the first ionisation enthalpy across the period.

Q.6. Use the periodic table to answer the following questions.

(i) Identify the element with five electrons in the outer sub-shell.

ii)Identify an element that would tend to lose two electrons.

iii)Identify an element that would tend to gain two electrons.

Ans- (i) Chlorine (ii) Magnesium (iii) Oxygen

Q.7. Explain why are cations smaller and anions larger in size than their parent atoms?

Ans- (a) The cations are smaller than their parent atoms due to the followingreasons:

Disappearance of the valence shell.

Increase of effective nuclear charge

b)The anions are larger than their parent atoms due to the following reason:

An increase in the number of electrons in the valence shell reduces the effective nuclear charge due to greater mutual shielding by the electrons. As a result, electron cloud expands, i.e., the ionic radius increases.

Q.8. Describe the theory associated with the radius of an atom as it

(a) gains an electron (b) loses an electron

Ans- **(a)**When an atom gains an electron, its size increases. When an electron isadded, the number of electrons goes up by one. This results in an increase in repulsion among the electrons. However, the number of protons remains the same. As a result, the effective nuclear charge of the atom decreases and the radius of the atom increases.

(b) When an atom loses an electron, the number of electrons decreases by onewhile the nuclear charge remains the same. Therefore, the interelectronic repulsions in the atom decrease. As a result, the effective nuclear charge increases. Hence, the radius of the atom decreases.

Q.9. How does atomic radius vary in a period and in a group? How do youexplain the variation?

Ans- Atomic radius generally decreases from left to right across a period. This isbecause within a period, the outer electrons are present in the same valence shell and the atomic number increases from left to right across a period, resulting in an increased effective nuclear charge. As a result, the attraction of electrons to the nucleus increases.

On the other hand, the atomic radius generally increases down a group. This is because down a group, the principal quantum number (n) increases which results in an increase of the distance between the nucleus and valence electrons.

Q.10. Consider the following species:

N^{3-}, O^{2-}, F^{-}, Na^{+}, Mg^{2+} and Al^{3+}

(a) What is common in them?

(b) Arrange them in the order of increasing ionic radii.

Ans- (a) the same number of electrons (10 electrons). Hence, the given speciesare isoelectronic.

(b) $Al^{3+} < Mg^{2+} < Na^{+} < F^{-} < O^{2-} < N^{3-}$

FIVE MARKS QUESTIONS

Q.1. What is the cause of the periodicity in the properties of the elements? How do the following properties vary in (a) a group and (b)in a period

(i) electronegativity (ii) ionisation enthalpy (iii) Atomic size

Ans- It is due to the similarity in the outer electronic configurations which gives rise to the periodic properties of the elements.

(a) In a group:
   (i) Electronegativity- It decreases down the group.
   (ii) Ionisation enthalpy- It decreases down the group.
   (iii) Atomic size- It increases down the group.

(b) In a period:
   (i) Electronegativity- Increases
   (ii) Ionisation enthalpy- Increases
   (iii) Atomic size- Dereases.

Q.2. The first ($\Delta_i H1$) and the second ($\Delta_i H$) ionization enthalpies (in kJ mol^{-1}) andthe ($\Delta_{eg} H$) electron gain enthalpy (in kJ mol^{-1}) of a few elements are given below:

Elements	$_iH1$	$_iH2$	$_{eg}H$
I	520	7300	–60
II	419	3051	–48
III	1681	3374	–328
IV	1008	1846	–295
V	2372	5251	+48
VI	738	1451	–40

Which of the above elements is likely to be :

(a) the least reactive element. (b) the most reactive metal.
(c) the most reactive non-metal. (d) the least reactive non-metal.
e)the metal which can form a stable binary halide of the formula MX_2, (X=halogen).

f) the metal which can form a predominantly stable covalent halide of the formula MX (X=halogen)?

Ans- **(a)**Element V is likely to be the least reactive element. This is because it hasthe highest first ionization enthalpy ($\Delta_i H_1$) and a positive electron gain enthalpy ($\Delta_{eg}H$).
b)Element II is likely to be the most reactive metal as it has the lowest first ionization enthalpy ($\Delta_i H_1$) and a low negative electron gain enthalpy ($\Delta_{eg}H$).
c)Element III is likely to be the most reactive non–metal as it has a high first ionization enthalpy ($\Delta_i H_1$) and the highest negative electron gain enthalpy ($\Delta_{eg}H$).
d)Element V is likely to be the least reactive non–metal since it has a very high first ionization enthalpy ($\Delta_i H_2$) and a positive electron gain enthalpy ($\Delta_{eg}H$).
e)Element VI has a low negative electron gain enthalpy ($\Delta_{eg}H$). Thus, it is a metal. Further, it has the lowest second ionization enthalpy ($\Delta_i H_2$). Hence, it can form a stable binary halide of the formula MX_2(X=halogen).
f)Element I has low first ionization energy and high second ionization energy. Therefore, it can form a predominantly stable covalent halide of the formula MX (X=halogen).

VALUE BASED QUESTIONS

1. In the long form of the periodic table, all the known elements have been divided into four blocks called s, p, d and f block elements. Till today elements with atomic number up to 112 have been fully characterized while momentary detection /isolation of elements with atomic numbers 113-118 have been successfully observed. The present set up of the long form of periodic table as on today can accommodate at the maximum, elements with atomic numbers up to 118. What would happen, if in future, elements with atomic numbers more than 118 are discovered? now answer the following questions:
   I) What modifications in the present set up of the long form of periodic table do you propose to accommodate elements with atomic numbers 119 and 120, if discovered in future? Which group and period would you assign to these two elements?
   II) If an element with atomic number 121 were discovered in future, can you still classify it into one of the four blocks; s ,p ,d and f. if not , then do you propose a new block for this element ? If yes, then what name do you propose for this block and how many new elements it can accommodate and where to place this block in the present set up of the long form of the periodic table?

Ans. (i) The maximum number of elements which can be accommodated in the present set up of the Long form of the periodic table is 118. Thereafter, filling of 8s orbital shall begin for this,

8th period will have to be incorporated in the present set of the long form of the periodic table. Since s-subshells can accommodate only 2 electrons therefore, elements with Z= 119 will be an alkali metal, placed in group 1 with $8s^1$ as the outer electronic configuration. Similarly, Element with Z= 120 will be an alkaline earth metals, placed in group 2 with $8s^2$ as the outer Electronic configuration.

(ii) An element with atomic number 121 , if discovered , cannot be placed into one of the already known four blocks, i.e., s ,p, d and f. it has to be placed in a new block called the g-block. It will have nine orbitals which can accommodate eighteen elements with Z=121-138. Like f-block, this newly conceived g-block will have to be placed at the bottom of the present set up of the long form of periodic table.

The main discrepancy of the long form of the periodic table is the placement of Lanthanoids and Actinoids at the bottom of the table. To include them in the main body of the periodic table, extended version of the long form of the periodic table has been proposed. Now answer the following questions :

I. Construct the extended version of the long form of the periodic table.
II. What are the drawbacks of the extended version of the periodic table, if any why?

Ans.(i) Place the representative elements of group 1 and 2 on the left hand side and those of group 13- 18 on the right hand side in the same way as they appear in the traditional long form of the Periodic table. Divide the space between these 2 sets of the representative elements into 2 parts. on the left Part, place the inner transition elements in 2 horizontal series of 14 elements each, the first Horizontal series which lies in the 6th period consist of 14 Lanthanoids (Z=57-70) while the second horizontal series which lies in the 7th period also consists of 14 Actinoids(Z=89 to 102).

On the right part, place the transition elements in 4 horizontal series of 10 elements each: first series with Z=21 to 30 lies in the fourth period, second series with Z=39 to 48, lies in fifth period, 3rd series, with Z= 71 to 80 lies in the sixth period and 4th series with Z=103 to 112 lies in the 7th period. The outlines of the extended version of the periodic table thus obtained are shown below.

	Representative Elements			Transition Elements		Noble Gases
1	Z= 1					Z=2
2	Z=3	Z=4			Z=5-9	Z=10
3	Z=11	Z=12			Z=13-17	Z=18
4	Z=19	Z=20	Inner Transition Elements	Z=21-30	Z=31-35	Z=36
5	Z=37	Z=38		Z=39-48	Z=49-53	Z=54
6	Z=55	Z=56	Lanthanoids (Z= 57-70)	Z= 71-80	Z=81-85	Z=86
7	Z=87	Z=88	Actinoids (Z= 89-102)	Z=103-112	Z=113-117	Z=118

(ii) Although in the extended version of the periodic table, elements are arranged in increasing order of their atomic numbers without any discontinuity, but it has the following two discrepancies. Elements La (Z= 57) and Ac (Z= 89) are considered as inner transition elements. But in traditional long form they are considered as transition elements. Elements Lu (Z= 71) and Lr(Z= 103) are considered as transition elements but in the long form, they are considered as inner transition elements.

2. The element radium was discovered by Madam Curie. She was successful in isolating 0.2 g of the element from mineral pitch blende. For this work, she was awarded the Nobel Prize in chemistry in 1911. In her honor, UNESCO declared the year 2011 as international year of chemistry. The discovery of radium opened a new field of radioactivity and thereafter many new radioactive elements have either been isolated from natural sources or have been prepared artificially in the laboratory by nuclear Reactions. Radioactive materials are extensively been used for power generation and in the diagnosis and treatment of many diseases. In spite of these benefits, radiation leakage from nuclear plants causes severe health hazards to the mankind. Further atomic bombs (which were dropped by USA in Hiroshima and Nagasaki in japan in august in 1945) and other weapons of mass destruction use radioactive materials. Now answer the following questions.

(i) Name the group and period of the periodic table in which radium lies.
(ii) Name a few other radioactive elements which have been prepared in the lab.
(iii) Name two radioactive materials which have been used for treatment of cancer.
(iv) In view of the nuclear power plant disasters which occurred in 3 Mile-Island (USA), Chernobyl (Ukraine), Fukushima (Japan) etc. should the nuclear power generation be stopped? Give your suggestions.

Ans. (i) Radium - Group 2, Period 7. It is the last element of the alkaline earth family.
(ii) Tc (Technetium), Po (Polonium), U (Uranium) and all the transuranic elements from $_{93}$Npto $_{103}$Lr. Notable amongst these is Plutonium ($_{94}$Pu) which is extensively being used in power plants.

(iii) $_{88}Ra^{226}$ and $_{27}Co^{60}$ are widely used for treatment of cancer.

(iv) The three main sources of power are : Fossil Fuels, Hydroelectric power, and atomic power, in view of the depleting stocks of fossil fuels and hydroelectric power limited by landscape, the only alternative is Atomic Power. But nuclear power plants have serious disadvantages. Elaborate and expensive safety, precautions must be taken to protect plant workers and the residents of the surrounding areas from radiation. The nuclear reactor must be heavily shielded and housed inside a building made of metal and concrete. Since loss of coolant water can result in the meltdown of the reactor core, backup emergency coolant systems are required. In view of these fears, the opening of new nuclear power plants in Tamil Nadu and Haryana are being opposed by the residents of the concerned areas.

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CHAPTER-4

CHEMICAL BONDING AND MOLECULAR STRUCTURE

OCTET RULE- During a chemical reaction the atoms tend to adjust theirelectronic arrangement in such a way that they achieve 8 e^- in their outermost electron. This is called octet rule.

CHEMICAL BOND- the chemical force which keeps the atoms in any moleculetogether is called a chemical bond.

IONIC BOND- The columbic force of attraction which holds the appositivelycharged ions together is called an ionic bond. An ionic bond is formed by the complete transfer of one or more electrons from the atom of a metal to an atom of non- metal.

LATTICE ENTHALPY- The molar enthalpy change accompanying the completeseparation of the constituent particles that compose of the solids (such as ions for ionic solid, molecules for molecular solids) under standard conditions is called lattice enthalpy ($\Delta_l H^o$). The lattice enthalpy is a positive quantity.

ELECTRO VALENCY: The number of electrons lost or gain by an atom of anelement is called as electrovalency.

The element which give up electrons to form positive ions are said to have positive valency, while the elements which accept electrons to form negative ions are said to have negative valency.

FORMATION OF AN IONIC BOND: It is favoured by, (i) the low ionisationenthalpy of a metallic element which forms the cations, (ii) High electron gain enthalpy of non- metallic element which forms the anions, (iii) Large lattice enthalpy i.e; the smaller size and the higher charge of the atoms.

COVALENCY: The number of electrons which an atom contributes towardsmutual sharing during the formation of a chemical bond called its covalency in that compound.

SINGLE COVALENT BOND: A covalent bond formed by the mutual sharing ofone pair of electrons is called a single covalent bond, or simply a single bond. A single covalent bond is represented by a small line (−) between the two atoms.

DOUBLE COVALENT BOND: A covalent bond formed by the mutual sharing oftwo pair of electrons is called a double covalent bond, or simply a double bond. A double covalent bond is represented by two small horizontal lines (=) between the two atoms. E.g. O=O, O=C=O etc.

TRIPLE COVALENT BOND: A covalent bond formed by the mutual sharing ofthree pair of electrons is called a triple covalent bond, or simply a triple bond. A triple covalent bond is represented by three small horizontal lines (≡) between the two atoms. E.g. N≡N, H-C≡C-H etc.

FORMATION OF A COVALENT BOND: Formation of a covalent bond isfavoured by

1. High ionisation enthalpy of the combining elements.
2. Nearly equal electron gain enthalpy and equal electro-negativities of combining elements.
3. High nuclear charge and small atomic size of the combining elements.

POLAR COVALENT BOND: The bond between two unlike atoms which differin their affinities for electrons is said to be polar covalent bond. E.g. H-Cl

COORDINATE BOND: The bond formed when one sided sharing of electronstake place is called a coordinate bond. Such a bond is also known as dative bond. It is represented by an arrow (→) pointing towards the acceptor atom. E.g. $H_{3N}\rightarrow BF_3$

Bond Length: Bond length is defined as the equilibrium distance between thenuclei of two bonded atoms in a molecule

Bond Angle: It is defined as the angle between the orbitals containing bondingelectron pairs around the central atom in a molecule/complex ion

Bond Enthalpy: It is defined as the amount of energy required to break one moleof bonds of a particular type between two atoms in a gaseous state.

Bond Order: In the Lewis description of covalent bond, the Bond Order is givenby the number of bonds between the two atoms in a molecule **Resonance**: Whenever a single Lewis structure cannot describe a moleculeaccurately, a number of structures with similar energy, positions of nuclei, bonding and non-bonding pairs of electrons are taken as the canonical structures of the hybrid which describes the molecule accurately

Dipole moment: The product of the magnitude of the charge and the distancebetween the centres of positive and negative charge.It is a vector quantity and is represented by an arrow with its tail at the positive centre and head pointing towards a negative centre. Dipole moment (μ) = charge (Q) × distance of separation (r)

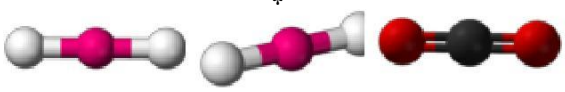
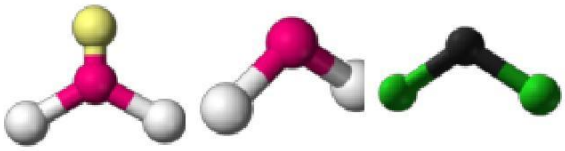
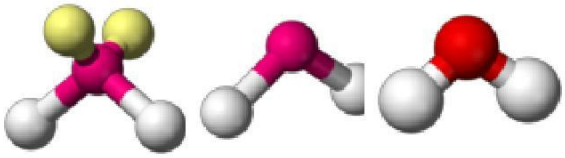
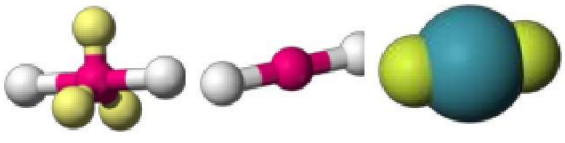
SIGMA BOND: A covalent bond formed due to the overlapping of orbitals of thetwo atoms along the line joining the two nuclei (orbital axis) is called sigma (σ) bond. For example, the bond formed due to s-s and s-p, p-p overlapping along the orbital axis are sigma bonds.

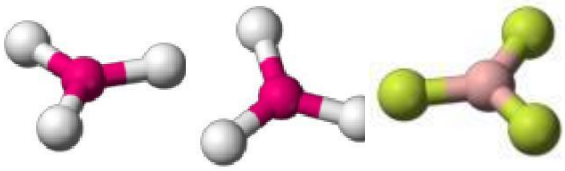
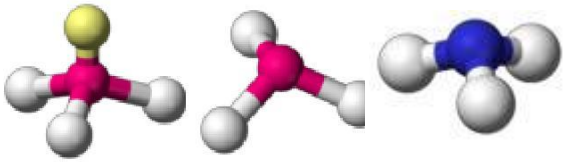
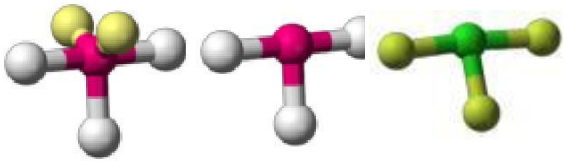
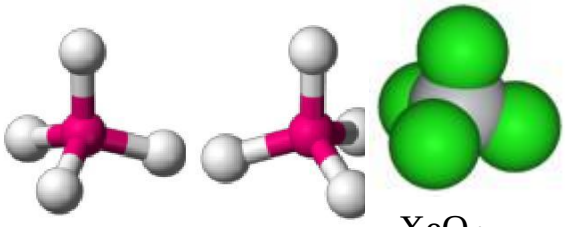
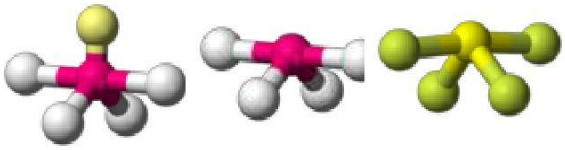
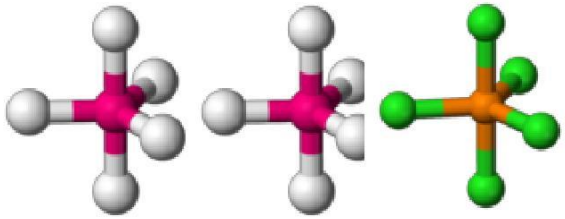
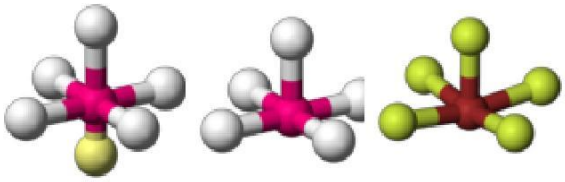
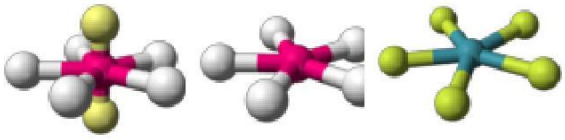
Pi- BOND: A covalent bond formed by the side wise overlapping of p- or d-orbitals of two atoms is called as pi (π) bond. For example, the bond formed due to the sideways overlapping of the two p- orbitals is a pi- bond.

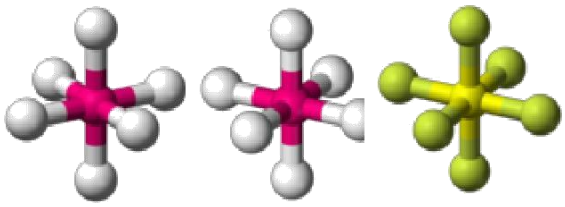
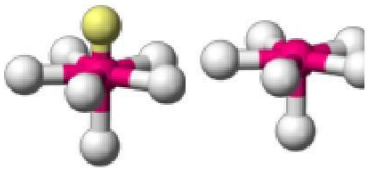
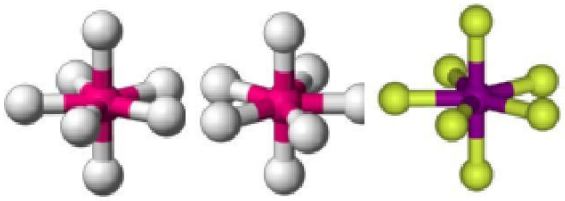
HYDROGEN BOND: The bond between the hydrogen atom of one molecule anda more electro- negative element of same or another molecule is called as hydrogen bond.

HYBRIDIZATION: The process of mixing of the atomic orbitals to form newhybrid orbitals is called hybridization. All hybrid orbitals of a particular kind have equal energy, identical shapes and are symmetrically oriented in shape.

The hybrid orbitals are designed according to the type and the atomic orbitals merging together, e.g.,

Molecule Type	**Shape**	**Electron arrangement**†	**Geometry**‡	**Examples**
AX_2E_0	Linear			$BeCl_2$, $HgCl_2$, CO_2
AX_2E_1	Bent			NO_2^-, SO_2, O_3, CCl_2
AX_2E_2	Bent			H_2O, OF_2
AX_2E_3	Linear			XeF_2, I_3^-, $XeCl_2$

AX_3E_0	Trigonal planar		BF_3, CO_3^{2-}, NO_3^-, SO_3
AX_3E_1	Trigonal pyramidal		NH_3, PCl_3
AX_3E_2	T-shaped		ClF_3, BrF_3
AX_4E_0	Tetrahedral		CH_4, PO_4^{3-}, SO_4^{2-}, ClO_4^-, XeO_4
AX_4E_1	Seesaw		SF_4
AX_4E_2	Square planar		XeF_4
AX_5E_0	Trigonal bipyramidal		PCl_5
AX_5E_1	Square pyramidal		ClF_5, BrF_5, $XeOF_4$
AX_5E_2	Pentagonal planar		XeF_5^-

AX_6E_0	Octahedral		SF_6, WCl_6
AX_6E_1	Pentagonal pyramidal		$XeOF_5^-$, IOF_5^{2-}[7]
AX_7E_0	Pentagonal bipyramidal		IF_7

ONE MARK QUESTIONS

Q.1. What is the total number of sigma and pi bonds in the following molecules?

(a) C_2H_2 (b) C_2H_4

Ans- there are three sigma and two pi-bonds in C_2H_2.

there are five sigma bonds and one pi-bond in C_2H_4.

Q.2. Write the significance of a plus and a minus sign shown in representing the orbitals.

Ans- Molecular orbitals are represented by wave functions. A plus sign in an orbital indicates a positive wave function while a minus sign in an orbital represents a negative wave function.

Q.3. How do you express the bond strength in terms of bond order?

Ans- Bond strength represents the extent of bonding between two atoms forming a molecule. The larger the bond energy, the stronger is the bond and the greater is the bond order.

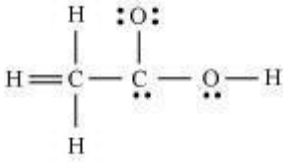
Q.5. Define the bond length.

Ans- Bond length is defined as the equilibrium distance between the nuclei of two bonded atoms in a molecule.

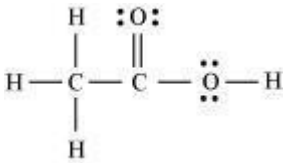
Q.6. Arrange the bonds in order of increasing ionic character in the molecules: LiF, K_2O, N_2, SO_2 and ClF_3.

Ans- $N_2 < SO_2 < ClF_3 < K_2O < LiF$.

Q.7. The skeletal structure of CH_3COOH as shown below is correct, but some of the bonds are shown incorrectly. Write the correct Lewis structure for acetic acid.



Ans- The correct Lewis structure for acetic acid is as follows:



Q.8. Define octet rule.

Ans- The elements tend to adjust the arrangement of their electrons in such a way that they (except H and He) achieve eight electrons in their outermost shell. This is called octet rule.

Q.9. Define lattice enthalpy.

Ans- The energy required when one mole of an ionic compound in crystalline form is split into the constituent ions is called lattice enthalpy.

Q.10. Which type of bond is formed when the atoms have zero difference in electronegativity?

Ans- Covalent bond.

TWO MARKS QUESTIONS

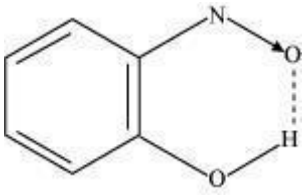
Q.1. Define hydrogen bond. Is it weaker or stronger than the van der Waals forces?

Ans- A hydrogen bond is defined as an attractive force acting between the hydrogen attached to an electronegative atom of one molecule and an electronegative atom of a different molecule (may be of the same kind).

There are two types of H-bonds:

4. Intermolecular H-bond e.g., HF, H_2O etc.

5. Intramolecular H-bond e.g., o-nitrophenol



Hydrogen bonds are stronger than Vander Waals forces since hydrogen bonds are regarded as an extreme form of dipole-dipole interaction.

Q.2. Write the favourable factors for the formation of ionic bond.

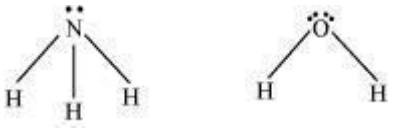
Ans-(i) Low ionization enthalpy of metal atom.

3. High electron gain enthalpy ($\Delta_{eg}H$) of a non-metal atom.

4. High lattice energy of the compound formed.

Q.3. Although geometries of NH_3 and H_2O molecules are distorted tetrahedral, bond angle in water is less than that of ammonia. Discuss.

Ans- The molecular geometry of NH_3 and H_2O can be shown as:



The central atom (N) in NH_3 has one lone pair and there are three bond pairs.
In H_2O, there are two lone pairs and two bond pairs.

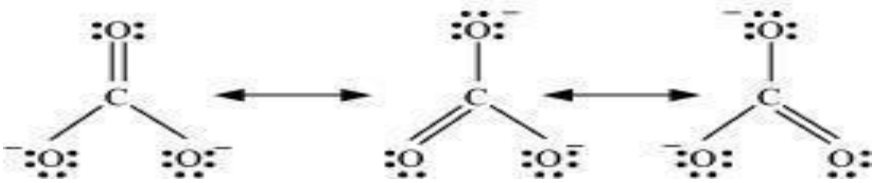
The two lone pairs present in the oxygen atom of H_2O molecule repel the two bond pairs. This repulsion is stronger than the repulsion between the lone pair and the three bond pairs on the nitrogen atom.

Since the repulsions on the bond pairs in H_2O molecule are greater than that in NH_3, the bond angle in water is less than that of ammonia.

Q.4. Explain the important aspects of resonance with reference to the CO_3^{2-} ion.

Ans- According to experimental findings, all carbon to oxygen bonds in CO_3^{2-} are equivalent. Hence, it is inadequate to represent CO_3^{2-} ion by a single Lewis structure having two single bonds and one double bond.

Therefore, carbonate ion is described as a resonance hybrid of the following structures:



Q.5. H_3PO_3 can be represented by structures 1 and 2 shown below. Can these two structures be taken as the canonical forms of the resonance hybrid representing H_3PO_3? If not, give reasons for the same.



(1) (2)

Ans- The given structures cannot be taken as the canonical forms of the resonance hybrid of H_3PO_3 because the positions of the atoms have changed.

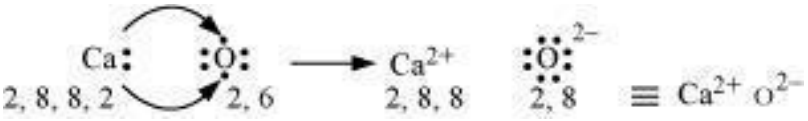
Q.6. Use Lewis symbols to show electron transfer between the following atoms to form cations and anions: (a) Ca and O (c) Al and N.

Ans**(a)**<u>Ca and O:</u>

The electronic configurations of Ca and O are as follows:

Ca: 2, 8, 8, 2 O: 2, 6

Oxygen requires two electrons more to complete its octet, whereas calcium has two electrons more than the nearest noble gas i.e., Argon. Hence, the electron transfer takes place as:

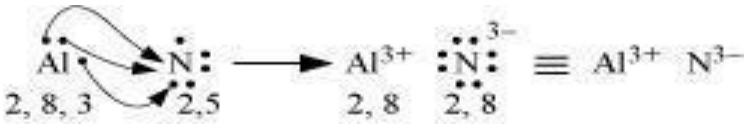


(b) <u>Al and N:</u>

The electronic configurations of Al and N are as follows:

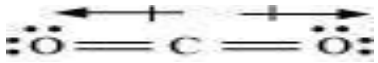
Al: 2, 8, 3 N: 2, 5

Nitrogen is three electrons short of the nearest noble gas (Neon), whereas aluminium has three electrons more than Neon. Hence, the electron transference can be shown as:



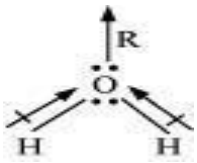
Q.7. Although both CO_2 and H_2O are triatomic molecules, the shape of H_2O molecule is bent while that of CO_2 is linear. Explain this on the basis of dipole moment.

Ans- According to experimental results, the dipole moment of carbon dioxide is zero. This is possible only if the molecule is linear so that the dipole moments of C– O bonds are equal and opposite to nullify each other.



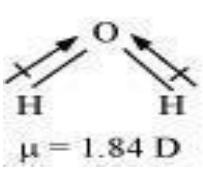
Resultant $\mu = 0$ D

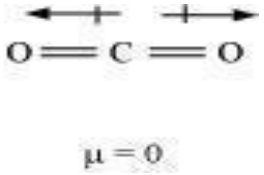
H_2O, on the other hand, has a dipole moment value of 1.84 D (though it is a triatomic molecule as CO_2). The value of the dipole moment suggests that the structure of H_2O molecule is bent where the dipole moment of O–H bonds are unequal.



Q.8. Write the significance/applications of dipole moment.

Ans- Dipole moment is the measure of the polarity of a bond. It is used to differentiate between polar and non-polar bonds since all non-polar molecules (e.g. H_2, O_2) have zero dipole moments. It is also helpful in calculating the percentage ionic character of a molecule.





Q.9. Use molecular orbital theory to explain why the Be_2 molecule does not exist.

Ans- The electronic configuration of Beryllium is $1s^2 2s^2$.

The molecular orbital electronic configuration for Be_2 molecule can be written as:

$\sigma_{1s}^2 \; \sigma_{1s}^{*2} \; \sigma_{2s}^2 \; \sigma_{2s}^{*2}$ Hence, the bond order for Be_2 is $\frac{1}{2}(N_b - N_a)$.

Where,

N_b= Number of electrons in bonding orbitals
N_a= Number of electrons in anti-bonding orbitals

$\therefore$ Bond order of Be_2 $= \frac{1}{2}(4-4) = 0$

A negative or zero bond order means that the molecule is unstable. Hence, Be_2 molecule does not exist.

Q.10. Distinguish between a sigma and a pi bond.

Ans- The following are the differences between sigma and pi-bonds:

Sigma (σ) Bond	**Pi (π) Bond**
(a) It is formed by the end to end overlap of orbitals.	It is formed by the lateral overlap of orbitals.
(b) The orbitals involved in the overlapping are *s–s*, *s–p*, or *p–p*.	These bonds are formed by the overlap of *p–p*orbitals only.
(c) It is a strong bond.	It is weak bond.
(d) The electron cloud is symmetrical about the line joining the two nuclei.	The electron cloud is not symmetrical.
(e) It consists of one electron cloud, which is symmetrical about the internuclear axis.	There are two electron clouds lying above and below the plane of the atomic nuclei.
(f) Free rotation about σ bonds is possible.	Rotation is restricted in case of pi-bonds.

Q.11. Explain with the help of suitable example polar covalent bond.

Ans- When two dissimilar atoms having different electronegativities combine to form a covalent bond, the bond pair of electrons is not shared equally. The bond pair shifts towards the nucleus of the atom having greater electronegativity. As a result, electron distribution gets distorted and the electron cloud is displaced towards the electronegative atom.

As a result, the electronegative atom becomes slightly negatively charged while the other atom becomes slightly positively charged. Thus, opposite poles are developed in the molecule and this type of a bond is called a polar covalent bond.

HCl, for example, contains a polar covalent bond. Chlorine atom is more electronegative than hydrogen atom. Hence, the bond pair lies towards chlorine and therefore, it acquires a partial negative charge.



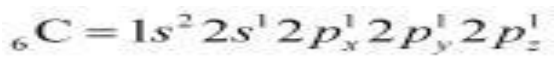
THREE MARKS QUESTIONS

Q.1. Write Lewis dot symbols for atoms of the following elements: Mg, Na, B, O, N, Br.

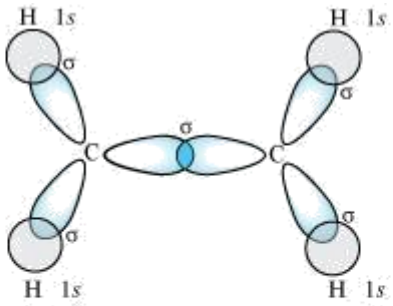
Ans- Mg: Na.  :   

Q.3. Draw diagrams showing the formation of a double bond and a triple bond between carbon atoms in C_2H_4 and C_2H_2 molecules.

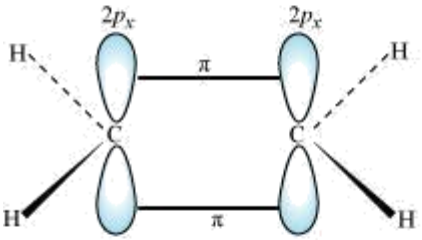
Ans- **C_2H_4:**The electronic configuration of C-atom in the excited state is:



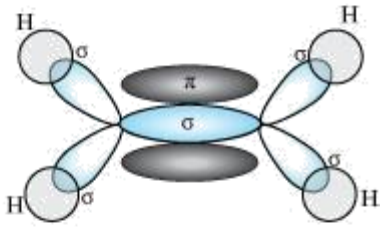
In the formation of an ethane molecule (C_2H_4), one sp^2 hybrid orbital of carbon overlaps a sp^2hybridized orbital of another carbon atom, thereby forming a C-C sigma bond. The remaining two sp^2 orbitals of each carbon atom form a sp^2-*s* sigma bond with two hydrogen atoms. The unhybridized orbital of one carbon atom undergoes sidewise overlap with the orbital of a similar kind present on another carbon atom to form a weak π-bond.



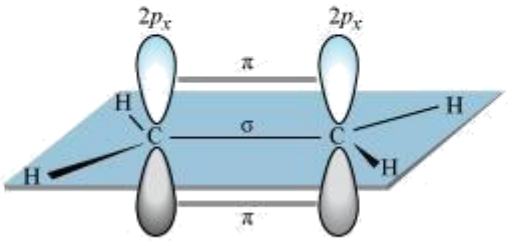
(a)



(b)



(c)



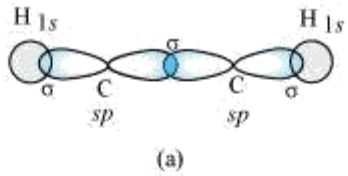
(d)

C_2H_2 :

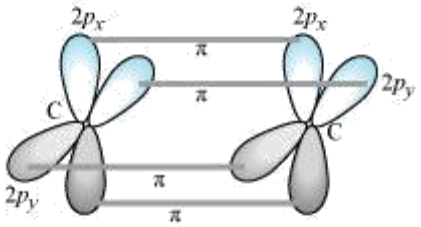
In the formation of C_2H_2 molecule, each C–atom is *sp* hybridized with two 2*p*-orbitals in an unhybridized state.

One *sp* orbital of each carbon atom overlaps with the other along the internuclear axis forming a C–C sigma bond. The second *sp* orbital of each C–atom overlaps a half-filled 1*s*-orbital to form a σ bond.

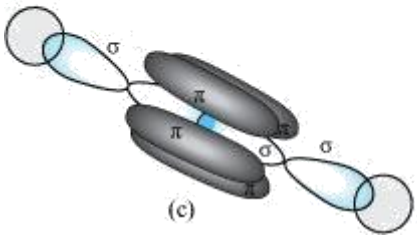
The two unhybridized 2*p*-orbitals of the first carbon undergo sidewise overlap with the 2*p* orbital of another carbon atom, thereby forming two pi (π) bonds between carbon atoms. Hence, the triple bond between two carbon atoms is made up of one sigma and two π-bonds.



(a)



(b)



Q.4. Explain the formation of H_2 molecule on the basis of valence bond theory.

Ans- Let us assume that two hydrogen atoms (A and B) with nuclei (N_A and N_B) and electrons (e_A and e_B) are taken to undergo a reaction to form a hydrogen molecule. When A and B are at a large distance, there is no interaction between them. As they begin to approach each other, the attractive and repulsive forces start operating.

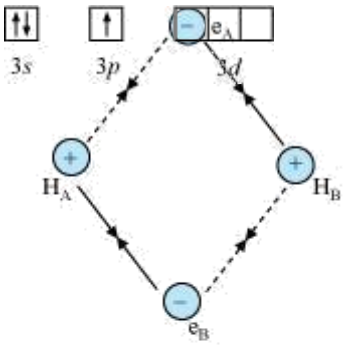
Attractive force arises between:

8. Nucleus of one atom and its own electron i.e., $N_A - e_A$ and $N_B - e_B$.
9. Nucleus of one atom and electron of another atom i.e., $N_A - e_B$ and $N_B - e_A$.

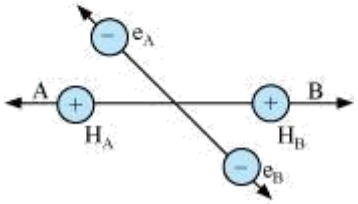
Repulsive force arises between:

5. Electrons of two atoms i.e., $e_A - e_B$.
6. Nuclei of two atoms i.e., $N_A - N_B$.

The force of attraction brings the two atoms together, whereas the force of repulsion tends to push them apart.



Attractive Forces



Repulsive Forces

The magnitude of the attractive forces is more than that of the repulsive forces.Hence, the two atoms approach each other. As a result, the potential energy decreases. Finally, a state is reached when the attractive forces balance the repulsive forces and the system acquires minimum energy. This leads to the formation of a dihydrogen molecule.

Q.5. Write the important conditions required for the linear combination of atomic orbitals to form molecular orbitals.

Ans- The given conditions should be satisfied by atomic orbitals to form molecular orbitals:

3. The combining atomic orbitals must have the same or nearly the same energy. This means that in a homonuclear molecule, the 1*s*-atomic orbital of an atom can combine with the 1*s*-atomic orbital of another atom, and not with the 2*s*-orbital.

4. The combining atomic orbitals must have proper orientations to ensure that the overlap is maximum.

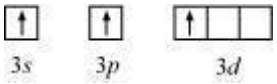
5. The extent of overlapping should be large.

Q.6. Describe the hybridisation in case of PCl_5. Why are the axial bonds longer as compared to equatorial bonds?

Ans- The ground state and excited state outer electronic configurations of phosphorus (Z = 15) are:

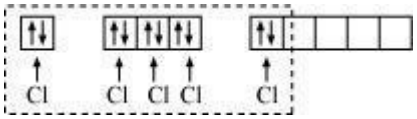
Ground state:

Excited state:

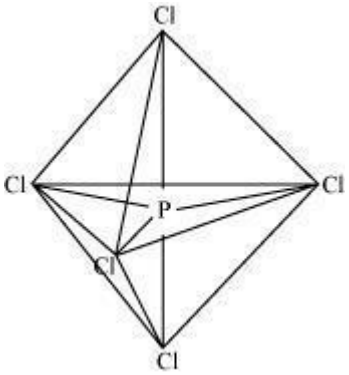


Phosphorus atom is sp^3d hybridized in the excited state. These orbitals are filled by the electron pairs donated by five Cl atoms as:

PCl_5



The five sp^3d hybrid orbitals are directed towards the five corners of the trigonal bipyramidals. Hence, the geometry of PCl_5 can be represented as:



There are five P–Cl sigma bonds in PCl_5. Three P–Cl bonds lie in one plane and make an angle of 120° with each other. These bonds are called equatorial bonds. The remaining two P–Cl bonds lie above and below the equatorial plane and make an angle of 90° with the plane. These bonds are called axial bonds.

As the axial bond pairs suffer more repulsion from the equatorial bond pairs, axial bonds are slightly longer than equatorial bonds.

Q.7. What is meant by the term bond order? Calculate the bond order of:

N_2, O_2, O_2^+ and O_2^-.

Ans- Bond order is defined as one half of the difference between the number of electrons present in the bonding and anti-bonding orbitals of a molecule.

Bond order = $\frac{1}{2}(N_b - N_a)$

Bond order of N_2

$$[\sigma(1s)]^2[\sigma^*(1s)]^2[\sigma(2s)]^2[\sigma^*(2s)]^2[\pi(2p_x)]^2[\pi(2p_y)]^2[\sigma(2p_z)]^2$$

Number of bonding electrons, $N_b = 10$

Number of anti-bonding electrons, $N_a = 4$

Bond order of nitrogen molecule $= \frac{1}{2}(10-4) = 3$

Bond order of O_2

$$[\sigma-(1s)]^2[\sigma^*(1s)]^2[\sigma(2s)]^2[\sigma^*(2s)]^2[\sigma(1p_z)]^2[\pi(2p_x)]^2[\pi(2p_y)]^2[\pi^*(2p_x)]^1[\pi^*(2p_y)]^1$$

Bond order $= \frac{1}{2}(N_b - N_a) \quad = \frac{1}{2}(8-4) = 2$

Hence, the bond order of oxygen molecule is 2.

Similarly, the electronic configuration of O_2^+ can be written as:

$$KK[\sigma(2s)]^2[\sigma^*(2s)]^2[\sigma(2p_z)]^2[\pi(2p_x)]^2[\pi(2p_y)]^2[\pi^*(2p_x)]^1$$

Bond order of $O_2^+ = \frac{1}{2}(8-3) = 2.5$

The electronic configuration of O_2^- ion will be:

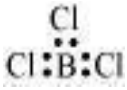
$$KK[\sigma(2s)]^2[\sigma^*(2s)]^2[\sigma(2p_z)]^2[\pi(2p_x)]^2[\pi(2p_y)]^2[\pi^*(2p_x)]^2[\pi^*(2p_y)]^1$$

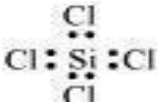
Bond order of $O_2^- = \frac{1}{2}(8-5) = 1.5$

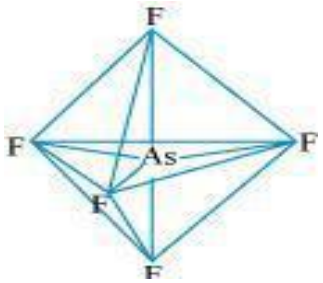
Q.8. Discuss the shape of the following molecules using the VSEPR model:

$BeCl_2$, BCl_3, $SiCl_4$, AsF_5, H_2S, PH_3

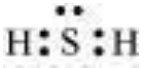
Ans- $BeCl_2$: Cl : Be : Cl The central atom has no lone pair and there are two bond pairs. i.e., $BeCl_2$ is of the type AB_2. Hence, it has a linear shape.

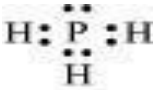
$\underline{BCl_3}$:  The central atom has no lone pair and there are three bond pairs. Hence, it is of the type AB_3. Hence, it is trigonal planar. 

$\underline{SiCl_4}$:  The central atom has no lone pair and there are four bond pairs. Hence, the shape of $SiCl_4$ is tetrahedral being the AB_4 type molecule.



$\underline{AsF_5}$: The central atom has no lone pair and there are five bond pairs. Hence, AsF_5 is of the type AB_5. Therefore, the shape is trigonal bipyramidal.

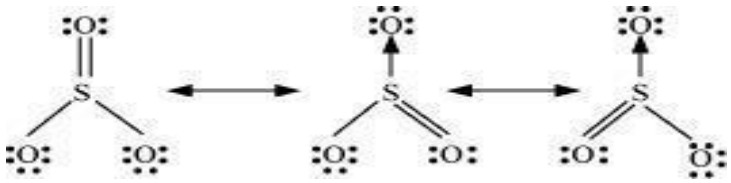
H_2S:  The central atom has one lone pair and there are two bond pairs. Hence, H_2S is of the type AB_2E. The shape is Bent.

$\underline{PH_3}$:  The central atom has one lone pair and there are three bond pairs. Hence, PH_3 is of the AB_3E type. Therefore, the shape is trigonal bipyramidal.

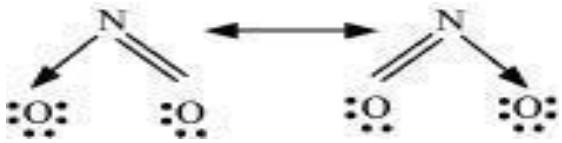
Q.9. Write the resonance structures for SO_3, NO_2 and NO_3^- .

Ans- The resonance structures are:

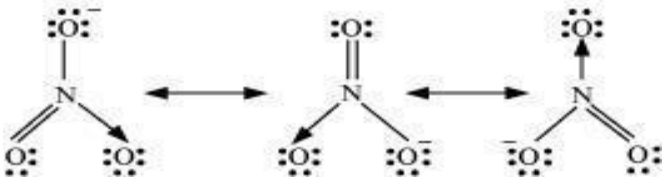
6. SO_3:



7. NO_2 :



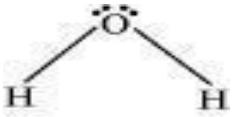
(c) NO_3^- :



Q.10. What do you understand by bond pairs and lone pairs of electrons? Illustrate by giving example.

Ans- The shared pairs of electrons present between the bonded atoms are called **bond pairs**. All valence electrons may not participate in bonding. The electron pairs that do not participate in bonding are called**lone pairs** of electrons.

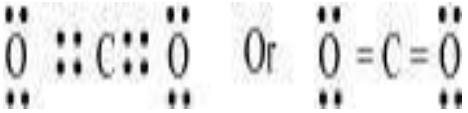
In H_2O, there are two bond pairs and two lone pairs on the central atom (oxygen).



FIVE MARKS QUESTIONS

Q.1. Define octet rule. Write its significance and limitations.

Ans-- The octet rule or the electronic theory of chemical bonding was developed by Kossel and Lewis. According to this rule, atoms can combine either by transfer of valence electrons from one atom to another or by sharing their valence electrons in order to attain the nearest noble gas configuration by having an octet in their valence shell.



The octet rule successfully explained the formation of chemical bonds depending upon the nature of the element.

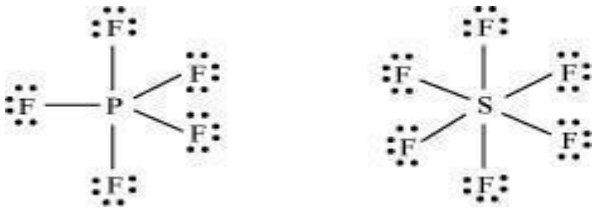
<u>Limitations of the octet theory:</u>

The following are the limitations of the octet rule:

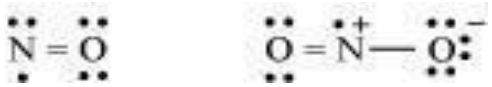
i)The rule failed to predict the shape and relative stability of molecules.

ii)It is based upon the inert nature of noble gases. However, some noble gases like xenon and krypton form compounds such as XeF_2, KrF_2 etc.

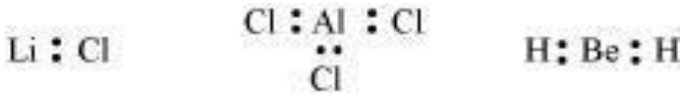
iii)The octet rule cannot be applied to the elements in and beyond the third period of the periodic table. The elements present in these periods have more than eight valence electrons around the central atom. For example: PF_5, SF_6, etc.



iv)The octet rule is not satisfied for all atoms in a molecule having an odd number of electrons. For example, NO and NO_2 do not satisfy the octet rule.



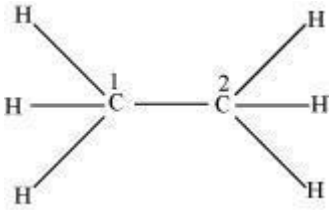
v)This rule cannot be applied to those compounds in which the number of electrons surrounding the central atom is less than eight. For example, LiCl, BeH_2, $AlCl_3$ etc. do not obey the octet rule.



Q.2. Which hybrid orbitals are used by carbon atoms in the following molecules?

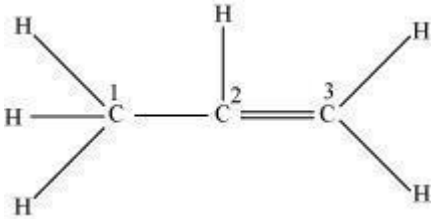
$CH_3–CH_3$; (b) $CH_3–CH=CH_2$; (c) CH_3-CH_2-OH; (d) CH_3-CHO (e) CH_3COOH

Ans- **(a)**



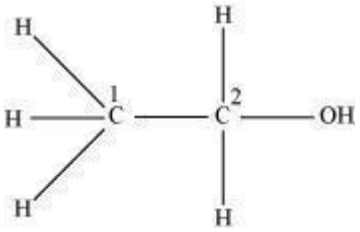
Both C_1 and C_2 are sp^3 hybridized.

(b)



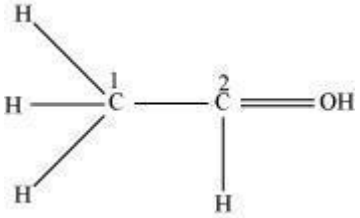
C_1 is sp^3 hybridized, while C_2 and C_3 are sp^2 hybridized.

(c)



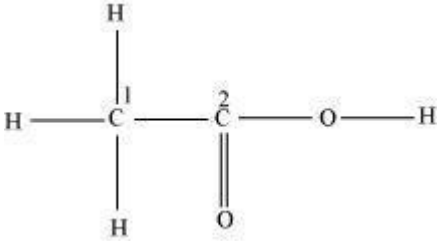
Both C_1 and C_2 are sp^3 hybridized.

(d)



C_1 is sp^3 hybridized and C_2 is sp^2 hybridized.

(e)



C_1 is sp^3 hybridized and C_2 is sp^2 hybridized.

Q.3. Compare the relative stability of the following species and indicate their magnetic properties;

O_2, O_2^+, O_2^- (superoxide), O_2^{2-} (peroxide)

Ans- There are 16 electrons in a molecule of dioxygen, 8 from each oxygen atom. The electronic configuration of oxygen molecule can be written as:

$$[\sigma-(1s)]^2[\sigma^*(1s)]^2[\sigma(2s)]^2[\sigma^*(2s)]^2[\sigma(1p_z)]^2[\pi(2p_x)]^2[\pi(2p_y)]^2[\pi^*(2p_x)]^1[\pi^*(2p_y)]^1$$

Since the 1*s* orbital of each oxygen atom is not involved in boding, the number of bonding electrons = 8 = N_b and the number of anti-bonding orbitals = 4 = N_a.

Bond order $= \frac{1}{2}(N_b - N_a)$

$= \frac{1}{2}(8-4)$

= 2

Similarly, the electronic configuration of O_2^+ can be written as:

$$KK[\sigma(2s)]^2[\sigma^*(2s)]^2[\sigma(2p_z)]^2[\pi(2p_x)]^2[\pi(2p_y)]^2[\pi^*(2p_x)]^1$$

N_b= 8

N_a= 3

Bond order of $O_2^+ = \frac{1}{2}(8-3)$

= 2.5

Electronic configuration of O_2^- ion will be:

$$KK[\sigma(2s)]^2[\sigma^*(2s)]^2[\sigma(2p_z)]^2[\pi(2p_x)]^2[\pi(2p_y)]^2[\pi^*(2p_x)]^2[\pi^*(2p_y)]^1$$

N_b= 8

N_a= 5

Bond order of $O_2^- = \frac{1}{2}(8-5)$

= 1.5

Electronic configuration of O_2^{2-} ion will be:

$$KK[\sigma(2s)]^2[\sigma^*(2s)]^2[\sigma(2p_z)]^2[\pi(2p_x)]^2[\pi(2p_y)]^2[\pi^*(2p_x)]^2[\pi^*(2p_y)]^2$$

N_b= 8

$N_a = 6$

Bond order of $O_2^{2-} = \frac{1}{2}(8-6)$

= 1

Bond dissociation energy is directly proportional to bond order. Thus, the higher the bond order, the greater will be the stability. On this basis, the order of stability is

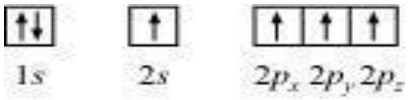
$$O_2^+ > O_2 > O_2^- > O_2^{2-}$$.

HOTS

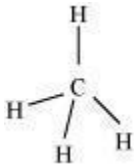
Q.1. Apart from tetrahedral geometry, another possible geometry for CH_4 is square planar with the four H atoms at the corners of the square and the C atom at its centre. Explain why CH_4 is not square planar?

Ans- Electronic configuration of carbon atom: $_6C: 1s^2 2s^2 2p^2$

In the excited state, the orbital picture of carbon can be represented as:



Hence, carbon atom undergoes sp^3 hybridization in CH_4 molecule and takes a tetrahedral shape.



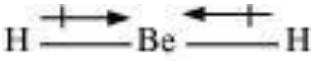
For a square planar shape, the hybridization of the central atom has to be dsp^2. However, an atom of carbon does not have *d*-orbitalsto undergo dsp^2 hybridization. Hence, the structure of CH_4 cannot be square planar.

Moreover, with a bond angle of 90° in square planar, the stability of CH_4 will be very less because of the repulsion existing between the bond pairs. Hence, VSEPR theory also supports a tetrahedral structure for CH_4.

Q.2. Explain why BeH_2 molecule has a zero dipole moment although the Be–H bonds are polar.

Ans- The Lewis structure for BeH_2 is as follows: H : Be : H

There is no lone pair at the central atom (Be) and there are two bond pairs. Hence, BeH_2 is of the type AB_2. It has a linear structure.

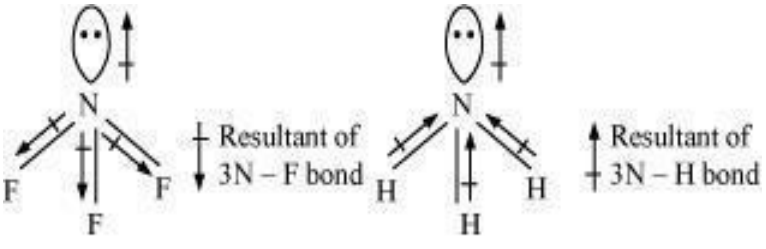


Dipole moments of each H–Be bond are equal and are in opposite directions. Therefore, they nullify each other. Hence, BeH_2 molecule has zero dipole moment.

Q.3. Which out of NH_3 and NF_3 has higher dipole moment and why?

Ans- In both molecules i.e., NH_3 and NF_3, the central atom (N) has a lone pair electron and there are three bond pairs. Hence, both molecules have a pyramidal shape. Since fluorine is more electronegative than hydrogen, it is expected that the net dipole moment of NF_3 is greater than NH_3. However, the net dipole moment of NH_3 (1.46 D) is greater than that of NF_3 (0.24 D).

This can be explained on the basis of the directions of the dipole moments of each individual bond in NF_3 and NH_3. These directions can be shown as:



Thus, the resultant moment of the N–H bonds add up to the bond moment of the lone pair (the two being in the same direction), whereas that of the three N – F bonds partly cancels the moment of the lone pair.

Hence, the net dipole moment of NF_3 is less than that of NH_3.

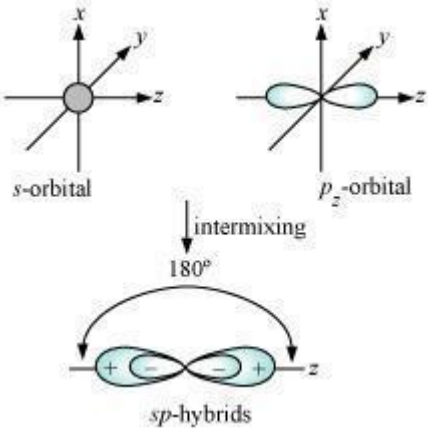
Q.4. What is meant by hybridisation of atomic orbitals? Describe the shapes of *sp*, sp^2, sp^3 hybrid orbitals.

Ans- Hybridization is defined as an intermixing of a set of atomic orbitals of slightly different energies, thereby forming a new set of orbitals having equivalent energies and shapes.

For example, one 2*s*-orbital hybridizes with two 2*p*-orbitals of carbon to form three new sp^2 hybrid orbitals.

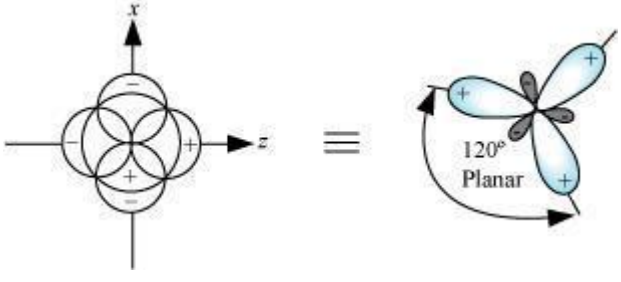
These hybrid orbitals have minimum repulsion between their electron pairs and thus, are more stable. Hybridization helps indicate the geometry of the molecule.

Shape of *sp* hybrid orbitals: *sp*hybrid orbitals have a linear shape. They areformed by the intermixing of *s* and *p* orbitals as:



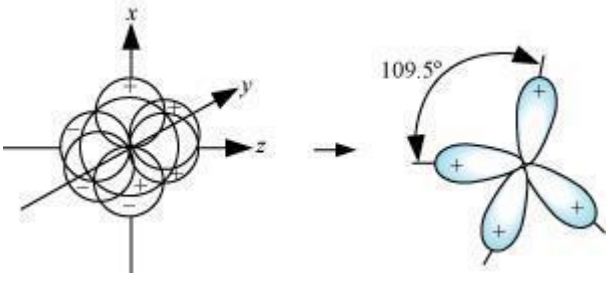
Shape of sp^2 hybrid orbitals:

sp^2hybrid orbitals are formed as a result of the intermixing of one *s*-orbital and two2*p*-orbitals. The hybrid orbitals are oriented in a trigonal planar arrangement as:



Shape of sp^3 hybrid orbitals:

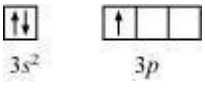
Four sp^3 hybrid orbitals are formed by intermixing one s-orbital with three *p*-orbitals. The four sp^3 hybrid orbitals are arranged in the form of a tetrahedron as:



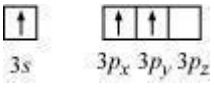
Q.5. Describe the change in hybridisation (if any) of the Al atom in the following reaction.

$$AlCl_3 + Cl^- \longrightarrow AlCl_4^-$$

Ans- The valence orbital picture of aluminium in the ground state can be represented as:



The orbital picture of aluminium in the excited state can be represented as:



Hence, it undergoes sp^2 hybridization to give a trigonal planar arrangement (in $AlCl_3$).

To form $AlCl_4^-$, the empty $3p_z$ orbital also gets involved and the hybridization changes from sp^2 to sp^3. As a result, the shape gets changed to tetrahedral.

Value Based Questions

Q1 There are large number of compounds used in the household besides a large number of others which we use in our everyday life. From the point of view of chemistry, these compounds are either ionic or covalent . quite often,

these compounds are given some common names instead of their chemical names.

Now answer the following questions :

i) Give one important use of each of the following compounds :
Magnesium hydroxide, magnesium sulphate, potassium nitrate, sodium carbonate.
ii) Which compound of Li is used for treatment of manic depression.
iii) Which compound of Zn is present in calamine lotion used for protection of skin.
iv)Which acid is present in the orange juic e & which acid is present in batteries.
v) Which compound of sodium is used as a a) bleach b) as drain cleaner c) in tooth paste.

A1) i) Magnesium hydroxide - milk of magnesia an antacid
magnesium sulphate - laxative & anti convulsant.
potassium nitrate - igniter in fireworks & matchstics.
sodium carbonate - as washing soda.

ii) lithium carbonate is used for treatment of manic depression (which is due to over anxiety & excitement)

iii) Zinc oxide.

iv) Citric acid & sulphuric acid.

v) a) Sodium hypochlorite.
b) Sodium hydroxide.
c) sodium flouride.

Q2) A large number of compounds contain molecules in which hydrogen atom is linked to oxygen, nitrogen or 85elocali.Such molecules undergo intermolecular H-bonding. As a result, they possess special properties which have a great significance in nature & have helped the living organisms in their survival.

Now answer the following questions :

i) Why ice floats on water, how has it helped the aquatic life,
ii) What is the rolewof hydrogen bonding in plants.
iii) What is the rolewof hydrogen bonding in protiens & DNA.

A2) I) H- bonding in water molecules of ice creates a lot of vacant spaces. As, a result density of ice is less than that of water . hence, it floats on the surface of water .In winters, when the temperature is very low, generally upper layer of water in the lake freezes. The aquatic life remains protected in the water underneath.

(ii) in the vascular plants, the movement of water through the plant;s xylem and other vessels takes place through hydrogen bonding.water molecules are not only hydrogen bonded to each other but also to the cellulose cell comprising the wall of the plant cells. As a result, capillary action is set in which allowsplant to pull up water into the roots. Furthermore, hydrogen bonding can create a long chain of water molecules which can overcome the force of gravity and travel upto the high altitude of the plants

(iii) the secondary structure of the protein involves interaction(mainly hydrogen bond) between the neighbouring polypeptide backbones which contain N-H bonded pairs and O- atoms. Consequently, H-atom bonded to N in one polypeptide backbone can form H-atom with O-atom in another chain and vice-versa. Though they are relatively weak, these bonds offer a great staibility to secondary protein structure because they repeata great number of times.hydrogen bonding plays an important role in DNA base pairing

Q.3. Metallic bonds are the chemical bonds that hold together in metals. They differ from covalent and ionic bonds because the electrons in the metallic bonding is delocalized, i.e., they are not shared between only two atoms. Instead, the electrons in metallic bonds float freely through the lattice of the metal nuclei. This type of bonding gives metals many unique material properties including excellent thermal and electrical conductivity, high melting melting points, malleability and ductility.

Now answer the following questions :

(i) Which metal is used in making the filament of electric bulbs and why?
(ii) Which metal is generally used for making target to be hit by cathode rays in an X- rays tube and why?
(iii) Why metals are used in the form of their alloys?
(iv) Why steel vessels with copper base/bottom are preferred over steel vessels for the heating on the flame?
(v) Iron is cheaper than copper. Then why copper wire are used in the electric wires but not iron wires?

(vi) Which alloy is used in making heating elements and why?

A3. (i) Tungsten is used because it has very high melting point.

(i) Mollebdenumis generally used because it is a hard metal.

(ii) Pure metals may not have the desired properties. By alloying them with a metal or a non-metal, we can have the desred properties depending upon the use.

(iii) Copper has excellent thermal conductivity as compared to steel. Hence, steel vessels with copper bottom are preferred for heating on the flame.

(iv) Copper has a very high electrical conductivity as compared to iron.hence copper wires are preffered over iron wires.

(v) Nicrome alloy (80% Ni, 20-% Cr) is used for making heating elements because it is very resistant to oxidation.

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CHAPTER 5

STATES OF MATTER

1. **Change in state** : It is over all effect of Intermolecularforces, molecular Interactional energy & thermal energy:
2. **Measurable properties of gases** : P,V, T, n, Viscosity,specific heat are some measurable properties.
3. **Gas Laws** : The quantitative relationship b/w any two ofthe variables (V, n, P,T) when other two are constant.
4. Boyle's Law : The pressure of fixed msss of gas varies inversely with the volume at constant T. P α 1/V(n,T const.)

   $P_1V_1 = P_2 V_2$
5. **Charle's Law** : At constant P, the volume of fixed amountof gas varies directly with its absolute temperature.

   V α T or $\frac{V}{T} = constant, \quad \frac{V_1}{T_1} \frac{V_2}{T_2}$ —
6. **Gay lussac's Law :** At constant V, The pressure of fixedamount of gas varies directly with its absolute temperature.

   P α T or $\frac{P}{T} = const, \quad \frac{P_1}{T_1} \frac{P_2}{T_2}$ —
7. Ideal gas equation : The relationship b/w P, V and T by Gas Laws PV= nRT.
8. Avogadro's Law: At given T and P, the volume of gas varies directly to the amount of gas . V α n (P, T constant)
9. Dalton's Law of partial persure : The pressure enerted by a mixture of non reacting gases is equal to the sum of their partial pressure at constant (V,T)

P (total) = P1 + P2 + P3 + ……. (T, V, constant)

10. Kinetic Molecular theory :

a. Gases consist of large number of identical particles (atoms or molecules) that are so small that the actual volume of the molecules is negligible in comparison to the empty space between them.
b. There is no force of attraction between the particles of a gas at ordinary temperature and pressure
c. Particles of a gas are always in constant and random motion
d. Pressure is exerted by the gas as a result of collision of the particles with the walls of the container
e. Collisions of gas molecules are perfectly elastic
f. At any particular time, different particles in the gas have different speeds and hence different kinetic energies
g. Average kinetic energy of the gas molecules is directly proportional to the absolute temperature

11. Real Gases : The gases which deviates from Ideal behavior at higher pressure and low temperature because of force of attraction between molecules increases .

12. Compressibility factor (Z) : It determine extent of devation of real gases from Ideal gas behavior : Z = $\frac{PV}{n.R.T.}$ for ideal gas Z=1, for Nonideal gas Z< 1, Z > 1

13. Vander waal's Equation :

$\left(P + \frac{n^2 a}{v^2}\right)(V - nb) = nRT$

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13. Critical Temperature : (Tc) The temperature above which a gas cannot be liquefied whatever high pressure may be

14. Critical Pressure : The minimum pressure required to liquity a gas at its critical temperature.

15. Critical Volume : The volume of 1 mole of gat at Tc, Pc.

16. Super cooled liquids: The liquids which are cooled to a temperature below its freezing point without freezing .

17. Elastic Collision : The collisions in which no loss of K.E. only there is transfer of energy.

18. Vapour pressure : The equilibrium pressure by vapour of liquid in a container at given temperature (T)

19. At higher altitude : The b.p. of water decreases b/c the atmospheric pressure is less than one atmosphere.

20. Surface Tension (V) : It is force acting per unit length perpendicular to the line drawn on the surface : (Nm^{-1}) : It decreases with increases in T, it increases with increase in external pressure, b/c of it falling drops of liquid are spherical, liquid in capillary tube rises.

21. Viscosity (η) : It is resistance offered to the flow of liquid due to friction b/w layer of fluids . $F = \eta . A . \frac{dv}{dx}$

23. Effect of T & P on viscosity : It decreases with increase in T, and increases with increase in P.

24. Low M.P. & B.P. of molecular liquids is due to low magnitude of molecular interaction energy.

One mark questions

1.At what condition surface tension vanishes ?

2.Why Helium is used in balloons in place of hydrogen?

3.At what temperature below which a gas does not obey ideal gas law?

4.At what temperature the volume of a gas is supposed to be zero?

5What is the molar volume at $0^{o}C$ and 1 bar pressure?

6Name the temperature above which a gas cannot be liquefied by any amount of pressure.

7What is the effect of increase of temperature on surface tension and viscosity in a liquid?

8How is the partial pressure of a gas in a mixture related to the total pressure of the gaseous mixture?

9Why vegetables are cooked with difficulty at hill station.

10What is the value of Z (compressibility factor) for an Ideal gas?

Answers to One mark questions

Ans 1 At critical temperature, meniscus b/w liquid and vapours disappears.

Ans 2. He is incombustible, though heavier than H_2.

Ans 3. Below Boyle Temperature.

Ans 4. At absolute zero (O K) temperature.

Ans 5. It 22400 ml.

Ans 6. It is critical temperature (Tc)

Ans 7. Both decreases with increase in temperature

Ans 8. $P_1 = X_1 \times P_{(total)}$

Ans 9. At hill station the atmospheric pressure is less and so boiling point decreases.

Ans10. For ideal gas Z=1.

Two Marks Questions

Q.1 How do you convert pressure in atmosphere in to SI unit.

Q.2 What type graph will you get when PV is plotted against P at constant temperature.

Q.3 What would have happened to the gas if the molecular

collisions were not elastic?

Q.4 At a particular temperature, why vapour pressure of acetone is less than ether?

Q.5 Why liquids diffuse slowly as compared to gases?

Q.6 What would be the SI unit for quantity PV^2T^2/n ?

Q.7 In terms of Charle's law explain why -273° C is the lowest temperature?

Q.8 For real gases the relation b/w P,V,T is given by vander Waal's equation, write it for n moles?

Q.9 What correction is applied to obtain pressure of dry gas with the total pressure & aqueous tension?

Q.10 Name two phenomena that can be explained on the basis of surface tension.

Answers to two marks questions

Ans 1. 1atm = 101325 Pa or Nm^{-2}, 1 bar = 10^5 Pa.

Ans 2 .A straight line parallel to pressure axis.

Ans 3. On every collision there is loss of energy, so molecules would have slowed down & settled down in vessel and pressure reduce to O.

Ans 4. b/c molecular force of attraction in acetone is stronger than those present in ether.

Ans 5. In liquids the molecules are less free than in a gas. i. e intermolecular forces in liquid are greater than in gas.

Ans 6. $\dfrac{P.V^2.T^2}{n} \quad \dfrac{N.m^2(m^3)^2(K)^2}{mal} = N.m^4\ k^2\ mal^1$

Ans 7 At -273° C, the Volume of gas becomes equal to zero i. e. gas ceases to exist.

Ans 8. vander Waal's equation

$$\left(P + \frac{n^2 a}{v^2}\right)(V - nb) = \mathrm{nRT}$$

Ans 9. $P_{(drygas)} = P_{(Total)} -$ Aqueous tension

Ans 10. Surface tension can explain (i) capillary action. (ii) Spherical shape of small drops of liquid.

Three Marks Questions

Q.1 Calculate temperature of 4.0 moles of gas occupying 5dm^3 volume at 3.32 bar. (R= 0.083 bar dm^3 k^{-1} mol^{-1})

Ans $PV = nRT$, or $T = \dfrac{PV}{nR}$

Or $T = \dfrac{3.32\ bar\ x\ 5dm^3}{4.0mol\ x\ 0.083bar\ dm^3\ \ k^1\ mol^1}$ 50.*Ans*

Q.2 34.05 ml of phosphorus vapours weight 0.0625g at 543° C and 1 bar pressure what is molar mass of phosphorous?

Ans. $\dfrac{P_1V_1}{T_1} = \dfrac{P_2V_2}{T_2}, \quad \dfrac{1\ x\ 34.05}{(54b\ \ 273)} \quad \dfrac{1\ x\ v_2}{273}\ at\ 0^o\ C$

Or $V_2 =$ 11.35 ml

Now 11.35 ml of vapours at 0°C and 1 bar pressure weight0.0625 gm

Now 22700 ml of vapors at 0° C and 1 bar pressure weight$\frac{0.0625}{11.35}$ x 22700

Q.3A mixture of dihydrogen and dinitrogen at 1 bar pressure contains 20% by weight of H_2. Calculate partial pressure of H_2.

AnsThe mixture of H_2 and O_2 contain 20% H_2 means 20 gm H_2& 80gm O_2

$$n(H_2) = \frac{20}{2} = 10 moles,\ n(O_2) = \frac{80}{32} = 2.5 contest.$$

$$P(H_2) = \frac{n(H_2)}{n(H_2) + n(V_2)} \times P(total) = \frac{10}{10 + 2.5} \times 1 bar = 0.8 bar$$

Q.4

Ans.

Q.5

Ans

Q.6

What will be minimum pressure required compressing $500dm^3$ of air at 1 bar to 200 dm^3 at 30° C?

P_1 = 1bar. V_1 = $500dm^3$

P_2 = ? V_2 =200 dm^3& temp. constant 30^oC So by Boyle's low

$P_1V_1 = P_2 V_2$

$$1 \times 500 = P_2 \times 200 \qquad \text{or } P_2 \frac{500}{200} = 2.5\ bar$$

Calcutate the total pressure in mixture of 4g of O_2 and 2gm. Of H_2 can fixed to a total volume of IL at 0° C. (R = 0.0821) Latm. Mol^{-1}

$$P \quad no_{2=}nH_2 \quad x \frac{RT}{V}$$

$$= \quad \frac{4}{32} \quad \frac{2}{2} - x \frac{0.0821 \times 273}{1} \frac{1}{8} \quad - \quad 1 \quad 22.4 \quad 25.2\ atm.\ Ans$$

Account for the following :

(b) The size of weather balloon becomes longer and larger as it ascend up into higher altitudes.

Ans (i) At higher altitudes atmospheric pressure decreases, but inside the balloon gas exerts pressure and size becomes larger.

Q.8 What do you mean by ideal gas and real gas? Why do real gases deviate from ideal behaviors?

Ans. Ideal gas follows gas laws at all P & T

Real gas donot " gas laws at all P & T.

Real gas deviate from ideal behaviour due to force of attraction and because of volume of molecules of gases are not negligible.

Q.9 Using the equation of state PV=nRT Show that at given T, density of gas proportional to gas pressure P.

Ans $PV = n\,RT, \quad P = \frac{n}{V} . RT$

$$P = \frac{m}{M} \cdot \frac{1}{V} \cdot R.T . = \frac{m}{V} . \frac{RT}{M}$$

$$P = P \frac{RT}{M} \quad (Where\ P = \frac{m}{v})$$

$$P \times P \qquad \frac{RT}{M} = const.$$

Q.10 Critical temperature of CO_2 and CH_4 gases are 31.1^0C and -81.9^0C respectively. Which of these has strong intermolecular forces & why?

Ans CO_2 has strong vander Waal's forces of attraction than CH_4. because of greater polarity and high molecular wt.

Hots Questions with Answers.

Q.1 (a)Why aerated water bottles kept under water during summer?

(b) Which property of liquid is responsible for spherical shape of drop?

(c) Why is moist air lighter than dry air?

(d) Define aqueous tension?

(e) What is units of a and b which are vander waal's constant?

Ans (a) To reduce temperature so to reduce pressure, other wise bottle may burst.

(b) Surface Tension.

(c) Moist air has water vapours which lowers vapour density, so it is lighter.

(d) It is pressure of water vapours at given T.

(e)Unit of a in atm. L^2 mol^{-2}, b is L $mol^{-1.}$

Q.2 Why does sharpened edge becomes smooth on heating up tomelting point?

Ans On heating the glass it melts and take up rounded shape at edges which has minimum surface area. b/c of surface tension. Q.3 Arrange following in order of increasing density:

$d = \frac{PM}{RT}$, O_2 at 25^0 C, 1atm, O_2 at O^0C, 2 atm, O_2 at273^0 C, 1atm.

Ans $d = \frac{PM}{RT}$, R, M are constant, so d depends upon $\frac{P}{T}$ so at 25^0C,

1Atm $\frac{P}{T} = \frac{1}{298}$, At 273^oC, 1atm $\frac{P}{T} = \frac{1}{546}$ Hence increasing order of

density will be: O_2 at 273^0 C, 1atm < O_2 at 25^0 C, 2atm.

Q.4 An O_2 cylinder has 10L O_2 at 200 atm. It patient takes 0.50ml of O_2 at 1 atm in one breath at 37^0C. How many breaths are possible?

Ans 10L at 200 atm = ? L at 1atm $P_1V_1 = P_2V_2$ $200 \times 10 = 1 \times V_2$ or

V_2 = 2000 L, No of breathes = $\frac{\text{Total Volume}}{\text{Volume for 1breath}} = \frac{2000\ L}{0.5 \times 10^3\ L} = 4 \times 10^6$

Value Based Questions

Q1.Boyle's law describes the inversely proportional relationship between the pressure and volume of a gas if the temperature is kept constant within a closed system.A large number of activities which we carry out in our day to day life such as pumping of air into the tube or a balloon, drawing medicine into a syringe and then injecting into the patient, etc. or a number of phenomena which occur such as dying of fish if brought to the surface of water or popping of ears at altitude when we fly in plane, etc. are also consequences of Boyle's law.

Now answer the following questions:

(i)Why do we use compressed natural gas (CNG) in our cars?

(ii)Why does tyre tube of car becomes harder and harder when more and more air is filled into it?

(iii)Why does the liquid medicine rush into the syringe when the needle is immersed in it and we pull out the piston of the syringe?

(iv)Fish survive inside the water but die when brought out of surface of water? Explain

(v)Why do we feel uneasy pain in our ears when our plane is ascending or descending? What should be done to overcome this problem?

(vi)The bubbles exhaled by a scuba diver grow in size as they approach the surface of ocean. Why?

(vii)Why mountain climbers and our jawans guarding the country's frontiers at high altitudes in Ladakh carry oxygen cylinders with them?

Ans.

(i)CNG is under very high pressure and hence occupies a small volume to be comfortably adjusted into the fuel tank.

(ii)When we fill more and more air into the tube, no. of molecules of air inside the tube increases. Initially, there is increase in volume of the tube so that increase in pressure is not very high. Further increase in volume is small; hence increase of pressure is large. This makes the tube harder.

(iii)When we pull the piston outwards, the volume inside the syringe increases and hence pressure decreases. Now, as the pressure outside is greater than the pressure inside, it pushes the medicine into the syringe.

(iv)Fish that live in low depths survive under high pressure due to large volume of water above them.When brought to the surface of the ocean, pressure is reduced dramatically. As a result, volume of the gases increases in their bodies. This causes the rupture of cells & other biological structures.

(v)This is due to the imbalance of air pressure inside and outside the ear drums.When the plane starts to rise, it is going from an area of high pressure to which our ears are accustomed to an area of low pressure. As the inside pressure of the ears is now higher, we feel uneasy pain.

To overcome this problem, we should swallow hard to clear passage for the opening tube in the throat to allow excess pressure to escape.

(vi)Deep low in water, pressure is high, therefore, volume of bubbles is small. As they rise up, pressure decreases. Hence, volume of bubbles increases, i.e. they become large in size.

(vii)At high altitudes, the pressure of air decreases and so does its density. As a result, amount of oxygen in the inhaled air decreases.

Q2.Charles law states that if the volume of a gas is kept constant, the pressure of the gas is directly proportional to temp. This implies that in an ideal situation where the vol. does not change, if either the pressure or the temp. is increased, the other one will increase in the same proportion . This aspect has a no. of applications in our everyday life.

Now answer the following questions:

(i)When we use a pressure deodorant can for a few seconds, the can tends to become cooler.Why?

(ii)When the ping pong ball, a toy of the children, gets dented without being punctured, what do we do to regain the original shape and why?

(iii)Beer or soda cans and bottles have a label on them stating 'Store in cool,dry place'.Why?

(iv)Inflated football gets deflated in winter. Why?

(v)Why hot air is filled in the balloons for metrological observations?

Ans.(i)When we spray the can, the liquid spray of the can is released. Hence, pressure inside decreases a bit. Since the can's volume does not change, the temp. falls.

(ii)We place the dented ping pong ball in warm water. As temp.inside increases, the pressure inside increases. As a result, dent gets removed.

(iii)The reason is that these cans have a lot of artificial pressure stored in them. When exposed to direct heat, the pressure inside the cans rises because vol. remains constant. When the pressure exceeds the certain limit, the can explodes.

(iv) When the ball is brought outside, the temp. dropped. As a result, the pressure of the air inside the football also dropped making the ball look deflated.

(v)Air expands on heating. Hence, its density decreases. Thus, hot air is lighter than atmospheric air. This helps the balloon rise up.

Q3.Viscosity of a liquid is its resistance to flow. Some liquids like water, petrol, etc. flow very easily. They are said to be less viscous. On the other hand, some liquids like honey,

syrup, ketchup, etc. flow difficultly and are said to be highly viscous. The viscosity of liquids has an important role to play in a no. of items that we use.

Now, answer the following questions:

(i)Applying the concept of viscosity explain the occurrence of cardiac arrest. Why there are greater no. of cardiac arrest in winter than in summer?

(ii)What is the role of viscosity in a hydraulic jack used to lift a heavy load like car, etc.?

(iii)What is the role of viscosity in automatic door closers?

(iv)What is the role of viscosity in shock absorbers of vehicles?

(v)Glass is considered as highly viscous liquid. If it were not, what have happened?

Ans.

(i)Blood is a viscous liquid that flows through the blood vessels. When a person takes diet rich in fats, they get deposited in the arteries. In winter, usually people overeat especially ghee products. As a result, deposit of fats in the arteries increases. As a result, the arteries are narrowed and the flow of blood through them is difficult.

(ii)A hydraulic jack relies on the viscosity of the oil to control the rate of fall when lowering the load.

(iii)Automatic doors closers are often oil filled. They use the viscosity of the oil to control the speed of the door as it closes.

(iv)Shock absorbers use the viscosity of their oil contents damp-out the shock motion imparted by an uneven road surface.

(v)If glass were not as viscous as it actually is, windows would not have lasted for very long.

Q4.Surface tension is one of the most important and interesting properties of the liquids. It is the force acting downwards on the surface of the liquid which tends to pull the surface downwards so that it has minimum surface area. The value of surface tension of a liquid depends upon the intermolecular forces and hence on the nature of the liquid. When a liquid comes in contact with another liquid or some surface, the behavior observed depends upon the adhesive forces between them. This property of liquids has a no. of applications in daily life.

Now, answer the following questions:
(i) How the property of surface tension helps in the clinical test for jaundice?

(ii)Why washing of clothes and dishes is more effective with hot water than with cold water?

(iii)Why rain water drops falling on the waxed car smoothly slide down to the ground?

(iv)Why water sticks to a glass surface but mercury does not?

Ans.(i)Normal urine has a surface tension of about 66 dynes/cm but if bile is present (a test for jaundice), it drops to about 55 dynes/cm. In this test, sulphur is sprinkled on the urine surface. If it floats, the test is negative but if it sinks, it is positive.

(ii)Hot water is more effective because surface tension is lowered and it becomes a better wetting agent.

(iii)Waxing changes the surface properties of the car. Water tends to adhere weakly to the wax and strongly to itself.

(iv)In case of water taken on the glass surface the adhesive force between water and glass are greater than cohesive forces among water molecules but in case of mercury taken in glass tube, it is the reverse.

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CHAPTER 6
THERMODYNAMICS"

Brief Summary of the chapter:

1. Thermodynamics: Science which deals with study of different forms of energy and quantitative relationship.

2. System & Surroundings: The part of universe for study is called system and remaining portion is surroundings.

3. State of system & state function: State of system is described in terms of T, P, V etc. The property which depends only on state of system not upon path is called state function eg. P, V, T, E, H, S etc.

4. Extensive & Intensive Properties: Properties which depends on quantity of matter called extensive prop. eg. mass, volume, heat capacity, enthalpy, entropy etc. The properties which do not depends on matter present depends upon nature of substance called Intensive properties. eg. T,P, density, refractive index, viscosity, bp, pH, mole fraction etc.

5. Internal energy: The total energy with a system.

   i.e. $U = E_e + E_n + E_c + E_p + E_k$ + ------

   $\Delta U = U_2 - U_1$ or $U_P - U_R$& U is state function and extensive properly. If $U_1 > U_2$ energy is released.

6. Heat (q): It I a form of energy which is exchanged between system and

surrounding due to difference of temperature. Unit is Joule (J) or Calorie (1 Calorie = 4.18 J).

7. First Law of Thermodynamics: It is law of conservation energy. Energy can neither be created not destroyed, it may be converted from one from into another.

Mathematically $U = q + w$, $w = -p.\ V$ (work of expansion)

$U = q - p.\ V$ or $q = U + p.$

V, q,w are not state function.ButU is state function.

8. Enthalpy (H): At constant volume $V = 0$, $q_v = U$ So $\Delta H =$

$\Delta U + p.\ \Delta V$, $q_p = H_2\ H_1 = H$

$H = U + P.\ V.$

17. Relationship between q_p, q_v i.e. $\Delta H \& \Delta U$

It is $\Delta H = \Delta U + RT$ or $q_p = q_v + \Delta ng.RT$

10. Exothermic and Endothermic reactions: $\Delta H = -Ve$ for exothermic and $\Delta H = +Ve$ for endothermic reaction i.e. evolution and absorption of heat.

Eg

$C + O_2 \rightarrow CO_2 + 393.5$ KJ, $\Delta H = -393.5$ KJ (exothermic)

$N_2 + O_2 \rightleftharpoons 2NO - 180.7$ KJ, $\Delta H = 180.7$ KJ (Endothermic)

- Enthalpy of reaction ($\Delta_r H$): The amount of heat evolved or absorbed when the reaction is completed.
- Standard Enthalpy of reaction ($\Delta_r H^0$) at 1 bar pressure and specific temp. (290K) i.e. standard state.
- Different types of Enthalpies of reactions:

(i) Enthalpy of combustion ($_cH$), (ii) Enthalpy of formation ($_fH$)

(iii) Enthalpy of neutralization (iv) Enthalpy of solution

(v) Enthalpy of atomization($_aH$), (vi)Enthalpy of Ionisation ($_iH$)

(vii) Enthalpy of Hydration ($_{hyol.}H$) (viii) Enthalpy of fusion ($_{fus.}H$)

(ix) Enthalpy of vaporization ($_{vap.}H$) (x)Enthalpy of sublimation ($_{sub.}H$)

$(\Delta_{sub.}H) = \Delta_{fus.}(H) - \Delta_{vap}(H)$

- Hess's Law of constant heat summation: The total amount of heat change is same whether the reaction takes place in one step or in several steps.

  i.e. $\Delta H = \Delta H_1 + \Delta H_2 + \Delta H_3 +$ --------

  - Bond enthalpy: It is amount of energy released when gaseous atoms combines to form one mole of bonds between them or heat absorbed when one mole of bonds between them are broken to give free gaseous atoms.

    Further $\Delta_r H =$ B.E. (Reactants) - B.E. (Products)

  - Spontaneous & Non Spontaneous Processes: A process which can take place by itself is called spontaneous process. A process which can neither take place by itself or by initiation is called non Spontaneous.

- Driving forces for spontaneous process: (i) Tendency for minimum energy state. (ii) Tendency for maximum randomness.

- Entropy (S): It is measure of randomness or disorder of system. i.e. Gas>Liquid>Solid.

  Entropy change (S) = $\frac{q_{(rev.)}}{T}$ $Jk.K^{-1}.mol^{-1}$

  - Spontaneity in term of (S)

    $S_{(total)} = S_{(universe)} = S_{(system)} + S_{(surrounding)}$

If $S_{(total)}$ is +ve, the process is spontaneous.

If $S_{(total)}$ is –ve, the process is non spontaneous.

- Second Law of thermodynamics: In any spontaneous process, the entropy of the universe always increases. A spontaneous process cannot be reversed.

21. Gibb's free energy (G): defined as $G = H - T.S$ & $\Delta G = \Delta H - T\Delta S$ (Gibb's Helmholts equation) it is equal useful work i.e. - $G = W_{(useful)} = W_{(max.)}$

If G = -ve, process is spontaneous.

22. Effects of T on spontaneity of a process: $\Delta G = \Delta H - T.\Delta S$.

(i) For endothermic process may be non spontaneous at law temp.

For exothermic process may be non spontaneous at high temp. and spontaneous at law temp.

(f) Calculation of ($\Delta_r G^0$)

$$\Delta_r G^0 = \Sigma_f G^0_{(p)} - \Sigma_f G^0_{(r)}$$

(c) Relationship between ($\Delta_r G^0$) & equilibrium constant (k)

$$\Delta G = \Delta G^0 + RT ln Q \; \& \; \Delta G^0 = -2.303RT \log k.$$

25. Calculation of entropy change:

$$\Delta_r S^0 = \Sigma S^0_{(p)} - \Sigma S^0_{(r)}$$

ONE MARK QUESTIONS:

1. State First Law of thermodynamics.
2. What is a thermodynamic state function?
3. Give enthalpy (H) of all elements in their standard state.
4. From thermodynamic point to which system the animals and plants belong?

5. Predict the sign of S for the following reactions.

$CaCO_{3(s)} + CO_{2(g)} \xrightarrow{heat} CaO_{(s)} + CO_{2(g)}$

6. For the reaction $2Cl_{(g)} \rightarrow Cl_{2(g)}$, What will be the sign of ΔH and ΔS?

- State Hess's Law for constant heat summation?
- What is Gibb's Helmhaltz equation?
- Define extensive properties.

10. Give relationship between ΔH, ΔU for a reaction in gaseous state.

ANSWERS FOR ONE MARK QUESTIONS

1. Energy can neither be created nor destroyed. The energy of an isolated system is constant. $\Delta U = q + w$.
2. A function whose value is independent of path. eg. P, V, E, H
3. In standard state enthalpies of all elements is zero.
4. Open system.
5. ΔS is positive (entropy increases)
6. H: (–ve) b/c energy is released in bond formation and ΔS:

   (–ve) b/c atoms combines to form molecules.
7. The change of enthalpy of reaction remains same, whether the reaction is carried out in one step or several steps.
8. $\Delta G = \Delta H - T.\ \Delta S$
9. Properties which depends upon amount of substance called extensive properties. Volume, enthalpy, entropy.
10. $\Delta H = \Delta U + \Delta ng.\ RT$.

TWO MARKS QUESTIONS:-

Q.1 In a process, 701J heat is absorbed and 394J work is done by system. What is change in Internal energy for process?

Q.2 Given: $N_{2(g)} + 3H_{2(g)} \rightarrow 2NH3(g)$, $\Delta_r H^0 = -92.4 KJ.mol^{-1}$. What is the standard enthalpy of formation of $NH_{3(g)}$.

Q.3 Calculate entropy change in surroundings when 1.0 mol of $H_2O(l)$ is formed under standard conditions? Given $\Delta H^0 = -286 KJmol^{-1}$.

Q.4 Give relationship between entropy change and heat absorbed/evolved in a reversible reaction at temperature T.

Q.5 What is spontaneous change? Give one example.

Q.6 A real crystal has more entropy than an Ideal Crystal. Why?

Q.7 Under what condition, the heat evolved/absorbed in a reaction is equal to its free energy change?

Q.8 Predict the entropy change in-

(i) A liquid crystallizes into solid

(ii) Temperature of a crystallize solid raised from OK to 115K Q.9 What is bond energy? Why is it called enthalpy of atomization? Q.10 Calculate entropy change for the following process.

$H_2O_{(s)} = H_2O_{(l)}$, is 6.0 KJ mol^{-1} at 0^0C.

ANSWER FOR TWO MARKS QUESTIONS:

1. q = 701J, w = 394J, so $\Delta U = q + w = 701 - 394 = 307J$.

2. ${}_fH\ NH_{3(g)} = -\frac{92.4}{2} 46.2 KJ.mol^{1}$

7. $q_{(rev.)} = -\ H^0 = -286\ KJmol^{-1} = 286000 Jmol^{-1}$

$\Delta S \quad \frac{q_{(rev.)}}{T} \quad \frac{286000\ J.mol_{1}}{298\ K} \quad 959\ J.K^{1}mol^{1}$

4. $\Delta S = \frac{q_{(rev)}}{T}$

5. A process which can take place of its own or initiate under some condition. eg. Common salt dissolve in water of its own.

6. A real crystal has some disorder due to presence of defects in their structural arrangement, and Ideal crystal does not have any disorder.

7. In $\Delta G = \Delta H - T\Delta S$, when reaction is carried out at OK or $\Delta S = 0$, then $\Delta G = \Delta H$.

8. (i) Entropy decreases b/c molecules attain an ordered state.

(ii) entropy increase b/c from OK to 115K particles begin to move.

9. It is the amount of energy required to dissociate one mole of bonds present b/w atoms in gas phase. As molecules dissociates into atoms in gas phase so bond energy of diatomic molecules is called enthalpy of atomization.

10. $H_2O_{(s)}$ = $H_2O_{(l)}$ at 0^0C, $\Delta_{fus}H = 6KJ\ mol^{-1}$

$= 6000J\ mol^{-1}$

$T_f = 0^0C = 0 + 273 = 273K$

Do $\Delta_{fus}s = \frac{\Delta_{fus}H}{T} \quad \frac{6000\ J.mol^{1}}{273\ K} \quad 21.98J.K^{1}mol^{1}$

THREE MARKS QUESTIONS:

Q.1 For oxidation of iron, $4Fe_{(s)} + 3O_{2(g)} \rightarrow$ 2Fe2O3(s)

S is $-549.4J.K^{-1}\ mol^{-1}$, at 298K. Inspite of –ve entropy change of this reaction,

Why the reaction is spontaneous? ($_rH^0 = -1648x10^3\ J.mol^{-1}$)

Q.2 Using the bond energy of H_r = 435 KJ mol^{-1}, Br_2 = 192 KJ mol^{-1}, HBr = 368 $KJmol^{-1}$. Calculate enthalpy change for the reaction $H_{2(g)} + Br_{2(g)}2HBr_{(g)}$

Q.3 Enthalpies of formation of $CO_{(g)}$, $CO_{2(g)}$, $N_2O_{(g)}$ and $N_2O_{4(g)}$ and –110, –393, 81 and 9.7 KJ mol^{-1} respectively. Find value $_rH$ for the reaction $N_2O_{4(g)} + 3CO_{(g)}$

Q.4 For the reaction at 298K, 2A+B= C, $\Delta H = 400\ KJ\ mol^{-1}$, $\Delta S = 0.2\ KJ\ mol^{-1}\ K^{-1}$. At what temperature will the reaction become spontaneous, considering H, S be constant at the temp.

Q.5 The equilibrium constant for a reaction is 10. What will be the value of G^0? R $= 8.314J.K^{-1}mol^{-1}$ T = 300K.

Q.6 What do you understand by state function? Neither q nor w is a state function but q+w is a state function ? Explain

Q.7 Justify the following statements:

(i) An endothermic reaction is always thermodynamically spontaneous.

(ii) The entropy always increases on going from liquid to vapour state at any temperature T.

Q.8 Calculate the temperature above which the reduction reaction becomes spontaneous:

$PbO_{(s)} + C_{(s)}$ $Pb_{(s)} + CO_{(g)}$, given [$\Delta H = 108.4$ KJ mol^{-1}, $\Delta S = 190 J.K^{-1}$ mol^{-1}].

Q.9 From the data given below at 298K for the reaction:

$CH_{4(g)} + 2O_{2(g)}$ $CO_{2(g)} + 2H_2O_{(l)}$ Calculate enthalpy of formation of $CH_{4(g)}$ at 298K. Given:[$\Delta_f H = -890.5$KJ,

[$\Delta_f H_{co2}$ = - 393.5KJ.mol , $\Delta_f H_{(H2O)}$ = - 286.0KJ.mol^{1}]

Q.10 For the reaction $NH_4Cl_{(s)=}$ NH3(g) + $HCl_{(g)}$ at 25^0C enthalpy change H = 177KJ.mol^{-1} and S = 285$J.K^{-1}$ mol^{-1}. Calculate free energy change G at 25^0C and predict whether the reaction is spontaneous or not.

ANSWER TO THREE MARKS QUESTIONS:

Ans.1 $\Delta S_{(surr.)} = \frac{{}_r H^0}{T}$ C $\frac{1648 \times 10^3 J.mol^{1})}{298 K}$ $^{5530}_{1}$ mol^{1}. JK

& $S_{(system)} = -549.4QJK^{-1}mol^{-1}$.

${}_rS_{(total)} = 5530 - 5494 = 4980.6\ J.K^{-1}mol^{-1}$

Since ${}_rS_{(total)}$ is +ve, so the reaction is spontaneous.

Ans.2 $\Delta_r H^0$ = Σbond enthalpies$_{(rect.)}$ - Σbond enthalpies$_{(prod.)}$

$= H_{H\ H}^{0\ 0}\ H_{Br\ Br}\ ^2\ H_{H\ Br}^{0}$

$= [435 - 192] - [2 \times 368]$ KJ mol^{-1}

$= 627 - 736 = -109$KJ. Mol^{-1} Ans.

Ans.3 ${}_rH = {}_fH^0{}_{(prod.)} - {}_rH^0{}_{(rect)}$

$= {}_f H^0{}_{(N_2O)} + 3\ {}_f H^0{}_{(CO_2)} - ({}_fH^0\ N_2O_4 + 3\ {}_f H^0{}_{(CO)})$

$= [81 + 3(-393)] - [9.7 + (-110)]$

$= [81 + 1179] - [9.7 - 330] = -777.7$KJ

Ans.4 $\Delta H = 400$ KJ mol^{-1}, S = 0.2 KJK^{-1} mol^{-1}.

$\Delta G = \Delta H - T.\ \Delta S$

$O = 400 - 0.2 \times T$ ($\Delta G = 0$ at equilibrium)

$T = \frac{400}{0.2}$ 2000K, so reaction will be spontaneous above 2000K.

Ans.5 $\Delta_r G^0 = -2.303\ RT \log K$

(iv) $-2.303 \times 8.314 \times 300 \times \log 10$

(v) $-19.147 \times 300 \times 1 = -5744.1 J$ $_rG^0 = -5.7441 KJ.mol^{-1}$

Ans.6 The property whose value depends upon state of system and is independent of path. $q + w = \Delta U$, which is a state function as value of ΔU does not depends upon path.

Ans.7 (a) It is false, exothermic reaction is not always spontaneous. If ΔS = +ve and T. $\Delta S > \Delta H$. The process will be non spontaneous even it. It is endothermic.

(b) The entropy of vapour is more than that of liquid, so entropy increases during vaporization.

Ans.8 $\Delta G = \Delta H - T.\ \Delta S$, at equilibrium $\Delta G = 0$, $\Delta H = T.\ \Delta S$

$$T = \frac{\Delta H}{\Delta S} \quad \frac{108.4 \times 10^{3}\ J.mol_{1}}{190\ J.K^{1} mol^{1}} \quad 570.526\ K$$

So the reaction will be spontaneous above 570.52K, as above this temperature G will be –ve.

Ans.9 $_rH = \Delta_f H_{(CO_2)} \quad \Delta_f H_{2\ 4(g)} \quad \Delta_f H_{(O_2)}$

$-890.5KJ = -393.5KJ + 2v - 286 - H_f(CH_4) - O$

$= H_f(CH_4) = -75.0\ KJ.mol^{-1}$.

$= H_{f_{4}(CH)}$ $75.0KJ.mol^{1}$.

Ans.10 $\Delta H = 177\ KJ\ mol^{-1}$, $\Delta S = 285\ JK^{-1}\ mol^{-1}$

$\Delta G = \frac{\Delta H - T.\ \Delta S}{1000} = 177KJ \frac{298 \times 285}{} KJ$

$= 177\ KJ - 84.93\ KJ = 92.07\ KJ.mol^{-1}$.

Since ΔG is +ve, so the reaction is non spontaneous.

FIVE MARKS QUESTIONS:-

Q.1 What is entropy? Why is the entropy of a substance taken as zero at 0K? Calculate the $_rG$ for the reaction?

$N_{2(g)} + 3H_{2(g)} \quad 2NH_{3(g)}$ at 298K

The value of equilibrium constant (K) is $6.6x10^5$, R = $8.314JK^{-1}mol^{-1}$. Ans: It is measure of randomness or disorder of system.

Because at O K there is complete order in the system.

$\Delta G^0 = -2.303\ RT\ logK = -2.303 \times 8.314 \times 298 \times log6.6 \times 105$

$= -5705.8[log6.6 + log\ 10^5]$

$= -5705.8[0.8195 + 5.0] = -5705.8 + 5.8195J$

$= -33204.903J$

$\Delta G^0 = -33.205\ KJ\ mol^{-1}$.

Q.2 (i) What are extensive property and intensive properties?
(ii)Calculate the value of equilibrium constant (K) at 400K for

$2\ NOCl_{(g)} \rightarrow 2NO_{(g)} + Cl_{2(g)}$.

$\Delta H^0 = 77.2KJ.mol^{-1}$, $S^0 = 122J.K^{-1}mol^{-1}$ at 400K, R = $8.314\ J.K^{-1}mol^{-1}$.

Ans. (i) An **extensive property** is a property whose value depends on the quantity or size of matter present in the system Those properties which do not depend on the quantity or size of matter present are known as **intensiveproperties**

(ii) $\Delta H^0 = \Delta H^0 - T.\ S^0 = 77.2KJ - \frac{400x122}{1000} KJ.mol^{1}$

$= 77.2 - 48.8 = 28.4\ KJ\ mol^{-1}$

and $\Delta G^0 = -2.303\ RT\ logK.$

$28400 = -2.303 \times 8.314 \times 400\ log\ K.$

$logK = \frac{7.1}{2.303x8.314} \quad 3.7081\ \bar{4}.2919K$

$K - antilog\ (\bar{4}.2919) \quad 1.95x10^{4}$Ans.

Q.3 Define standard enthalpy of formation. Calculate the enthalpy of formation of

benzene from data

$C_6H_{6(l)} + \frac{15}{2}O_{2(g)} \rightarrow 6CO_{2(g)} + 3H_2O_{(l)}, \quad {}_cH^0 = 3266.0KJ$

$C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)} \qquad {}_fH^0 = 393.1KJ$

$H_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow H_2O_{(l)} \qquad {}_fH^0 = 286.0KJ$

Ans. Ans. The standard enthalpy change for the formation of one mole of a compound from its elements in their most stable states of aggregation (also known as reference states) is called Standard Molar Enthalpy of Formation**.**

$${}_cH^0 = 6\,{}_fH^0_{(CO_2)} + 3\,{}_fH^0_{(H_2O)} - {}_fH^0_{(C_6H_6)} - \frac{15}{2}\,{}_fH^0_{(O_2)}$$

$= -3266KJ = 6 \times -393.1 + 3 \times -286 - {}_fH^0_{(C_6H_6)} - 0$

$= -3218$ kJ/mol

HOTS QUESTIONS

1. Why standard entropy of an elementary substance is not zero whereas standard enthalpy of formation is taken as zero?

   Ans. A substance has a perfectly ordered arrangement only at absolute zero. Hence , entropy is zero only at absolute zero. Enthalpy of formation is the heat change involved in the formation of one mole of the substance from its elements. An element formed from it means no heat change.
2. The equilibrium constant for a reaction is one or more if $\Delta G°$ for it is less than zero. Explain

   Ans. $-\Delta G° = RT \ln K$, thus if $\Delta G°$ is less than zero. i.e., it is negative, then ln K will be positive and hence K will be greater than one.
3. Many thermodynamically feasible reactions do not occur under ordinary conditions. Why?

   Ans. Under ordinary conditions, the average energy of the reactants may be less than threshold energy. They require some activation energy to initiate the reaction.

Value based questions

Q.1 All kinds of laws govern our everyday life. Traffic laws tell that you have to stop at stop signs. Government laws demand that you have to pay income tax on your salary/income, etc. Even scientific laws apply to everyday life. For example, law of gravity forces us to move on earth and we cannot move in the air. Another set of scientific laws that affect our everyday life are the laws of thermodynamics. For example, first law of

thermodynamics states that energy can neither be created nor be destroyed but it can be converted from one form to another. This is also sometimes referred to as law of conservation of energy.

Now answer the following questions.

(i) How is first law of thermodynamics applicable to an electrical fan or a heater?

(ii) How does first law of thermodynamics apply to a hydroelectric power plant?

(iii) If a stone at a height falls, it hits the objects or breaks our head. How does first law of thermodynamics apply here?

Ans.

(i) In an electrical fan, the electrical energy is converted into mechanical work that moves the blades. In a heater, electrical energy is converted into heat energy.

(ii) Running water of the river or lake has kinetic energy. By stopping the flow in a dam, the kinetic energy is converted into potential energy. When water is allowed to fall on the turbine, it is converted into kinetic energy which is then converted into mechanical work that runs the turbine and ultimately converted into electrical energy.

(iii) The stone at a height has potential energy. When it falls its potential energy is converted into kinetic energy and hence is associated with a force that hits the object or breaks our head.

Q.2 Second law of thermodynamics states that complete conversion of energy of one kind into another is not possible as some energy is always lost in the form of some other energy. For example, in a car, the heat energy produced by the combustion of a fuel is not completely converted into mechanical work as some energy may be used up in overcoming friction of wheels, etc. Besides other factors, the efficiency depends upon the nature of the working substance (fuel) used in the engine (called heat engines). These studies were made by Sadi Carnot in 1820.

Now answer the following questions.

(i) How does second law of thermodynamics explain the working of heat engine (steam, gasoline, diesel) used in all types of vehicles (motorcycles, cars, trucks, ships, aeroplanes, etc)?

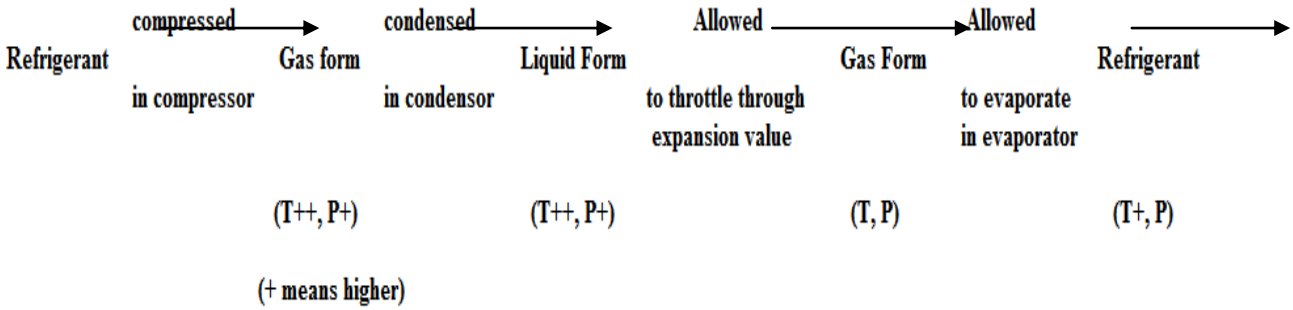
(ii) How does the second law of thermodynamics explain the working of refrigeration (refrigerators and air conditioners)?

(iii) If we push a car even with a great force, it stops after some time. Do you think it moves the distance we expected? Why or why not? How does the second law of thermodynamics explain it?

Ans.

(i) All standard heat engines (steam, gasoline, diesel) work by supplying heat to a gas. The gas then expands in a cylinder and pushes a piston to do its work. But the heat and/or the gas must somehow then be dumped out of the cylinder to get ready for the next cycle. As a result, some heat is always lost and is not completely converted into work. That is why efficiency is never 100%.

(ii) In refrigeration, heat flows from higher temperature to lower temperature in a compressor with mechanical work which is done by the compressor on the system. The system consists of four main components, compressor, condenser, expansion valve and evaporator. The following cycle is repeated over and again :



(iii) No, it does not move the distance we except because the energy spent by us in pushing the car is not completed converted into mechanical work but a part of it is is used up in overcoming friction between the tyres and the pavements .

Q3. A lady asked a doctor why she is enable to lose weight though she takes care of diet and exercise. Out of joke, the doctor said, ″Law of conservation of mass hold good as if you will lose weight, someone else will gain weight.” However, it was all fun as law of thermodynamics really does not work that way. Laws of thermodynamics help us to understand some of the biological processes occurring in our body.

Now answer the following questions.

(i) Why sometimes we are not able to lose weight inspite of proper diet and regular exercise? Explain in terms of first law of thermodynamics.

(ii) How doest the concept of entropy (a measure of randomness) used in thermodynamics explain the death of cell if we do not eat food?

Ans.

(i) No doubt, to manage our weight, calories in should be equal to calories out, i.e., there has to be balancing between calories eaten and calories burnt. The first law of thermodynamics deals with the energy balanced but applies to a closed system. Human body is not of a closed system but an open system. In real life, balancing energy includes a lot more than just the calories we eat and calories we burn. This is because some calories are stored as fat and as glucose in lever and calories are also used in growth, tissue restoration and number of metabolic process. Thus, calories eaten and calories used in exercise and work are only two small parts of energy balance and hence meaningless unless all others variables are controlled which they can never be as they are not under our control.

(ii) Food gives heat energy to the cell. As environment is at lower temperature the cell lose their heat to the environment. By losing heat energy, we (our cells) in effect lower our entropy. If the cell fails to obtained its entropy increases indefinitely leading to the destruction of its highly ordered low entropy structure leading to the cell (our organism) death.

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CHAPTER-7

EQUILIBRIUM

Equilibrium state- When rate of formation of a product in a process is in competition with rate of formation of reactants, the state is then named as

"Equilibrium state" .

Equilibrium in physical processes: solid ⇌ liquid ⇌ gas

$H_2O(s) \rightleftharpoons H_2O(l) \rightleftharpoons H_2O(vap)$

Law of chemical equilibrium: At a given temperature, the productofconcentrations of the reaction productsraised to the respective stoichiometriccoefficient in the balanced chemicalequation divided by the product ofconcentrations of the reactants raised totheir individual stoichiometric coefficientshas a constant value. This is known asthe Equilibrium Law or Law of ChemicalEquilibrium.

$aA + bB \rightleftharpoons cC + dD$

$K_c = [C]^c [D]^d / [A]^a [B]^b$

Chemical equation	**Equilibriumconstant**
$aA + b B \rightleftharpoons c C + D$	K
$cC + d D \rightleftharpoons a A + b B$	$K'c=(1/Kc)$
$na A + nb B \rightleftharpoons ncC + ndD$	$K''c= (Kc^n)$

Concentrations or partial pressure of pure solids orliquids do notappear in the expression of the equilibriumconstant. In the reaction,

$Ag_2O(s) + 2HNO_3(aq) \rightleftharpoons 2AgNO_3(aq) + H_2O(l)$ $Kc=\frac{[AgNO3]}{[HNO3]^2}$

If *Qc>Kc*, the reaction will proceed in thedirection of reactants (reverse reaction).If *Qc<Kc*, the reaction will proceed in thedirection of the products (forward reaction)

Kp is equilibrium constant in terms of partial pressure of gaseous reactants and products.

Kc is equilibrium constant in terms of molar concentration of gaseous reactants and products.

$Kp = Kc (RT)^{\Delta n}$ here **R** is gas constant, **T** is temperature at which the process is carried out &Δ**n** is no. of moles of gaseous product minus no. of moles of gaseous reactants.

If $Kc > 10^3$; Kc is very high i.e. the reaction proceeds nearly to completion.

If $Kc < 10^3$; Kc is very small i.e. the reaction proceeds rarely.

- If Kcis ranging in the range of 10^3 to 10^{-3}; i.e. reactants and products are just in equilibrium.
- $G^0 = -RT \ln K$ or $G^0 = -2.303RT \log K$

Factors affecting equilibrium constant:- temperature, pressure, catalyst and molar concentration of reactants and products.

Le Chatelier's principle:-It states that a**change in any of the factorsthatdetermine the equilibrium conditions of asystem will cause the system to changein such a manner so as to reduce or tocounteract the effect of the change.**

Arrhenius acids are the substances that ionize in water to form $\mathbf{H^+}$.

Arrhenius bases are the substances that ionize in water to form $\mathbf{OH^-}$.

Lewis acids are lone pair (of e-) accepters while Lewis bases are lone pair donators.

Proton donor are acids while proton accepters are bases(Bronsted-Lowry concept).

The acid-base pair thatdiffers only by one proton is called a **conjugateacid-base pair**. IfBrönsted acid is a strong acid then **itsconjugate base is a weak base and *viceversa*.**

Ionic product of water.$K_W = [H^+][OH^-]$

pH = -log [H^+] ; here[H^+] is molar concentration of hydrogen ion.

pH + pOH =14

pKa + pKb =14

Ka x Kb = Kw = ionic product of water**=1 x 10^{-14}**

Buffer solution :The solutions which resist change in pH on dilution or withthe addition of small amounts of acid or alkali are called Buffer Solutions**.**

common ion effect: It can be defined as a shift in equilibrium on adding asubstance that provides more of an ionic species already present in the dissociation equilibrium.

Hydrolysis of Salts: process of interaction between water andcations/anions or both of salts is calledhydrolysis.

The cations (e.g., Na^+, K^+,Ca^{2+}, Ba^{2+}, etc.) of strong bases and anions(e.g., Cl^-, Br^-, NO_3^-, ClO_4^- etc.) of strong acids simply get hydrated but do not hydrolyse, andtherefore the solutions of salts formed fromstrong acids and bases are neutral i.e., theirpH is 7.

Salts of weak acid and strong base e.g.,CH_3COONa are basic in nature.

Salts of strong acid and weak base e.g.,NH_4Cl, are acidic

Salts of weak acid and weak base, e.g.,CH_3COONH_4. The pH is determined by the formula $pH = 7 + ½ (pK_a – pK_b)$

Solubility product- product of the molar concentrations of the ions in a saturated solution, each concentration term raised to the power equal to the no. of ions produced.

ONE MARK QUESTIONS

Q.1. Mention the factors that affect equilibrium constant.

Ans. Temperature, pressure, catalyst and molar concentration of reactants andproducts.

Q.2. Whatis ionic products of water? Ans. Kw = $[H^+]$ $[OH^-]$

Q.3. Write conjugate acids of H_2O & NH_3.

Ans. H_3O^+ & NH_4^+.

Q.4. Define Arrhenius acids.

Ans. Arrhenius acids are the substances that ionize in water to form$\mathbf{H^+}$.

Q.5. Define the term degree of ionization.

Ans.Extent up to which an acid/base/salt ionize to form ions.

Q.6. What are Buffer solutions?

Ans.The solutions which resist change in pH on dilution or with the addition of smallamounts of acid or alkali are called Buffer Solutions.

Q.7. Write Kc for the gaseous reaction- $N_2 + 3H_2 \rightleftharpoons 2NH_3$

Ans. $Kc=[NH_3]^2/[N_2]\,[H_2]^3$

Q.8. Out of H_2O & H_3O^+which is stronger acid?

Ans. H_3O^+.

Q.9. What is common ion effect?

Ans. Shift in equilibrium on adding a substance that provides more of an ionic species already present in the dissociation equilibrium.

Q.10. Write relationship between Kp and Kc for the gaseous reaction - $N_2 + O_2 \rightleftharpoons 2NO$

Ans.Kp = KcasΔn is zero for the above said reaction.

TWO MARKS QUESTIONS

1. What is effect of catalyst on equilibrium constant Kc?

Ans . A catalyst does not affect equilibrium constant because it speeds up both forward and backward reactions to the same extent.

- State Le Chatelier's principle.

  Ans.It states that a change in any of the factors thatdetermine the equilibrium conditions of asystem will cause the system to changein such a manner so as to reduce or tocounteract the effect of the change.

- What is meant by conjugate acid –base pairs? Explain.

  Ans:- $\underset{\text{base}}{H_2O} + \underset{\text{acid}}{HCl} \rightleftharpoons \underset{\text{conjugate acid}}{H_3O^+} + \underset{\text{conjugate base}}{Cl^-}$

- Classify the following bases as strong and weak bases: $NaHCO_3$, NaOH, KOH, $Ca(OH)_2$, $Mg(OH)_2$.
  Ans:-strong base NaOH, KOH ; weak bases $NaHCO_3$,$Ca(OH)_2$, $Mg(OH)_2$.

- The concentration of hydrogen ion in a sample of soft drink is 3.8×10^{-3}M. What is its pH ?
  Ans:-$pH = -\log[3.8 \times 10^{-3}]$
  $-\{\log[3.8] + \log[10^{-3}]\}$
  $-\{(0.58) + (-3.0)\} = -\{-2.42\} = 2.42$
  Therefore, the pH of the soft drink is 2.42and it is acidic.

- The species: H_2O, HCO_3^-, HSO_4^- and NH_3can act both as Bronsted acids and bases.For each case give the correspondingconjugate acid and conjugate base.

Ans:-

Species	Conjugate acid	Conjugate base
H_2O	H_3O^+	OH^-
HCO_3^-	H_2CO_3	CO_3^{2-}
HSO_4^-	H_2SO_4	SO_4^{2-}
NH_3	NH_4^+	NH_2^-

(c) Explain Lewis acids and bases with suitable examples.
Ans:-Lewis acids are lone pair (of e-) accepters while Lewis bases are lone pair donators.
$AlCl_3$ is a Lewis acid while NH_3 is a Lewis base.

(d) What is difference between alkali and bases? Give examples.
Ans:- An alkali is a water soluble base. All the alkalis are bases but all the bases are not alkali.$Ca(OH)_2$ is a base but Ex- NaOH is an alkali/base. not an alkali.

F) Explain homogeneous and heterogeneous equilibrium giving examples.

Ans:- If all the reactants and products present in an equilibrium mixture are in same phase→homogeneous equilibrium.

If all the reactants and products present in an equilibrium mixture are in different phase→ heterogeneous equilibrium.

$N_{2\,(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$ homogeneous equilibrium

$CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$ heterogeneous equilibrium

THREE MARK QUESTIONS

G)The pH of some common substances is given bellow. Classify the substances as acidic/basic

Name of fluid	pH
Lime water	10
Milk of magnesia	10
Human saliva	6.4
Lemon juice	2.2
Sea water	7.8
Vinegar	3
milk	6.8

Ans.:- acidic-Human saliva, Lemon juice, milk, vinegar
Basic- Lime water, sea water, milk of magnesia.

2. Explain general characteristics of acids and bases.

Ans.:- Most of the acids taste sour.Acids are known to turn blue litmus paper into red and liberate dihydrogen on reacting with some metals.

Bases are known to turn red litmus paper blue, tastebitter and feel soapy.

3. Water is amphoteric in nature. Explain.

Ans.:- Water can react with acid as well as base

$H_2O + HCl \rightarrow H_3O^+ + Cl^-$ water is basic

$H_2O + NH_3 \rightarrow OH^- + NH_4^+$ water is acidic

4. Describe the effect of :

A)addition of H_2

B)addition of CH_3OH

C)removal of CO

d) removal of CH_3OHon the equilibrium of the reaction:

$2H2(g) + CO\ (g) \rightleftharpoons CH3OH\ (g)$

Ans.:- a) addition of H_2 equilibrium will shift on RHS

b) addition of CH_3OH equilibrium will shift on LHS

c) removal of CO equilibrium will shift on LHS

d) removal of CH_3OH equilibrium will shift on RHS

5. Classify the following species into Lewisacids and Lewis bases and showhowthese act as such:

11. HO^- (b)F^- (c) H^+ (d) BCl_3

Solution

a)Hydroxyl ion is a Lewis base as it candonate an electron lone pair ($:OH^-$).

b)Flouride ion acts as a Lewis base asit can donate any one of its fourelectron lone pairs.

c)A proton is a Lewis acid as it canaccept a lone pair of electrons frombases like hydroxyl ion and fluorideion.

d) BCl_3 acts as a Lewis acid as it canaccept a lone pair of electrons fromspecies like ammonia or aminemolecules.

7. For the equilibrium,$2NOCl(g) \rightleftharpoons 2NO(g) + Cl_2(g)$the value of the equilibrium constant, *Kc*is 3.75×10^{-6} at 1069 K. Calculate the *Kp*for the reaction at this temperature?

Solution

We know that,$Kp = Kc(RT)^{\Delta n}$

For the above reaction,$\Delta n = (2+1) - 2 = 1$

$Kp = 3.75 \times 10^{-6}(0.0831 \times 1069)$ $Kp =$ 0.033.

8. Hydrolysis of sucrose gives,Sucrose + $H_2O \rightarrow$Glucose + Fructose Equilibrium constant *Kc*for the reaction is 2×10^{13} at 300K. Calculate ΔG^0at 300K.

Solution

$\Delta G^0 = - RT \ln Kc$

$\Delta G^0 = - 8.314 J\ mol^{-1} K^{-1} J \times 300K \times \ln(2 \times 10^{13})$

$\Delta G^0 = - 7.64 \times 10^4\ J\ mol^{-1}$

8. Explain the following :

(i) Common ion effect (ii) solubility products (iii) pH

Ans. (i) Suppression of ionization of weak electrolyte by adding a strong electrolyte having an ion common.

(ii) Product of the molar concentrations of the ions in a saturated solution, each concentration term raised to the power equal to the no. of ions produced.

iii) Negative logarithm of hydrogen ion concentration.

10. The values of *K*sp of two sparingly solublesalts $Ni(OH)_2$ and AgCN are $2.0 \times 10^{-15}$and 6×10^{-17} respectively. Which salt ismore soluble? Explain.

Solution

$AgCN \rightleftharpoons Ag^+ + CN^-$

$Ksp = [Ag^+][CN^-] = 6 \times 10^{-17}$

$Ni(OH)_2 \rightleftharpoons Ni^{2+} + 2OH^-$

$Ksp = [Ni^{2+}][OH^-]^2 = 2 \times 10^{-15}$

Let $[Ag^+] = S_1$, then [CN-] = S_1 Let $[Ni^{2+}]$ = S_2, then $[OH^-] = 2S_2$

$S_1^2 = 6 \times 10^{-17}$, $S_1 = 7.8 \times 10^{-9}$ $(S_2)(2S_2)^2 = 2 \times 10^{-15}$, $S_2 = 0.58 \times 10^{-4}$

$Ni(OH)_2$ is more soluble than AgCN.

FIVE MARKS QUESTIONS

1. At 473 K, equilibrium constant *Kc*for decomposition of phosphorus pentachloride,PCl_5 is 8.3×10^{-3}. If decomposition is depicted as,
   PCl_5 (g) ⇌PCl_3 (g) + Cl_2 (g) $\Delta r H^0 = 124.0$ kJ mol^{-1}
   a) Write an expression for *Kc*for the reaction.
   b) What is the value of *Kc*for the reverse reaction at the same temperature?
   c)what would be the effect on *K*c if (i) more $PCl_5$is added (ii) pressure is increased(iii) the temperature is increased ?

   Ans: (a) $Kc=\frac{[PCl_3][Cl_2]}{[PCl_5]}$
   (b)120.48
   (i) equilibrium will shift on RHS
   (ii) equilibrium will shift on LHS
   (iii) equilibrium will shift on RHS

4. Dihydrogen gas is obtained from natural gas by partial oxidation with steam asper following endothermic reaction:CH_4 (g) + H_2O (g) ⇌CO (g) + $3H_2$ (g)
   i Write as expression for *Kp*for the above reaction.
   ii How will the values of *Kp*and composition of equilibrium mixture be affectedby(i) increasing the pressure(ii) increasing the temperature(iii) using a catalyst?
   Ans. (a) $K_p = p(CO).p(H_2)^3 / p(CH_4).p(H_2O)$
   (b)(i) value of Kp will not change, equilibrium will shift in backward direction.
   = Value of Kp will increase and reaction will proceed in forward direction.
   (iii)no effect.

5. What is meant by the conjugate acid-base pair? Find the conjugate acid/basefor the following species:HNO_2, CN^-, $HClO_4$, F^-, OH^-, CO_3^{2-}, and S^{2-}
   Ans.The acid-base pair thatdiffers only by one proton is called a conjugate acid-base pair

Species	Conjugate acid/base
HNO_2	NO_2^-
CN^-	HCN
$HClO_4$	ClO_4^-
F^-	HF
OH^-	H2O
CO_3^{2-}	HCO_3^{2-}
S^{2-}	HS^-

HOTS QUESTIONS

3.39 The value of *Kc*for the reaction2A ⇌B + C is 2×10^{-3}. At a given time,the composition of reaction mixture is[A] = [B] = [C] = 3×10^{-4} M. In whichdirection the reaction will proceed?

Solution

For the reaction the reaction quotient *Qc*is given by,$Qc = [B][C]/ [A]^2$

as [A] = [B] = [C] = 3×10^{-4}M

$Qc = (3 \times 10^{-4})(3 \times 10^{-4}) / (3 \times 10^{-4})2 = 1$

as*Qc*>*Kc*so the reaction will proceed inthe reverse direction.

2. PCl_5, $PCl_3$and $Cl_2$are at equilibrium at500 K and having concentration 1.59MPCl_3, 1.59M Cl_2 and 1.41 M PCl_5. Calculate *Kc*for the reaction,$PCl_5 \rightleftharpoons PCl_3 + Cl_2$

Solution

The equilibrium constant *Kc*for the abovereaction can be written as,

$$Kc = \frac{[PCl_3][Cl_2]}{[PCl_5]}$$

$= (1.59)^2 / 1.41 = 1.79$

3. Why is ammonia termed as a base though it does not contain OH^- ions? Ans.ammona is termed as a base on the basis of Lewis concept it can donate a lone pair of electrons.

Equilibrium-Value Based Questions

1. **We come across a number of phenomena in everyday life where the materials are in equilibrium with the surroundings or the reactants are in equilibrium with the products. If the amount of any of the reactants or products is disturbed, the equilibrium shifts to one side or the other. This shifting of equilibrium may be useful or harmful depending upon what type of equilibrium we are dealing with.**
   Now answer the following questions:
   i. **Why our clothes dry more quickly when there is a breeze or we shake the clothes as done by a dyer?**
   ii. **We sweat more on a humid day but it evaporates when we sit under fan. Why?**

Ans.)

i. This is because water vapors of nearby air are removed and cloth loses more water vapor to re-establish equilibrium with air.
ii. More sweating takes place because the surrounding air has large amount of water vapor and our skin cannot lose more to it. The fan removes the humid air and evaporation starts from the skin.

2. **Chemical equilibrium has an important role to play in a number of industrial processes. The aim of every industry is to get maximum yield of the product in minimum time. Le Chatelier principle helps to know the suitable conditions to achieve the goal.**
   Now answer the following questions:
   i. **Lime is prepared from limestone by heating in lime kilns that are well ventilated. Why?**

ii. **How does Le Chatelier principle help the industry to know the suitable conditions for the manufacture of ammonia, a large amount of which is needed for making fertilizers?**

iii. **Methanol is manufactured from carbon monoxide and hydrogen. Predict the most suitable conditions for its manufacture.**

Ans.)

i. The reaction involved is

$$CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}\ ,\ \Delta H = +178\ kJ\ mol^{-1}$$

Well ventilated kiln is used so that CO_2 is able to escape out and forward reaction is favored. Further, as the reaction is endothermic, therefore, formation of CaO is favored by high temperature (at least 900°C).

ii. The reaction involved is

$$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}\ ,\ \Delta H = -92\ kJ\ mol^{-1}$$

As reaction is accompanied by decrease in the number of moles (from 4 to 2), high pressure would favor formation of ammonia. But very high pressure means more dangerous and more costly engineering. Hence, an optimum pressure of about 200 atm is used.

Further, as reaction is exothermic, low temperature will favor the formation of NH_3. But too low temperature makes the reaction very slow. Hence an optimum temperature of 750 K is employed because it is better to have lower yield than wasting too much time unnecessarily. Further, to speed up the reaction, finely divided iron is added as catalyst with molybdenum as promoter.

iii. The reaction involved is

$$CO(g) + 2H2(g) \rightleftharpoons CH3OH(g)\ ,\ \Delta H = -90\ kJ\ mol^{-1}$$

Theoretically, the reaction is favored by high pressure and low temperature. In practice, a high pressure is used to give an acceptable yield but a moderate temperature along with a catalyst is employed to get an economic production rate. These conditions are: 250°C, Cu-ZnO-Al_2O_3 as catalyst and a pressure of 50-100 atm.

3. **We use a number of acids and bases in our life. Some of these are used as medicine or in household whereas some others are used in the industry. Besides, many fruits, vegetables and other eatables also contain acids, useful for the living system.**
Now answer the following questions:

i. **Name three acids used in the eatables. Give one use of them.**

ii. **Name three acids used in the industry. Give one important use of each of them.**

iii. **Name five fruits or vegetables and the acid present in each of them.**

iv. **Name three bases used in the industry or in household or as medicine and give one important use of each of them.**

Ans.)

i. Benzoic acid to preserve food, Carbonic acid to make aerated drinks and ethanoic acid used in vinegar.

ii. HCl to clean metals before electroplating or household cleaning, nitric acid in the production of fertilizers and explosives and sulphuric acid as a battery acid and in making fertilizers.

iii. Apples-malic acid, Orange-citric acid, Bananas-malic acid and citric acid, grapes-malic acid and tartaric acid(3:2), Potatoes-malic acid and citric acid, Tomatoes-citric acid and malic acid.

iv. Ammonia in the manufacture of fertilizers, NaOH in the manufacture of soaps, detergents and cleaners and Magnesium hydroxide as antacid to neutralize acidity in stomach.

4. **Acids are useful on one hand but sometimes they are troubleshooters on the other hand. Therefore, in a number of cases, acids have to be neutralized with the help of suitable bases. In nature, a number of plants grow some of which contain acids while some other contain bases.**

   Now answer the following questions:

   i. Which acid is called stomach acid? What is its use in the body and how does it harm our system? What is the treatment taken?

   ii. What is acid rain? How does it affect aquatic life and soil? What treatment has to be given to the soil?

   iii. Why do we get pain when we are stung by a honey bee or an ant? What is the treatment given?

   iv. If you go on hiking and trekking and by chance, you touch nettle plant whose leaves have stinging hair, you get pain. Why? What treatment will you take there?

Ans.)

i. HCl. It helps in digestion of food. Sometimes stomach produces too much acid and this causes pain and irritation. To neutralize the acidity antacids such as magnesium hydroxide are taken.

ii. When acidic gases present in the air dissolve in the rain water, such that the pH of the rain water falls below 5.6, it is called acid rain. When it flows into the rivers, it lowers the pH of river water; hence the survival of the aquatic life becomes difficult. When such water is absorbed by the soil, the soil becomes acidic and unfertile. The acidity is neutralized by adding slaked lime (Calcium Hydroxide).

iii. Honey bee or ant injects methanoic acid (formic acid) into our body causing pain and irritation. Use of mild base like baking soda on the stung area gives relief as it neutralizes the acid.

iv. Leaves of nettle plants contain methanoic acid causing pain and irritation. Rubbing with the leaf of dock plant (which also grows nearby) gives relief as it contains mild base and neutralizes the acid.

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CHAPTER-8
REDOX REACTIONS

oxidation	**reduction**
1. Addition of oxygen	1. Removal of oxygen
2. Removal of hydrogen	2. Addition of hydrogen
3. Addition of an electronegative element	3. Removal of an electronegative element
4. Removal of an electropositive element	4. Addition of an electropositive element
5. Loss of electron	5. Gain of electron

Oxidation number denotes theoxidation state of an element in a compound ascertained according to a setof rules formulated on the basis thatelectron in a covalent bond belongsentirely to more electronegative element.

Calculation of oxidation number-

- O. S. of all the elements in their elemental form (in standard state) is taken as zero O. S. of elements in Cl_2, F_2, O_2, P_4, O_3, Fe(s), H_2, N_2, C(graphite) is zero.
- Common O. S. of elements of group one (1st) is one. Common O. S. of elements of group two (2nd) is two.
- For ions composed of only one atom, theoxidation number is equal to the chargeon the ion.
- The oxidation number of oxygen in most compounds is –2.While in peroxides (e.g., H_2O_2, Na_2O_2), eachoxygen atom is assigned an oxidationnumber of –1, in superoxides (e.g., KO_2,RbO_2) each oxygen atom is assigned anoxidation number of –(½).
- In oxygendifluoride (OF_2) and dioxygendifluoride (O_2F_2), the oxygen is assignedan oxidation number of +2 and +1,respectively.
- The oxidation number of hydrogen is +1 but in metal hydride its oxidation no. is–1.
- In all its compounds, fluorine has anoxidation number of –1.
- The algebraic sum of the oxidation numberof all the atoms in a compound must bezero.
- In polyatomic ion, the algebraic sumof all the oxidation numbers of atoms ofthe ion must equal the charge on the ion.

Stocknotation:the oxidationnumber is expressed by putting a Romannumeralrepresenting the oxidation numberin parenthesis after the symbol of the metal

inthe molecular formula. Thus aurous chlorideand auric chloride are written as Au(I)Cl andAu(III)Cl_3. Similarly, stannous chloride andstannic chloride are written as Sn(II)Cl_2andSn(IV)Cl_4.

Oxidation: An increase in the oxidationnumber
Reduction: A decrease in the oxidationnumber

Oxidising agent: A reagent which canincrease the oxidation number of anelementin a given substance. These reagents are calledas **oxidants** also.
Reducing agent: A reagent which lowers the oxidation number of an elementin a givensubstance. These reagents are also called as**reductants**.

Redox reactions: Reactions which involvechange in oxidation number of theinteractingspecies

Balancing of redox reactions:Oxidation Number Method:

Write the net ionic equation for the reaction of potassiumdichromate(VI), $K_2Cr_2O_7$ with sodium sulphite,Na_2SO_3, in an acid solution to give chromium(III) ion and the sulphate ion.

Step 1: The skeletal ionic equation is:

$$Cr_2O_7^{2-}(aq) + SO_3^{2-}(aq) \rightarrow Cr^{3+}(aq) + SO_4^{2-}(aq)$$

Step 2: Assign oxidation numbers forCr and S

$$\overset{+6\ -2}{Cr_2O_7^{2-}}(aq) + \overset{+4\ -2}{SO_3^{2-}}(aq) \rightarrow \overset{+3}{Cr^{3+}}(aq) + \overset{+6\ -2}{SO_4^{2-}}(aq)$$

Step 3: Calculate the increase anddecrease of oxidation number, and makethem equal:

$$\overset{+6\ -2}{Cr_2O_7^{2-}}(aq) + 3\overset{+4\ -2}{SO_3^{2-}}(aq) \rightarrow 2\overset{+3}{Cr^{3+}}(aq) + 3\overset{+6}{SO_4^{2-}}(aq)$$

Step 4: Balance the charge by adding H^+as the reaction occurs in theacidic medium,

$$Cr_2O_7^{2-}(aq) + 3SO_3^{2-}(aq)\ \ 8H^+ \rightarrow 2Cr^{3+}(aq) + 3SO_4^{2-}(aq)$$

Step 5: Balance the oxygen atom by adding water molecule.$Cr_2O_7^{2-}(aq) + 3SO_3^{2-}(aq)\ 8H^+ \rightarrow 2Cr^{3+}(aq) + 3SO_4^{2-}(aq) + 4H_2O(l)$

Half Reaction Method

balance the equation showing the oxidation of Fe^{2+} ions to Fe^{3+} ions by dichromate ions $(Cr_2O_7)^{2-}$ in acidic medium,wherein, $Cr_2O_7^{2-}$ ions are reduced to Cr^{3+} ions.

Step 1: Produce unbalanced equation for thereaction in ionic form

$$Fe^{2+}(aq) + Cr_2O_7^{2-}(aq) \rightarrow Fe^{3+}(aq) + Cr^{3+}(aq)$$

Step 2: Separate the equation into halfreactions:

$$\overset{+2}{Fe^{2+}}(aq) \rightarrow \overset{+3}{Fe^{3+}}(aq)$$

Oxidation half :

$$\overset{+6\ -2}{Cr_2O_7^{2-}}(aq) \rightarrow \overset{+3}{Cr^{3+}}(aq)$$

Reduction half :

Step 3: Balance the atoms other than O andH in each half reaction individually.

$$Cr_2O_7^{2-}(aq) \rightarrow Cr^{3+}(aq)$$

Step 4: For reactions occurring in acidicmedium, add H_2O to balance O atomsand H^+ to balance H atoms.$Cr_2O_7^{2-}$ (aq) +14 $H^+ \rightarrow Cr^{3+}$(aq) + $7H^2O$ (l)

Step 5: Add electrons to one side of the halfreaction to balance the charges. Ifneed be,make the number of electrons equal in the twohalf reactions by multiplying one or both half reactions by appropriate coefficients. Fe^{2+} (aq) $\rightarrow Fe^{3+}$ (aq) + e–

$Cr_2O_7^{2-}$ (aq) + $14H^+$(aq) + 6e– $\rightarrow 2Cr^{3+}$(aq) +$7H_2O$

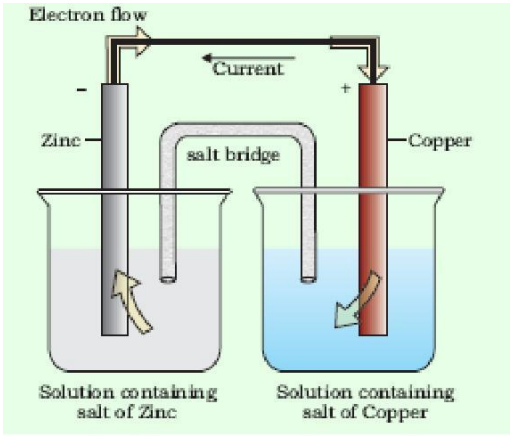
(l) $6Fe^{2+}$ (aq) $\rightarrow$6 Fe^{3+} (aq) +6 e–

Step 6: We add the two half reactions toachieve the overall reaction and cancel theelectrons on each side. This gives the net ionicequation as :

$6Fe^{2+}$(aq) + $Cr_2O_7^{2-}$ (aq) + $14H^+$(aq) $\rightarrow$ 6 Fe^{3+}(aq) +$2Cr^{3+}$(aq) + $7H_2O$(l)

A **redox couple** is defined as havingtogether the oxidised and reduced forms of asubstance taking part in an oxidation orreduction half reaction. Represented as Zn^{2+}/Zn and Cu^{2+}/Cu.

- Electrochemical cells are the devices which are used to get electric current by using chemical reaction.



Daniell cell having electrodes of zinc and copper dipping in the solutions of their respective salts.

The potential associated with eachelectrode is known as **electrode potential**. Ifthe concentration of each species taking partin the electrode reaction is unity (if any gasappears in the electrode reaction, it is confinedto 1 atmospheric pressure) and further thereaction is carried out at 298K, then thepotential of each electrode is said to be the**Standard Electrode Potential**.

SHE is used to measure electrode potential and its standard electrode potential is taken as 0.00 V.

ONE MARK QUESTIONS

- Define oxidation and reduction in terms of oxidation number.

  Ans Increase in Oxidation Number is Oxidation and decrease in Oxidation Number is called reduction.
- What is meant by disproportionation? Give one example.

  Ans : In a disproportionation reaction an element simultaneously oxidized and reduced.

  $P_4 + 3OH^- + 3H_2O \rightarrow PH_3 + 3H_2PO_2^-$
- What is O.N. of sulphur in H_2SO_4?Ans: +6

- Identify the central atom in the following and predict their O.S. HNO_3
  Ans: central atom:- N; O.S. +5
- Out of Zn and Cu which is more reactive? Ans: Zn.
- What is galvanization?
  Ans: Coating of a less reactive metal with a more reactive metal e.g.- coating of iron surface with Zn to prevent rusting of iron.
- How is standard cell potential calculated using standard electrode potential?
  Ans: $E^0_{cell} = E^0_{cathode} - E^0_{anode}$
- What is O.S. of oxygen in H_2O_2? Ans: -1.
- The formation of sodium chloride from gaseous sodium and gaseous chloride is a redox process justify.
  Ans: Na atom get oxidize and Cl is reduced.

TWO MARKS

QUESTIONS 1. Write the balanced redox reaction .

(I) $MnO_4^-(aq) + Fe^{2+}(aq) \rightarrow Mn^{2+}(aq) + Fe^{3+}(aq)$ [acidic medium]

(II) $Cr_2O_7^{2-} + Fe^{2+} \rightarrow Cr^{3+} + Fe^{3+}$ [Acidic medium]

Ans:- (i) $MnO_4^-(aq) + 5Fe^{2+}(aq) + 8H^+(aq) \rightarrow Mn^{2+}(aq) + 5Fe^{3+}(aq) + 4H_2O_{(l)}$ (ii) $Cr_2O_7^{2-} + 6Fe^{2+} + 14H^+ \rightarrow 2Cr^{3+} + 6Fe^{3+} + 7H_2O$

- Identify the strongest & weakest reducing agent from the following metals: .Zn, Cu, Na, Ag, Sn
  Ans: Strongest reducing agent: Na, weakest reducing agent: Ag.
- Determine the oxidation no. of all the atoms in the following oxidants:$KMnO_4$, $K_2Cr_2O_7$ and $KClO_4$
  Ans :
  In $KMnO_4$ K = +1, Mn = +7, O = -2
  In $K_2Cr_2O_7$ K = +1, Cr = +6, O = -2
  In $KClO_4$ K = +1, Cl = =+7, O = -2
- Determine the oxidation no. of all the atoms in the following species:Na_2O_2 and OF_2.
  Ans: In Na_2O_2 Na = +1, O = -1
  In OF_2, F = -1, O = +2
- Is it possible to store :
  (i) H_2SO_4 in Al container?(ii) CuSO4 solution in Zn vessel? Ans : (i) yes. (ii) No.
- Calculate the standard e.m.f. of the cell formed by the combination of $Zn/Zn^{2+} \| Cu^{2+}/Cu$.
  Solution- : $E^0_{cell} = E^0_{cathode} - E^0_{anode}$
  $= 0.34 - (-0.76) = 1.10V.$
- Identify the oxidizing and reducing agents in the following equations:

(i) $MnO_4^-(aq) + 5Fe^{2+}(aq) + 8H^+(aq) \rightarrow Mn^{2+}(aq) + 5Fe^{3+}(aq) + 4H_2O_{(l)}$ (ii) $Cr_2O_7^{2-} + 6Fe^{2+} + 14H^+ \rightarrow 2Cr^{3+} + 6Fe^{3+} + 7H_2O$

Ans : (i) O.A. = MnO_4^- ; R.A.= Fe^{2+}

(ii)O.A.=$Cr_2O_7^{2-}$; R.A.= Fe^{2+}

- Predict all the possible oxidation states of Cl in its compounds. Ans:- 0, -1, +1, +3, +5, +7
- Formulate possible compounds of „Cl" in its O.S.is: 0, -1, +1, +3, +5, +7
  Ans: Cl_2, HCl, HOCl, HOClO, $HOClO_2$, $HOClO_3$ respectively.
- List three measures used to prevent rusting of iron.
  Ans: (i) galvanization(coating iron by a more reactive metal)
  - greasing/oiling
  - painting.

THREE MARK QUESTIONS

- Write short notes on :
  (a) Electrochemical series(b) redox reactions (c) oxidizing agents
  Ans :(a) Electrochemical series :- arrangement of metals(non-metals also) in increasing order of their reducing power or vice versa.

(b) Reactions in which both Oxidation and reduction take place simultaneously are REDOX REACTIONS.

(c)oxidizing agents : chemical specie which can oxidize the other one or can reduce itself.

- Calculate O. S. of sulphur in the following oxoacids of Sulphur :
  H_2SO_4 ,$H_2SO_3H_2S_2O_8$and $H_2S_2O_7$
  Ans : +6, +4, +6 and +6 respectively.
  (calculate by considering x Ox. No.of S and taking +1 of H, -2 0f "O" and -1 of "O" in peroxide bond.)
- Explain role of salt bridge in Daniell cell.
  Ans : (a) it completes the electric circuit in the cell.
  it maintains the electric neutrality in the cell.
- Account for the followings :
  (i) sulphur exhibits variable oxidation states. Ans.
  Due to the presence of vacant „d orbitals in „S
  (ii) Fluorine exhibits only -1 O.S.

Ans . It is most electronegative element

(iii) oxygen can't extend its valency from 2.

Ans. Small size/unavailabilityof vacant 'd' orbitals in O

5. Balance the equation $MnO_4^- + I^- \rightarrow Mn^{2+} + I_2 + H_2O$by ion electron method in acidic medium.

Ans :<u>Step-I</u> Balancing of reduction half reaction by adding protons and electrons on LHS and more water molecules on RHS: $8H^+ + MnO_4^- + 5e^- \rightarrow Mn^{2+} + 4H_2O$

Step-II Balancing of oxidation half reaction by adding electrons on RHS:

$2I^- \rightarrow I_2 + 2e^-$

Step-III To multiply the OHR by 5; RHR by2 andto add OH & RH reactions to get overall redox reaction(cancellation of electrons of RH & OH reactions):

$$[8H^+(aq) + MnO_4^-(aq) + 5e^- \rightarrow Mn^{2+}(aq) + 4H_2O(l)] \times 2$$

$$[2I^- \rightarrow I_2 + 2e^-] \times 5$$

$$MnO_4^-(aq) + 5Fe^{2+}(aq) + 8H^+(aq) \rightarrow Mn^{2+}(aq) + 5Fe^{3+}(aq) + 4H_2O_{(l)}$$

complete and balance the following equations:

` $H^+ + Cr_2O_7^{2-} + Br^- \rightarrow 2Cr^{3+} + Br_2 +$ -----

- $H_2O_2 + Cl^- \rightarrow OH^- + Cl_2$
- $Zn + Cu^{2+} \rightarrow$?

Ans :(i) $14H^+ + Cr_2O_7^{2-} + 6\,Br^- \rightarrow 2Cr^{3+} + 3Br_2 + 7H_2O$

- $H_2O_2 + 2Cl^- \rightarrow 2OH^- + Cl_2$

$$Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$$

= Identify the oxidizing and reducing agents in the following equations:

(i) $Fe + H_2SO_4 \rightarrow FeSO_4 +$

H_2 (ii)$H_2 + Cl_2 \rightarrow 2HCl$

(iii) $MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$

Ans :(i) O.A. $=H_2SO_4$; R.A.= Fe

(ii) O.A. = Cl_2; R.A.=H_2

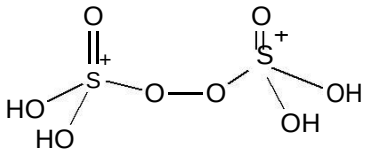
(iii)O.A. = MnO_2; R.A. =HCl

8. . Arrange the following in increasing order of their reducing power:

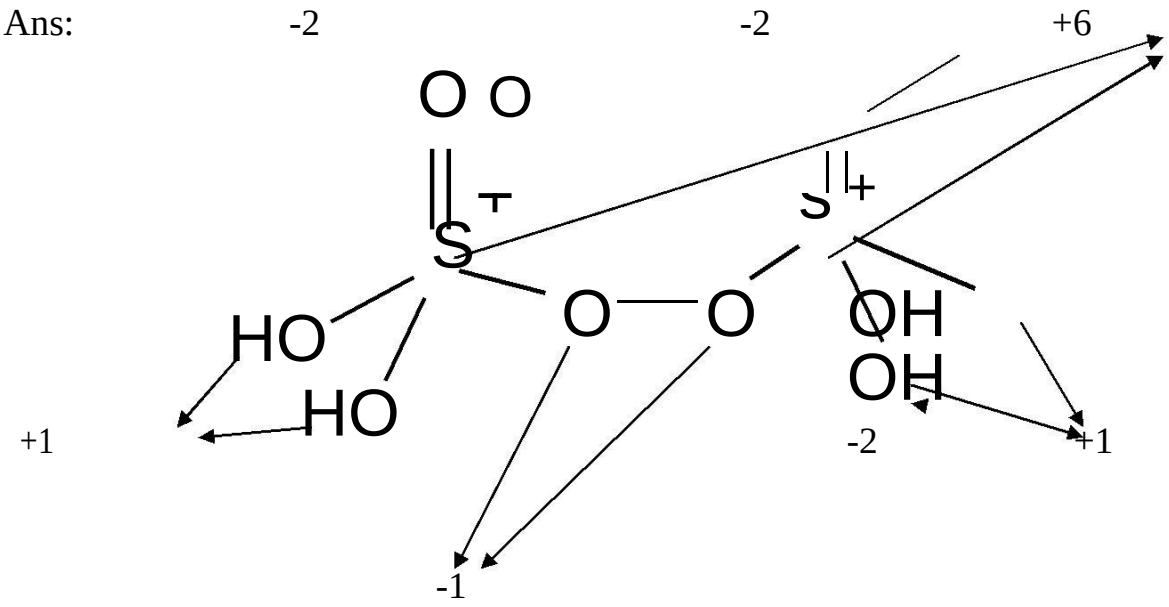
Cu, Ag, Au, Zn, Fe, Al, Na, Mg, Pt(SHE), Hg, Ca, K

Ans : Au, Hg, Ag, Cu, Pt(SHE), Fe, Zn, Al, Mg, Na, Ca, K

9. Indicate O.S. of each atom present in given structure of peroxodisulphuric acid



Ans:



10.What is SHE? What is its use?

Ans :Standard Hydrogen Electrode (SHE) has beenselected to have zero standard potential at alltemperatures. It consists of a platinum foilcoated with platinum black (finely dividedplatinum) dipping partially into an aqueous solution in which the activity (approximateconcentration 1M) of hydrogen ion is unity andhydrogen gas is bubbled through the solutionat 1 bar pressure.The potential of the other half cell is measuredby constructing a cell in which reference electrode is standard hydrogen electrode. The potential of the other half cell is equal to thepotential of the cell.

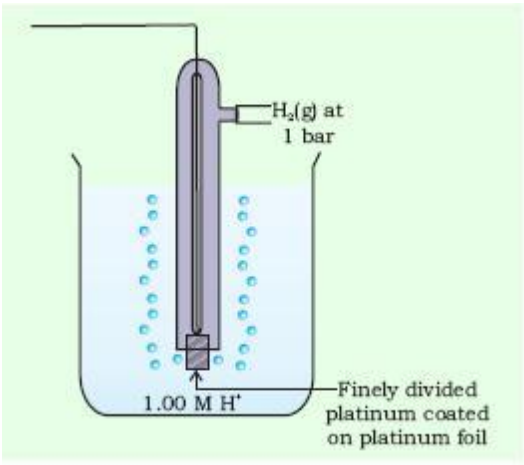


Fig: SHE

HOTS QUESTIONS

1. Is rusting of iron an electrochemical phenomenon? How ?explain.

Ans : Yes. Rusting of iron is an electrochemical phenomenon because this is possible due to formation of a small electrochemical cell over rough surface of iron and the following redox reaction takes place there in that cell-

Oxidation $Fe(s) \rightarrow Fe^{2+}(aq) + 2e^-$

Reduction $O_2 + 4H^+ \ 4e^- \rightarrow 2H_2O$

$e^- + ½\ O_2 + 2H_2O + 2Fe^{2+} \rightarrow Fe_2O_3 + 4H^+$

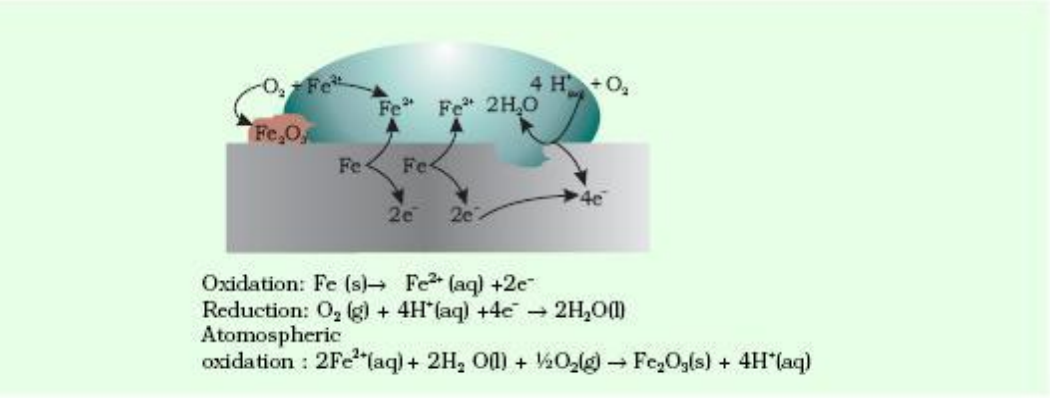


Fig. 5.14 Corrosion of iron in atmosphere.

- We expand croreof Rupees and even thousands of lives every year due to corrosion. How can be preventing it. Explain.

Ans : (i) By Galvanization: Coating of a less reactive metal with a more reactive metal e.g.- coating of iron surface with Zn to prevent rusting of iron.

(ii) By greasing /oiling (to keep away the object from the contact of air & moisture.)

(iii)By painting (to keep away the object from the contact of air & moisture.)

Value Based Questions

Q.1 All living beings require energy and most of that energy comes from the sun. Photosynthesis is the series of reactions by which green plants capture the energy of sunlight and store it as chemical energy compounds such as glucose. In contrast, most living beings use chemical energy through a process called respiration.

Now answer the following questions

(i) What is photosynthesis? Is it a redox reaction? During photosynthesis, name the substance which is oxidised and the substance which is reduced?

(ii) What is respiration? Is it a redox reaction? During respiration, name the substance which is oxidised and the substance which is reduced?

(iii) Are photosynthesis and respiration processes exactly reverse of each other? Comment.

Ans.

(i) Chlorophyll is the green pigment present in the leaves of the plants. It absorbs sunlight and converts CO_2 and H_2O present in the atmosphere into glucose. In this way, energy of sunlight is stored as chemical energy in the form of glucose.

$$6CO_2 + 6H_2O + \text{energy (sunlight)} \xrightarrow{\text{Chlorophyll}} \underset{\text{Glucose}}{C_6H_{12}O_6} + 6O_2$$

(ii) During respiration, glucose is oxidised to CO_2 and H_2O and the stored energy is released which is used by the living beings to carry out the various life processes such as movement, growth and synthesis of other life sustaining molecules.

$$C_6H_{12}O_6 + 6O_2 \longrightarrow 6CO_2 + 6H_2O + \text{energy}$$

Like photosynthesis, respiration is also a redox reaction in which, glucose is oxidised to CO_2 and O_2 is reduced to H_2O.

(iii) Although respiration seems to be exactly reverse of photosynthesis, but actually, they are not so. Both are catalyzed by a different set of enzyme-catalyzed reactions.

Q2. Police often use a device called breath analyzer to measure the amount of ethyl alcohol(C_2H_5OH) in the blood stream of a person suspected of driving under influence of alcohol.

Now answer the following questions

i) On what device does a breath analyzer works?
ii) Discuss the working of a breath analyser.

Ans.

i) A breath analyzer uses a fuel cell. A fuel cell is a device which converts the energy produced by combustion of a fuel such as H_2, C_2H_5OH, natural gas, etc, into electrical energy. The fuel cell used in breath analyser consists of two platinum electrodes. A concentrated solution of KOH or NaOH is placed between the two electrodes to act as the electrolyte.

ii) When a suspected person blows into the breath analyser, ethyl alcohol is oxidised to acetic acid at the anode.

$$\underset{\text{ethyl alcohol}}{C_2H_5OH_{(g)}} + 4OH^-_{(aq)} \longrightarrow \underset{\text{acetic acid}}{CH_3COOH_{(aq)}} + 3H_2O_{(l)} + 4e^-$$

At the cathode, oxygen (obtained from air) is reduced

$$O_2\,(g) + 2H_2O_{(l)} + 4e^- \longrightarrow 4\,OH^-_{(aq)}$$

The overall reaction is simply the oxidation of ethyl alcohol to acetic acid and water.

$$C_2H_5OH_{(g)} + O_2\,(g) \rightarrow CH_3COOH_{(aq)} + H_2O_{(l)}$$

The amount of electric current produced depends upon the amount of alcohol in the breath; higher the amount of alcohol in the blood higher the current produced. When the fuel cell breath analyser is correctly calibrated, it can accurately measure the amount of blood alcohol of a suspected drunk driver.

Q.3 In electrochemical cells, a spontaneous redox reaction is used to generate electric current. Conversely, in electrolysis, electric current is used to drive an otherwise non-spontaneous redox reaction. Electrolysis has numerous applications in industry.
Now answer the following questions

(i) Metals occur in the Earth's crust as minerals/ores. Name a few metals which can be extracted from their purified ores by electrolysis.

(ii) What is electroplating? What are its uses?

Ans.

(i) The sodium metal is extracted by electrolysis of fused sodium chloride. Sodium is obtained at the cathode while chlorine is produced at the anode.

$$NaCl\ (melt) \xrightarrow{\text{Dissocation}} Na^{+}(melt) + Cl^{-}(melt)$$

Cathode: $Na^{+}(melt) + e^{-} \longrightarrow Na_{(l)}$

Anode: $2Cl^{-}(melt) \longrightarrow Cl_{2(g)} + 2e^{-}$

Similarly, Mg is extracted from fused $MgCl_2$ and Ca from fused $CaCl_2$.

Likewise, aluminium is extracted by electrolysis of fused alumina (Al_2O_3) using carbon electrodes. Since alumina is a bad conductor of electricity, therefore cryolite (Na_3AlF_6) is added to the fused alumina to make it conducting. The reactions taking place during electrolysis are:

Cathode: $Al^{3+}(melt) + 3e^{-} \longrightarrow Al_{(l)}$

Anode: $C_{(s)} + O^{2-}(melt) \longrightarrow CO_{(g)} + 2e^{-}$

$C_{(s)} + 2O^{2-}(melt) \longrightarrow CO_{2(g)} + 4e^{-}$

(ii) *The decomposition of one metal over another through electrolysis is called electroplating.* Thepurpose of electroplating is

a. To protect metals from rusting
b. To decorate/beautify articles of iron, copper etc., by depositing silver or gold on them.

To carry out electroplating, the article to be electroplated is made the cathode while the anode consists of the pure metal to be deposited and the electrolyte consists of some soluble salt of the metal to be deposited. During electrolysis, the pure metal from the solution gets deposited on the article and an equivalent amount of pure metal from the anode goes into the solution. By the process of electroplating, silver and gold plated artificial jewelry is prepared.

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CHAPTER 9
HYDROGEN

✪ **Position of Hydrogen in Periodic Table**

- Lightest element known having atomic number 1.
- Dihydrogen
- It resembles both alkali metals and halogens and therefore, its position is anomalous.
- In modern periodic table it is located separately

☆ **Resemblance with alkali metals:-**

1. Electronic configuration

$_1H = 1s^1$ $_{11}Na = 1s^2, 2s^2, 2p^6, 3s^1$ $_{19}K = 1s^2, 2s^2, 2p^6, 3s^2 3p^6, 4s^1$

- **Electropositive character:** H^+, Na^+, K^+ etc.
- **Oxidation state:** +1
- **Combination with electronegative elements:** form binary compounds withelectronegative elements like alkali metals.

Halides: HClNaCl, KCletc

Sulphides: H_2S Na_2S, K_2S etc

☆ **Resemblance with halogens:-**

1. Electronic configuration:

Both contain one electron less than the nearest noble gas configuration $_1H = 1s^1$ (near to $_2He$)

$_9F = 1s^2, 2s^2, 2p^5$ (near to $_8Ne$)

$_{17}K = 1s^2, 2s^2, 2p^6, 3s^2 3p^5$ (near to $_{18}Ar$)

Non-metallic character: like halogens, hydrogen is non-metallic in nature.

Atomicity: Diatomic molecules.

Formation of similar types of compounds:

i. Halides: CCl_4, $SiCl_4$, $GeCl_4$

ii. Hydrides: CH_4, SiH_4, GeH_4

5. Oxidation state: –1

☆ $Na^{+1}H^{-1}$ $Na^{+1}Cl^{-1}$

Difference from alkali metals:-

1) **Ionization enthalpy: -** the ionization enthalpy of hydrogen is very high incomparison to alkali metals.
2) **Non- metallic character:** alkali metals are typical metals while hydrogen isnon-metal
3) **Atomicity**: hydrogen is diatomic while alkali metals are monoatomic.
4) **Nature of compounds:** the compounds of hydrogen are predominantlycovalent while those of alkali metals are ionic. For example: HCl is covalent while NaCl is ionic.

   The oxides of alkali metals are basic while hydrogen oxide is neutral.

Difference from halogens:-

1) **Less tendency for hydride formation:** Hydrogen has less tendency to take upelectron to form hydride ion (H^-) as compared to the halogens which from halide ions (X^-) very easily.
2) **Absence of unshared pairs of electrons :**
3) **Nature of oxides:** The oxides of halogens are acidic while hydrogen oxide isneutral.

✪ **Occurrence of Hydrogen:**

➢ Hydrogen, the most abundant element in the universe and the third most abundant on the surface of the globe, is being visualised as the major future source of energy

✪ **Isotopes of hydrogen:-**

S.N.	Property	Protium	Deuterium	Tritium
1	Relative abudance	99.985 %	0.015 %	10^{-15}%
2	Relative atomic mass	1.007825	2.014102	3.016
3	Radioactive stability	Non-radioactive	Non-radioactive	Radioactive $t_{1/2}$ = 12.334 yrs

✪ **Preparation:**

☆ **Methods for commercial production of dihydrogen**

1. Electrolysis of water

$$2H_2O(l) \xrightarrow[\text{traces of acid/base}]{\text{Elctrolysis}} 2H2(g) + O_2(g)$$

➢ The hydrogen prepared by this method is of very high purity. However, this method is not commonly used because it is very expensive. This method is can be used only at those places where the electricity is cheap.

2. By the reaction of steam on coke :-

$$C + H_2O(g) \xrightarrow{1270\ k} CO + H_2$$

Water gas

- Since the mixture of CO and H_2 is used for the synthesis of methanol and a number of hydrocarbons, it is also called synthesis gas or syn gas.
- The process of producing syn gas from coal or coke is called coal gasification.

$$CO + H_2 + H_2O \xrightarrow[770\ k]{Fe2O3/Cr2O3} CO_2 + 2H_2$$

Water gassteam

➢ This reaction is called water gas shift reaction.

✪ **Properties of Hydrogen:-**

☆ **Physical Properties:-**

- It is slightly soluble in water (about 2 %)
- It is highly combustible and therefore should be handled carefully.
- It lightest substance. The weight of one litre hydrogen at NTP is only 0.0899 g.

☆ **Chemical properties:-**Not very reactive due to high bond dissociation energy(435.88 kJ mol^{-1} at 298.2 K)

(i)Combustion: - It burns with pale blue flame

$$2H_2\,(g) + O_2\,(g) \longrightarrow 2H_2O(l)$$

(ii) Reaction with metals:-Reactive metals like Na, K, Ca, Li and form hydrides.

$$Ca + H_2 \xrightarrow{525\,k} CaH_2$$

➢ Metals like Pt, Pd, Ni (elements of d block) form interstitial hydrides by absorbing large volume of hydrogen. Such hydrogen is called ‘occluded hydrogen and this property of adsorption of a gas by a metal is called occlusion.

(iii) Reaction with metal oxides:-Hydrogen reduces oxides of less active metals tocorresponding metal.

$$Fe_3O_4 + 4H_2 \xrightarrow{Heat} 3Fe + 4H_2$$

$$CuO + H_2 \xrightarrow{Heat} Cu + H_2O$$

(iv) Reaction with non-metals:-

$$3H_2\,(g) + N_2\,(g) \xrightarrow[673k,200\ atm]{Fe,\ Mo} 2NH_3(g)$$ { Haber process}

$\triangle H = -92.6$ kj/mole

$$2H_2\,(g) + O_2\,(g) \xrightarrow[\triangle H = -285.9]{970\,k} 2H_2OH$$

(v) Reaction with carbon monoxide:-

$$CO + H_2 \xrightarrow[700\ k,200\ atm]{ZnO,\ Cr2O3} CH_3OH$$

(vi) Reaction with unsaturated Hydrocarbons:-

(a) $$H_2C = CH_2 + H_2 \xrightarrow[473\ k,]{Ni,\ Pd\ or\ Pt} H_3C - CH_3$$

(b) Hydroformylation of olefins to aldehydes: Hydroformilation or Oxo process

$$RCH = CH_2 + H_2 + CO \xrightarrow[high\ temp]{[Co(CO)4]2} RCH_2CH_2CHO$$ and

- **Hydrogenation of oils:-**Vegetable oils are polyun-saturated in nature. The C=C bonds in oils can easily undergo oxidation and the oil becomes rancid i.e.,

unpleasant in taste. Hydrogenation reduces the number of double bonds but completely.

Vegetable Oil + H_2 $\xrightarrow[473\ k,]{Ni}$ *Solid*

✪ **Uses of Hydrogen:-**it itusedas a reducing agent.

In the manufacture of vanaspati fat, ammonia, metal hydrides, methanol, fertilizers such as urea etc.

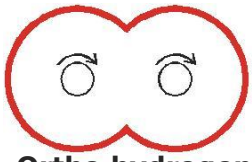
In the manufacture of synthetic petrol.

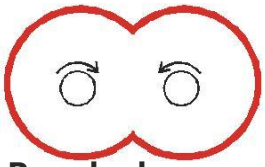
In the atomic hydrogen torch and oxy hydrogen torches for cutting and welding. Dihydrogen is dissociated with the help of an electric arc and the hydrogen atoms produced are allowed to recombine on the surface to be welded. High temperature of about 4000 k is generated.



In the fuel cell for generating electrical energy.

✪ **Ortho and parahydrogens:-**A molecules of dihydrogen …….**abc**


Ortho hydrogen


Para hydrogen

➢ They show different physical properties. For example :

- The thermal conductivity of para hydrogen is about 50 % greater than that of ortho hydrogen.
- The melting point of para hydrogen is 0.15 k below that of hydrogen containing 75% ortho hydrogen.
- They show similar chemical properties.

✪ **Atomic hydrogen:-**

- Because of high H—H bond enthalpy, atomic hydrogen is produced only at high temp in an electric arc or under ultraviolet radiation.

$$H_2\ (g) \xrightarrow{3270\ k} 2H\ (g) \quad \Delta H = +\ 435.9\ kj$$

(Atomic hydrogen)

◘ Highly reactive.

❑ Half life period is 0.3 sec and therefore, it immediately gets converted into the molecular form liberating a large amount of energy which is used for cutting and welding purposes.

✪ **Nascent hydrogen:-**The hydrogen produced in contact with the substance to bereduced is known as 'nascent hydrogen'. It is very reactive form of hydrogenBetter reducing agent than ordinary dihydrogen.

✪ **Hydrides:-**Under certain conditions H_2combines with almost all the elements ,exceptnoble gases to form compounds called hydrides.

There are three types of hydrides ,they are

= Ionic or saline hydrides

= Covalent or molecular hydrides (iii) Metallic or non-stoichiometric hydrides**(i) Ionic or saline hydrides:-**

These are the compounds of H_2 formed with most of the s-block elements which are highly electro positive.

(ii) Covalent or molecular hydrides:-These are the compounds of hydrogenformed with most of the p-block elements

[a]Electron deficient:- The hydrides which do not have sufficient number ofelectrons to form normal covalent bonds is called electron deficient hydride. For example, hydride of group 13 (BH_3, AlH_3, etc.).They are known as Lewis acids i.e., electron acceptors. To make up their deficiency they generally exist in polymeric forms such as B_2H_6, Al_2H_6, etc.

- **Electron precise:-**The hydrides which have sufficient number of electrons required for forming covalent bonds is called electron precise hydride. For example, hydrides of group 14 (CH_4, SiH_4, GeH_4, SnH_4, PbH_4 etc.) they have tetrahedral geometry.
- **Electron rich hydrides:-**The hydrides which have excess electrons as required to form normal covalent bonds is called electron rich hydride. For example, hydrides of group 15 to 17 (NH_3, PH_3, H_2O, H_2S, H_2Se, H_2Te, HF etc.)

(iii) Metallic or non-stoichiometric hydrides:-

➢ *These are formed by many d-block and f-block elements*

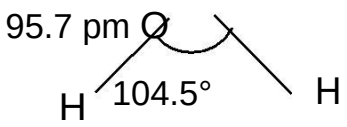
➢ *These hydrides conducts heat and electricity though not efficient.*

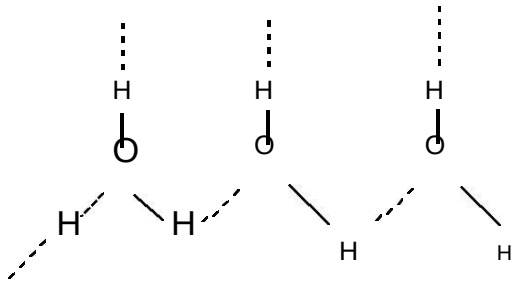
✪ **Water: -**Water! It is the major part of all living organisms.water is also knownas the river of life.

➢ Human body has about 65%and some plants haveasmuch as 95%water.

❖ **STRUCTURE OF WATER:-**

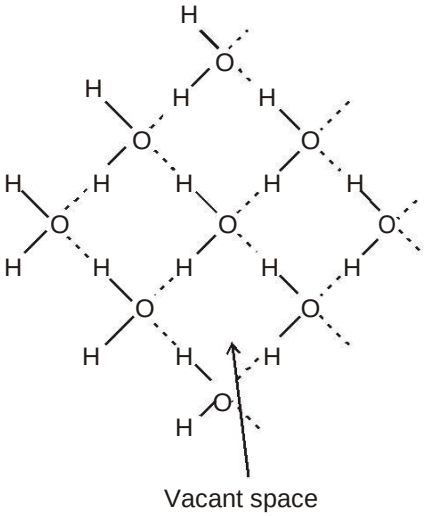
➢ In a gas phase water is bent molecule with a bond angle of 104.5 and O-H bond length of 95.7pm It is highly polar molecule.





Solid state

❖ **Structure of ice**:-Ice has a highly ordered 3D hydrogen bonded structure. Each oxygen atom is surrounded tetrahedrally by four other four other oxygen atoms at a distance of 276 pm



☆**Chemical Properties of water:-**

[i] AMPHOTERIC NATURE:-It has the ability .it acts as an acid with NH_3 and as a base with H_2S

$H_2O(l) + NH_3(aq) \rightarrow OH^-(aq) + NH_4^+(aq)$

$H_2O(l) + H_2S(aq) \rightarrow H_3O^+(aq) + HS^-(aq)$

[ii] REDOX REACTIONS INVOLVING WATER:-

➢ Water can be easily reduced to H_2 by highly electropositive metals

$2H_2O(l) + 2Na(s) \rightarrow 2NaOH(aq) + H_2(g)$

[iii] HYDROLYSIS REACTION:-Due to high dielectric constant, it has a verystrong hydrating tendency .it dissolves many ionic compounds

$P_4O_{10}(s) + 6H_20(l) \rightarrow 4H_3PO_4(aq)$ $SiCl_4(l) + 2H_2O(l) \rightarrow SiO_2(s) + 4HCl(aq)$

[iv] HYDRATES FORMATION:-From the aqueous solutions many salts can becrystallized as hydrated salts. It of different types.

(1) Coordinated water e.g., $[Cr(H_2O)_6]^{3+}\ 3Cl^-$

(2) Interstitial water e.g.,$BaCl_2.2H_2O$

(3) hydrogen-bonded water e.g. $[Cu(H_2O)_4]^{2+}SO_4^{2-}$ .H_2Oin $CuSO_4.5H_2O$

✪ **Hard & Soft Water:-**

➢ The water which contains dissolved salts of bicarbonates, sulphates and chlorides of calcium and magnesium is called hard water. Hard water does not produce lather with soap solution.

➢ Soft water is free from bicarbonates, sulphates and chlorides of calcium and magnesium. It produces lather with soap solution easily. e.g., distilled water, rain water..

☆ **Types of hardness:-**The hardness of water is of two types

(i)Temporary hardness

➢ Due to presence of soluble bicarbonates of calcium and magnesium.

➢ Can be removed by simple boiling.

(ii)Permanent hardness

➢ Due to presence of chlorides and sulphates of calcium and magnesium.

➢ Requires treatment of water to remove this type hardness.

➢ ***Do you know?***

Temporary hardness is also called carbonate hardness &
Permanent hardness is also called non-carbonate hardness

☆ **Softening of water:-**The process of removal of Ca^{2+} and Mg^{2+} ions from wateris called softening of water.

❖ **Removal of temporary hardness:-**

10. By boiling :

$$M(HCO_3)_2 \xrightarrow{Heat} MCO_3 + H_2O + CO_2 (M = Ca \text{ or } Mg)$$

(Soluble) (Insoluble)

2. Clark's method or calcium hydroxide method

$$Ca(HCO_3)_2 + Ca(OH)_2 \rightarrow 2CaCO_3 + 2H_2O$$

(Soluble) (Insoluble)

$$Mg(HCO_3)_2 + 2Ca(OH)_2 \rightarrow 2CaCO_3 + Mg(OH)_2 + 2H_2O$$

(Soluble) (Insoluble)(Insoluble)

❖ **Removal of permanent hardness:-**

(i) By washing soda ($Na_2CO_3.10H_2O$) treatment:

$$CaCl_2 + Na_2CO_3 \rightarrow 2CaCO_3 + 2NaCl$$

(Insoluble)

$$MgSO_4 + Na_2CO_3 \longrightarrow MgCO_3 + Na_2SO_4$$

(Insoluble)

(ii) By using inorganic cation exchanger (permutit method or Zeolite method):

$$Na_2Al_2Si_2O_8 + CaCl_2 \rightarrow Ca(Al_2Si_2O_8)_2 + 2NaCl$$

Zeolite Settles at bottom

The zeolite can be regenerated by treatment with sodium chloride solution. $Ca(Al_2Si_2O_8)_2 + 2NaCl \rightarrow Na_2Al_2Si_2O_8 + CaCl_2$

2 By organic ion exchanger or synthetic resins (ion exchange resins):-

➢ Synthetic resins are the insoluble polymeric solids having giant hydrocarbon network containing reactive acidic or basic groups. These are superior to Zeolitebecause they can remove all types of cations as well as anions present in water. This resulting water is known as demineralised or deionised water.

➢ These are two types:

2. Cation exchanger resins: they have acidic groups such as COOH or SO3H.they may be represented as **resin—H^+**

$$Mg^{2+} \text{ (In hard water)} \quad 2H\text{-resin} \text{ (Cation exchanger)} \rightarrow Mg(resin)_2 + 2H^+$$

$$Ca^{2+} \text{ (Hard water)} \quad 2H\text{-resin} \text{ (Cation exchanger)} \rightarrow Ca(resin)_2 + 2H^+$$

5. **Anion exchanger resins:** they have basic groups such as–OH^-or–NH_2.they may be represented as **resin—OH^-** or **resin—$NH_3^+OH^-$**

SO^{-2}_4 + 2HO resin $\longrightarrow$ SO_4 resin $_2$+ 2OH

Hard water Anion exchanger

Cl^- + HO resin $\longrightarrow$ Cl Resin + OH^-

Hard water Anion exchanger

Regeneration of resiners:

Cl re sin + NaOH $\longrightarrow$ HO re sin NaCl

Exchausted resin Re generated resin

✪ **Hydrogen peroxide [H_2O_2]:-** discovered by French chemist J.L. Thenard

❖ **Methods of preparation**

1) From sodium peroxide (Merck's process):-

Na_2O_2 + H_2SO_4 → Na_2SO_4 + H_2O_2

(20% ice cooled solution) (30% solution)

2. From Barium peroxide:-

Hydrogen peroxide was first prepared by J. L. Thenard in 1818 by acidifying barium peroxide and removal of excess water by evaporation under reduced pressure.

$BaO_2.8H_2O + H_2SO_4 \longrightarrow BaSO_4 + 8H_2O + H_2O_2$

➢ Barium sulphate is filtered off leaving behind H_2O_2.

Store of Hydrogen peroxide:-

3.40 Itmust be kept in wax lined coloured bottles because the rough glass surface causes its decomposition.

3.41 A small amount of phosphoric acid , glycerol or acetanilide is generally added which retard the decomposition of H_2O_2. These are also called negative catalysts.

Physical properties of Hydrogen peroxide:-

3. In the pure state H_2O_2 is an almost colourless(very pale blue) liquid.
4. H_2O_2 is miscible with water in allproportions and forms a hydrate $H_2O_2.H_2O$ (mp 221K).
5. A 30% solution of H_2O_2 is marketedas '100 volume' hydrogen peroxide. It means thatone millilitre of 30% H_2O_2 solution will give 100 Vof oxygen at STP. Commercially, it is marketedas 10 V, which means it contains 3% H_2O_2.

Chemical properties of Hydrogen peroxide:-

Oxidising properties

(iv) $2K_4[Fe(CN)_6] + H_2SO_4 + H_2O_2$

$2K_3[Fe(CN)_6] + 2H_2O + K_2SO_4$

(v) $C_6H_6 + H_2O_2 \longrightarrow \underset{\text{Phenol}}{C_6H_5OH} + H_2$

(vii) $NaNO_2 + H_2O_2 \longrightarrow NaNO_3 + H_2O$

(viii) $K_2Cr_2O_7 + H_2SO_4 + 4H_2O_2 \longrightarrow K_2SO_4 + CrO_5 + 5H_2O$

Reducing properties

(i) $H_2O_2 + O_3 \longrightarrow H_2O + O_2$

(ii) $Ag_2O_2 + H_2O_2 \longrightarrow 2Ag + H_2O + O_2$

(iii) $PbO_2 + 2HNO_3 + H_2O_2 \longrightarrow Pb(NO_3)_2 + 2H_2O + O_2$

Reducing properties

(iv) $2KMnO_4 + 3H_2SO_4 + 5H_2O_2$

$K_2SO_4 + 2MnSO_4 + 8H_2O + 5O_2$

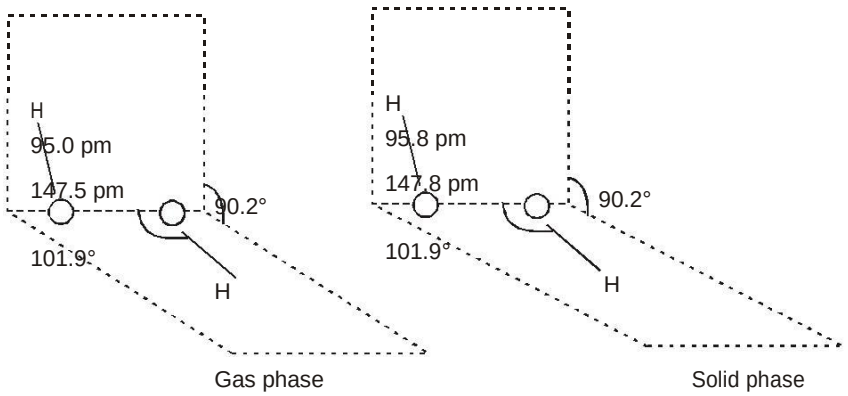
(v) $Cl_2 + H_2O_2 \longrightarrow 2HCl + O_2$

(vi) $6KAuCl_4 + 3H_2O_2 \longrightarrow 2Au + 2KCl + 6HCl + 3O_2$

❖ **Uses of hydrogen peroxide**

1) For bleaching silk, wool, hair and leather
2) As rocket fuel

❖ **Structure of hydrogen peroxide**



Structure of hydrogen peroxide

Hydrogen economy (Hydrogen as fuel)

☞ The electricity cannot be stored to run automobiles. It is not possible to store and transport nuclear energy. Hydrogen is an alternative source of energy and hence called as 'hydrogen economy'. Hydrogen has some advantages as fuel

☞ Available in abundance in combined form as water.

☞ On combustion produces H_2O. Hence pollution free.

☞ H_2-O_2 fuel cell give more power.

☞ Excellent reducing agent. Therefore can be used as substitute of carbon in reduction for processes in industry.

❖ **Obstacles in hydrogen economy**

☞ **Transportation:**

☞ Hydrogen gas is explosive and hence it is difficult to store and transport.

☞ Formation of hydrogen from H_2O:

☞ The cheaper production of the hydrogen is basic requirement of hydrogen economy which is not possible now.

☞ The main aim and advantage of hydrogen economy is to transmit energy in four of hydrogen.

One mark question:

1. What is meant by 10 volume hydrogen peroxide?
2. Why is dihydrogen gas not preferred in balloons?
3. Name the constituents of water gas.
4. Name one compound each in which hydrogen exists in (i) positive oxidation state, and (ii) Negative oxidation state.
5. What type of elements form interstitial hydrides?
6. How many hydrogen bonded water molecule(s) are present in $CuSO_4.5H_2O$?
7. What happens when heavy water is added to calcium carbonate?
8. Concentrated sulphuric acid cannot be used for drying H_2. Why?
9. Complete the following reactions?

   (i) $C_3H_8\,(g) + 3H_2O($ $\xrightarrow[\text{catalyst,}]{\text{heat}}$

   (ii) $Zn(s) + NaOH(aq)$ $\xrightarrow{heat}$

10. How heavy water is is obtained from ordinary water?

Solutions:

1. It means that 1 ml of H_2O_2 will give 10 ml of oxygen at N.T.P.
2. Dihydrogen gas is combustible in nature. Therefore, it may react with oxygen highly violently. Thus, it is not used in balloons.
3. Carbon monoxide and hydrogen.

4. (i) HCl (ii) NaH
5. Elements of d-and f-block.
6. In $CuSO_4.5H_2O$, there is one hydrogen bonded water molecule which is outside the coordination sphere. The other four molecules of water are coordinated.
7. Deutero acetylene is formed.

$$CaC_2 + 2D_2O \longrightarrow Ca(OD)_2 + C_2D_2$$

8. Conc. H_2SO_4 on absorbing water from moist H_2 produces so much heat that H_2 catches fire.
9.

(i) $C_3H_8(g) + 3H_2O(\xrightarrow[\text{catalyst,}]{\text{heat}} 3CO + 7H_2(g)$

(ii) $Zn(s) + NaOH(aq) \xrightarrow{heat} Na_2ZnO_2(aq) + H_2(g)$

10. Heavy water is obtained from ordinary water by repeated electrolysis in the presence of 3% NaOH.

Two mark question

1. Can we use concentrated sulphuric acid and pure zinc in the preparation of dihydrogen?
2. Write the chemical reactions to show the amphoteric nature of water.
3. Why is hydrogen peroxide stored in wax-lined plastic coloured bottles?
4. H_2O_2 acts as an oxidizing agent as well as reducing agent. Why?
5. What causes the temporary and permanent hardness of water?

Solutions:

1. (a) Conc. H_2SO_4 cannot be used because it acts as oxidizing agent also and gets reduced to SO_2.

$$Zn + \text{dil } H_2SO \longrightarrow ZnSO_4 + 2H_2O + SO_2$$

(b) Pure Zn is not used because it is non-porous and reaction will be slow. The impurities in Zn help in constitute of electrochemical couple and speed up reaction.

2. Water is amphoteric in nature and it behaves both as an acid as well as a base. With acids stronger than itself (eg., H_2S) it behaves as a base and with bases stronger than itself (eg. NH3) it acts as an acid.

(i) As a base: $H_2O(l) + H_2S(aq) \rightarrow H_3O^+(aq) + HS^-(aq)$

(ii) As an acid: $H_2O(l) + NH_3(aq) \rightarrow OH^-(aq) + NH_4^+(aq)$

3. The decomposition of H_2O_2 occurs readily in the presence of rough surface (acting as catalyst). It is also decomposed by exposure of light. Therefore, wax-lined smooth surface and coloured bottles retard the decomposition of H_2O_2.

4. In H_2O_2, oxygen has -1 oxidation state which lies between maximum (0 or +2 in OF2) and minimum -2. Therefore, oxygen can be oxidized to O_2 (zero oxidation state) acting as reducing agent or can be reduced to H_2O or OH- (-2 oxidation state) acting as an oxidizing agent.

$$\overset{-1}{O_2} \rightarrow \overset{0}{O_2} + 2e^-$$

(Reducing agent)

$$\overset{-1}{O_2} + 2e^- \rightarrow \overset{-2}{2O}$$

(Oxidizing agent)

5. Temporary hardness is due to presence of soluble bicarbonates of calcium and magnesium. On the other hand, permanent hardness is due to presence of chlorides and sulphates of calcium and magnesium.

Three mark question:

1. Hard water is unsuitable for laundry, washing and dyeing. Explain.
2. What do you understand by (i) electron-deficient, (ii) electron-precise, and (iii) electron-rich compounds of hydrogen? Provide justification with suitable examples.
3. Compare the structures of H_2O and H_2O_2.

Solutions:

1. Ans: - since we know that the soap are the sodium salts of higher fatty acids like stearic acid ($C_{17}H_{35}COOH$), oleic acid ($C_{17}H_{33}COOH$) or palmitic acid ($C_{17}H_{31}COOH$). When soap is added to hard water, the anions of soap combine with Ca^{+2} and Mg^{+2} ions to form calcium and magnesium salt which are insoluble in water.

$$\underset{\text{From hard Water}}{M^{+2}} + \underset{\text{sodium stearate (Soap)}}{2C_{17}H_{35}COONa} \rightarrow \underset{\text{Metal stearate (precipitate)}}{(C_{17}H_{35}COO)_2M\downarrow} + 2Na$$

Therefore, no lather is produce until all the calcium and magnesium ions are precipitated. This also results into wastage of lot of soap. So hard water is unsuitable for laundry, washing and dyeing.

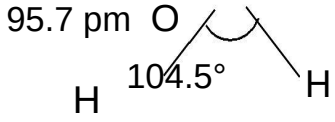
2. **[a]Electron deficient**:- The hydrides which do not have sufficient number of electrons to form normal covalent bonds is called electron deficient hydride. For example, hydride of group 13 (BH_3, AlH_3, etc.).

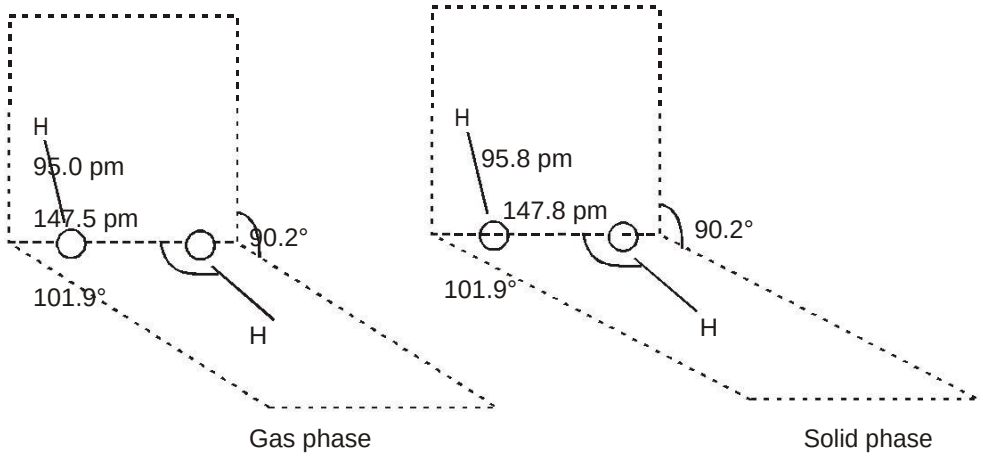
[b] Electron precise:-The hydrides which have sufficient number of electronsrequired for forming covalent bonds is called electron precise hydride. For

example, hydrides of group 14 (CH_4, SiH_4, GeH_4, SnH_4, PbH_4 etc.) they have tetrahedral geometry.

[c] Electron rich hydrides:-The hydrides which have excess electrons asrequired to form normal covalent bonds is called electron rich hydride. For example, hydrides of group 15 to 17 (NH_3, PH_3, H_2O, H_2S, H_2Se, H_2Te, HF etc.)

3. In water, O atom is sp^3 hybridised and there are two O—H bonds and two sp^3 hybrid orbitals occupy lone pairs of stronger lone pair-lone pair repulsions than bond pair-bond pair repulsions, the H-O-H bond decreases from 109.5 to 104.5^0. Therefore, water molecule is a bent or angular molecule. H_2O_2 has non-planar structure. In this structure, two O—O oxygen atoms are bonded to each other by a single covalent bond and each O atom is further bonded to a hydrogen atom by a single covalent bond. The two O—H bonds are in different planes in 111.50 in the gas phase.





Structure of hydrogen peroxide

VALUE BASED QUESTIONS

Q1.Earlier syngas (mixture of CO &H_2) was produced only by coal gasification. However, due to limited coal reserves and environmental concerns about greenhouse gases emitted during coal gasification, now a days, biomass gasification is used for production of syngas in many developed countries.Syn gas is used for the production of hydrogen gas.

Now answer the following questions

I. What is syngas?

II. What is the main difference in coal gasification and biomass gasification?

III. What are the industrial uses of syngas?

Q2. Use of excessive pesticides for crop protection and defective sewage disposal systems has contaminated the underground water. Consequently, water-borne diseases are on the rise. To safeguard the health of the general public many methods are used by Public Health Department to purify drinking water. In our houses, every family uses UV light equipped RO systems for purifying drinking water. Demineralised water(as good as distilled water) can also be prepared by the use of ion exchange method.

Now answer the following questions

I. What is RO? What happens during RO purification?

II. How does UV light help in purifying drinking water?

III. What is demineralized water? Is it fit for drinking purposes?

Q.3The coal and petroleum reserves of the world are limited and are fast dwindling. Therefore, the mankind needs an alternative source of energy. Nuclear power has recently posed severe health and environmental problems and even if these problems are solved, it cannot be used for running cars, ships and flying aeroplanes. One proposal is to burn hydrogen as a fuel. This proposal which is called hydrogen economy has some advantages as well as limitations.

I. Name some advantages of hydrogen economy.

II. What are the limitations of hydrogen economy? Suggest some means to overcome these limitations.

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Chapter 10
The s- Block Element

Important points

Groups (1 & 2) belong to the s-block of the Periodic Table.

Group 1 consists of : lithium, sodium, potassium, rubidium, caesium and francium and collectively known as the *alkali metals.*

Group 2 include : beryllium, magnesium,calcium, strontium, barium and radium. Except Beryllium they are known as *alkaline*

- **Physical properties-**

a) **Large atomic radii:** The atomic radii of alkali metals are the largest in theirrespective periods. These increase as we travel down the group.

b) **Large ionic radii:** The ionic radii increase as we move down the group due to theaddition of a new energy shell with each succeeding element.

c) **Low ionization enthalpy:** The ionization enthalpies decrease as we move downthe group.The ionization enthalpies of the alkali metals are the lowest due to loosely held s- electron.

d) **Hydration enthalpy:** It decreases with the increase in ionic radii.The hydrationenthalpy of Li ion is the maximum and the hydration enthalpy of Cs ion is the minimum.

e) **Oxidation state:** The alkali metals exhibit oxidation state of +1 in

theircompounds and are strongly electropositive in character. The electropositive character increases from Li to Cs.

f) **Metallic character:** The metallic character increases down the group.

g) **Melting point and boiling point::** The m p and b p of alkali metals are very lowand decrease with increase in atomic number.

h) **Nature of bonds formed:** These metals form ionic bonds. The ionic characterincreases as we down the group.

i) **Flame colouration:** All the alkali metals impart a charactersistic colour to theflame.

j) **Photoelectric effect:** Alkali metals (except Li) exhibits photoelectric effect.

❖ **Chemical features of alkali metals:**

a) **Reducing character:** As the ionization enthalpies of the alkali metals decreasedown the group their reducing character or reactivity in the gaseous state increases down the group. i.e., Li < Na < K < Rb < Cs .

b) **Reaction with dihydrogen:** Alkali metals react with dry hydrogen at about673 K to form crystalline hydrides which are ionic in nature and have high melting points.

$$2\,M + H_2 \xrightarrow{\text{Heat}} 2M^{+}H^{-}$$

c) **Oxides and hydroxides:** Alkali metals when burnt in air form different compounds, for example the alkali metals on reaction with limited quantity of oxygen form normal oxides (M_2O) M= Li, Na, K, Rb, Cs

5) **Reaction with halogens:** The members of the family combine with halogento form corresponding halides which are ionic crystalline solids.
Reactivity of alkali metls with particular halogen increases from Li to Cs.

6) **Reaction with water:** Alkali metals react with water and other compoundscontaining acidic hydrogen atoms such as hydrogen halides, acetylene etc. to liberate hydrogen gas.

7) **Solubility in liquid ammonia:** All alkali metals dissolve in liquid ammoniagiving deep blue solutions which are conducting in nature.

8) **Reaction with sulphur and phosphorus:** Alkali metals react with sulphur andphosphorus on heating to form sulphides and phosphides respectively.

❖ **Diagonal relationship between Li and Al**

Li resembles Mg mainly due to similarity in sizes of their atoms and ions. The main points of similarity are:

4) Both are quite hard.
5) Both LiOH and $Mg(OH)_2$ are weak bases.
6) Carbonates of both on heating decompose to produce oxides and carbondioxide.
7) Both react with nitrogen to give ionic nitrides.
8) Nitrates of both decompose on heating to give oxides.
9) Both Li and Mg do not form solid bicarbonates.
10) Because of covalent character LiCl and $MgCl_2$ are soluble in ethanol.
11) The hydroxides, bicarbonates and fluorides of both Li and Mg are sparingly soluble in water.

❖ **Biological importance of Na and K**

- Sodium ions participate in the transmission of nerve signals.
- Sodium ions also regulate flow of water across the cell membranes and in transport of sugars and amino acids into the cells.
- Potassium ions are the most abundant cations within cell fluids, where they activate many enzymes, participate in oxidation of glucose to produce ATP.
- Potassium ions in combination with sodium ions are responsible for transmission of nerve signals.
- The functional features of nerve cells depend upon the sodium potassium ion gradient that is established in the cell.

❖ **Group 2 elements: Alkaline earth metals**

 Atomic radii : The atomic radii of alkaline earth metals are fairly large thoughsmaller than the corresponding alkali metals and they increase down the group. This is because on moving down the group, atomic radii increase primarily due to the addition of an extra shell of electrons in each succeeding element.

 Ionic radii: the atoms of these elements form divalent ions which show the sametrend of increase in their size down the group.

 Ionization enthalpy: The alkaline earth metals have fairly low Ionizationsenthalpies though greater than those of the corresponding elements of group 1 and this value decreases down the group.

(d) Hydration enthalpy: the Hydration enthalpies of alkaline earth metal ion decrease as the size of the metal ion increases down the Group $Be^{2+} > Mg^{2+} > Ca^{2+} > Sr^{2+} > Ba^{2+}$

(e) Oxidation State: All the members of the family exhibit +2 oxidation state in theircompounded and the form divalent cations (M^{2+})

(f) Electro negativity : The electro negativity values of alkaline earth metals arequite close to those of alkali metals, though slightly more.

(g) Metallic Character : Alkaline earth metals have stronger metallic bonds ascompared to the alkali metals present in the same period.

(h) Melting and boiling point : The melting and Boiling points of these metals arehigher than those of alkali metals present in the same period.

(i) Colouration to the flame : With the exceptio9n of beryllium and magnesium, therest of the elements impart characters in colour to the same flame. For example,

Be	Mg	Ca	Sr	Ba	Ra
-	-	Brick Red	Crimson Grassy	Green	Crimson

J) Complex formation: Generally the members do not form complexes. However,smaller ions (Be & Mg Ions) form complexes with the electron donor species

(v) **Formation of organo-metallic compounds:** Both beryllium and magnesiumform a number of organo-metallic compounds containing M-C bond with certain organic compounds. For example, magnesium reacts with alkyl halide in the presence of dry ether to give Grignard reagent.

(vi) **Reducing character:** Alkaline earth metals are weak reducing agent than thecorresponding alkali metals which have lower ionization enthalpies and comparatively bigger atomic sizes.

(vii) **Reaction with oxygen:** With the exception of Ba and Ra which form peroxides (MO_2) rest of the metals form normal oxides (MO) on heating with excess of oxygen.

(viii) **Reaction with halogen**s: The members of the family combine directly withhalogen at appropriate temperature to form corresponding halides.

(ix) **Reaction with water:** The members of this group are less reactive towards wateras compared to the corresponding alkali metals because these are less electropositive in nature.

(x) **Reaction with hydrogen:** The members except Be combine with hydrogendirectly upon heating to form metal hydrides.

Uses of some important compounds:-

(i) Caustic soda:

It is used: in soap, paper, textile, petroleum industry **ii) Sodium carbonate**

It is used:

a) in glass and soap industry
b) in paper making and textile manufacturing
c) in paint and dye stuffs
d) in metal refining

(c) in production of sodium compounds such as borax, caustic soda, sodium phosphate etc.

iii**) Quick lime:**

It is used:

1 in the preparation of cement, glass and calcium carbide.
2.In the purification of sugar
3. In softening of hard water d. As a flux in the extraction of metal

26. Lime stone: It is used

as building material
in the manufacture of quick lime
in Solvay process to prepare Na_2CO_3 as it is a source of CO_2
in metallurgy for the extraction of iron
in toothpaste and certain cosmetics

27. Cement: It is an important building material. It is used in concrete and reinforcedconcrete, in plastering and in the construction of bridges, dams and buildings.

28. Plaster of paris: It is used

in making moulds for pottery and ceramics etc.
in surgical bandages for setting broken bones of the body
for making statues, models, decorative materials and black board chalk.

❒ **Biological importance of Ca and Mg**

Magnesium ions are concentrated in animal cells and Calcium ions are concentrated in body fluids, outside the cell.

All enzymes that utilize ATP in phosphate transfer require magnesium ion as cofactor.

In green plants magnesium is present in chlorophyll.

Calcium and magnesium ions are also essential for the transmission of impulses along nerve fibres.

Calcium ions are important in blood clotting and are required to trigger the contraction of muscles.

Calcium ions also regulate the beating of the heart.

One mark questions:

1. Why are halides of beryllium polymeric?

Ans:- the halides of Be are electron deficient as their octets are incomplete. Therefore, to complete their octets, the halides polymerize.

2. Name the groups which constitute s-block elements. Ans:- group-1 and 2

3.Arrange the alkaline earth metal carbonates in the decreasing order of thermal stability.

Ans:- $BaCO_3$> $SrCO_3$> $CaCO_3$> $MgCO_3$> $BeCO_3$

4.Write the general electronic configuration of s-block elements. Ans:- [Noble gas] ns^{1-2}

5.What is the chemical formula of Plaster of Paris? Ans:- $CuSO_4.1/2H_2O$

6.Name the compound which can be obtained by Solvay's process.

Ans:- Sodium carbonate

7.How does the basic character of hydroxides of alkali metals vary down the group? Ans:- Increases down the group

8.Which out of $MgSO_4$ or $BaSO_4$ is more soluble in water? Ans:- $MgSO_4$

9.Name radioactive elements of group 1 and 2. Ans:- Francium and Radium.

10.Which elements of alkaline earth metals family do not give characteristic flame colouration?

Ans:- Be and Mg

Two marks questions

1. Among the alkali metals which has

(i) Highest melting point

(ii) Most electropositive character

(iii) Lowest size of ion

(iv) Strongest reducing character.

Ans:- (i) Li (ii) Cs (iii) Li (iv) Li

(iii) Complete the following reactions:

$Mg(NO_3)_2 \xrightarrow{Heat}$

(ii) $LiOH \xrightarrow{Heat}$

(iii) $Na_2O + H_2O$

(iv) $Na + O_2 \longrightarrow$

Ans:-

(i) $2Mg(NO_3)_2 \xrightarrow{Heat} 2MgO + 4NO_2 + O_2$

(ii) $2LiOH \xrightarrow{Heat} Li_2O + H_2O$

(iii) $Na_2O + H_2O \longrightarrow Na_2CO_3$

(iv) $2Na + O_2 \longrightarrow Na_2O_2$

- **Name the chief factors responsible for anomalous behaviour or lithium.**

Ans:- the anomalous behaviour of lithium is because of its:

Small size of atom and ion,
High ionization enthalpy, and
Absence of d-orbitals in its Valence shell.

- **Which out of Li and Na has greater value for the following properties:**
  Hydration enthalpy
  Stability of hydride
  Stability of carbonate
  Basic character of hydroxide

Ans:- (i) Li (ii) Li (iii) Na (iv)Na

5. Why are alkali metals not found in nature?

Ans. Alkali metals are highly reactive in nature due to low ionization enthalpy andstrong electropositive character. They do not occur in free state and are always combined with other elements. As a result alkali metals are not generally found in nature.

6.Why are lithium salts commonly hydrated and those of the other alkali ions usually anhydrous?

Ans. In the lithium salt, the Li^+ion due to very small size gets readily hydrated oncoming in contact with moisture (water). Therefore, lithium salts are commonly hydrated. But the other alkali metal ions are comparatively big in size. They have therefore, lesser tendency to get hydrated. These salts are usually anhydrous.

7. Beryllium and magnesium do not give colour to flame whereas other alkaline earth metals do so why?

Ans: Beryllium and magnesium atoms in comparison to other alkaline earth metals are comparatively smaller and their ionisation enthalpies are very high. Hence, the energy of the flame in not sufficient to excite their electrons to higher energy levels. These elements, therefore, do not give any colour in Bunsen flame.

8.Why are alkali metals soft and have low melting points?

Ans: Alkali metals have only one valence electron per metal atom. As a result, the binding energy of alkali metal ions in the close-packed metal lattices are weak. Therefore, these are soft and have low melting point.

9.Which out of the following and why can be used to store an alkali metal?

H_2O, C_2H_5OH and Benzene

Ans:- Benzene can be used to store an alkali metal because other substance react with alkali metal as:

$$Na + H_2O \longrightarrow NaOH + 1/2H_2$$
$$Na + C_2H_5OH \longrightarrow C_2H_5ONa + 1/2H_2$$

10. Why are alkali metals not found free in nature?

Ans:- alkali metals are highly reactive and therefore, are not found free in nature, they are present in the combined state in the form of halides, oxides, silicates, nitrates, etc.

Three marks questions

1. When an alkali metal dissolves in liquid ammonia the solution can acquire different colours. Explain the reasons for this type of colour change.

Ans. The dissolution of the metal in liquid ammonia is accompanied by theirformation of ammoniated electrons that give rise to dark colour. This is because ammoniated electrons absorb energy corresponding to the red region of the visible light. However, if the concentration increases above 3M, the colour changes to copper-bronze and the solution acquires metallic luster due to the formation of metal ion clusters.

$$\mathbf{M+(x+y)NH_3 \rightarrow [M(NH_3)_x]^+ [e(NH_3)_y]^-}$$

2. In what ways lithium shows similarities to magnesium in its chemical behaviour?

Ans. Li resembles Mg mainly due to similarity in sizes of their atoms and ions. Themain points of similarity are:

Both are quite hard.

1 Both LiOH and $Mg(OH)_2$ are weak bases.

2 Carbonates of both on heating decompose to produce oxides and carbondioxide.

3 Both react with nitrogen to give ionic nitrides.

3. Discuss the various reactions that occur in the Solvay process.

Ans. In Solvay ammonia process.

When carbon dioxide is passed through a concentrated solution of brine saturated with NH_3, $NaHCO_3$ gets precipitated. $NaHCO_3$ on subsequent heating gives Na_2CO_3.

$$NaCl + NH_3 + CO_2 + H_2O \rightarrow NaHCO_3 + NH_4Cl$$
$$2\ NaHCO_3 \rightarrow Na_2CO_3 + CO_2 + H_2O$$

CO_2 needed for the reaction is prepared by heating calcium carbonate and the quick lime, CaO is dissolved in water to form slaked lime, $Ca(OH)_2$

$$CaCO_3 \rightarrow CaO + CO_2$$
$$CaO + H_2O \rightarrow Ca(OH)_2$$

NH_3 needed for the purpose is prepared by heating NH_4Cl and $Ca(OH)_2$

$$2\,NH_4Cl + Ca(OH)_2 \rightarrow 2\,NH_3 + CaCl_2 + H_2O$$

4. What happen when (i) magnesium is burnt in air (ii) quick lime is heated with silica (iii) chlorine reacts with slaked lime (iv) calcium nitrate is heated?

Ans. (i) A mixture of magnesium oxide and magnesium nitride is formed

$$5Mg + \underset{\textbf{In air}}{\mathbf{O_2 + N_2}} \rightarrow 2\,MgO + Mg_3N_2$$

(ii) Calcium silicate is formed.

$$CaO + SiO_2 \rightarrow CaSiO_3$$

(iii) Calcium oxychloride (bleaching powder) is formed

$$Ca(OH)_2 + Cl_2 \rightarrow CaOCl_2 + H_2O$$

(iv) Nitrogen dioxide is evolved.

$$Ca(NO_3)_2 \xrightarrow{Heat} CaO + 2\,NO_2 + O_2$$

5. Describe the importance of the following (i) limestone (ii) cement (iii) plaster of paris.

Ans. i) Lime stone: It is used

(ii) as building material

(iii) in the manufacture of quick lime

(iv) in Solvay process to prepare Na_2CO_3 as it is a source of CO_2

(v) in metallurgy for the extraction of iron

(vi) in toothpaste and certain cosmetics

ii) Cement: It is an important building material. It is used in concrete and reinforcedconcrete, in plastering and in the construction of bridges, dams and buildings.

iii) Plaster of paris: It is used

d) in making moulds for pottery and ceramics etc.

e) in surgical bandages for setting broken bones of the body

f) for making statues, models, decorative materials and black board chalk.

6. What happens when:

(iii) Sodium metal is dropped in water?

(iv) Sodium metal is heated in free supply of air?

(v) Sodium peroxide dissolves in water?

Ans. a) Sodium metal catches fire and hydrogen gas is evolved

$$2Na + 2H_2O \longrightarrow 2NaOH + H_2 + Heat$$

b) Sodium peroxide is formed

$$2Na + O_2 \longrightarrow Na_2O_2$$

c) (i) Sodium peroxide reacts with water at ordinary temperature to liberate oxygen gas

$$Na_2O_2 + 2H_2O \longrightarrow 4\,NaOH + O_2$$

ii) With ice cold water, H_2O_2 is formed

$$Na_2O_2 + 2H_2O \longrightarrow 2\,NaOH + H_2O_2$$

6. State as to why

i)a solution of Na_2CO_3 is alkaline?

ii)alkali metals are prepared by electrolysis of their fused chlorides?

iii)sodium is found to be more useful than potassium?

Ans. (a) Sodium carbonate being a salt of strong base (NaOH) and weak acid(H_2CO_3) forms alkaline solution upon hydrolysis

$$Na_2CO_3 + 2H_2O \rightarrow 2NaOH + H_2CO_3$$

(b) Since the discharge potential of alkali metals is much higher than that of hydrogen, therefore, when the aqueous solution of any alkali metal chloride is subjected to electrolysis, H_2 instead of the alkali metal is produced at the cathode. Therefore, to prepare alkali metals, electrolysis of their fused chlorides is carried out. (c) Sodium is relatively more abundant than potassium. At the same time, it is also less reactive and its reactions with other substances can be better controlled.

8. Why are potassium and cesium, rather than lithium used in photoelectric cells?

Ans. The ionization enthalpy of lithium is quite high. The photons of light are not ina position to eject electrons from the surface of lithium metal. Therefore photoelectric effect is not noticed. However, both potassium and cesium have comparatively low ionization enthalpies. This means that the electrons can quite easily be ejected from the surface of these metals when photons of certain minimum frequency (threshold frequency) strike against their surface

9. Why is Li_2CO_3 decomposed at a lower temperature whereas Na_2CO_3 at higher temperature?

Ans. Li^+ ion is very small in size. It is stabilized more by smaller anions such asoxide ion rather than large anions such as carbonate. Therefore Li_2CO_3 decomposes into Li_2O on mild heating. On the other hand, Na^+ ion is larger in size. It is stabilized more by carbonate ion than oxide ion. Hence, Na_2CO_3 does not undergo thermal decomposition easily.

10. Explain why can alkali and alkaline earth metals not be obtained by chemical reduction methods?

Ans. The metals belonging to both these families are very strong reducing agents. Itis therefore not possible to reduce their oxides by reacting with common reducing agents like carbon (coke), zinc etc. These are normally isolated by carrying out the electrolysis of the salts of these metals in the molten state.

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Five marks questions:

- Compare the solubility and thermal stability of the following compounds of the alkali metals with those of the alkaline earth metals.(a) Nitrates (b) Carbonates (c) Sulphates.

Ans. Solubility:

In case of alkali metals: Nitrates, carbonates and sulphates of alkali metals are soluble in water. In alkali metals lattice energies decrease more rapidly than the hydration energies, therefore their solubility increases down the group.

In case of alkaline earth metals: Nitrates of all alkaline earth metals are soluble in water but their solubility decreases down the group because their hydration energies decrease more rapidly than their lattice energies.

Since the size of CO_3^{2-} anions is much larger than the cations, therefore lattice energies remain almost constant with in a particular group. Since, the hydration energies decrease as we move down the group, therefore the solubility of alkaline earth metal carbonates and sulphates decrease down the group. However, the hydration energy of Be^{2+} and Mg^{2+} ions overcome the lattice energy factor and therefore $BeSO_4$ and $MgSO_4$ are readily soluble in water while the solubility of other sulphates decreases down the group from $CaSO_4$ to $BaSO_4$.

Thermal Stability:

a) Nitrates: Nitrates^{2-} and SO_4^{2-} of both alkali and alkaline earth metals decompose on heating. All alkaline earth metal nitrates decompose to form metal oxide, NO_2 and O_2.

$2M(NO_3)_2 \longrightarrow 2MO + 4NO_2 + O_2$

M= Be, Mg, Ca, Sr, or Ba

The nitrates of Na, K. Rb and Cs decompose to form metal nitrites and O_2. $2MNO_3 \longrightarrow 2MNO_2 + O_2$

However, due to diagonal relationship between Li and Mg, lithium nitrate decomposes like $Mg(NO_3)_2$ to form metal oxide, NO_2 and O_2.

$4LiNO_3 \xrightarrow{Heat} 2LiO_2 + 4NO_2 + O_2$

3. Carbonates: Carbonates of alkaline earth metals decompose on heating to form metal oxide and carbon di oxide.

$2MCO_3 \longrightarrow 2MO + CO_2$ M= Be, Mg, Ca, Ba

Further as the electropositive character of the metal increases down the group the stability of these metal carbonates increases or the temperature of their decomposition increases.

6. Sulphates: Sulphates of alkaline earth metals decompose on heating to form metal oxide and SO_3.

$MSO_4 \xrightarrow{Heat} 2MO + SO_3$ M= Be, Mg, Ca, Ba

The temperature of decomposition of these sulphates increases as the electropositive character of the metal or the basicity of the metal hydroxide increases down the group.

Among the alkali metals due to diagonal relationship, Li_2SO_4 decomposes like $MgSO_4$ to form the corresponding metal oxide and SO_3.

$$Li_2SO_4 \xrightarrow{Heat} Li_2O + SO_3$$

$$MgSO_4 \xrightarrow{Heat} 2MgO + SO_3$$

Other alkali metals are stable to heat and do not decompose easily.

2. Compare the alkali metals and alkaline earth metals with respect to (i) ionization enthalpy (ii) basicity of oxides and (iii) solubility of hydroxides..

Ans.

2 Ionization enthalpy (I E): I E of alkaline earth metals are higher than those of alkali metals of group 1. This is because the atoms of alkaline earth metals have smaller size (due to higher nuclear charge) as compared to the alkali metals.

3 Basicity of oxides: The oxides of alkali and alkaline earth metals dissolve in water to form their respective hydroxides. These hydroxides are strong bases. The hydroxides of alkaline earth metals are less basic than of alkali metals of the corresponding periods. This is due to their (i) high ionization enthalpy (ii) small ionic size and (iii) dipositive charge on the ions.

As a result M-O bond in these hydroxides is relatively stronger than that of corresponding alkali metals and therefore does not break.

4 Solubility of hydroxides: Because of smaller size and higher ionic charge, the lattice enthalpies of alkaline earth metals are much higher than those of alkali metals and hence the solubility of alkali metal hydroxides is much higher than that of alkaline earth metals. However the solubility of the hydroxides of both alkali and alkaline earth metals increase down the group due to large decrease in their lattice enthalpies as compared to their hydration enthalpies.

3.42 Explain the significance of sodium, potassium, magnesium and calcium in biological fluids.

Ans. Significance of sodium and potassium:

= Sodium ions participate in the transmission of nerve signals.

= Sodium ions also regulate flow of water across the cell membranes and in transport of sugars and amino acids into the cells.

= Potassium ions are the most abundant cations within cell fluids, where they activate many enzymes, participate in oxidation of glucose to produce ATP.

(iv) Potassium ions in combination with sodium ions are responsible for transmission of nerve signals.

(v) The functional features of nerve cells depend upon the sodium potassium ion gradient that is established in the cell.

Significance of Magnesium and Calcium:

1. Magnesium ions are concentrated in animal cells and Calcium ions are concentrated in body fluids, outside the cell.
2. All enzymes that utilize ATP in phosphate transfer require magnesium ion as cofactor.
3. In green plants magnesium is present in chlorophyll.
4. Calcium and magnesium ions are also essential for the transmission of

impulses along nerve fibres.

5. Calcium ions are important in blood clotting and are required to trigger the contraction of muscles.
6. Calcium ions also regulate the beating of the heart.

HOTS QUESTIONS

1. Potassium carbonate cannot be prepared by Solvay process. Why?

Ans. This is due to the reason that potassium bicarbonate ($KHCO_3$) formed as an intermediate (when CO_2 gas is passed through ammoniated solution of potassium chloride) is highly soluble in water and cannot be separated by filtration.

2. The hydroxides and carbonates of sodium and potassium are easily soluble in water while the corresponding salts of magnesium and calcium are sparingly soluble in water. Explain.

Ans. All the compounds are crystalline solids and their solubility in water is guided by both lattice enthalpy and hydration enthalpy. In case of sodium and potassium compounds, the magnitude of lattice enthalpy is quite small as compared to hydration enthalpy since the cationic sizes are large. Therefore, the compounds of sodium and potassium that are mentioned, readily dissolve in water. However, in case of corresponding magnesium and calcium compounds, the cations have smaller sizes and more magnitude of positive charge. This means that their lattice enthalpies are more as compared to the compounds of sodium and potassium. Therefore, the hydroxides and carbonates of these metals are only sparingly soluble in water.

3. Why is LiF almost insoluble in water whereas LiCl soluble not only in water but also in acetone?

Ans. The low solubility of LiF in water is due to its very high lattice enthalpy (F^- ion is very small in size). On the other hand, in lithium chloride (LiCl) the lattice enthalpy is comparatively very small. This means that the magnitude of hydration enthalpy is quite large. Therefore lithium chloride dissolves in water. It is also soluble in acetone due to dipolar attraction. (Acetone is polar in nature)

VALUE BASED QUESTIONS

1. Alkali metals are very reactive. They readily react with water and air but lithium is the least reactive. Many of its properties are different from rest of the elements of this group. It makes alloys with many metals but its alloys with aluminum have many useful applications. Now answer the following questions
   (I) Unlike other alkali metals, which are placed in kerosene or some other hydrocarbon, solvent (to protect them from air and moisture) lithium is covered with grease or paraffin wax and placed in air. Explain why?
   (II) Should the alloys of lithium and aluminum be called as Lithium Aluminum alloys or Aluminum Lithium alloys? Comment upon their useful applications?

Ans. (i) Lithium is a very light metal; its density being 0.53 g cm^{-3} which is less than any hydrocarbon solvent such as hexane, benzene, kerosene, etc. As a result, it will float on these solvents. That is why it is not kept in these solvents. Instead it is wrapped in grease or paraffin wax and placed as such in a bottle.

(ii) Since lithium is the least dense metal its alloys with aluminum are much less dense than aluminum. Commercial alloys of lithium with aluminum contain a maximum of 2.45% by weight of lithium. Since w/W percentage of aluminum is much higher than that of lithium, therefore these are called Al-Li rather than Li-Al alloys. These alloys have a high tensile strength and hence are used in the construction of aeroplane frames.

2. All Ca^{2+}, Na^{+} and K^{+} are the most common ions in biological system. Although Na^{+} and K^{+} ions have similar chemical properties but their biological function are quiet different.
   Now answer the following questions:-
   (i) Sodium is found to be more useful than potassium. Explain
   (ii) What are the health benefits of potassium?
   (iii) Should potassium supplements be used to increase the uptake of potassium ions?
   (iv) Mention some rich sources of potassium foods?

Ans. (i) Sodium ions are primarily found in the blood plasma and in the interstitial fluid which surrounds the cells while potassium ions are primarily present within the cell fluids. Sodium ions primarily help in the transmission of nerve signals, in regulating the flow of water across cell membranes and in the transport of sugars and amino acids into the cell. Thus sodium is found to be more useful than potassium.

(ii) Potassium is important for healthy functioning of our heart, kidneys and other vital organs. Deficiency of potassium in diet causes heart diseases, high blood pressure, cancer, arthritis, digestive disorders and muscle cramps. The benefits of potassium are :

It reduces blood pressure.

Kidney stones are formulated by calcium being in the kidneys, the potassium prevents calcium from forming stones. The kidney filters the potassium and eliminates it in urine.

(a) Potassium helps in carrying oxygen to the brain and is important for healthy nerve functions. It is also good for relieving headache due to allergies.

(iii) Potassium supplements should only be taken with the recommendation and supervision of the doctor since potassium does interact with other medicines.

(iv) Some rich sources of potassium foods are: baked potatoes with skin, bananas, spinach, orange juice, tomato sauce, barley, beans, almonds etc.

3. Calcium ions play an important role in biological processes. About 90% of body calcium is present in bones and teeth. Calcium ions are also important in blood clotting and are required to trigger the contraction of muscles and to maintain the regular beating of heart. In spite of these benefits, calcium is also part of kidney stones which account for 80% of the renal stone diseases. Now Answer these following questions

(i) Name compounds of which the bones and the teeth are made of. Also name the compound of the enamel on the teeth.
(ii) What are calcium kidney stones and how are they formed?
(iii) What type of foods should be avoided to prevent the formation of kidney stones?
(iv) Should the intake of calcium be reduced to avoid formation of kidney stones?

Ans. (i) The bones and the teeth are made up of calcium apatite. $Ca_3(PO_4)_2$ and the enamel on teeth is made up of fluoropatite, $[Ca_3(PO_4)_2.CaF_2]$.

(ii) Calcium kidney stones are made up of calcium oxalate. Calcium ions combine with oxalate ions during digestion to form calcium oxalate. It is not readily absorbed into the body but gets deposited in the kidney as stones.

(iii) A diet rich in oxalate containing foods which can promote stone formation should be avoided. Some of the oxalate rich foods are : spinach, beets, chocolate, black tea and soybean crackers.

(iv) Diets rich in calcium have little effect on stone formation or may actually protect against the kidney stone formation. The use of calcium supplements or vitamin D or vitamin C supplements may, however , promote kidney stone formation and should be avoided by those who are at risk of kidney stone formation. Regular use of calcium carbonate antacids also increases stone formation.

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CHAPTER 11

The p-block elements

Elements in which the last electron enters in the any one of the three p- orbital of their outermost shells – p-block elements

- Gen. electronic configuration of outer shell is ns^2np^{1-6}

The inner core of e-config.may differ which greatly influences their physical & to some extent chemical properties.

- The block of elements in the periodic table consisting of the main groups :
- Group 13 (B to Tl)
- Group14 (C to Pb)
- Group15 (N to Bi)
- Group 16 (O to Po)
- Group17 (F to At)
- Group18 (He to Rn)

(1) Members at the top and on the right of the *p*-block are nonmetals (C, N, P, O, F, S, Cl, Br, I, At).

- Those on the left and at the bottom are metals (Al, Ga, In,Tl, Sn, Pb, Sb Bi, Po).

- Between the two, from the top left to bottom right, lie an ill-defined group of metalloid elements (B, Si, Ge, As, Te)

GROUP 13 : The boron group

Outer Electronic Configuration:-ns^2np^1

• group members: boron (B), aluminum (Al), gallium (Ga), indium (In)& thallium (Tl) . All, except boron, are metals.

- Boron show diagonal relationship with Silicon; both are semiconductors metalloids & forms covalent compounds.

- Boron compounds are electron deficient, they are lack of an octet of electrons about the B atom .

- diborane B_2H_6, is simplest boron hydride
- Structure: three-center two-electron: the H atoms are simultaneously bonded to two B atoms the B-H bridging bond lengths are greater than B-H terminal.

- - Boron oxide is acidic (it reacts readily with water to form boric acid)

- aluminium compounds:aluminium oxide is amphoteric

- aluminum halides, e.g., $AlCl_3$ is dimer, an important catalyst in organic chemistry have anincomplete octet, acts as Lewic acid by accepting lone pairs from Lewic bases, forming adduct

- aluminum hydride, e.g., $LiAlH_4$, a reducing agent

- Atomic Properties - Electronic Configurations

Element	Symbol	Atomic No.	Electronic Configuration	Abundance in Earth's Crest (in ppm)
Boron	B	5	$[He]2s^2\ 2p^1$	8
Aluminium	Al	13	$[Ne]3s^2\ 3p^1$	81,300
Galium	Ga	31	$[Ar]3d^{10}4s^2\ 4p^1$	15
Indium	In	49	$[Kr]\ 4d^{10}5s^2\ 5p^1$	1
Thallium	Tl	81	$[Xe]\ 5d^{10}6s^2\ 6p^1$	0.3

. Atomic and ionic radii

- The atomic and ionic radii of group 13 elements are compared to corresponding elements of group 2. From left to right in the period, the magnitude of nuclear charge increases but the electrons are added to, the same shell. These electrons do not screen each other, therefore, the electrons experience greater nuclear charge.

- In other words, effective nuclear charge increases and thus, size decreases. Therefore, the elements of this group have smaller size than the corresponding elements of second group.

- On moving down the group both atomic and ionic radii are expected to increase due to the addition of new shells. However, the observed atomic radius of Al (143 pm) is slightly more than that of Ga (l35 pm).

Ionization energies

The first ionization energies of group 13 elements are less than the corresponding members of the alkaline earths.

The sharp decrease in I.E. from B to Al is due to increase in size. In case of Ga, there are ten d-electrons in its inner electronic configuration.

The very high value of 3rd I. E. of thallium indicates that +3 O.N. state is not stable, rather +1 is more stable for thallium .

Electropositive (or metallic) character

the elements of group 13 are less electropositive as compared to elements of group 2. On moving down the group the electropositive (metallic) character increases because ionization energy decreases. For e.g., Boron is a non-metal white the other elements are typical metals.

Oxidation states

The common oxidation states of group 13 elements are +3 and + l .The stability of the + 1 oxidation state increases in the sequence Al <Ga< In <Tl, Due to Inert pair effect.

Element	B	Al	Ga	In	Tl
Oxidation state	+3	+3	+3, +1	+3, +1	+3, +1

Chemical reactivity of Gr.13 Elements

All elements in their compounds exhibit the oxidation state of + 3 and +1. Hydrides

- None of the group 13 elements reacts directly with hydrogen. However, a no. of hydrides of these elements have been prepared by indirect methods. The boron hydrides are called boranes& classified in two series: (a) B_nH_{n+4} called nidoboranes (b) B_nH_{n+6} called arachnoboranes
- INUDUSTRIAL PREPERATION :-

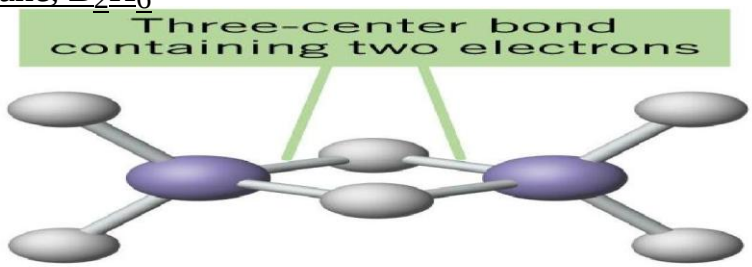
  $2BF_3(g) + 6LiH(s) \rightarrow B_2H_6(g) + 6LiF(s)$

- Laboratory method:
  - By the reaction of iodine with sodium borohydride in a high boiling solvent.

    $2NaBH_4 + I_2 \rightarrow B_2H_6 + 2NaI + H_2$
  - By reduction of BCl_3 with $LiAlH_4$

    $4BCl_3 + 3LiAlH_4 \rightarrow 2\ B_2H_6 + 3AlCl_3 + 3\ LiCl$

Structure of Diborane, B_2H_6





Some important characteristics of boranes:

i) Lower boranes are colourless gases while higher boranes are volatile liquids or solids.

- They undergo spontaneous combustion in air due to strong affinity of boron for oxygen.

$B_2H_6 + 3O_2 \rightarrow B_2O_3 + 3H_2O + Heat$

iii) Boranes react with alkali metal hydrides in diethyl ether to form

borohydride complexes.

$B_2H_6 + 2MH \rightarrow 2M^+[BH_4]^-$ (M= Li or Na)

Metal borohydride

- (iv) Diborane reacts with ammonia to give borazine at 450 K. $B_2H_6 + 6NH_3 \rightarrow 3B_3N_3H_6 + 12H_2$

- Borazine has a cyclic structure similar to benzene and thus is called inorganicbenzene

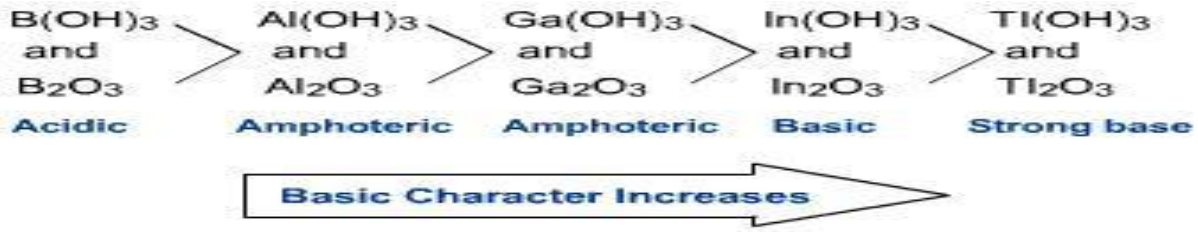
- The other elements of this group form only a few stable hydrides. The thermal stability decreases as we move down the group.

- AlH_3 is a colourless solid polymerized via Al - H - Al bridging units. These hydrides are weak Lewis acids and readily form adducts with strong Lewis base (B:) to give compounds of the type MH_3 (M = Al or Ga). They also form complex-tetrahydrido anions, $[MH_4]$-. The most important tetrahydrido compound is $Li[AlH_4]$

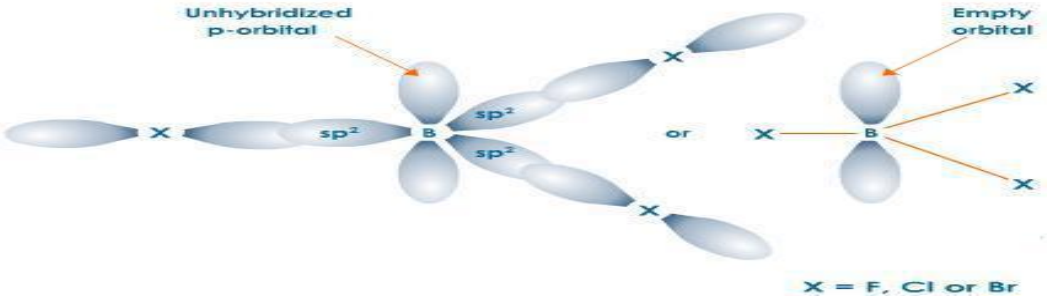
$$4LiH + AlCl_3 \xrightarrow{\text{ether}} LiAlH_4 + 3LiCl$$

OXIDES & HYDROXIDES

- M_2O_3& $M(OH)_3$

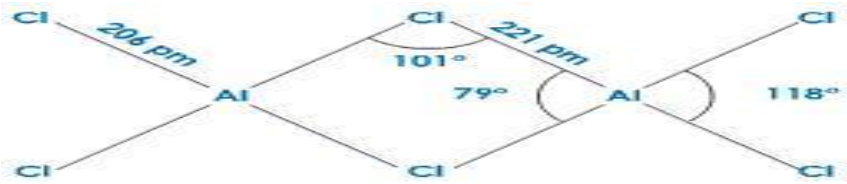


HALIDES: Structure of boron trihalides



Dimeric structure of aluminium chloride

– Boron halides do not form dimers because the size of boronis so small that it is unable to coordinate four large-sized halide ions.



Anomalous properties of boron

• Boron is a non-metal & bad conductor of electricity whereas aluminium is a metal & good conductor. B is hard but Al is a soft metal.

• Boron exists in two forms-crystalline and amorphous. But Al does not exist in different forms.

• The melting and boiling point of boron are much higher than that of Al .

` Boron forms only covalent compounds whereas Al forms even some ionic compounds.

The hydroxides and oxides of boron are acidic in nature whereas those of aluminium are amphoteric.

The trihalides of boron exist as monomers. On the other hand, aluminium halides exist as dimers .

• The hydrides of boron are quite stable while those of aluminium are unstable

Boron and silicon exhibit the typical properties of non-metals. These do not form cations. Both exist in amorphous as well as crystalline forms.

Boron oxide (B_2O_3) and silica (SiO_2) both are acidic and dissolve in alkali solutions to form borates and silicates respectively.

$B_2O_3 + 6NaOH \rightarrow 2Na_2BO_3 + 3H_2O$

$SiO_2 + 2NaOH \rightarrow Na_2SiO_3 + H_2O$

- The chlorides of both B and Si get hydrolyzed by water to boric acid and silicic acid respectively.

$BCl_3 + 3H_2O \rightarrow H_3BO_3 + 3HCl$ $SiCl_4 + 3H_2O \rightarrow H_2SiO_3 + 4HCl$

The hydrides of Boron and Silicon are quite stable. Numerous volatile hydrides are also known which catch fire on exposure to air and are easily hydrolyzed.
Both elements are semiconductors.

Behavior in Aqueous Solutions

1 Al, Ga, In and Tl exhibit a well-defined aqueous chemistry in their tripositive states. Species like $[M(OH)_4]^-$, $[M(H_2O)_2(OH)_4]^-$, $[M(H_2O)_6]^{3+}$ for M = Al, Ga, In, exist in aqueous solution.

- Al, Ga. In and T1 ions exist as octahedral aqua ions, $[M(H_2O)_6]^{3+}$ in aqueous solution and many salts like halides, sulphates, nitrates and perchlorates exist as hydrates.

- Aluminiumsulphate forms double salts - called alum, having the general formula $M_2SO_4.Al_2(SO_4)_3.12H_2O$, where $M=Na^+$ or K^+.

USES OF BORON & ALUMINIUM

- Aluminium is used extensively in industry and everyday life. It forms many useful alloys with Cu. Mn, Mg, Si and Zn. Hence, aluminium and its alloys find use in packaging, utensil making, construction, aerospace and other transportation industries. It is used as a conductor for transmission of electricity. Aluminium is alsoused in the alumino-thermite process for production of chromium and manganese from their ores.

Group 14 Elements:-The Carbon Family

Group 14 includes carbon (C), silicon (Si), germanium (Ge), tin (Sn) and lead (Pb).

General electronic configuration of carbon family is ns^2np^2.

Covalent radius:-Covalent radius expected to increase from Cto Si,

From Si to Pb small increase is found.

Ionization Enthalpy:-The first ionization enthalpies of group 14 elements are higher than those of the corresponding group 13 elements.

Electronegativity:-Group 14 elements are smaller in size as compared to group 13 elements that's why this group elements are slightly more electronegative than group 13

Chemical properties:-

Carbon and silicon mostly show +4 oxidation state. Germanium forms stable compounds in +4 state and only few compounds in +2 state.

Tin forms compounds in both oxidation states. Lead compounds in +2 state are stable and in +4 state are strong oxidizing agents.

Exception:-Pb_4 and SnF_4 are ionic in nature.

Except CCl_4 other tetrachlorides are easily hydrolysed by water.

Since carbon does not have d-orbitals and hence cannot expand its coordination number beyond 4

$CCl_4 + H_2O$ No Reaction

$SiCl_4 + 4H_2O$ $Si(OH)_4 + 4HCl$

Silicic acid

Allotropes of Carbon:-The three types of allotropes are –

1-Diamond 2-Graphite 3-Fullerence

Diamond:-In diamond each carbon atom undergas SP^3hybridisation.

Each carbon is tetrahedrally linked to four other carbon atoms.

Graphite:-In graphite, carbon is SP^2-hyberdized graphite has a two-dimensional sheet like structure consisting of a number of hexagonal rings fused together.

Graphite conducts electricity along the sheet.It is very soft and Slippery

FullerenceFullerence was discovered collectively by three scientists namely R.E Smalley,R.F Curl and H.W Kroto

SOME Important Compounds Of Carbon and Silicon

Carbon monoxide:-It I prepared by direct oxdisation of C in limited supply of oxygen.

$$2C + O_2(g) \rightarrow 2CO(g)$$

Commercially it is prepared by the passage of steam over hot coke

Carbon Dioxide:-It is prepared by complete combustion of carbon and carbon fuels in excess of air.

$$C(s) + O_2(g) \rightarrow CO_2(g)$$

Laboratory method:-

In laboratory it is prepared by the treatment of dilHCl on $CaCO_3$

$$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$$

Silicon dioxide:-Silicon dioxide is a COVALENT THREE DIMENSIONAL NETWORK SOLID.

Each silicon atom is covalently bonded in a tetrahedral manner to four oxygen atoms.

Silicones:-Silicones are the synthetic organo-siliconpolymers having general formulae $(R_2SiO)_n$ in which R = alkyl (methyl,ethyl or phenyl)

Silicates:-Silicates are exist in nature in the form of feldspar, zeolites,mica and asbestos etc.

The basic structure of silicates is SiO_4^{4-}

Zeolites:-Zeolites is aalumino-silicate of metal. Metal cations participating in formationof Zeolite are use usually Na^+,K^+,or Ca^{2+}.

Zeolites are used to remove permanent hardness of water.

ONE MARK QUESTIONS

11. Why is boron used in nuclear reactions?

Ans:-Because Boron can absorb neutrons.

12. By giving a balanced equation show how $B(OH)_3$ behaves as an acid in water.

Ans:-$B(OH)_3 + 2HOH \longrightarrow [B(OH)_4]^- + H_3O^+$

- Name the element of group 14 which exhibits maximum tendency for catenation?

Ans:-Carbon

- What is the basic building unit of all silicates?

Ans:-SiO_4^{4-} is the basic unit of all silicates.

- What happens when $NaBH_4$ reacts with iodine?

Ans:-$2NaBH_4 + I_2 \rightarrow B_2H_6 + 2NaI + H_2$.

6. What happens when boric acid is heated

Ans:-$4H_3BO_3 \longrightarrow 4HBO2 \longrightarrow H_2B_4O_7$.

7. What is producer gas?

Ans:-Producer gas is a mixture of CO and N_2 in the ratio of 2:1.

8.Write the state of hybridization of 'B' in

BF_3. ANS:-Hybridisation of 'B' in BF_3 is Sp^2.

9.Mention the state of hybridization in B in BH_4^-

. Ans:-Sp^3.

10. Which oxide of carbon is regarded as anhydride of carbonic acid. Ans:-CO_2 is regarded as a hydride of carbonic acid .

Because $H_2CO_3 \rightarrow H_2O + CO_2$

TWO MARKS QUESTIONS

1. Give the chemical reaction as an evidence for each of the following observations.

(i) Tin (II) is a reducing agent where as lead (II) is not.

(ii)Gallium (I) undergoes disproportionation reaction.

Ans:- (i) Due to inert pair effect pb^{2+} is more stable than Pb^{4+}. Whereas Sn^{4+} is more stable than Sn^{2+}.

(ii) $3Ga^+ \rightarrow 2Ga + Ga^{3+}$

This is because Ga^{3+} is more stable than Ga^+.

2.What happens when

Quick lime is heated with coke?

Carbon monoxide reacts with Cl_2

Ans:- (i) $CaO + C \rightarrow CaC_2 + CO$

(iii) $CO + Cl_2 \rightarrow COCl_2$

3.Give reason

(vi) C and Si are always tetravalent but Ge,Sn,Pb show divalency.

(vii) Gallium has higher ionization enthalpy than Al. Explain.

Ans:-(i) Ge, Sn, Pb show divalency due to inert pair effect, Pb^{2+} is more stable than Pb^{4+}.

(ii) Due to poor shielding effect of d-electrons in Ga effective nuclear charge increases as compared to Al thus the I.E is higher than Al.

4.Give reason why boron and aluminium tend to form covalent compounds.

Ans:-Sumof three ionization of both the element are very high. Thus they have no tendency to lose electrons to form ionic compound.Instead they form covalent compounds.

5.If B-Cl bond has a dipole moment, Explain why BCl_3 molecule has zero dipole moment.

Ans:- B-Cl bond has dipole moment because of polarity.In BCl_3 since the molecule is symmetrical thus the polarities cancel out.

6.Suggest a reason as to why CO is poisonous.

Ans:-CO reacts with haemoglobin to form carboxy-haemoglobin which can destroy the oxygen carrying capacity of haemoglobin and the man dies of suffocation.

7. What do you understand by-

(a) Inert pair effect:-The pair of electron in the valence shell does not take part in bond formation it is called inert pair effect.

(b) Allotropy:-It is the property of the element by which an element can exists in two forms which have same chemical properties but different physical properties due to their structures.

8.How is excessive content of CO_2 responsible for global warming?

Ans:-Excess of CO_2 absorbs heat radiated by the earth.Some of it dissipated into the atmosphere while the remaining part is radiated back to the earth.Temperature of the earth increases.

9.Describe two similarities and two dissimilarities between B and Al.

Ans:-Similarities:-

3 Both have same number of valence electrons.
4 Both have similar electronic configuration.

Dissimilarities:-

- Bis a non- metal where Al is a metal
- B forms acidic oxide whereas Al forms atmospheric oxides.

10.What are fullerene? How they were prepared?

Ans:- Fullerene are the allotropes of carbon.Its structure is like a soccer ball.

They are prepared by heating graphite in electric arc in presence of inert gases such as helium or argon.

THREE MARKS QUESTIONS

1.What happens when

(a)Borax is heated strongly

(b)Boric acid is added to water

(c)Aluminium is treated with dilute NaOH

Ans:-(a) $Na_2B_4O_7 . 10H_20 \rightarrow Na_2B_4O_7 \rightarrow 2NaBO_2 + B_2O_3$

(b) $B(OH)_3 + H_2O \rightarrow [B(OH)_4]^- + H^+$

(C) $2Al + 2NaOH + H_2O \rightarrow 2NaAlO_2 + 3H_2$

2.Explain the following reactions.

(a)Silicon is heated with methyl chloride at high temperature in the presence of copper.

(b)Silicon dioxide is treated with hydrogen fluoride.

(c) CO is heated with ZnO.

Ans:- (a) A mixture of mono-,di- and trimethylchlorosilianes along with a small amount of tetramethylsilane is formed.

$CH_3Cl + Si \rightarrow CH_3SiCl_3 + (CH_3)_2 SiCl_2 + (CH_3)_3SiCl + (CH_3)_4 Si$

(b) The initially formed silicon tetrafluroide dissolves in HF to form hydrofluorosilicic acid

$SiO_2 + 2HF \rightarrow SiF_4 + 2H_2O$

$SiF_4 + 2HF \rightarrow H_2SiF_6$

(c) ZnO is reduced to zinc metal

$ZnO + CO \rightarrow Zn + CO_2$

3.Give reasons:-

(a)Diamond is used as an abrasive.

4. Aluminium alloys are used to make aircraft body.

5. Aluminium utensils should not be kept in water overnight.

Ans:- (a) Diamond is used as an abrasive because it is an extremely hard substance.(b)Alloys of alumimium likeduralium is used to make aircraft body due to Someof its property .

(c)Generally aluminium metal does not react with water quickly but when it is kept overnight.It reacts slowly with water in presence of air.

$$2Al(s) + O_2(g) + H_2O(l) \rightarrow Al_2O_3(s) + H_2(g)$$

4.A certain salt X,gives the following results.

(i)Its aqueous solution is alkaline to litmus.

(ii)It swells up to a glassy material Y on strong heating.

(iii)when conc.H_2SO_4 is added to a hot solution of X,whitecrystalof an acid Z separates out.

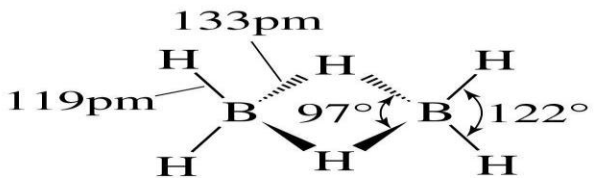
Ans:- (i) $Na_2B_4O_7 + 10H_2O \rightarrow 2NaOH + H_2B_4O_7 + 8H_2O$

(ii) $Na_2B_4O_7 \rightarrow 2NaBO_2 + B_20_3$

(iii)$Na_2B_4O_7.10H_2O + H_2SO_4 \rightarrow 4H_3BO_3 + Na_2SO_4 + 5H_2O$

5. draw structure of diborane .

Ans.



FIVE MARKS QUESTIONS

7. Explain the formation of (i) Water gas (ii) Producer gas. Give their uses. What happens when CO_2 is passed through lime water (i) for short duration (ii) for long duration.

Ans:-(i) $C(s) + H_2O(g) \rightarrow CO(g) + H_2(g)$
(Water gas)

(ii) $2C(s) + O_2 + 4N_2(g) \rightarrow 2CO(g) + 4N_2(g)$
(Producer gas)

Water gas and Producer gas are used as fuel.

$Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$

(White ppt.)

(i) $CaCO_3 + CO_2 + H_2O \rightarrow Ca(HCO_3)_2$
(Soluble)

5 (a) Why do Boron halides from addition compound with NH_3 ?
(b) Assign appropriate reason for each of the following observations :-

i)Anhydrous $AlCl_3$ is used as a catalyst in many organic reactions.
ii)No form of elemental silicon is comparable to graphite.

Ans:- (a) It is because BX_3 is electron deficient whereas NH_3 is electron rich.

(i) It is Lewis acid.

ii)It cannot form $p\pi - p\pi$ bond due to large size.

8. (i) Give reason for the following observations:-
i)The tendency for catenation decreases down the group in Group 14.
ii)The decreasing stability of +3 oxidations state with increasing atomic number in group 13.
iii)PbO_2 is a stronger oxidizing agent than SnO_2.
iv)Molten aluminium bromide is a poor conductor of electricity.

Ans:- (i)(a) It is due to decrease in bond dissociation energy which is due to increase in atomic size.

C-C > Si-Si >Ge-Ge>Sn-Sn>Pb-Pb.

ii)It is due to inert pair effect.

iii)PbO_2 is stronger oxidizing agent than SnO_2 because Pb^{2+} is more stable than Pb^{4+} whereas Sn^{4+} is more stable than Sn^{2+}.

iv) Molten $AlBr_3$ is poor conductor of electricity because it is covalent compound.

VALUE BASED QUESTIONS

Q. 1. Kevlar is a nylon obtained by condensation polymerization of 1, 4-diaminobenzene and 1, 4-benenedicarboxylic acid. Its fibres are five times stronger than steel & hence are used for making bullet-proof vests. There are two compounds of boron which are also used for making bullet-proof vests.

Now answer the following questions.

(i) Name the two compounds of boron which are used for making bullet-proof vests and how are they prepared ?

(ii) What are the other advantages of using boron based compounds over Kevlar for making bullet-proof vests ?

Ans. (i) The compounds of boron which are used for making bullet-proof vests are : boron nitride (BN) and boron carbide (B4C). These are prepared from boron oxide as follows :

$B_2O_3 + 2\ NH_3 \rightarrow 2\ BN + 3\ H_2O$ (T = 1173 K)

$2\ B_2O_3 + 7\ C \rightarrow B_4C + 6\ CO$ (T = 1873 K)

(iii) Boron has high ability to absorb neutrons and hence boron carbide or boron nitride based bullet-proof vests can also act as protective shield against radioactive radiations.

Q. 2. One of the elements of group 13 was not known at the time when Mendeleev gave his periodic table. This element has some interesting/remarkable properties much different from other elements of this group.

Now answer the following questions

(i) Name the element which was not known when Mendeleev gave his periodic table.

(ii) Mention its few interesting and remarkable properties.

(iiI) Give one use of this element

Ans. (i) The element not known when Mendeleev gave his periodic table is gallium (Ga).

(ii) Some of its remarkable properties are :

(a) It has the greatest liquid range of all known elements. It can melt (303 K) in your hand but it does not boil until the temperature reaches 2676 K.

(b) Like water, gallium is one of the few known materials which expands upon freezing.

(c) The greatest use of gallium is in the preparation of semiconductor gallium arsenide (GaAs) . Integrated circuits made up of GaAs have achieved operating speeds up to five times that of the fastest silicon chips currently available, & they will operate over a wider temperature range than silicon circuits. However, these advantages come at a price because As it is a volatile and toxic element and hence GaAs is difficult to make.

Q. 3. Carbon has two main categories of allotropes – crystalline & amorphous. Among crystalline allotropic forms of carbon, diamond & graphite are known since long while the third category of carbon called the carbon nanotubes was is discovered around 1991.

Now answer the questions :

(i) What are carbon nanotubes ? In what respects their structures differ from buckminster fullerence ?

(ii) What properties have made them so interesting?

(iii) Name same applications which are envisioned for these interesting molecules.

Ans. (i) A carbon nanotube consists of a two-dimensional array of hexagonal rings of carbon just as in a layer of graphite or a chicken wire.

The layer (or the chicken wire) is then rolled into a cylinder and capped at each end with half of a C60-fullerene.

(ii) Carbon nanotubes are very tough, about 100 times as strong as steel. They are electrically conducting only along the length of the tube.

(iiI) Carbon nanotubes can be used to produce very strong composite materials. They are also being used as probe tips for analysis of DNA and proteins by atomic force microscopy (AFM). It is also thought that carbon nanotubes ca be used to make molecular size test tubes or capsules for rug delivery.

Q. 4. For those who like the taste of coffee but do not want the caffeine, decaffeinated coffee is available. Decaffeinated coffee was first made from roasted green coffee beans with the solvent chloroform. Later, safer organic solvent methylene chloride was used. Today, most of the Decaffeinated coffee is extracted by supercritical carbon dioxide.

Now answer the questions.

(i) What is supercritical carbon dioxide and how is it produced?

(ii) Why supercritical fluids have recently gained importance as extracting solvents?

(iii) What are the solvents which are presently used for dry cleaning clothes? Can supercritical carbon dioxide be used as a solvent for dry cleaning clothes?

(iv) Would this excessive use of carbon dioxide not contribute to global warming? Explain.

Ans. (i) The critical temperature CO_2 is 304 K and its critical pressure is 73 atm. Therefore, in a tank of CO_2 above its critical temperature and pressure, the liquid & gas phases of CO_2 are replaced by a single phase called the supercritical carbon dioxide. Although supercritical CO_2 is a gas, it being very dense can be used as a solvent.

(ii) Supercritical carbon dioxide and other supercritical fluids have gained importance in recent times because of the possibility of replacing toxic and environmentally unfriendly solvents for extraction of organic compounds from natural sources.

(iii) These days the most common solvent used for dry cleaning clothes is perchloroethylene, $Cl_2C = CCl_2$ but is an air pollutant. Therefore, clothes can be dry cleaned with supercritical carbon dioxide if we use a special detergent.

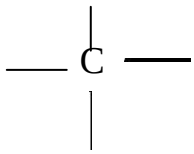
(iv) Although carbon dioxide is a greenhouse gas, but the gas once used can be recirculated and not allowed to escape into the atmosphere. Consequently, its impact on gloal warming will be negligible.

CHAPTER 12

Organic Chemistry : Some Basic Principles and Techniques

Organic compounds are the hydrocarbons and their derivatives and organic chemistry is that branch of chemistry that deals with the study of these compounds Tetravalency of carbon

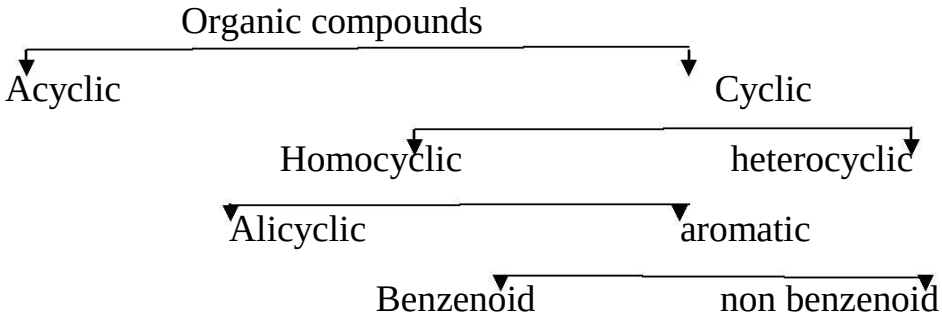
The atomic number of Carbon is 6 and its electronic configuration is 2,4 i.e. it has 4 valence electrons. Thus carbon is always tetracovalent, i.e. it forms 4 covalent bonds with other atoms



Due to tetravalency of carbon it has a tetrahedron shape.

Catenation- The self linking property of carbon is known as catenation. This is the main reason of existence of such large number of compounds

Classification of organic compounds



Functional groups:A functional group may be defined as an atom or a group ofatoms present in a molecule which largely determines the chemical properties.

CLASS OF ORGANIC COMPOUNDS	**NAME OF FUNCTIONAL GROUP**	**STRUCTURE**
Alkenes	double bond	=C=C=
Alkynes	triple bond	- C ≡ C -
Halogens	halogen	- X (F,Cl,Br,I)
Alcohols	hydroxyl	-OH
Aldehydes	aldehydic(formyl)	-CHO
Carboxylic acids	carboxyl	-COOH
Acid amides	amides	$-CONH_2$
Primary amines	amino	$-NH_2$

HOMOLOGOUS SERIES

Homologous series is defined as a family or group of structurally similar organic compounds all members of which contain the same functional group, show a gradation in physical and similarity in chemical properties and any two adjacent members of which differ by $-CH_2$ group. The individual members of this group are called homologues and the phenomenon is called homology.

NOMENCLATURE OF ORGANIC COMPOUNDS

Organic chemistry deals with millions of compounds. In order to clearly identify them, a systematic method of naming known as IUPAC system of nomenclature is adopted. The names are such that the listener can deduce the structure from it. The IUPAC name consists of three parts:

	Prefix	Word root	Suffix
EX:	3-methlyoctane		

NOMENCLATURE OF ALKANES

Straight chain alkanes:

The names of such compounds is based on their chain structure,and end with suffix '-ane' and carry a prefix indicating the number of carbon atoms present in the chain.

Branched chain hydrocarbons:

1.) The longest carbon chain in the molecule is identified.
2.) The numbering is done in such a way that the branched carbon atoms get the lowest possible value.
3.) The names of the alkyl groups attached as a branch are then prefixed to the name of the parent alkane and its position is indicated by numbers.
4.) The lower number is given to the first in alphabetical order.
5.) The carbon atom of the branch that attaches to the root alkane is numbered 1.

Organic compounds having Functional Groups:

The longest chain of carbon atoms containing the functional groups is numbered in such a way that the functional group attached to the carbon atom gets the lowest possible number in the chain.

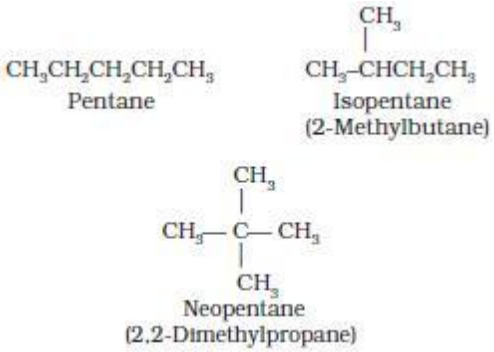
When there are more functional groups then a priority order is followed as:

$-COOH$, $-SO_3H$, $-COOR$, $COCl$, $-CONH_2$, $-CN$, $-HC=O$, $=C=O$, $-OH$, $-NH_2$, $=C=C=$, $-C\equiv C-$.

ISOMERISM

Two or more compounds having the same molecular formula but different physical and chemical properties are called isomers and this phenomenon is called isomerism.

Chain isomerism: When two or more compounds having same molecular formulabut different carbon skeletons are referred to as chain isomers.



Position Isomerism : Compounds which have the same structure of carbon chain butdiffer in position of double or triple bonds or functional group are called position isomers and this phenomenon is called Position isomerism. e g

CH_3-CH_2-CH=CH_2 CH_3-CH = CH – CH_3

Functional Isomerism : Compounds which have the same molecular formula butdifferent functional group are called functional isomers and this phenomenon is called functional Isomerism. e g

$CH_3 – CH_2 – OH$ $CH_3 – O – CH_3$

Metamerism: It is due to the presence of different alkyl groups on either side offunctional group in the molecule. Ex. $C_4H_{10}O$ represents $C_2H_5OC_2H_5$ and $CH_3OC_3H_7$.

FISSION OF COVALENT BOND

Heterolytic cleavage: In this cleavage the bond breaks in such a way that the sharedpair of electron remains with one of the fragments.

$H_3C – Br \longrightarrow {}^{+}CH_3 + Br^{-}$

Homolytic Cleavage: In this cleavage the shared pair of electron goes with each ofthe bonded atom.

$R – X \longrightarrow R^{\bullet} + X^{\bullet}$

Alkyl free radical

Nucleophiles : A reagent that brings an electron pair is called nucleophile ie nucleusseeking e g OH^- , CN^-

Electrophiles: A reagent that takes away electron pair is called electrophile i. e.electron seeking e g > C= O , $R_3C – X$

Inductive Effect: The displacement of the electron along the chain of the carbonatoms due to presence of an atom or group at the end of the chain.

$\partial{+}{+}{+}\ \partial\ {+}{+}\ \ \partial{+}$

$CH_3 \longrightarrow C\,H_2 \rightarrow CH_2Cl\rightarrow$

Resonance Effect : The polarity produced in the molecule by the interaction of twopi bonds or between a pi bond and lone pair of electron present on an adjacent atom.

There are two types of resonance effect:

- Positive resonance effect: In this effect the transfer of electrons is away from an atom or substituent group attached to the conjugated system.

The atoms or groups which shows +R effect are halogens,-OH , -OR,- NH_2

2) Negative resonance effect: In this effect the transfer of electrons is towards the atom or substituent group attached to the conjugated system.

The atoms or groups which shows -R effect are –COOH , -CHO , -CN

METHODS OF PURIFICATION OF ORGANIC COMPOUNDS:

Sublimation: This method is used to separate the sublimable compounds from nonsublimable compounds.

Crystallisation: This method is based on the difference in the solubilities ofcompound and impurities in a suitable solvent. The impure compound is dissolved in solvent at heated at higher temperature, on cooling the hot and concentrated solution pure compounds crystallizes out.

Distillation: This method is used to separate volatile liquids from non volatile liquidsand liquids having sufficient difference in their boiling points.

Fractional distillation: If the boiling points of two liquids having much difference, they areseparated by this method.

Distillation under reduced pressure: This method is used to purify liquids havinghigh boiling points and decomposes at or below their boiling points.

Steam distillation: This method is used to separate substances which are steamvolatile and are immiscible with water.

Differential Extraction: When an organic compound is present in an aqueousmedium it is separated by shaking it with organic solvent in which it is more soluble than in water. The aqueous solution is mixed with organic solvent in a separating funnel and shaken for sometimes and then allowed to stand for some time .when organic solvent and water form two separate layers the lower layer is run out by opening the tap of funnel and organic layer is separated. The process is repeated several times and pure organic compound is separated.

Chromatography: This technique is used to separate mixtures in to theircomponents, purify the compounds and test the purity of compounds.It is classified as

Adsorption Chromatography: It is based on the fact that different compounds are adsorbed on an adsorbent to different degrees. Silica jel or alumina is used as adsorbents.

Partition Chromatography: It is based on the continuous differential portioning of components of a mixture between stationary and mobile phase.

QUALITATIVE ANALYSIS OF ORGANIC COMPOUNDS

Detection of Carbon and Hydrogen: The Carbon and Hydrogen present in theOrganic compound is detected by heating the compound with Copper II oxide in a hard glass tube when carbon present in the compound is oxidized to CO2 which can be tested with lime Water and Hydrogenis converted to water which can be tested with anhydrous copper sulphate which turns blue.

$C + CuO \longrightarrow 2Cu + CO_2$

$2H + CuO \longrightarrow Cu + H2O$

$CO_2 + Ca(OH)_2 \longrightarrow CaCO_3 + H_2O$

$5H_2O + CuSO_4 \longrightarrow CuSO_4.5H_2O$

DETECTION OF OTHER ELEMENTS

Sodium Fusion Extract: A small piece of dry Sodium metal is heated with a organiccompound in a fusion tube for 2 -3 minutes and the red hot tube is plunged in to distilled water contained in a china dish. The contained of the china dish is boiled ,cooled and filtered. The filtrate is known as Sodium fusion extract.

Test for Nitrogen : The sodium fusion extract is boiled with iron II sulphate andthen acidified with Concsulphuric acid , the formation of Prussian blue colour confirms the presence of nitrogen.

$6CN^- + Fe^{2+} \longrightarrow [Fe(CN)_6]^{4-}$

$3[Fe(CN)_6]^{4-} + 4Fe^{3+} \xrightarrow{xH2O} Fe_4[Fe(CN)_6]_3.xH_2O$

Test for Sulphur: the sodium fusion extract is acidified with acetic acid and leadacetate is added to it. A black precipitate of lead sulphide indicates the presence of

sulphur. $S^{2-} + Pb^{2+} \longrightarrow PbS$

Black

Test for halogens: The sodium fusion extract is acidified with nitric acid and thentreated with silver nitrate. A white precipitate, soluble in ammonium hydroxide shows the presence of chlorine, a yellowish ppt. sparingly soluble in ammonium hydroxide shows the presence of bromine, a yellowish ppt. insoluble in ammonium hydroxide shows the presence of iodine.

$X^- + Ag^+ \longrightarrow AgX$

QUANTITIVE ANALYSIS(Carbon and Hydrogen)

Let the mass of organic compound be m g. Mass of water and carbon dioxide produced be m_1 and m_2 g respectively;

$$\% \text{ of carbon} = \frac{12 \times m_2 \times 100}{44 \times m}$$

$$\% \text{ of hydrogen} = \frac{2 \times m_1 \times 100}{18 \times m}$$

Nitrogen

DUMAS METHOD: A known mass of organic compound is heated with excess ofCuO in an atmosphere of CO_2, when nitrogen of the organic compound is converted into N_2 gas. The volume of N_2 thus obtained is converted into STP and the percentage of nitrogen determined by applying the equation:

$$\text{Volume of Nitrogen at STP} = \frac{P_1V_1 \times 273}{760 \times T_1}$$

$$\%N = \frac{28 \times \text{vol of } N_2 \text{ at STP} \times 100}{22400 \times \text{mass of the substance taken}}$$

KJELDAHL'S METHOD: A known mass of organic compound is heated withconc. H_2SO_4 in presence of K_2SO_4 and little $CuSO_4$ or Hg in a long necked flask called Kjeldahl's flask when nitrogen present in the organic compound is quantitatively converted into $(NH_4)_2SO_4$. $(NH_4)_2SO_4$ thus obtained is boiled with excess of NaOH solution to liberate NH_3 gas which is absorbed in a known excess of a standard acid such as H_2SO_4 or HCl.

The vol of acid unused is found by titration against a standard alkali solution. From the vol of the acid used, the percentage of nitrogen is determined by applying the equation,

$$\%N = \frac{1.4 \times \text{Molarity of the acid} \times \text{Basicity of the acid} \times \text{Vol of the acid used}}{\text{Mass of the substance taken}}$$

Halogens

Carius method:

A known mass of an organic compound is heated with fuming nitric acid in the presence of silver nitrate contained in a hard glass test tube known as carius tube in a furnace. Carbon and hydrogen present in the compound are oxidized to carbon dioxide and water. The halogen present forms the corresponding silver halide. It is filtered, dried, and weighed.

Let the mass of the organic compound taken = m g

Mass of AgX formed = m_1 g

1 mol of AgX contains 1 mol of of X

Mass of halogen in m_1 g of AgX

$$= \frac{\text{at mass of X x } m_1 \text{ g}}{\text{Molecular mass of AgX}}$$

% of halogen

$$= \frac{\text{at mass of X x } m_1 \text{ g x 100 \%}}{\text{Molecular mass of AgX x m}}$$

Sulphur

Let the mass of the organic compound taken = m g

Mass of $BaSO_4$ formed = m_1 g

$$\% \text{ of sulphur} = \frac{32 \times m_1 \times 100 \%}{233 \times m}$$

Phosphorous

Let the mass of the organic compound taken = m g

Mass of ammonium phosphomolydate = m_1 g

$$\% \text{ of phosphorous} = \frac{31 \times m_1 \times 100 \%}{1877 \times m}$$

Oxygen

Let the mass of the organic compound taken = m g Mass of CO_2= m_1 g

$$\% \text{ of oxygen} = \frac{32 \times m_1 \times 100 \%}{44 \times m}$$

One Mark Questions

Q1 Suggest a method to purify a liquid which decomposes at its boiling point.

A 1 The process Distillation Under reduced pressure is used to purify a liquid which decomposes at its boiling point.

Q 2 How will you separate a mixture of O-nitrophenol and p- nitrophenol ?

A 2 O-nitrophenol is steam volatile therefore it can be separated by Steam distillation.

Q 3 Lassaigne's test is not shown by diazonium Salt. Why?

A 3 On heating diazonium Salts loses Nitrogen and could not fuse with the Sodium metal therefore diazonium Salt do not show Positive Lassaigne's test for nitrogen.

Q 4 Alcohols are weaker acids than Water, Why ?

A 4 The alkyl group in alcohols has + I effect due to which electron density is increases on Oxygen atom which makes the release of hydrogen ion more difficult from alcohol.$R \rightarrow O \rightarrow H$

Q 5 Why is nitric acid is added to Sodium extract before adding Silver nitrate for testing halogens ?

A 5 Nitric acid is added to decompose NaCN and Na_2S

$NaCN + HNO_3 \rightarrow NaNO_3 + HCN$

$Na_2S + 2HNO_3 \rightarrow 2NaNO_3 + H_2S$

Q 6 which of the two $O_2NCH_2CH_2^-$ or $CH_3CH_2O^-$ is expected to be more stable and why ?

A 6 NO_2 group has –I effect and disperse the negative charge on Oxygen atom

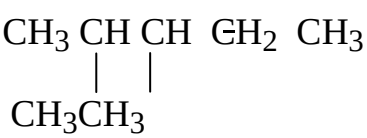
$O_2N \leftarrow CH_2 \leftarrow CH_2O^-$

Q 7 Arrange the following in increasing Order of Stability ;

$(CH_3)_3C^+$, $CH_3CH_2CH_2C^+H_2$, $CH_3CH_2C^+HCH_3$,$CH_3C^+H_2$, $CH_3CH_2C^+H_2$ A7

$CH_3C^+H_2 < CH_3CH_2C^+H_2 < CH_3CH_2CH_2C^+H_2 < CH_3CH_2C^+HCH_3 < (CH_3)_3C^+$

Q 8 Write the IUPAC name of the following



A 8 2,3Dimethylpentane

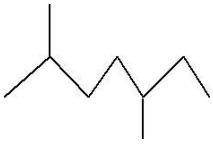
Q 9 Write the hybridized state of C atoms in the following

$CH_2 = CH - C \equiv N$

A 9 $sp^2 sp^2 sp$

$CH_2 = CH - C \equiv N$

Q 10 Give the IUPAC name of the following compound.



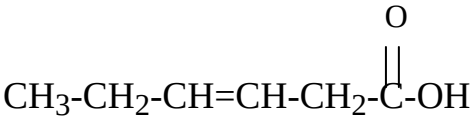
A 10 2,5Dimethylheptane

Two Marks Questions

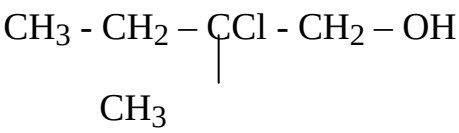
Q 1 Draw the Structures of the following compounds.

A) Hex-3-enoic acid b) 2-chloro-2-methylbutan-1-ol

A 1 a)

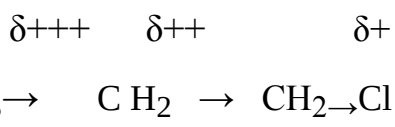


c)



Q 2 Explain Inductive effect with example.

A 2 Inductive Effect: The displacement of the electron along the chain of the carbon atoms due to presence of an atom or group at the end of the chain.



Q 3 Explain why $(CH_3)_3C^+$ is more stable than $CH_3C^+H_2$.

A 3 $(CH_3)_3C^+$ has nine alpha hydrogens and has nine hyperconjugation structures while $CH_3C^+H_2$has three alpha hydrogens and has three hyperconjugation structures, therefore $(CH_3)_3C^+$ is more stable than $CH_3C^+H_2$

Q 4 Give the number of Sigma and pi bonds in the following molecules

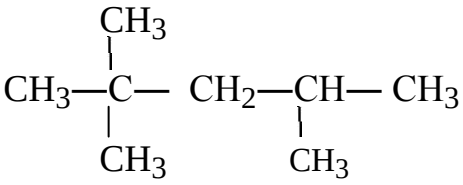
a) CH_3NO_2 b)$HCONHCH_3$

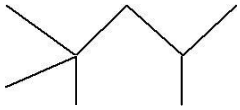
A 4 a) 6 Sigma and 1 pi bond

b) 8 Sigma and 1 pi bond

Q 5 Write the condensed and bond line formula of 2,2,4-Trimethylpentane.

A 5





Q 6 How Sodium fusion extract is prepared ?

A 6 A small piece of dry Sodium metal is heated with a organic compound in a fusion tube for 2 -3 minutes and the red hot tube is plunged in to distilled water contained in a china dish. The contained of the china dish is boiled ,cooled and filtered. The filtrate is known as Sodium fusion extract.

Q 7 Explain the principle of paper chromatography.

A 7 Paper chromatography is based on the difference in the rates at which the components of a mixture are adsorbed. The material on which different components are adsorbed is called Stationary phase which is generally made up of alumina, silica jel or activated charcoal. The mixture to be separated is dissolved in a suitable medium and it is called moving phase. The moving phase is run on the Stationary phase; the different compounds are adsorbed on stationary phase at different rates.

Q 8 Why is an organic compound fused with Sodium for testing nitrogen,halogens and sulphur ?

A 8 On fusing with sodium metal the elements presents in an organic compounds are converted in to sodium salts which are water soluble which can be filtered and detected by the respective tests.

Q 9 It is not advisable to use sulphuric acid in place of acetic acidfor acidification while testing sulphur by lead acetate test. Give reason

A 9 Lead acetate on reacting with sulphuric acid will give a white ppt of lead sulphatewhih interfere in the detection of sulphur.

$$(CH_3COO)_2Pb + H_2SO \longrightarrow PbSO_4 + 2\ CH_3COOH$$

Q 10 Under what conditions can the process of steam distillation is used ?

A 10 Steam distillation is used to purify the liquids which are steam volatile and water and the liquid are not miscible with each other.

Three Marks Questions

Q 1 In an estimation of sulphur by carius method 0.468 g of an organic compound gave 0.668 g of barium sulphate. Find the percentage of sulphur in the compound.

A 1 Mass of the compound = 0.468 g

Mass of the barium sulphate = 0.668 g

% of sulphur = $\frac{32 \text{ X Mass of barium sulphate X } 100}{233 \text{ X Mass of the compound}}$

- $\frac{32 \times 0.668 \times 100}{233 \times 0.468}$
- 19.60 %

Q 2 Which bond is more polar in the following pairs of molecules.

a) H_3C-H, H_3C-Br b) H_3C-NH_2, H_2C-OH c) H_3C-OH, H_3C-SH

A 2 a) C-Br because Br is more electronegative than H

- C-O because O is more electronegative than N
- C-O because O is more electronegative than S

Q 3 Define Isomerism.Explain position Isomerism and Functional Isomerism with examples.

A 3 Two or more compounds having the same molecular formula but different physical and chemical properties are called isomers and this phenomenon is called isomerism.

Position Isomerism : Compounds which have the same structure of carbon chain but differ in position of double or triple bonds or functional group are called position isomers and this phenomenon is called Position Isomerism. e g

CH_3-CH_2-CH=CH_2 CH_3-CH = CH – CH_3

Functional Isomerism :Compounds which have the same molecular formula but different functional group are called functional isomers and this phenomenon is called functional Isomerism. e g

$CH_3 – CH_2 – OH$ $CH_3 – O – CH_3$

Q 4 write the IUPAC names of the following compounds.

A. $CH3 – CH2 – \overset{\overset{O}{\|}}{C} – CH2 – \overset{\overset{O}{\|}}{C} – CH3$

- HC Ξ C – CH = CH – CH – CH_2
- Cl $CH_2CH_2CH_2CH_2Br$

A 4 A hexane 2,4dione

- hexa-1,3-dien-5-yne
- 1-bromo-4-chlorobutane

Q 5 Define Homologous series. Write the general formula of alkane, alkene and alkynes.

A 5 Homologous Series : It is defined as group of similar organic compounds which contains the similar functional groups and the two adjacent members of the series is differ by a $-CH_2$ group.

Alkanes C_nH_{2n+2}

Alkenes C_nH_{2n}

Alkynes C_nH_{2n-2}

Q 6 How many Sigma and pi bonds are present in the following molecules .

- $HC \equiv CCH = CHCH_3$
- $CH_2 = C = CHCH_3$

A 6 A Sigma bonds = 10 pi bonds = 3

B Sigma bonds = 9 pi bonds = 2

Q 7 Define functional groups. Write the general formula of Carboxylic acids acid chlorides.

A 7 Functional Groups :It is an atom or group of atoms bonded together in a unique manner which is usually the site of chemical reactivity in an organic molecule. e g CH_3OH

General formula of Carboxylic acids : $C_nH_{2n+1}COOH$

General formula of acid chlorides :RCOCl

Q 8 Write a shirt note on differential extraction.

A 8 When an organic compound is present in an aqueous medium it is separated by shaking it with organic solvent in which it is more soluble than in water. The aqueous solution is mixed with organic solvent in a separating funnel and shaken for sometimes and then allowed to stand for some time .when organic solvent and water form two separate layers the lower layer is run out by opening the tap of funnel and organic layer is separated. The process is repeated several times and pure organic compound is separated.

Q 9 How carbon and Hydrogen is detected in a organic compounds.

A 9 The Carbon and Hydrogen present in the Organic compound is detected by heating the compound with Copper II oxide in a hard glass tube when carbon present

in the compound is oxidized to CO2 which can be tested with lime Water and Hydrogenis converted to water which can be tested with anhydrous copper sulphate which turns blue.

$C + CuO \rightarrow 2Cu + CO_2$

$2H + CuO \rightarrow Cu + H_2O$

$CO_2 + Ca(OH)_2 \longrightarrow CaCO_3 + H_2O$

$5H_2O + CuSO_4 \rightarrow CuSO_4.5H_2O$

Q 10 Write a short note on Resonance effect .

Resonance Effect : The polarity produced in the molecule by the interaction of two pi bonds or between a pi bond and lone pair of electron present on an adjacent atom.

There are two types of resonance effect:

- Positive resonance effect: In this effect the transfer of electrons is away from an atom or substituent group attached to the conjugated system.
  The atoms or groups which shows +R effect are halogens,-OH , -OR,-NH_2
2. Negative resonance effect: In this effect the transfer of electrons is towards the atom or substituent group attached to the conjugated system.

   The atoms or groups which shows -R effect are –COOH , -CHO , -CN

Five Marks Questions

Q 1 Differentiate between the principle of estimation of nitrogen in an organic compound by i) Dumas method ii) Kjeldahl's method.

Ans: DUMAS METHOD: A known mass of organic compound is heated with excess of CuO in an atmosphere of CO_2, when nitrogen of the organic compound is converted into N_2 gas. The volume of N_2 thus obtained is converted into STP and the percentage of nitrogen determined by applying the equation:

$$\text{Volume of Nitrogen at STP} = \frac{P_1V_1 \times 273}{760 \times T_1}$$

$$\%N = \frac{28 \times \text{vol of } N_2 \text{ at STP} \times 100}{22400 \times \text{mass of the substance taken}}$$

KJELDAHL'S METHOD: A known mass of organic compound is heated with conc. H_2SO_4 in presence of K_2SO_4 and little $CuSO_4$ or Hg in a long necked flask called Kjeldahl's flask when nitrogen present in the organic compound is quantitatively converted into $(NH_4)_2SO_4$. $(NH_4)_2SO_4$ thus obtained is boiled with excess of

NaOHsolution to liberate NH_3 gas which is absorbed in a known excess of a standard acid such as H_2SO_4.

The vol of acid unused is found by titration against a standard alkali solution. From the vol of the acid used, the percentage of nitrogen is determined by applying the equation,

$$\%N = \frac{1.4 \times \text{Molarity of the acid} \times \text{Basicity of the acid} \times \text{Vol of the acid used}}{\text{Mass of the substance taken}}$$

Q 2 A sample of 0.50g of organic compound was treated according to Kjeldahl's method. The ammonia evolved was absorbed in 50mL of 0.5M H_2SO_4. The residual acid required 60mL of 0.5M solution of NaOH for neutralization. Find the percentage composition of nitrogen in the compound.

Ans: the vol ofH_2SO_4 used.

Vol of acid taken=50mL of 0.5M H_2SO_4= 25mL of 1M H_2SO_4

Vol of alkali used for neutralization of excess acid= 60 mL of 0.5m NaOH=30mL of 1M NaOH

Now 1 mole of H_2SO_4 neutralizes 2 moles of NaOH

(i.e. $H_2SO_4 + 2\ NaOH \rightarrow Na_2SO_4 + 2H_2O$)

… 30 mL of 1M NaOH = 15mL of 1M H_2SO_4

% of nitrogen.

1 mole of H_2SO_4 neutralizes 2 moles of NH_3 … 10mL of 1M H_2SO_4 = 20mL of 1M NH_3

But 1000mL of 1M NH_3 contain N=14g.

20 ml of 1M NH_3 will contain nitrogen = $\frac{14 \times 20}{1000}$

But this amount of nitrogen is present in 0.5 g of organic compound

$$\therefore \%\ \text{of N} = \frac{14 \times 20 \times 100}{1000 \times 0.5} = 56.0$$

Q 3 You have a mixture of three liquids A, B , C. there is a large difference in the boiling point of A and the rest two liquids. Boiling points of liquids B and C are quite close. Liquid A boils at higher temperature than B and C and boiling point of B is lower than C. How will you separate the components of the mixture.

Ans Since the boiling point of liquid A is much higher than those of liquids B and C , therefore separate liquid A by simple distillation. Since boiling points of liquids B and C are quite close but much lower than liquid A therefore mixture of B and C will distil together leaving behind A. on further heating A will distil over.

Now place the mixture of liquids B and C in a flask fitted with fractionating column. Since the b.p. of liquid B is lower than that of C , on fractional distillation first liquid B will distil over and than liquid C.

HOTS QUESTIONS

Q 1 Explain hyperconjugation effect. How does hyperconjugationeffect explain the stability of alkenes?

Ans The relative stability of various classes of carbonium ions may be explained by the number of no-bond resonance structures that can be written for them. Such structures are obtained by shifting the bonding electrons from an adjacent C-H bond to the electron deficient carbon so the positive charge originally on carbon is dispersed to the hydrogen. This manner of electron release by assuming no bond character in the adjacent C-H bond is called Hyperconjugation. Greater the hyperconjugation greater will be the stability of alkenes.

$$CH_3 - CH = CH - CH_3 < CH_3 - \underset{\displaystyle CH_3}{\underset{|}{C}} = CH - CH_3 \quad < CH_3 - \underset{\displaystyle CH_3}{\underset{|}{C}} = \underset{\displaystyle CH_3}{\underset{|}{C}} - CH_3$$

Q 2In DNA and RNA nitrogen is present in the ring system. Can kjeldahl method be used for the estimation of nitrogen present in these ?give reasons

AnsIn DNA and RNA nitrogen is present in hetrocyclicrings.Kjeldahl method can not be used to estimate nitrogen present in the ring because cannot be completely converted in to $(NH_4)_2SO_4$ during digestion. Therefore Kjeldahl method can not be used to estimate nitrogen present in DNA and RNA.

Q 3 1.216 g of an organic compound was Kjeldahlised and the ammonia evolved was absorbed in 100 mL 1N H_2SO_4. The remaining acid solution was made upto500ml by addition of water. 20ml of this dilute solution required 32mL of N/10 caustic soda solution for complete neutralization. Calculate the percentage of nitrogen in the organic compound.

Ans 20 ml of dil. Unreacted H_2SO_4 = 32mL of N/10 NaOH sol.

$$500\text{ml of dilunreacted } H_2SO_4 = 32 \times \frac{500\text{ mL}}{20} \text{ of } \frac{\text{NNaOH}}{10} = \frac{32 \times 500 \text{ ml of 1 N NaOH}}{20 \times 10}$$

$$= 80\text{ml 1 N NaOH}$$

But 80ml 1 N NaOH= 80ml 1 N NaOH So, acid left unused = 80ml 1 N H_2SO_4

Acid used =(100 – 80) = 20ml 1 N H_2SO_4

$$\%N = \frac{1.4 \times \text{Normality of the acid} \times \text{Vol of the acid used}}{\text{Mass of the substance taken}}$$

$$= \frac{1.4 \times 1 \times 20}{1.216} = 23.026$$

Value based questions

Q. 1. Chemistry plays a very important role in our everyday life. Beginning from morning when we use toothpaste to brush our teeth and throughout the day we use a large number of materials which have been provided to us by R & D (Research and development) carried out in the field of chemistry.

Now, answer the following questions :

(i) Why should we use iodized salt ?

(ii) Why should our toothpaste contain fluoride?

(iii) What are the relative advantages and disadvantages of using soaps and detergents?

(iv) Why earlier leaded petrol was used but now a days unleaded petrol is used?

Ans. (i) Iodine is needed to protect us from thyroid disorders.

(ii) Fluoride protects our gums and prevents tooth decay.

(iii) Soap is a good cleansing agent & is 100% biodegradable. As a result, it does not cause any pollution problem. However, soaps have two disadvantages. Firstly, they cannot be used in hard water. Secondly, they cannot be used in acidic solution. However, they have one disadvantage that they are not completely biodegradable and hence cause water pollution.

(iv) Earlier, tetraethyl lead was added to petrol by the oil companies to act as anti-knocking agent. But on decomposition, it produces lead which is poisonous. Hence, its use has been stopped. Quality of petrol has been improved by other methods.

Q. 2. Chemistry has made a significant contribution in the medical field by providing a large number of drugs and medicines to fight our diseases. It is an endless effort by pharmaceutical companies to develop better and better medicines. The day is not far off when treatment of disease like cancer will be considered as common as bad cold.

Now, answer the following questions :

(i) Why should we avoid the use of drugs?

(ii) What for are analgesics used?

(iii) What for are antibodies used?

(iv) What for are anaesthetics used?

(v) What for are tranquilizers used?

(vi) What for Dettol and savlon are used?

(vii) What for do we use Gammexane?

(viii) Which compunds are generally present in the life saving drugs used for curinjg cancer and AIDS?

(ix) What is the role of vitamins in our life?

(x) Which vitamin is produced in our body when we sit in the sun or which vitamin helps in the absorption of calcium in our body?

Ans. (i) A medicine is a chemical substance which cures the disease, is safe to use, has negligible toxicity and does not cause addiction. In contrast, a drug is a chemical substance which also cures the disease but is habit forming, causes addiction and has serious side effects. Hence, its use should be avoided.

(ii) Analgesics are medicines used to get relief from pain.

(iii) Antibiotics are medicines used to curb infection.

(iv) Anaesthetics are medicines given during surgery (operation) to stop you feeling pain in the whole of your body when you are unconscious or in a part of your body when you are awake.

(v) Tranquilizers are drugs used for the treatment of mental stress/disease.

(vi) Dettol and savlon are antiseptics used to stop infection of wounds.

(vii) Gammexane is an insecticide used to kill rats, mosquitos and flies.

(viii) cis-platin and taxol are used for cancer and AZT (Azido- thymidine) is used for AIDS.

(ix) Vitamins are chemical substances, the deficiency occurs, they have to be taken in the form of pills or capsules.

(x) Vitamin D.

Q. 3. Chemistry has played a vital role in the field of agriculture. It has provided us with chemical fertilizers which have helped to increase the yield of fruits, vegetables & other crops. It has given us insecticide and pesticides which protect the plants from insects and pests. It has given us preservation to preserve food products like jam, butter etc. for a longer period.

Now, answer the following questions :

(i) How is the use of chemical fertilizers harming the aquatic life and mankind: what alternative do you suggest?

(ii) How is the use of pesticides and insecticides harming the mankind? What alternative do you suggest?

(iii) Huge amounts of synthetic milk and milk products are flowing into the market. Suggest what should be done to check it?

Ans. (i) During rainfall, nitrogenous fertilizers was away into rivers, lakes & oceans. It help the algae to bloom. When short live algae die, decomposing bacteria consume dissolved oxygen. As a result, aquatic life begins to perish. Secondly, artificial fertilizers deplete the soil of organic matter. As a result, it loses ability to hold water and is more subject to erosion. Moreover, these fertilizers tavel from soil to plant and then from plant to human being causing health hazards. The solution to the problem is that natural fertilizers like cow-dung should be encouraged, i.e., production of organic food should be increased.

(ii) Pesticides have neurological effects on humans such as headache and hand tremors. A number of them are responsible for cancer. The solution to the problem is that their use should be minimized and such chemicals should be produced which kill pests but have no effect on humans.

(iii) The supply of milk and milk products should not flow direct to the consumer through the vendor. It should be regulated through a centre (set up by the Govt. or licenced by the Govt. to private agency) where all milk and milk products are received and tested before they are sold to consumer.

Q. 4. India is a fast developing nation when a very large number of chemical industries are coming evry year. These include industries like those of glass, paper, textile, plastics, leather, dye paints, sugar, pharmaceuticals etc. The rat- race for earning more profits and also due to competition of rates in the market, some industries are producing poor quality products and chemicals which are harming the society at large.

Now, answer the following questions :

(i) Production of spurious drugs is harming the mankind. What steps should be taken to check it.

(ii) How do paints sometimes prove to be harmful?

(iii) Why should the production of polythene bags be banned?

(iv) Name a few industrial products which are recyclable. What should we do with their products?

(v) What are the relative advantage and disadvantages of using synthetics fibres versus natural fibres (cotton, silk, wool etc.)?

Ans. (i) Strict quality control law should be enforced. Not only such factories should be closed, people running such factories should be severely punished. These factories should be black-listed and photographs of owners should be published in all leading newspapers.

(ii) Many paints contain lead which is poisonous.

(iii) Use of polythene bags should be banned because it is not biodegradable. It may, therefore, pile up as waste and swallowed by animals causing death or choke the sewer.

(iv) Advantages of synthetic fibres.

(a) Strong (b) East to wash and wear (need no ironing)

(c) can be stretched easily to different sizes like nylon socks

(d) Available in variety of colours

(e) Generally cheaper than natural fibres

(f) do not shrink.

Disadvantages of synthetic fibres.

(a) does not absorb moisture/swat keeping the heat trapped in the body.

(b) Give uncomfortable feel for being used as underwears

(c) Catch fire or melt more easily than natural fibres.

Advantages of natural fibres.

(a) Clothes made from natural fibres are more comfortable because they an absorb moisture/sweat.

(b) As they are natural materials, they are harmless to environment.

(c) They are fire-resstant.

Disadvantages of natural fibres,

(a) More expensive

(b) Need ironing as they are not crease-resistant.

(c) Might shrink on aggressive washing.

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CHAPTER 13

HYDROCARBON

Hydrocarbons are composed of Carbon and hydrogen.

The important fuels like Petrol, kerosene, coal gas, CNG, LPG etc. are all hydrocarbons or theirmixture. Sources:

Petroleum and natural gas are the major sources of aliphatic hydrocarbon while coal is an important source of aromatic hydrocarbons. The oil trapped inside the rocks is known as petroleum. PETRA –ROCK, OLEUM –OIL. The oil in the petroleum field is covered with a gaseous mixture known as natural gas. The main constituents of the natural gas are methane, ethane, propane and butane.

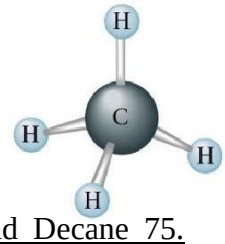
CLASSIFICATION OF HYDROCARBONS:

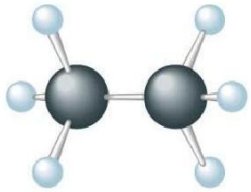
HYDROCARBON

- Acyclic or Aliphatic (Open chain)
  - Alkanes
  - AlkenesAlkyne
- Carbocyclic or Cyclic
  - Alicyclic
    - Cycloalkanes
    - Cycloalkenes
    - Cycloalkynes
  - Aromatic

Alkanes:-

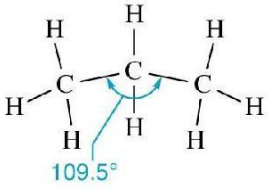
- Paraffins
- General formula C_nH_{2n+2}
- sp^3 hybridisation
- C–C bond length 1.15 4 A^0
- Chemically unreactive
- Show chain, position and optical isomerism.
- Heptane has 9 isomer, Octane 18 and Decane 75.

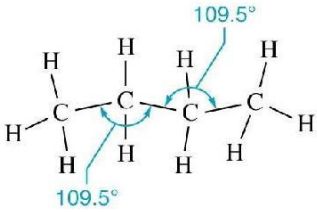


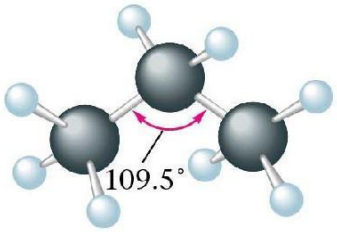


(c)

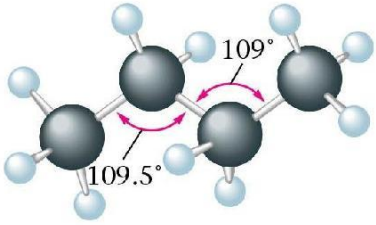
Nomenclature:







(a) Propane



(b) Butane

Preparation:-

❶ Wurtz reaction:-

$2CH_3CH_2Br + 2Na \xrightarrow[\text{ether}]{\text{Dry}} CH_3CH_2CH_2CH_3 + 2NaBr$

- Follow mainly free radical mechanism
- Useful in preparingan alkane containingeven number of carbonatoms
- Stepping up reaction Frankland reaction

$RX + Zn + Rx = R-R + ZnX_2$

❷ From Grignard reagent (RMgX)

$RMgX + HOH \rightarrow RH + Mg(OH)X$

$RMgX + R'OH \rightarrow RH + Mg(OR')X$

$RMgX + R'NH_2 \rightarrow RH + Mg(NHR')X$

❸ From unsaturated hydrocarbons:-
Sabatier-Senderens reduction

$R\ CH\ CH_2 + H_2 \xrightarrow{Ni} R\ CH_2\text{-}CH_3$

$R\ C\ CH + H_2 \xrightarrow{Ni} RCH_2\text{-}CH_3$

4. From carboxylic acids-
Decarboxylation.-

$$CH_3COO^-Na^+ + NaOH \xrightarrow[\Delta]{CaO} CH_4 + Na_2CO_3$$

Sodium ethanoate

Kolbe's electrolytic- method

$$2CH_3COO^-Na^+ + 2H_2O$$

Sodium acetate

$\downarrow$ Electrolysis

$$CH_3-CH_3 + 2CO_2 + H_2 + 2NaOH$$

❖ **Physical Properties:-**

5. Nature:- Non-Polar due to covalent nature of C—C bond and C—H bond. C—C bond enrgy = 83 kj/mole and C—H bond energy = 99 kj/mole.
C_1—C_4 = gases, C_5—C_{17} = colourless odourless liquid and > C_{17} = Solid.
6. Solubility:- Like dissolve like

Viz, Polar compounds dissolve in polar solvent and Non-Polar compound dissolve in non polar solvent.

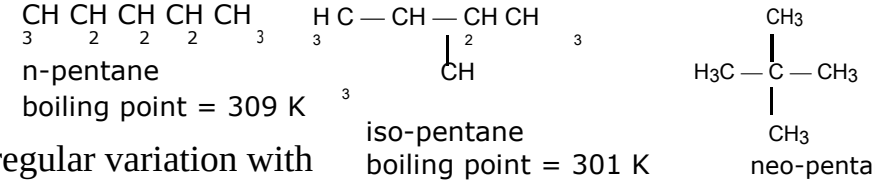
10. Boiling point:- Low boiling point due to non polar in nature.

The molecules are held together only by Since we known that the magnitude of proportional to the molecular size. Therefore, the boiling point increases with

increase the molecular size i.e. with increase in number of carbon atoms.

Noted:- the boiling points of the branched chain Alkanes are less than the straightchain isomers.

This is due to the fact that branching of the chain makes the molecule more compact and thereby decreases the surface aria and consequently, the magnitudes of Van der Waals' forces also decrease.



regular variation with even number of carbon atoms having higher melting point Alkanes having immediately next lower and immediately next higher odd number of carbon atoms.

❖ **Chemical properties**

❶ Combustion:- $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

❷ Oxidation:-

$$CH_4 \quad O_2 \xrightarrow[573\ K]{Cu} 2CH_3OH$$

$$CH_4 \quad O_2 \xrightarrow{Mo_2O_3} HCHO \quad H_2O$$

Methanal

❸ Substitution:-

① Halogenation:-

$$CH_4 + Cl_2 \xrightarrow{UV} CH_3Cl + HCl$$

$$CH_3Cl \xrightarrow{UV} CH_2Cl_2 \xrightarrow{UV} CHCl_3 \xrightarrow{UV} CCl_4$$

Noted:- Iodination is a reversible reaction. So it is carried out by heating alkane in the presence of some oxidizing agent like iodic acid (HIO_3) or nitric acid (HNO_3) or mercuric oxide (HgO) which oxidizes HI formed during the reaction.

$$CH_4 + I_2 \rightarrow \quad CH_3I + HI$$

$$5HI + HIO_3 \longrightarrow 3H_2O + 3I_2$$

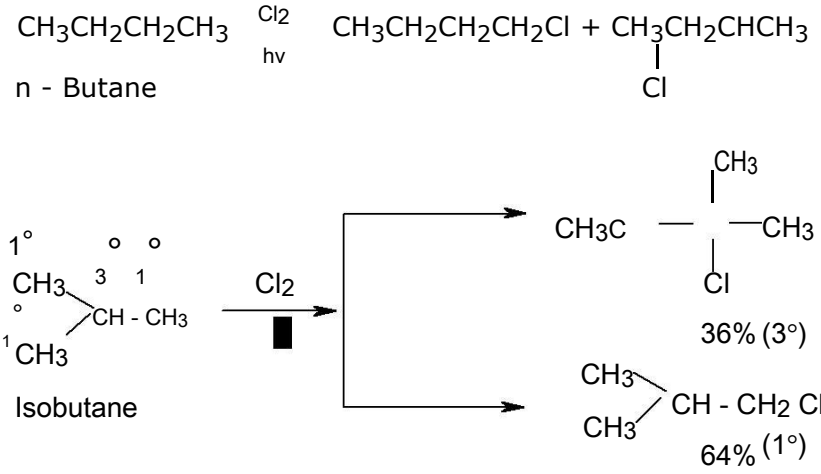
$$2HI + 2HNO_3 \longrightarrow 2H_2O + I_2 + 2NO_2$$

Noted:- Fluorination of alkane takes place explosively resulting even in the rupture of C—C bond in higher alkanes.

Features of Halogenations:-

(i) The reactivity of Halogens:- $F_2 > Cl_2 > Br_2 > I_2$. (ii) The rate of replacement of Hydrogens of alkanes is:

$$3° > 2° > 1°$$



Mechanism:- halogenations reaction take place by free radical mechanism. The reaction proceeds in the following steps:

Initiation

7. Chain initiation step:-

$$Cl — Cl \xrightarrow{hv} 2Cl^{*}$$

18. **Chain Propagation step:-**

$$CH_4 + Cl^{*} = CH_3^{*} + HCl$$

$$CH_3^{*} + Cl_2 = CH_3Cl + Cl^{*}$$

19. **Chain Termination step:-**

$$Cl^{*} + Cl^{*} = Cl_2$$

$$CH_3^{*} + CH_3^{*} = CH_3 - CH_3$$

$$CH_3^{*} + Cl^{*} = CH_3Cl$$

Nitration:-

The reaction takes places by free radicals mechanism at high temp (450^0C).

At high temp C—C bond is also broken so that mixture of nitroalkanes is obtained.

$$CH_3CH_2CH_3 \xrightarrow[\text{Conc. } HNO_3]{450°C} CH_3CH_2CH_2NO_2 + CH_3CHCH_3 + CH_3NO_2 + CH_3CH_2NO_2$$

(with NO_2 on the CH_3CHCH_3)

25% 40% 10% 25%

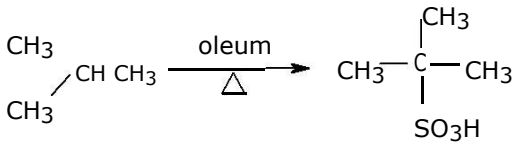
The reaction occurs as:

$$HO\text{-}NO_2 \xrightarrow[\text{Homolytic fission}]{450^0C} HO^{o} + {}^{o}NO_2$$

$$RH + {}^{0}OH \longrightarrow R^{o} + HOH$$

$$R^{o} + {}^{o}NO_2 \longrightarrow RNO_2$$

③ Sulphonation:- replacement of hydrogen atom of alkane by $-SO_3H$ group.



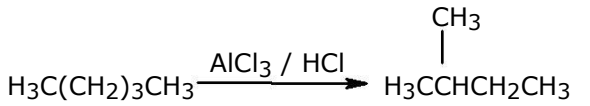
isobutane tert butyl sulphonic acid

The reaction occurs as:

$$HO\text{-}SO_3 \xrightarrow[\text{Homolytic fission}]{450^0C} HO^{o} + {}^{o}SO_3H$$

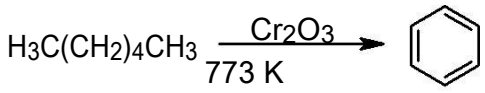
$$RH + {}^{0}OH \longrightarrow R^{o} + HOH$$

$$R^{o} + {}^{o}SO_2H \longrightarrow RSO_2H$$



n-Pentane 2-Methyl butane

❺ Aromatization:-



Hexane 10-20 atm Benzene

This method is also called dehydrogenation or hydroforming

Similarly, heptane gives toluene, n-Octane give o-xylene and 2, methyl heptane give m-xylene.

❻ Thermal decomposition or Pyrolysis or cracking or Fragmentation: - when higher alkanes are heated at high temp (about 700-800k) in the presence of alumina or silica catalysts, the alkanes break down to lower alkanes and alkenes.

$$CH_3\text{-}CH_2\text{-}CH_3 \longrightarrow CH_3\text{-}CH\text{-}CH_2 + CH_3\text{-}CH_3 + C_2H_4 + CH_4$$

❼ Action of steam:- catalyst: nickel, alumina Al_2O_3

$$CH_4 + H_2O(\text{Steam}) \xrightarrow{1000\ ^{o}C} CO + 3H_2$$

This reaction is used for the industrial preparation of hydrogen from natural gas. 8. Isomerisation:-

$$CH_3(CH_2)_4CH_3 \xrightarrow{\text{Anhy. } AlCl_3/HCl}$$

n-Hexane

$$CH_3CH(CH_3)-(CH_2)_2-CH_3 + CH_3CH_2-CH(CH_3)-CH_2-CH_3$$

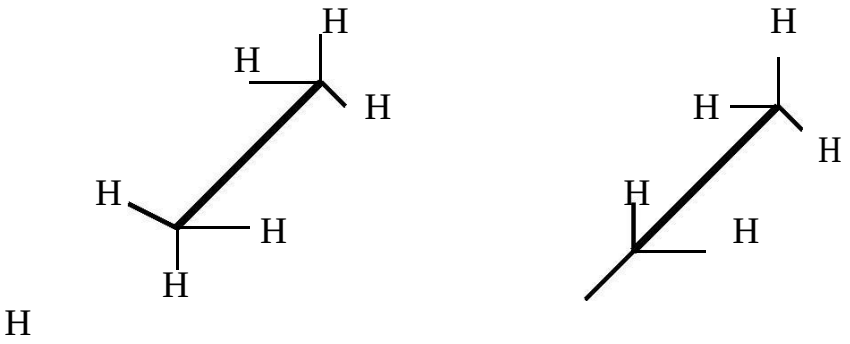
2-Methylpentane 3-Methylpentane

❖ CONFORMATIONAL ISOMERISM:

The different molecular arrangements arising as a result of rotation around carbon carbon single bonds are called conformational isomers or rotational isomers and the phenomenon is called conformational isomerism.

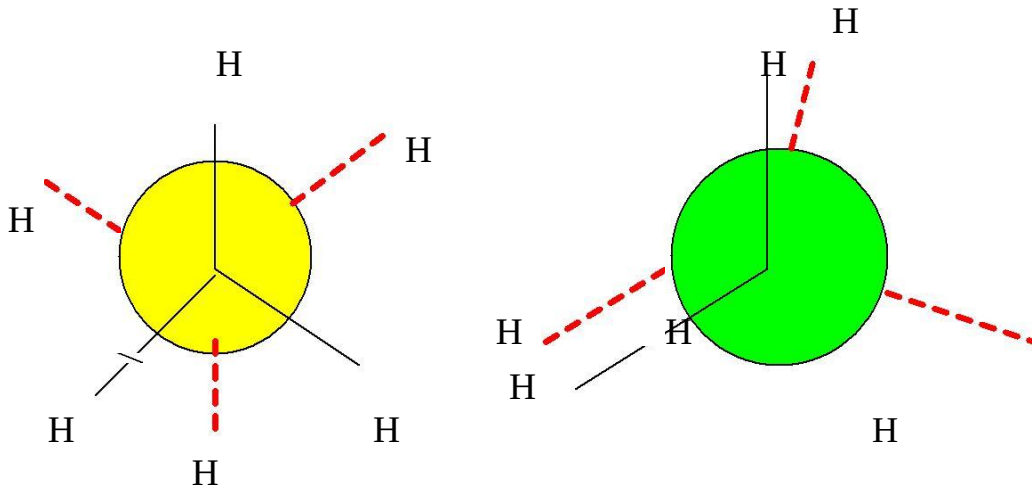
Numerous possible arrangements of ethane are possible. Two extreme conformations are known. These are eclipsed conformation and staggered conformation.

SAWHORSE REPRESENTATION



STAGGERED ECLIPSED

NEWMAN PROJECTION

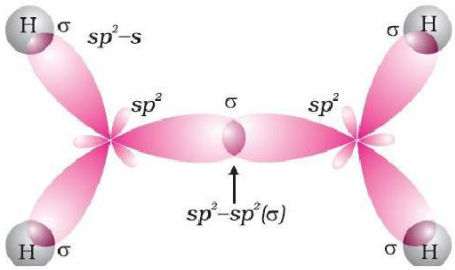


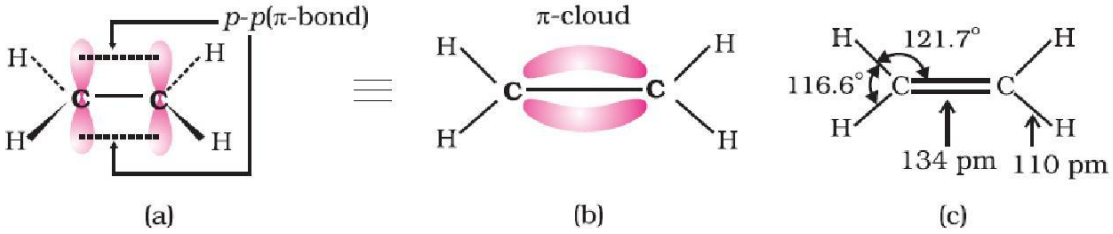
STAGGERED → SKEW ← ECLIPSED

Alkenes

- Unsaturated hydrocarbon which have double bond.
- General molecular formula C_nH_{2n}
- C–C bond hybridization 1.34 A^0
- sp^2 hybridization
- When we treated Alkene with chlorine, oily products are obtained. So Alkenes are also known as Olefins. (Greek olefiant meaning oil forming).
- Show chain, positional and geometrical isomerism

Structure of double bond:-





- **Preparation:-**

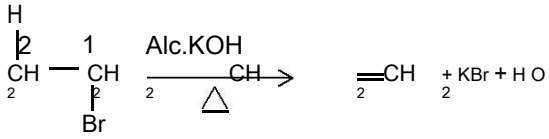
From Alkynes:- Alkynes on partial reduction with Partially deactivatedpalladised charcoal known as *Lindlar's*give alkynes*catalyst*.

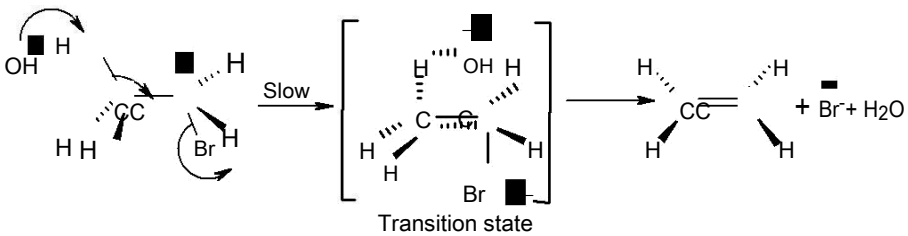
$$CH{\equiv}CH + H_2 \xrightarrow{Pd/C} CH_2{=}CH_2$$

Ethyne → Ethene

From Haloalkanes: - dehydrohalogenation

(E_2 or 1,2-elimination or Bita-elimination)



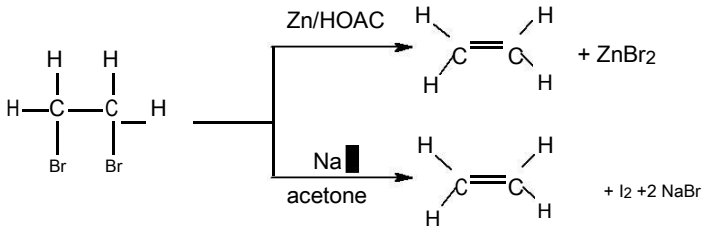


- predominantformationof a substituted alkene is formed according to Saytzeff's rule

$$CH_3 - CH_2 - CH_2 - CH_3 \xrightarrow{Alc.\ KOH} CH_3 - CH{=}CH - CH_3 + CH_3 - CH_2 - CH{=}CH_2$$

Major Minor

3. From Dihaloalkanes: - dehalogenation



4. From Alcohols:- Dehydration
(E1 - elimination)

$$CH_3CH_2CH_2OH \xrightarrow[160°]{Conc.H_2SO_4} CH_3CH = CH_2 + H_2O$$

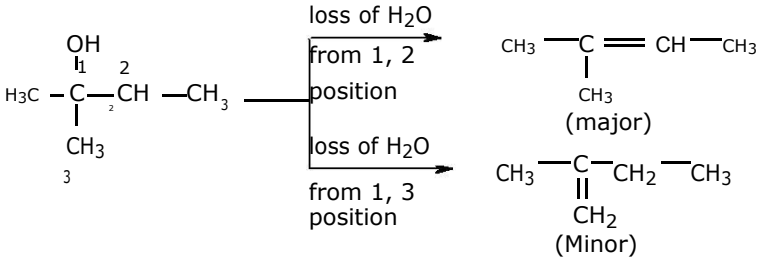
$$CH_3CH_2CH_2CH_2OH \xrightarrow[300°\ C]{Al_2O_3} CH_3CH_2CH = CH_2$$

Mechanism

$$CH_3 - CH_2 - CH_2 - OH \xrightarrow{H^+} CH_3 - CH_2 - CH_2 - \overset{+}{O}H_2 \xrightarrow{-H_2O} CH_3 - CH_2 - \overset{+}{C}H_2$$

$$CH_3\underset{H}{\underset{|}{C}}HCH_2 \xrightarrow[1]{H} CH_3 - CH = CH_2$$

propene



❖ Chemical Properties:-

Addition Reaction:- Alkene show electrophilic addition reaction.

Addition of Hydrogen:-

$$RCH=CH_2 \xrightarrow{H_2/Ni} RCH_2CH_3$$

Addition of Halogens:-

$$CH_2=CH_2 + Br_2 \xrightarrow[Solvent]{CCl_4} \underset{Br}{\underset{|}{CH_2}} - \underset{Br}{\underset{|}{CH_2}}$$

$$CH_2 - CH_2 - Br \xrightarrow[Solvent]{H_2O} Br - CH_2 - CH_2 - OH$$

(Brown colour) (Colourless)

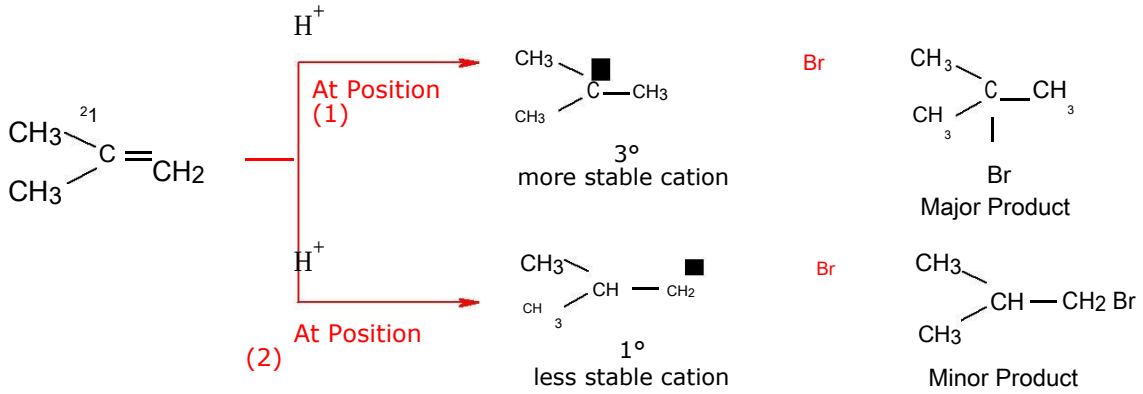
Addition of hydrogen halides-

Addition reaction of HBr to symmetrical alkenes

$$CH_2=CH_2+H-Br\longrightarrow CH_3-CH_2-Br$$

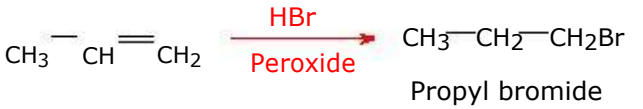
Addition reaction of HBr to unsymmetrical alkenes takes place according to Markovnikov Rule

Markownikov rule:- negative part of the addendum (adding molecule) gets attachedto that carbon atom which possesses lesser number of hydrogen atoms. e g



Peroxide effect or Kharasch-In1933(*Anti*KharaschandMark

Mayo observed that when HBr is added to an unsymmetrical double bond in the presence of organic peroxide, the reaction take places opposite to the Markovnikov rule.



(i) $C_6H_5-C(=O)-O-O-C(=O)-C_6H_5 \xrightarrow{\text{Homolysis}}$ Benzoyl peroxide

$$2C_6H_5-C(=O)-\dot{O}: \rightarrow 2\dot{C}_6H_5+2CO_2$$

(ii) $\dot{C}_6H_5+H-Br\xrightarrow{\text{Homolysis}}C_6H_6+\dot{B}r$

(iii) $CH_3—CH=CH_2 + \dot{Br}$

Homolysis

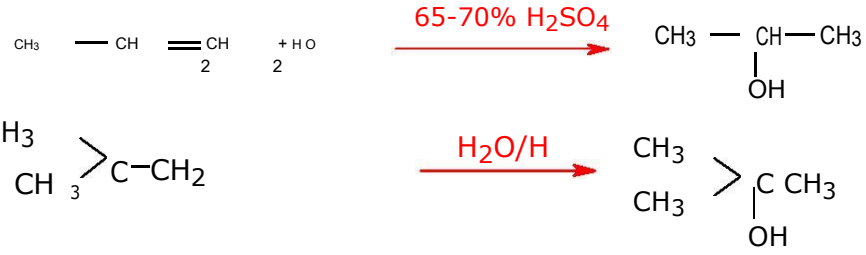
$CH_3—\dot{C}H—CH_2Br$ (a) (less stable primary free radical)

$CH_3—CH—\dot{C}H_2$ (b) (more stable secondary free radical)

(iv) $CH_3-\dot{C}H-CH_2Br + H-Br \xrightarrow{\text{Homolysis}} CH_3-CH_2-CH_2Br + \dot{Br}$ (major product)

Noted:- peroxide effect is applicable only to HBr and not to HF, HCl and HI. Addition of HF, HCl and HI takes place presence of peroxide.

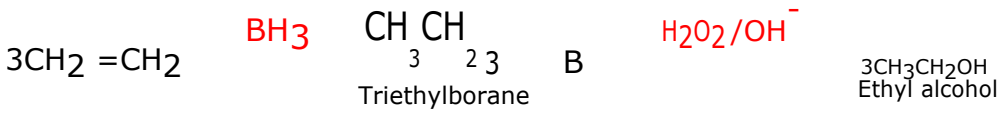
4. Addition of water (Hydration):- Acid catalyzed addition of water



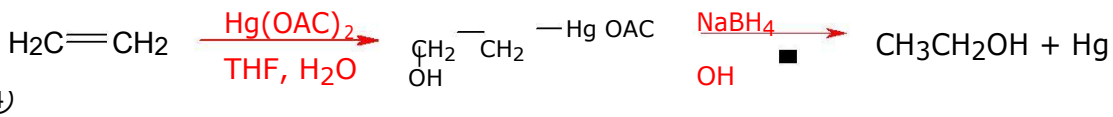
❷ Oxidation:-

① Combustion:- $CO_2 + H_2O$

② Hydroboration–oxidation:- Alkanes react with diborane to form trialkyl boranes which on oxidation with alkaline H_2O_2 give alcohols.

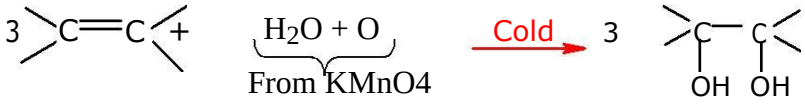


③ Oxymercuration–demercuration:-



④ Oxidation with potassium permanganate:-

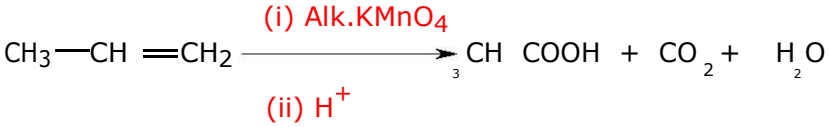
$2KMnO_4 + H_2O \longrightarrow 2KOH + 2MnO_2 + 3[O]$



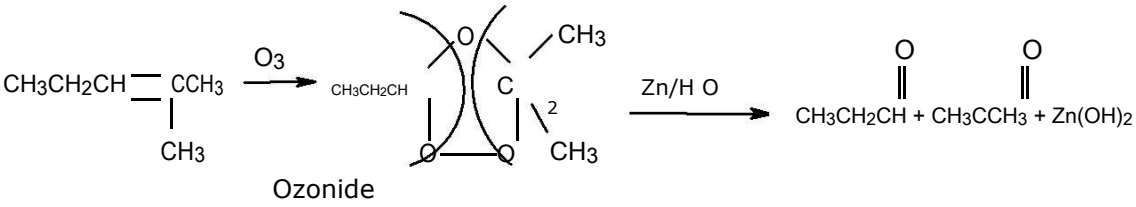
(pink Colour) (Colourless)

- This reaction is also called *Hydroxylation*
- *Cis product I.e. cis*-diol is obtained.

Noted:- The alkaline potassium permanganate solution is known as Ba It has bright pink colour. It oxidizes alkenes to glycols which is colourless. This reaction is used as a test for the presence of double bond in a molecule. This is also known as Baeyer test.



☺ Oxidation with Ozone:- Ozonolysis –give carbonyls compounds



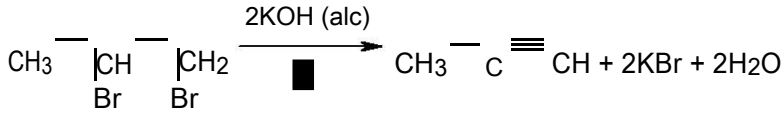
Noted:- Bromine water test and Baeyer's te double bond while ozonolysis is used to detect the position of double bond.

Alkynes

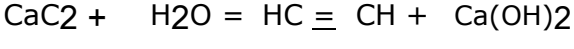
- Unsaturated hydrocarbon which have triple bond.
- General molecular formula C_nH_{2n-2}
- sp hybridization
- Shows chain, positional and functional isomerism

❖ Preparation:-

From vicinal dihalides: - dehalogenation



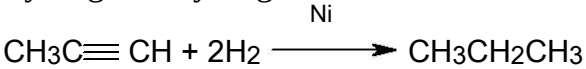
By the action of water on calcium carbide:-



❖ Chemical Properties:-

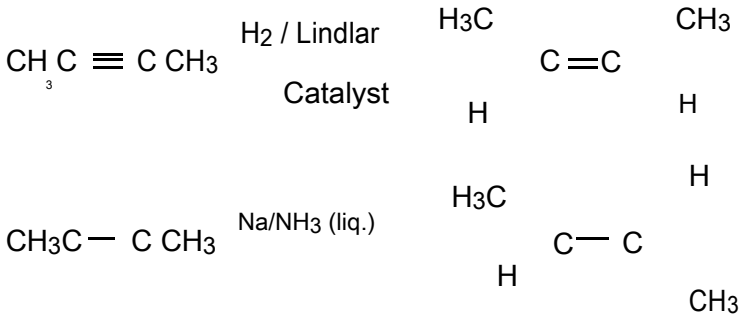
❶ Addition Reaction:- Alkyne show electrophilic addition reaction. ①

Addition of Hydrogen:- Hydrogenation.

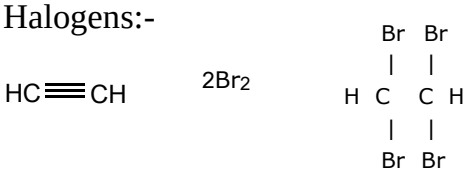


Noted:- It may be noted that the hydrogenation can be controlled at the alkene stage only. This is possible by using a *Lindlar's*or*sodiumcatalystsinliquidNH*$_3$200k temp..

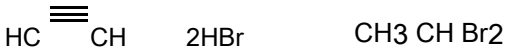
Noted:- It may be again noted that the catalytic reduction of alkynes in the presence of Lindlar's-alkenescatalystwhileinthepresencegivesofsodiumcisinliquid NH_3 (Birch reduction) gives trans-alkenes.



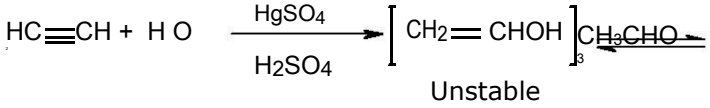
② Addition of Halogens:-

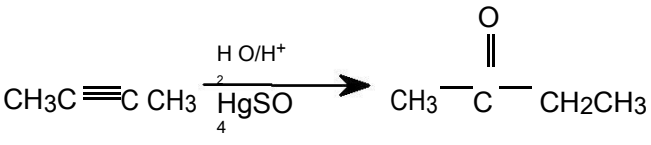


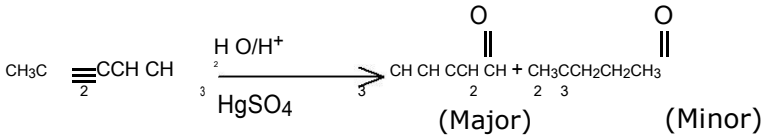
③ Addition of hydrogen halides:-



④ Addition of water (Hydration):- Acid catalyzed addition of water



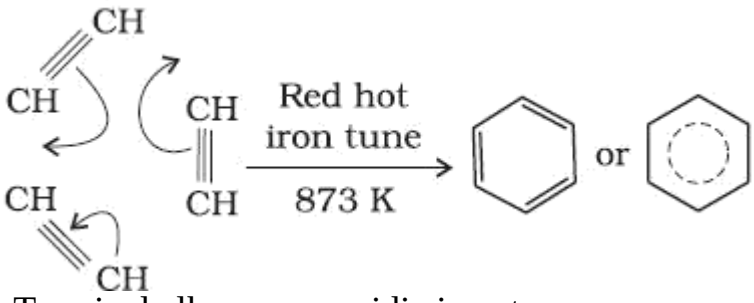




s Polymerisation-

***Linear polymerisation*:**of ethyne gives polyacetylene or polyethynewhich is a high molecular weight polyene containing repeating units of (CH = CH –CH = CH) and can be represented as — (CH = CH –CH = CH)n —

Cyclic polymerization- results in the formation of aromatic compound.



Acidity of Alkynes- Terminal alkynes are acidic in nature.

$$HC \equiv CH + Na \rightarrow HC \equiv C^{-}Na^{+} + \tfrac{1}{2}H_2$$

Monosodium ethynide

$$CH_3 - C \equiv C - H \quad + \quad Na^{+}NH_2^{-}$$

$$\downarrow$$

$$CH_3 - C \equiv C^{-}Na^{+} \quad + \quad NH_3$$

Sodium propynide

Alkanes, alkenes and alkynes follow the following trend in their acidic behaviour :

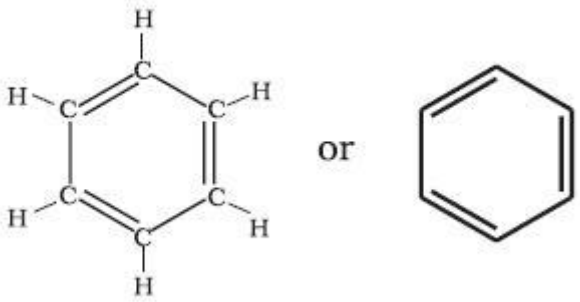
i) $HC \equiv CH > H_2C = CH_2 > CH_3 - CH_3$

ii) $HC \equiv CH > CH_3 - C \equiv CH >> CH_3 - C \equiv C - CH_3$

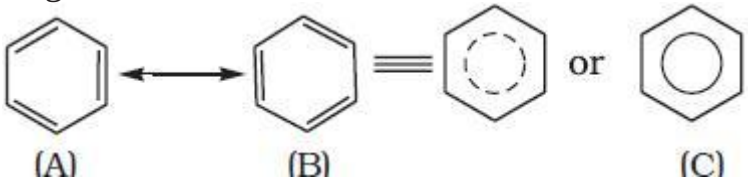
AROMATIC HYDROCARBON

Aromatic compounds containing benzene ring are known as **benzenoids** and those not containing a benzene ring are known as **non-benzenoids**.

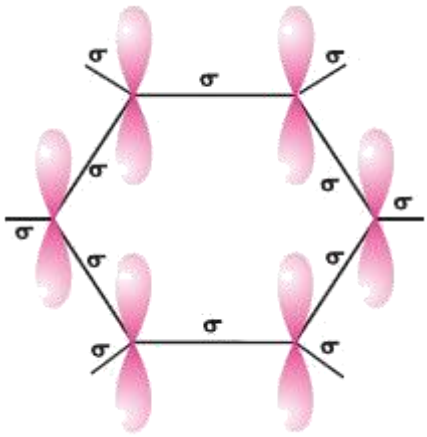
Structure of Benzene- Kekul*é* structure



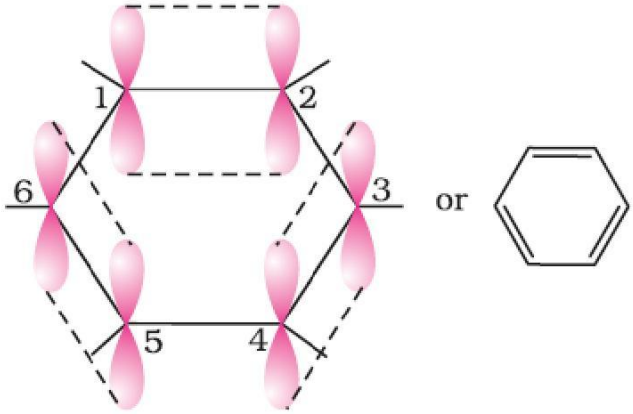
Resonance and stability of benzene-Benzene is a hybrid of various resonatingstructures.

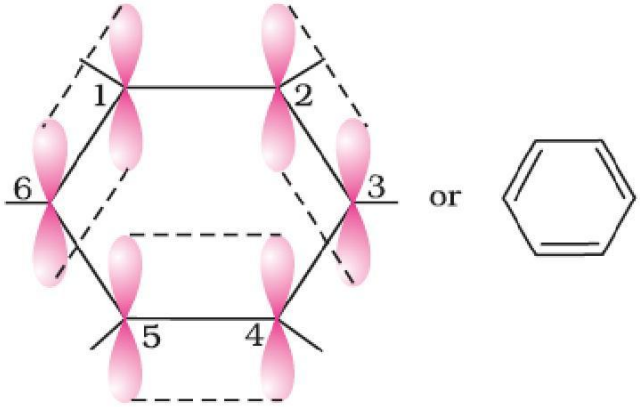


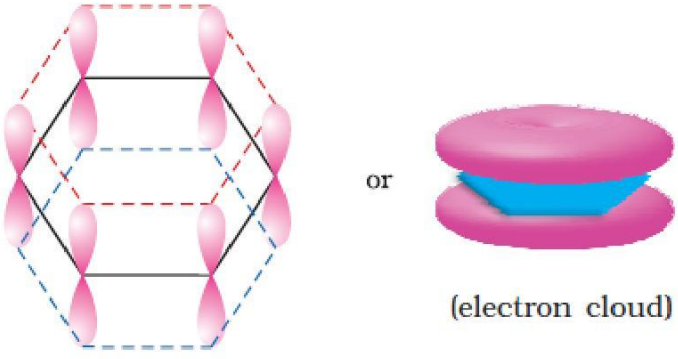
The orbital overlapping picture benzene- All the six carbon atoms in benzene aresp^2hybridized and these hybrid orbitals form sigma bonds.



The unhybridised *p* orbital of carbon atoms are close enoughbondby to f lateral overlap.







The sixctronsπare elethusdelocalised and can move freely about the six carbon nuclei. The delocalisedelectroncloudattractedπ more strongly by the nuclei of the carbon atoms than the electron cloud localized between two carbon atoms. Therefore, presence of delocalisedelectronsinbenzeneπ makes it more stable .

Aromaticity:- The compounds that follow the following features are to be considered aromatic.

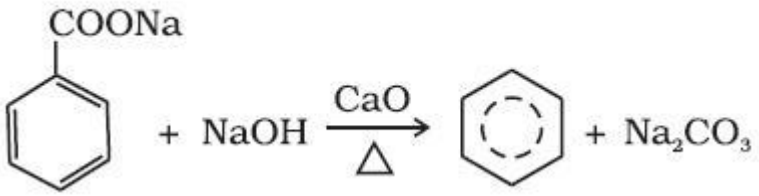
2. Planarity
3. Complete delocalization inthering of the

Presence of $(4n + 2)$ π electronswhereisan integer (*n*the=0,1, ring 2, . . .). This is often referred to as **Hückel Rule**.

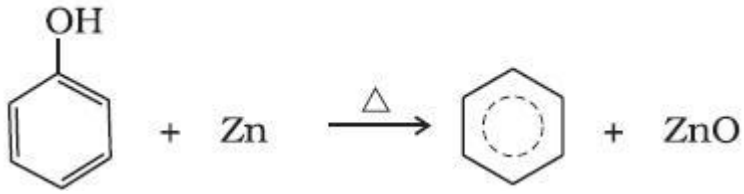
Preparation of Benzene:

(i) Cyclic polymerisation of ethyne:

= Decarboxylation of aromatic acids*:*



8. *Reduction of phenol:* Phenol is reduced to benzene by passing its vapours overheated zinc dust



Physical properties:

- Aromatic hydrocarbons are non- polar molecules and are usually colourless liquids or solids with a characteristic aroma.
- Aromatic hydrocarbons are immiscible with water but are readily miscible with organic solvents.
- They burn with sooty flame.

Chemical properties

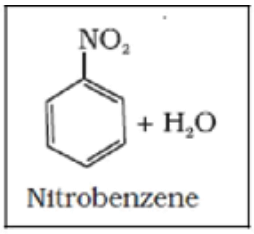
Arenes are characterised by electrophilic substitution reactions proceed via thefollowing three steps:

- Generation of the eletrophile
- Formation of carbocation intermediate
- Removal of proton from the carbocation intermediate



Nitration

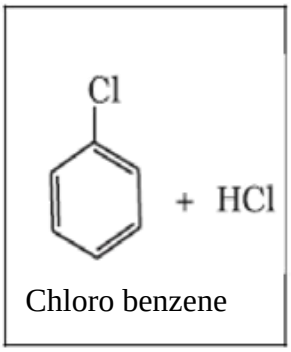
Conc.HNO_3+Conc.H_2SO_4



Halogenation

+Cl_2

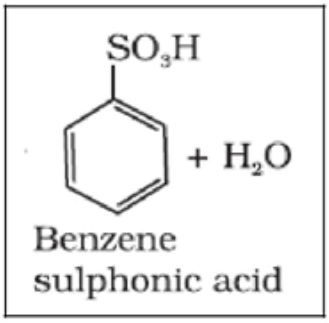
Anhyd. $AlCl_3$



Sulphonation

Fuming sulphuric acid

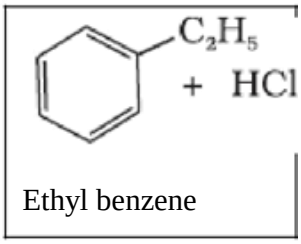
$H_2SO_4(SO_3)$



Friedel-Crafts alkylation

+C_2H_5Cl

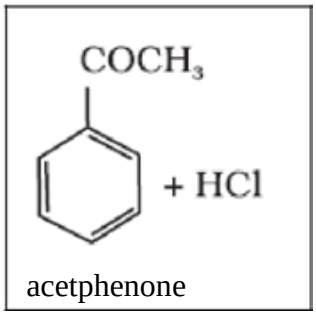
Anhyd. $AlCl_3$



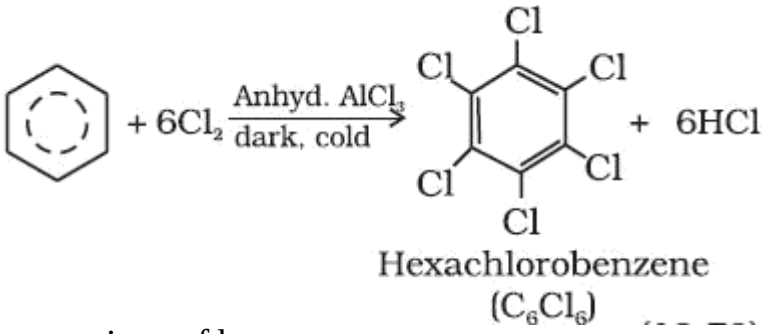
Friedel-Crafts acylation

+CH_3COCl

Anhyd. $AlCl_3$

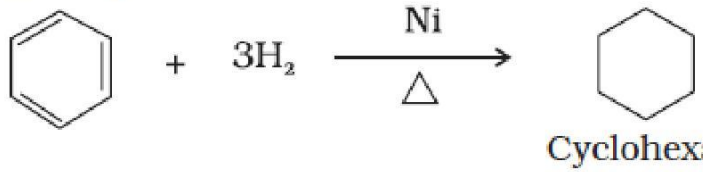


benzene ontreatment with excess of chlorine in the presence of anhydrous AlCl3 in dark yields hexachlorobenzene (C_6Cl_6)

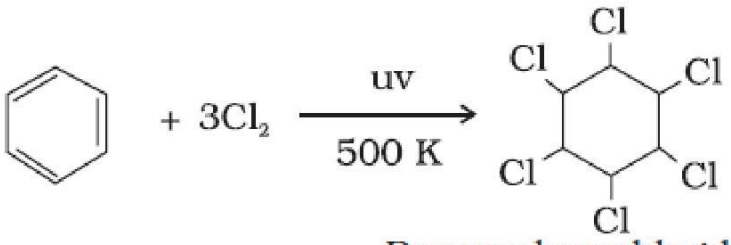


Hexachlorobenzene
(C_6Cl_6)

Addition reactions of benzene-



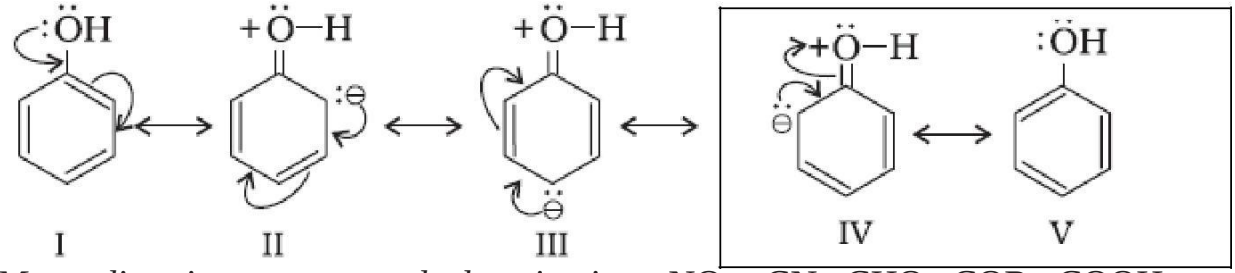
Cyclohexane



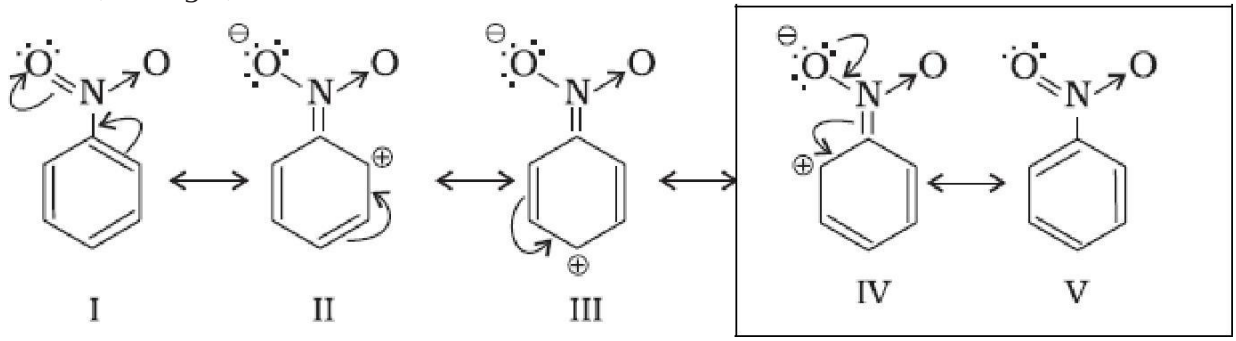
Benzene hexachloride,
(BHC)

Directive influence of a functional group in monosubstituted benzene:-

13. Ortho and para directing groups and activating- –OH, $–NH_2$, –NHR, –$NHCOCH_3$, $–OCH_3$, $–CH_3$, $–C_2H_5$, etc.



14. *Meta directing group and deactivating:* –NO_2,–CN,–CHO,–COR,–COOH,–COOR, $–SO_3H$, etc.



- Ortho and para directing groups and deactivating- Halogens because of their strong –I effect, overall electron density on benzene ring decreases. However,

due to resonance the electron density on *o*–and *p*–positions is greater than that at the *m*-position. Hence, they are also *o*–and *p*–directing groups.

CARCINOGENICITY AND TOXICITY-Benzene and polynuclear hydrocarbonscontaining more than two benzene rings fused together are toxic and said to possess cancer producing (carcinogenic) property.

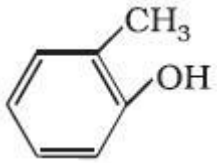
ONE MARK QUESTIONS

5. What are hydrocarbons?
   Ans. Compounds of hydrogen and carbon.
6. What is the general formula of alkanes? Ans. C_nH_{2n+2}
7. Write the general formula of alkenes. Ans. C_nH_{2n}
8. What is the general formula of alkynes? Ans. C_nH_{2n-2}
9. Give the IUPAC name of lowest molecular weight alkane that contains a quaternary carbon.
   Ans. 2,2dimethylpropane.
10. Arrange the following in the increasing order of C-C bond length-
    C_2H_6 C_2H_4 C_2H_2
    Ans. $C_2H_2 < C_2H_4 < C_2H_6$
3. Out of ethylene and acetylene which is more acidic and why?
   Ans. Acetylene, due to greater electonegativity of the sp hybrid carbon.
4. Name two reagents which can be used to distinguish between ethene and ethyne.
   Ans.Tollen's reagent and ammonical Cuprous chloride
5. Arrange the following in order of decreasing reactivity towards alkanes. HCl, HBr, HI, HF
   Ans.HI> HBr> HCl >HF
10. How will you detect the presence of unsaturation in an organic compound?
    Ans. Generally Unsaturated organic c and Bromine water.

11. What is Grignard reagent?
    Ans. Alkyl magnesium halides

TWO MARKS QUESTIONS

1. Write the IUPAC names of the following-

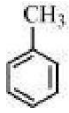
a. $CH_2=CH-C\equiv C-CH_3$ b. 

Ans. a .Pent-en-3-yne b. 2-methylphenol

4. Write chemical equations for combustion reaction of (i) Butane (ii) Toluene Ans.

(i) $2C_4H_{10(g)} + 13O_{2(g)} \longrightarrow 8CO_{2(g)} + 10H_2O_{(g)} + \text{Heat}$

Butane

(ii)  $+9O_{2(g)} \longrightarrow 7CO_{2(g)} + 4H_2O_{(g)} + \text{Heat}$

Toluene

3. What are the necessary conditions for any system to be aromatic?

Ans. A compound is said to be aromatic if it satisfies the following three conditions: **(i)** It should have a planar structure.

(ii) The–electronsπof the compound are completely delocalized in the ring.(iii)The total–electronsnumberpresent inoftheringπshould be equal to $(4n + 2)$, where $n = 0, 1, 2 \ldots$ etc. This is kn

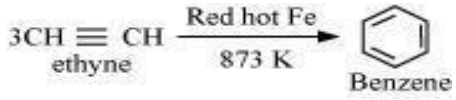
8. What effect does branching of an alkane chain has on its boiling point?

Ans. As branching increases, the surface area of the molecule decreases whichresults in a small area of contact. As a result, the Van der Waals force also decreases which can be overcome at a relatively lower temperature. Hence, the boiling point of an alkane chain decreases with an increase in branching.

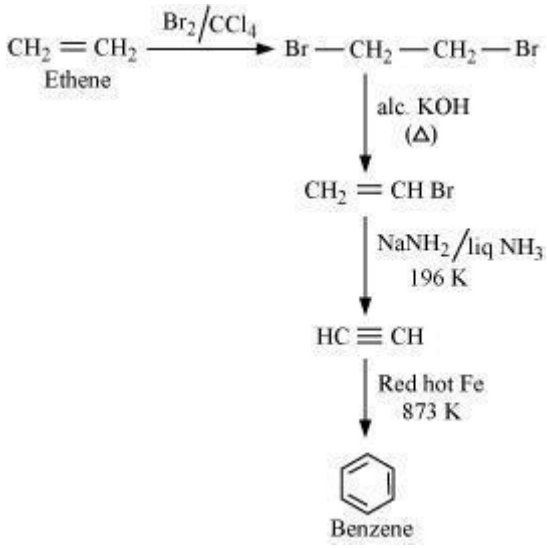
9. How would you convert the following compounds into benzene?

= Ethyne (ii) Ethene

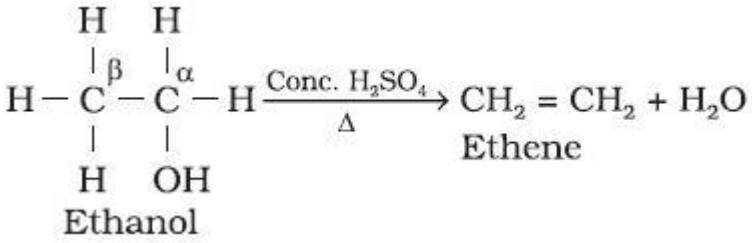
Ans. **(i)**Benzene from Ethyne:



(ii) Benzene from Ethene:



6. Suggest the name of Lewis acids other than anhydrous aluminium chloride which can be used during ethylation of benzene.
   Ans. anhydrous $FeCl_3$, $SnCl_4$, BF_3etc.
7. Write the name of all the possible isomers of C2H2Cl2 and indicate which of them is non-polar.
   Ans.(i) cis-1,2-dichloroethene (ii) trans-1,2-dichloroethene (iii) 1,1-dichloroethene. trans-1,2-dichloroethene is non-polar.
8. Although benzene is highly unsaturated, it does not undergo addition reactions, why?
   Ans. Because of extra stability due to delocalization of π-electrons.
9. What are alkanes? Why are they called paraffins?
   Ans. Those hydrocarbons which contain single bond between carbon- carbon are called alkanes. They are called paraffins because they are very less reactive (Latin- Parum= little, affins = affinity)
10. How can ethene be prepared from (i) ethanol (ii) ethyl bromide?
    Ans. (i) Ethene from ethanol- by acidic dehydration of alcohols



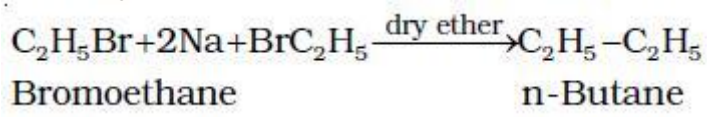
6 Ethene from ethyl bromide- by dehydrohalogenation of ethyl bromide
$CH_3CH_2Br + KOH\ (alc) \rightarrow {}_2HC = CH_2 + KBr + H_2O$

THREE MARKS QUESTIONS

3.43 What is Wurtz reaction? How can it be used to prepare butane?

Ans- When alkyl halides is treated with metallic Na in presence of dry ether, alkanes are formed. This reaction is called Wurtz reaction.

Butane is prepared by the reaction of bromoethane with metallic Na in presence of dry ether



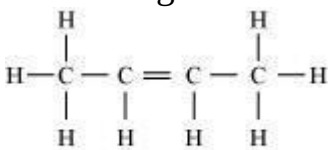
3.44 An alkene 'A'–C,containseightC–Hσ bondsthree–CandCπ one bond. 'A' on ozonolysis givesolarmass44 twou. mo
Deduce IUPAC name of 'A'.

Ans.. The formation of two moles of an aldehyde indicates the presence ofidentical structural units on both sides of the double bond containing carbon atoms. Hence, the structure of 'A' c

XC = CX

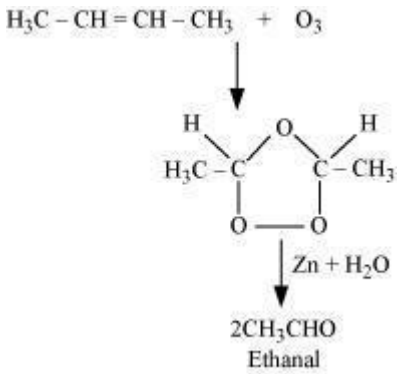
There are eight C–H σ bondsence,. Hthere are 8 hydro there are three C–C bonds. Hence, there are four carbon atoms present in the structure of 'A'.

Combining the inferences, the struct



(A)

the IUPAC name-2-ene .of ‘A’ is But Ozonolysis of ‘A’ takes place as:

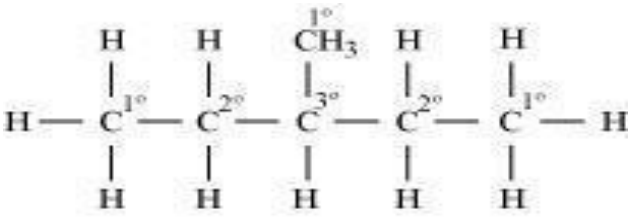


The final product is ethanal with molecular mass

$$=\left[(2\times12)+(4\times1)+(1\times16)\right]$$

$$=44\text{ u}$$

= In the alkane H3C –CH2 –C(CH3)2 –CH2 –CH(CH3)2, identify 1°,2°,3° carbon atoms and give the number of H atoms bonded to each one of these. Ans.



The given structure has five 1° carbon atoms and fifteen hydrogen atoms attached to it.

The given structure has two 2° carbon atoms and four hydrogen atoms attached to it.

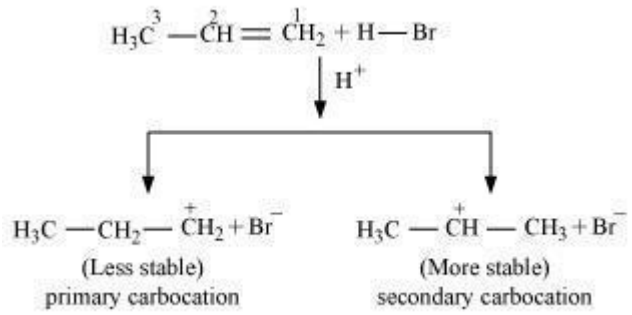
The given structure has one 3° carbon atom and only one hydrogen atom is attached to it

FIVE MARKS QUESTIONS

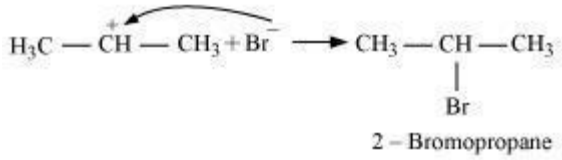
(vi) Addition of HBr to propene yields 2-bromopropane, while in the presence of benzoyl peroxide, the same reaction yields 1-bromopropane. Explain and give mechanism

Ans. Addition of HBr to propene is an example of an electrophilic substitution reaction.

Hydrogen bromide provides an electrophile, H^+. This electrophile attacks the double bond to form 1° and 2° carbocations as shown:

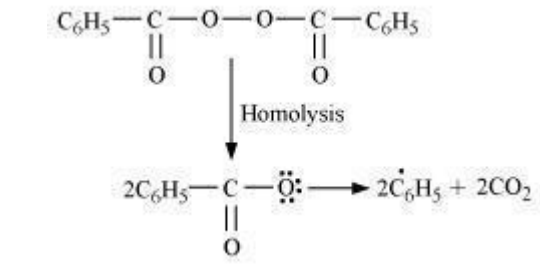


Secondary carbocations are more stable than primary carbocations. Hence, the former predominates since it will form at a faster rate. Thus, in the next step, Br^- attacks the carbocation to form 2 –bromopropane as the major product.

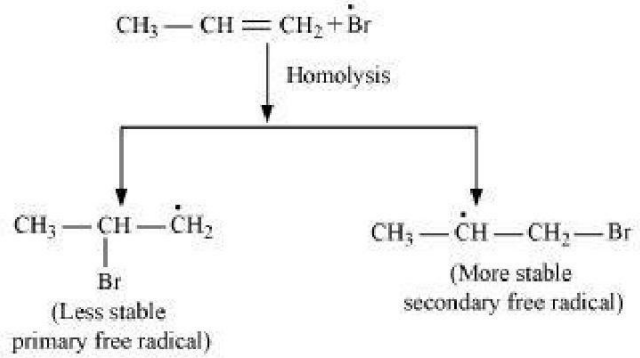


This reaction follows Markovnikov's

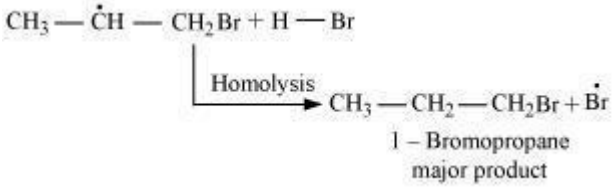
In the presence of benzoyl peroxide, an addition reaction takes place anti to Markovnikov's rule. The reaction fol







Secondary free radicals are more stable than primary radicals. Hence, the former predominates since it forms at a faster rate. Thus, 1 – bromopropane is obtained as the major product.



HOTS QUESTIONS

9. How will you demonstrate that double bonds of benzene are somewhat different from that of olefins?
   Ans.The double bonds of olefins decolourize bromine water and discharge the pink colour of Bayer's reagent while
10. How will you separate propene from propyne?
    Ans. By passing the mixture through ammonical silver nitrate solution when propyne reacts while propene passes over.
11. Write is the structure of the alkene which on reductive ozonolysis gives butanone and ethanol,
    Ans.-
    $CH_3CH_2C(CH_3)=CHCH_3$

VALUE BASED QUESTIONS

Q.1 Saturated aliphatic hydrocarbons are called alkanes. They are mainly used as fuels. Depending upon their physical properties such as boiling points, density, solubility, they have many other uses. Now answer the following questions.

i) Out of propane and butane why is propane used as a fuel for barbeque grill?
ii) Which alkane is used in cigarette lighters?
iii) What is mineral oil? Give its uses.
iv) What is the use of solid alkanes?
v) What is Vaseline? Mention some of its uses.

Ans.

i) Barbeque is usually done in the open. In cold countries, although both propane and butane can be used but propane is preferred because it works better in low temperature. The reason being that due to lower boiling point (i.e., 231 K), propane always remains as a gas in the cylinder inspite of cooling produced by Joule Thomson effect during its use and thus ensures a constant supply of the gas. In contrast, butane due to its higher boiling point (272.5 K) and cooling produced by Joule - Thomson effect, it becomes a liquid in the cylinder and thus the supply of the gas becomes slower and slower as the gas is being used.

ii) Butane is used in cigarette lighters since it can be easily liquefied at room temperature.

iii) Mineral oil is a mixture of liquid alkanes containing 16-18 carbon atoms and is used as a laxative and as a lubricant.

iv) Alkanes with 18 or more carbon atoms are waxy solids. These are used as waxy coatings on fruits and vegetables to help them retain moisture, inhibit mould growth and enhance appearance.

v) Vaseline or petroleum jelly, is a mixture of solid alkanes having low melting points. It can prevent moisture loss and chapped lips and soften nail cuticles. It is also used in ointments and cosmetics.

Q.2 What are monomers and polymers?

Ans. Polymerization is a process in which a large number of simple molecules combine together to form big molecules. The simple molecules are called monomers and the big molecules are called polymers.

Q.3 To prevent damage and facilitate easy shipping, many fruits such as bananas, mangoes, pineapples, etc. And some vegetables such as tomatoes are often plucked before they are ripe. Before the fruit or the vegetable is brought to the market, it is exposed to some chemicals which accelerates the ripening process. If, however, the fruit or the vegetable is allowed to stay on the plant, natural ripening process starts.
Now answer the following questions.

(i) What happens during natural ripening process?
(ii) Name two chemicals which are used to ripen the fruits/vegetables artificially.

(iii) Do the artificially ripen fruits/vegetables safe to use? Do they taste the same as naturally ripen fruits?

Ans

(i) Ethylene is an important plant hormone. The plant emits ethylene which promotes the ripening process.

(ii) The two chemicals commonly used are: ethylene and calcium carbide. Prior to sending the fruits/vegetables to the market, they are exposed to ethylene gas. Alternatively, the fruits/vegetables are placed on the floor and calcium carbide is place in between them. They are then covered with a tarpaulin and water is sprinkled on it. Calcium carbide reacts with water to give acetylene gas which like ethylene accelerates the ripening process.

$$\underset{\text{Calc. Carbide}}{CaC_2} + H_2O \longrightarrow Ca(OH)_2 + \underset{\text{Acetylene}}{HC \equiv CH}$$

Commercial calcium carbide has impurities of calcium arsenide and calcium phosphide. These also react with water producing arsine (AsH_3) and phosphine (PH_3).

$$Ca_3As_2 + 6H_2O \longrightarrow 3Ca(OH)_2 + 2AsH_3$$
$$Ca_3P_2 + 6H_2O \longrightarrow 3Ca(OH)_2 + 2PH_3$$

Since both arsine and phosphine are poisonous gases, therefore, it is not always safe to use artificially ripen fruits. Since the process of artificial ripening cannot be banned, thereore, fruits/vegetables must be properly washed before using. Further, it is a common observation that artificially ripen fruits/vegetables do not taste the same as naturally ripen fruits/vegetables. The reason is however not well understood.

Q.4 Di substituted alkenes containing same or different substituents on each carbon atom of the double bond show geometrical or cis-trans isomerism. These isomers have different structures and hence have different physical properties and some chemical properties as well. Heat or light can bring about the conversion of cis-isomers to the trans-isomers. cis-trans-isomerism is responsible for vision.

Now answer the following questions.

(i) Name the organic compound whose cis-trans isomerism is responsible for vision.

(ii) Discuss briefly the chemistry of vision?

(iii) What is night blindness? What is its cause? How can the night blindness be cured?

Ans.

(i) The organic compound whose cis-trans isomerism is responsible for vision is called retinal.

$$\textit{Cis}\text{-Retinal} \underset{}{\overset{h\nu}{\rightleftharpoons}} \textit{trans}\text{-Retinal}$$

(ii) The retinas of eyes consist of two types of cells called rods and cones. The rods on the edge of the retina allow us to see in dim light or at night while the cones in the centre produce vision in bright light. In the rods ,there is a protein called rhodopsin that absorbs light. Rhodopsin also contains the unsaturated compound *Cis*-Retinal. When rhodopsin absorbs light, *Cis*-Retinal is converted into *trans*-Retinal. Since *trans*-Retinal has different shape, it no longer fits in the protein and hence separates from the protein. This process of conversion of *Cis*-Retinal to *trans*-Retinal and its subsequent separation from the protein generates an electrical signal that the brain converts into an image.

Later on, an enzyme converts the *trans*-Retinal back to *Cis*-Retinal and rhodopsin is reformed.

(iii) The inability of a person to see in dim light or at night is called night blindness. This is due to deficiency of the protein rhodopsin in the rods of retina. One common cause of night blindness is deficiency of vitamin A in the diet. In our diet, we get vitamin A from plant pigment containing β- carotene which is found in our foods such as carrots and green vegetables (cabbage, spinach, squash, etc.). In the intestine, β-carotene is converted to *Cis*-Retinal or stored in the liver for future use. If sufficient quantity of retinal is not available, enough rhodopsin is not produced to enable us to see adequately in dim light or at night.

Q.5 Benzene is the simplest and ideal aromatic compound. It is also one of the most basic petrochemicals. It is mainly used to prepare a number of important chemicals such as toluene, phenol, aniline, biphenyl, naphthalene and anthracene which are used in the manufacture of dyes, detergents, drugs, explosives, pesticides, rubbers and lubricants. In spite of its numerous benefits to mankind it is found to be carcinogenic.

Now answer the following questions.

(i) Mention three uses of benzene in petrochemical industry.

(ii) Benzene is used as an additive in gasoline. Why so?

(iii) Comment upon bad health effects and carcinogenicity of benzene.

(iv) In view of toxicity and carcinogenicity of benzene, should its production be banned? Give your suggestions.

Ans.

(i) Benzene is mainly used as an intermediate to make other chemicals. About 8% of benzene is consumed in the production of three chemicals, ethylbenzene, cumene, and cyclohexane. **Ethylbenzene** is a precursor to styrene, which is used to make polymers and plastics. **Cumene** is converted into phenol (for resins and adhesives) and acetone (a useful solvent for laboratories and industries). Cyclohexane is used to manufacture nylon.

(ii) Because of its high octane number and antiknocking properties benzene is used as an additive in gasoline. However, due to its harmful health effects and the possibility of benzene entering the ground water, USA in 2011 has reduced the benzene content in gasoline from 1% to 0.62%.

(iii) Exposure of benzene vapours to humans is a global problem. It affects liver, kidney, lungs and heart. Benzene increases the risk of cancer and other illness. In the body, it gets oxidised to benzene oxide which is not excreted readily and can interact with DNA to produce harmful mutations. Benzene is also a notorious cause of bone marrow failure.

(iv) Although benzene is a potential carcinogen, its production cannot be banned because its wide spread application in the manufacture of dyes, drugs, detergents, nylons, explosives and pesticides. Instead, benzene should be used in fuming cupboards and well ventilated laboratories.

Further, in case, benzene is being used only as a solvent, it should be replaced by toluene which has similar solvent properties but is not carcinogenic and is also less toxic.

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CHAPTER 14

ENVIRONMENTAL CHEMISTRY

Environmental chemistry deals with the study of the origin,transport,reactions,effects,fates of chemical species in the environment.

ENVIRONMENTAL POLLUTION:-Environmental pollution is the effect of undesirable changes in our surroundings that have harmful effects on plants, animals and human beings.A substance which causes pollution is called a pollutant.they can be solid,liquid or in the gaseous state.

ATMOSPHERIC POLLUTION:-The atmosphere that surrounds the earth is not of the same thickness at different heights.Atmospheric pollution is generally studied as tropospheric and stratospheric pollution.The ozone layer prevents about 99.5%of the sun's UV rays.

TROPOSPHERIC POLLUTION:-Tropospheric pollution occurs due to the presence of undesirable solid or gaseous particles in the air. The following are the major gaseous and particulate pollutants present in the troposphere;

- Gaseous air pollutants:These are oxides of sulphur, nitrogen and carbon, hydrogen sulphide, hydrocarbons, ozone and other oxidants.
- Particulate pollutants; these are dust, mist, fumes, smoke, smog etc

- GLOBAL WARMING AND GREENHOUSE EFFECT:-About 75% of the solar energy reaching the earth is absorbed by the earth's surface,which increases it's temperature.The rest of the heat radiates back to the atmosphere.Some of the heat is trapped by the gases such as carbon dioxide,methane,ozone,CFCS and Water vapour.they add to the heating of the atmosphere causing Global warming

  In a greenhouse,visible light passes through the transparent glass and heats up the soil and the plants.The warm soil and plants emit infrared rays,it partly reflects and partly absorbs these radiations,this mechanism keeps the energy of the sun trapped in the greenhouse.

ACID RAIN:When the pH of the rain water drops below5.6, it is called acid rain.Acid rain is harmful for agriculture, trees and plants as it dissolves and washes away nutrients needed for their growth. It causes respiratory ailments

in human beings and animals. When acid rain falls and flows as ground water to reach rivers, lakes etc. it affects plants and animal life in aquatic ecosystem

SMOG:The word smog is derived from smoke and fog.There are two types of smog:classical and photochemical smog. Classical smog occurs in cool humid climate. It is a mixture of smoke, fog and sulphur dioxide. It is also called reducing smog. Whereas photochemical smog occurs in warm and dry sunny climate. It has high concentration of oxidizing agents and therefore ,it is also called as oxidizing smog

OZONE HOLE:Depletion of ozone layer is known as ozone hole.

EFFECTS OF DEPLETION OF THE OZONE LAYER: With the depletion of ozone layer, more UV radiation filters into troposphere. UV radiations lead to ageing of skin, cataract, sunburn, skin cancer, killing of many phytoplanktons, damage to fish productivity etc

WATER POLLUTION:-contamination of water by foreign substances which make it harmful for health of animals or plants or aquatic life and make it unfit for domestic, industrial and agriculture use.

SOURCES/ CAUSES OF WATER POLLUTION-

- Sewage and domestic wastes
- Industrial effluents
- Agriculture effluents
- Siltation-mixing of soil or rock into water
- Thermal pollutants
- Radioactive discharge

EUTROPHICATION:The process in which nutrientenriched water bodies support a dense plantpopulation, which kills animal life by deprivingit of oxygen and results in subsequent loss ofbiodiversity is known asEutrophication

BOD: The amount of oxygen required by bacteria to break down the organic matter present in a certain volume of a sample of water, is called Biochemical Oxygen Demand (BOD)

SOIL POLLUTION:Insecticides, pesticides and herbicides cause soil pollution.

GREEN CHEMISTRY:Green chemistry us a way of thinking and is about utilizing the existing knowledge and principles of chemistry and other sciences to reduce the adverse impact on environment.Green chemistry is a production process that would bring out minimum pollution or deterioration to the environment..Utilization of existing knowledge base for reducing the chemical hazards along with the development of activities is the foundation of green chemistry.

ONE MARK QUESTION

1. What is the name of the compound formed when CO combines with blood? Ans: - Carboxyhaemoglobin.
2. Which zone is known as ozonosphere?
   Ans: - Stratosphere.
3. Which main gas esis responsible for damage in ozone layer?
   Ans:-NO and CFCs(freons).
4. What is the nature of classical smog?
   Ans:-Reducing
5. Name the acids which are responsible for acid rain?
   Ans:-H_2SO_4,HNO_3 and HCl.
6. List out the gasses which are considered as major source of air pollution? Ans:- Carbon monoxide(CO),sulphur dioxide(SO_2)and oxides of nitrogen(NO_2).
7. What is PAN stands for?
   Ans:-It is peroxyacetyl nitrate.
8. Give the examples of insecticides?
   Ans:-DDT,BHC.
9. Which gas is mainly responsible for BHOPAL gas tragedy?
   Ans:-Methyl isocyanate.
10. What should be the tolerable limit of F^- ions in drinking water?
    Ans:-1ppm or 1mg dm^{-3} .

TWO MARKs QUESTION

1. What is 'acid rain'? How is it harmful to the environment?
   Ans:-Acid rain is the rain water mixed with small amount of sulphuric acid, nitric acid along with hydrochloric acid which are formed from the oxides of sulphur and nitrogen present in air as pollutants.It has a pH of 4-5.
   Harmful effects of acid rain:-
   It is toxic to vegetation and aquatic life.
   It damages buildings and status

2. What do youmean byGreen house effect?What is the role of CO_2 in the greenhouse effect.
   Ans:-It is the phenomenon in which earth's atmosphere traps the heat from the sun and prevents it from escaping in outer space. Gases such as CO_2 ,methane,ozone,CFCs are believed to be responsible for this effect.
   Heat from the sun after being absorbed by the earth absorbed by CO_2 and then radiated back to the earth.Thus making the environment of the earth warm.

3. Which gases are responsible for greenhouse effect?List some of them. Ans:-CO_2 is mainly responsible for greenhouse effect.Other greenhouse gases are methane nitrous oxide, water vapours.

4. What is smog?How is classical smogdifferent from photochemical smog? Ans:-The word smog is a combination of smoke andfog.It is a type of air pollution that occurs in many cities throughoutthe world.Classical smog occurs in cool humid climate.It is also called reducing smog.
   Whereas photochemical smog occurs in warm and dry sunny climate. It has high concentrationof oxidizing agents and therefore,it is also called as oxidizing smog.

5. What are the reactions involved for ozone layer depletion in the stratosphere? Ans:-
   $CF_2Cl_2(g) + UV$ ---------- ▶ $Cl(g) + CF_2Cl(g)$
   $Cl(g) + O_3(g)$ —$ClO(g) + O_2(g)$
   $ClO(g) + O(g)$ ----- $Cl^* + O_2(g)$

6. What is the full form of BOD and COD?
   Ans:-BOD stands for Biochemical Oxygen Demand whereas COD stands for Chemical Oxygen Demand.

7. What are viable and non-viable particulates?
   Ans:-Viable particulates:-They are minute living organisms that are dispersed in the atmosphere including bacteria,fungi,moulds,algae etc.
   Non-viable particulates:-They are formed by the breakdownof larger materials or by the condensation of minute particles and droplets.

8. What is B.H.C?Give its IUPAC name?
   Ans:-B.H.C is BENZENE HEXACHLORIDE Its IUPAC name is 1,2,3,4,5,6- hexa chlorocyclohexane.

9. What is meant by PCBs?
   Ans:-PCBs arepolychlorinated biphenyls.They are contaminants of water. They are used as fluids in transformers and capacitors.

10. What is the compound formed when COcombines with blood?
    Ans:-When CO combines with blood, the following reaction occurs forming carboxyhaemoglobin:-
    $Hb + CO \rightarrow$ (HbCO) Carboxyhaemoglobin

THREE MARKS QUESTION

1. What do you understand by- (i) Mist (ii) Smoke (iii) Fumes Ans(i) Mists:-Mists are produced by particles of spray liquids and the condensation of vapours in air.
   (ii) Smoke:- They are very small soot particles produced by burning of organic matter.
   (iii) Fumes:- These are condensed vapours;fumes of metals are well known particulates of this type.
2. Define the term pesticides? What are three categories of pesticides? Ans:-Pesticides are substances which are used to kill or block the reproductive process of unwanted organisms.
   The three main categories of pesticides are:-
   (i)Insecticides:-These are used to control insects and curb diseasesand protect crops.

(ii) Herbicides:-These are used to kill weeds .Example- sodium chlorate ($NaClO_3$), sodium arsinite(Na_3AsO_3)

(iii) Fungicides:- These are used to check the growth of fungi. Example-methyl, mercury.

3. What do you mean by ozone hole? What are its consequences?
   Ans:-Depletion of ozone hole creates some sort of holes in the blanket of ozone which surround as. This is known as ozone hole.
   (i) With the depletion of the ozone layer UV radiation filters into the troposphere which leads to aging of skin,cataract,sunburn etc.
   (ii) By killing many of the phytoplanktons it can damage the fish productivity.

4. What are harmful effects of photochemical smog and how can they becontrolled?
   Ans:- (i)Photochemical smog causes eye irritation.
   (ii) It damages plants (the leaves develop a metallic sheen)
   (iii) Rubber on exposure to photochemical smog loses its elasticity and becomes inflexible and brittle.
   Usually catalyticconverters are used in the automobiles, whichprevent the release of nitrogen oxide andhydrocarbons to the atmosphere. Certainplants e.g., Pinus, Juniparus, Quercus, Pyrusand Vitis can metabolise nitrogen oxide andtherefore, their plantation could help in thismatter.
5. Give three examples in which green chemistry has been applied.
   Ans:-
   (i) In dry cleaning ,use of liquefied CO_2 in place of tetrachloroethene($Cl_2C=CCl_2$)
   (ii)In bleaching of paper using H_2O_2 in place of chlorine.
   (iii)In the manufacture of chemicals like ethanal using environment-friendly chemicals and conditions.

VALUE BASED QUESTIONS

Q.1.Environmental pollution is causing a serious threat to living as well as non-livng organisms on the earth.Due to combustion of fossil fuel(wood,petrol,oil,etc.),a no. Of gas like CO, oxide of nitrogen,oxides of sulphur,hydrocarbons,etc.enter into the atmospherewhich are poisonous and harmful.every country has made strict laws for industries as well as for individual citizen to keep pollution under control.

(i)why it is advised not to sleep with burning coke angithi in a closed room on winter nights?How is poisoning caused and what step should be taken immediately as a part of the treatment?

(ii)How does the combustion of motor fuels cause pollution of the atmosphere?What step is taken by the manufactures of these vehicles to check this pollution?

(iii)name one natural and one human activity by which SO_2 enter into the atmosphere?How does it affect the plants and damages the building?What step should be taken to check this pollution?

Ans.1 (i).in the presence of insufficient oxygen in the closed room,coke burns to produce a lot of carbon monoxide.It combine with haemoglobin to forn carboxyhaemoglobin.Thus,the function of haemoglobin to carry oxygen to diff. Body cell is inhibited this causes oxygen starvation leading to comaand death.

(ii)when motor fuel burns, they produces high temp. At which N_2 and O_2 of the air combine to form oxides of n2which are toxic and affect respiratory system and damages the lungs.

(iii)SO_2 enters into atmosphere through volcanic eruption and through combustion of sulphur containing coal and fuel oil in thermal power plants.Its presence around the plants slows down the formation of chhlorophyllresulting in loss of green colour.it dissolves in rain water froming sulphuric acid which attack marbel building like taj in Agra

Q.2.Nature has provided us with an ozone layer at a height of about 25-30 Km from the surface of the earth so that harsh UV rays coming from the sun do not reaches us.However,human activities on the earth have been damaging this ozone layed and it is believed that a hole has been created in this layer.Similarly, due to certain other human activities on the earth such as burning of fossil fuels,etc..,a no. Of pollution gases such as oxides of N_2,oxides of sulphur and unburnt hydrocarbon are entering ith the atmosphere besides particles of soot,fly ash and other chemicals produced or used by the industries.

(i)Which activieed on the earth has been depletion of ozone layer?What is the most serious effect?what has been done to protect ozone layer?

(ii)Presence of polluting gases,hydrocarbons and particulates result into formation of smog.What should be done to prevent the formation of smog?

(iii)CO_2 gas is normal concentration is not a pollutant.Rather it is needed for photosynthesis in plants.However,if present in larger amounts, it has serious consequences.How?what should done to maintain a proper balance of CO_2 in atmosphere?

Ans.2.(i)Use of CFC used in refrigeration equipments as coolants have a very long life time and ultimately reach the ozone layer and result into its decomposition.The most serious effect is that UV Rays coming From the sun can pass through this hole and reach the surface of the earth causing skin cancer.The use of CFCs has,Therefore,been completely banned.

(ii) Refer to `Control of photochemical smog`

(iii) presence of excess CO_2 in the atmosphere result in `greenhouse effect`This will result mealting of Glacier and polar ce caps and level of sea water may rise thereby flooding the coastal land.For steps taken, To maintain a proper balance of CO_2 .

Q.3.Water is most imp for our survival.Plants need water for their growth.We need water for drinking,taking bath,etc,.But water in the lake,river,etc. Is not clean for all purposes.ocean though convert about 70% of the earth surfacebut being salty is unfit for human consumption. In a number of ways, we are responsible for causing water pollution good citizen; we should stop doing things leading to water pollution.

(i) There was a time when people in rural area used to use well water for drinking but it is not advisible now?how is it getting polluted?

(ii)Our religion suggests taking bath in Gangs or even Drinking it.What are your views as a student of science?What do you suggest should be done?

(iii)How sometimes sea water gets polluted to the extent that aquatic life including fisg die?

(iv)How does the use of fertilizer affectthe aquatic life in the lake and other water bodies?

(v)why 1ppm of fluride is added to drinking water?What will happen if concentration is higher?

(vi)Why lead pipes should not be used for transport of water?

(vii)What is the safest pH range for drinking water?

Ans.(i)Do yourself

(ii)The water that of river including that of ganga is getting polluted Due to industrial discharges from paper,textile,and many other industries and also due to Domestic Sewage discharge by people living on the bank of this river.The use of water from these river can cause water borne diseases like Jaundice,Cholera,etc.

(iii)Pollution in sea water by oil spill due to accident of ship carrying oil or leakage of pipe line or deliberately done as a golf war covers the surface of sea water by a layer of oil .As a result ,aquatic life perishes.

(iv)During rain excess fertilizer present on the soil are washed away into lakes and rivers. These fertilizers accelerate the growth of algae. This covers the water surface. Hence, concentration of dissolved oxygen decreases.Further,when short lived algae die or the excessive phytoplankton present in the water start decomposing, the decomposing bacteria consume dissolved oxygen.Thus,the amount of dissolved oxygen in water Decreases and the fish starts dying.

(v)Fluorida protects teeth against decay. Concentration above 2 ppm causes brown mottling of teeth.

(vi)Lead poisoning occurs which damages Kidney, Liver,Brain and Reproductive system.

(vii)5.5-9.5

Q.4.If we manage our household waste and industrial waste, we can save ourselves from pollution to considerable extent. By doing so, we will not only save ourself from pollution but in a number of cases, these wastes can be used for productive purposes.

(i)How should we manage our house hold waste so that it causes no pollution and also some of it can be put to useful purposes?

(ii)Industrial waste should not be thrown into the river,sea,etc.as it cause water pollution

Suggest two method s to dispose industrial waste so that it causes no pollution and instead it becomes useful.

(iii)How the biological waste of the hospitals should be handled which runs into tonnes? Give the merits and demerits of the method used.

(iv)Give at least three examples where industrial waste is being utilized for productive purposes.

Ans.(i)we should keep garbage bins in our house, one for recyclables waste and other for non-recyclable waste. The recyclable waste like plastic, broken glass, scrap iron, used paper, etc.should be sold to the vendor who in return tell of the factory for reproduction. The non recyclable waste should be picked up by the municipal van which in turn should dump the biodegradables waste into pits so that it is converted into manure.

(ii)(a)Burning and incineration:-The combustable industrial waste should be burnt at a far off place so that it is reduced to ash which can be used as a land filling material.

(b)Dumping:-The biodegradable waste including sewage sludge and waste of chemical industries containing compounds of nitrogen and phosphorous should be dumped into the land so that with the passage of time, it is converted into manure.

(iii)The best method to decompose the hospital waste is burning and incineration as the hospital waste contains a number of chemical compounds like polychlorinated diphenyls which at high temp. Breaks into simple harmless products. The disadvantages of this method is that if the combustion is incomplete, the products formed may be toxic

(iv)(a)fly ashes produced by combustion of industrial waste is used in the cement industry.

(b)The plastic waste is being converted into fuel which has high octane number. It does not contain any lead. It is known as green fuel.

(c)The biodegradable waste can be used for production of electricity. For this purpose, the waste is cultured with suitable bacteria, As a result, methane gas is produced, called biogas. The biogas is producing electricity whereas the left out material is used as manure.

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